



US012414364B2

(12) **United States Patent**
Kim et al.

(10) **Patent No.:** **US 12,414,364 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **SEMICONDUCTOR DEVICE**

(71) Applicant: **SAMSUNG ELECTRONICS CO., LTD.**, Suwon-si (KR)

(72) Inventors: **Ju Youn Kim**, Suwon-si (KR); **Gi Gwan Park**, Hwaseong-si (KR)

(73) Assignee: **SAMSUNG ELECTRONICS CO., LTD.**, Suwon-si (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 513 days.

(21) Appl. No.: **17/731,316**

(22) Filed: **Apr. 28, 2022**

(65) **Prior Publication Data**

US 2022/0262793 A1 Aug. 18, 2022

Related U.S. Application Data

(60) Continuation of application No. 16/117,065, filed on Aug. 30, 2018, now Pat. No. 11,355,492, which is a (Continued)

(30) **Foreign Application Priority Data**

Jan. 25, 2016 (KR) 10-2016-0008981
Mar. 10, 2016 (KR) 10-2016-0028719
(Continued)

(51) **Int. Cl.**
H10D 84/83 (2025.01)
H10B 10/00 (2023.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10D 84/834** (2025.01); **H10B 10/12** (2023.02); **H10D 1/00** (2025.01); **H10D 30/6213** (2025.01);
(Continued)

(58) **Field of Classification Search**

CPC H01L 27/0886; H01L 21/823431; H01L 21/82345; H01L 21/823456; H01L 27/088;
(Continued)

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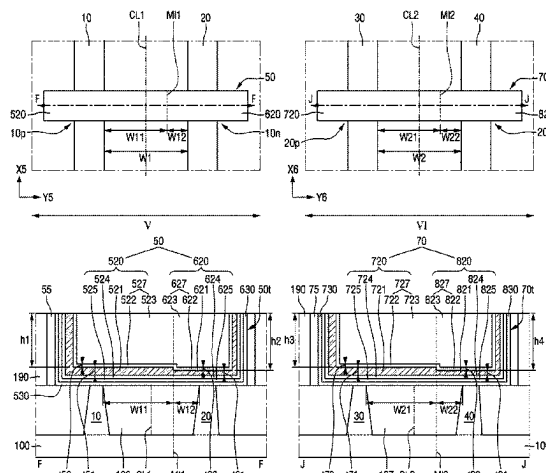
Primary Examiner — Shahed Ahmed

(74) *Attorney, Agent, or Firm* — Muir Patent Law, PLLC

(57) **ABSTRACT**

A semiconductor device including a substrate includes a first region and a second region and first and second transistors in the first and second regions, respectively. The first transistor includes a first gate insulating layer on the substrate, a first lower TiN layer on and in contact with the first gate insulating layer, a first etch-stop layer on the first lower TiN layer and a first upper gate electrode on the first etch-stop layer. The second transistor includes a second gate insulating layer on the substrate, a second lower TiN layer on and in contact with the second gate insulating layer, a second etch-stop layer on the second lower TiN layer and a second upper gate electrode on the second etch-stop layer. A thickness of the first lower TiN layer is less than a thickness of the second lower TiN layer.

11 Claims, 88 Drawing Sheets



Related U.S. Application Data

division of application No. 15/413,680, filed on Jan. 24, 2017, now Pat. No. 10,068,901.

64/518; H10D 64/667; H10D 84/014;
H10D 84/0158; H10D 84/038; H10D
84/83; H10D 64/665–669; H10D
84/0151; H10D 84/0188

See application file for complete search history.

(30) Foreign Application Priority Data

Mar. 10, 2016 (KR) 10-2016-0028822
Mar. 11, 2016 (KR) 10-2016-0029542

(51) Int. Cl.

H10D 1/00 (2025.01)
H10D 30/62 (2025.01)
H10D 64/01 (2025.01)
H10D 64/27 (2025.01)
H10D 64/66 (2025.01)
H10D 84/01 (2025.01)
H10D 84/03 (2025.01)

(52) U.S. Cl.

CPC **H10D 64/017** (2025.01); **H10D 64/517**
(2025.01); **H10D 64/518** (2025.01); **H10D**
64/667 (2025.01); **H10D 84/014** (2025.01);
H10D 84/0142 (2025.01); **H10D 84/0158**
(2025.01); **H10D 84/038** (2025.01); **H10D**
84/83 (2025.01)

(58) Field of Classification Search

CPC H01L 28/00; H01L 29/42372; H01L
29/42376; H01L 29/66545; H01L
29/7854; H01L 29/495–4958; H01L
29/4966–4975; H01L 29/435; H01L
21/28061; H01L 21/76224–76237; H01L
21/28079–28088; H10B 10/12; H10D
84/834; H10D 1/00; H10D 64/517; H10D

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FIG. 1

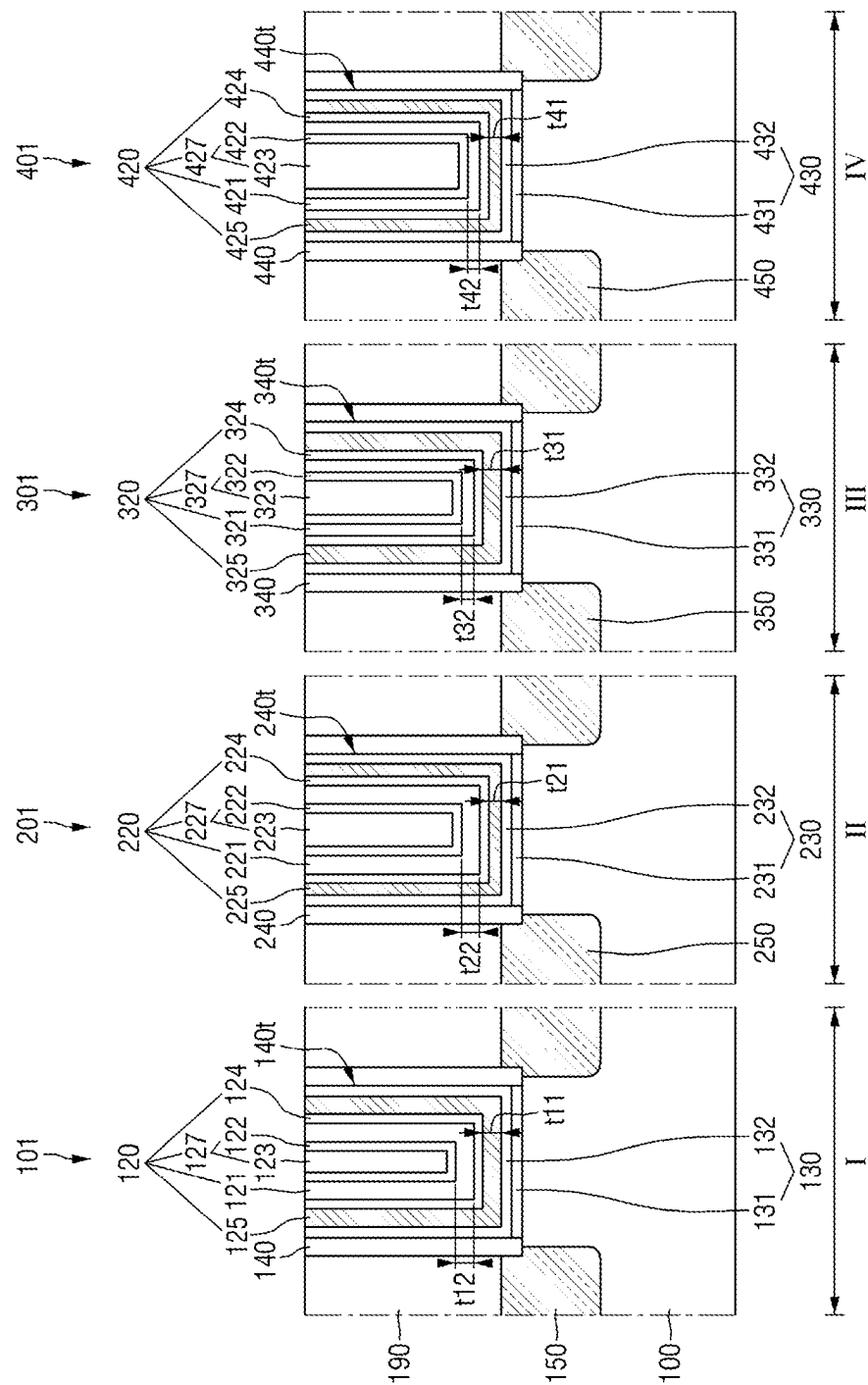


FIG. 2

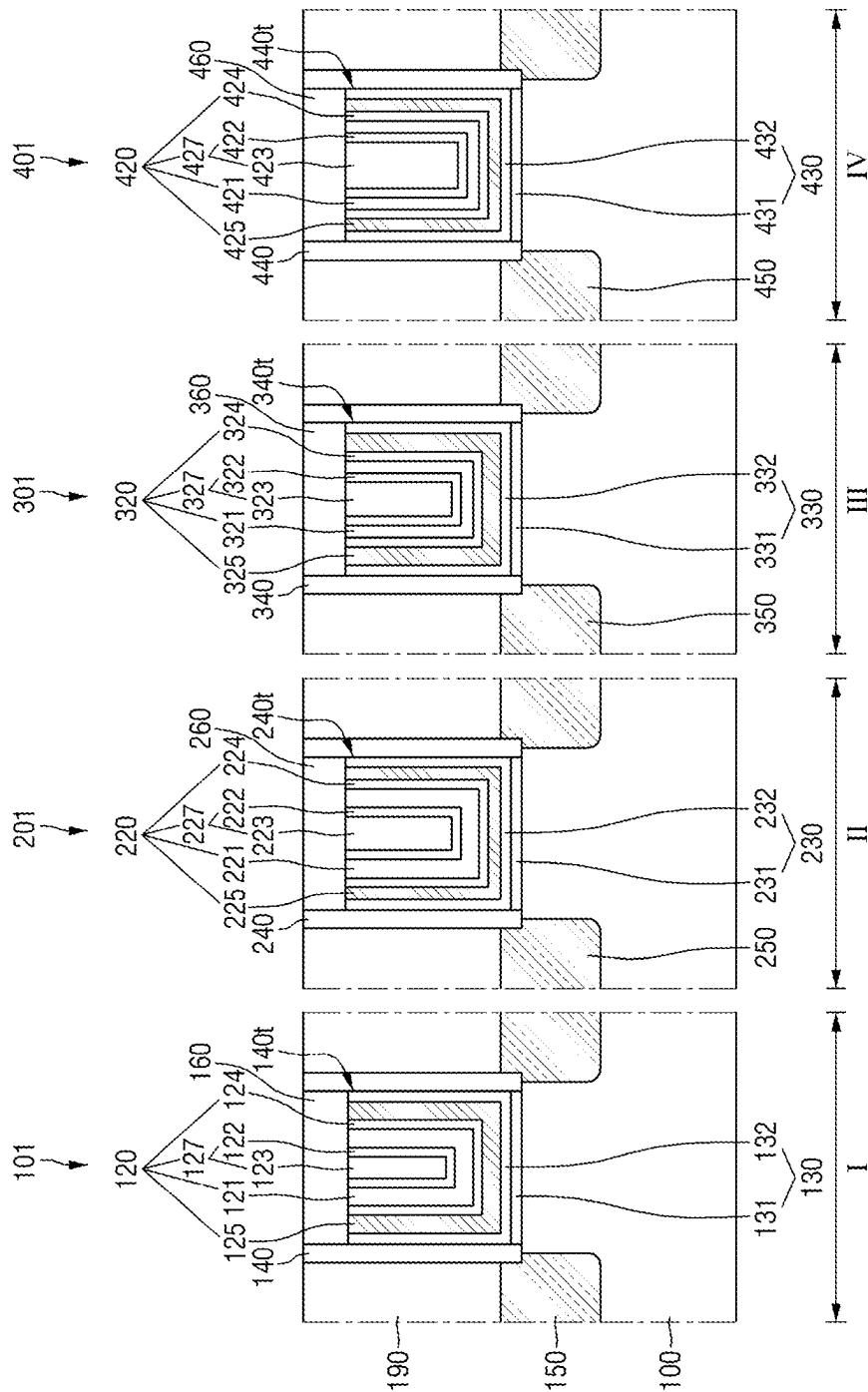


FIG. 3

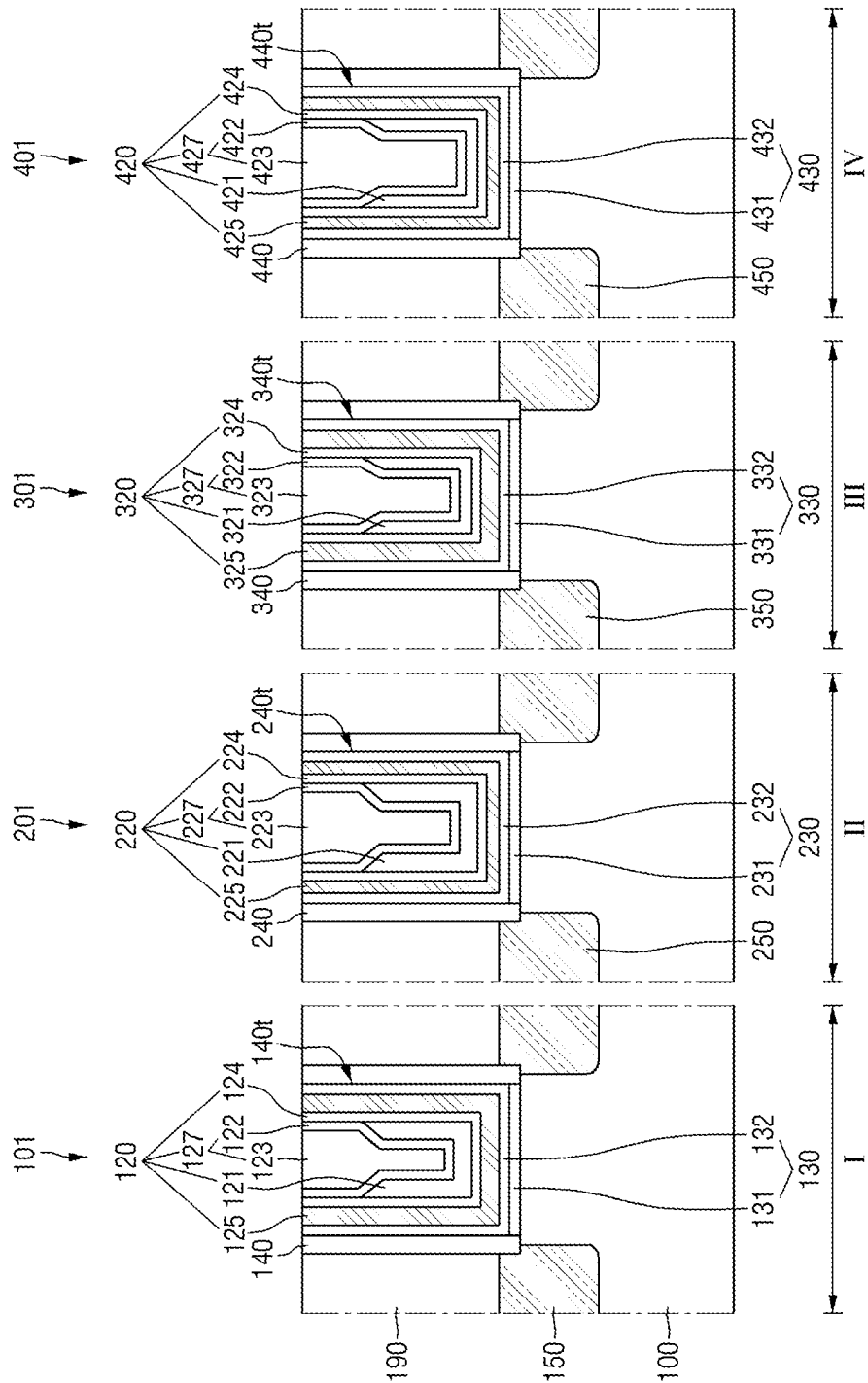


FIG. 4

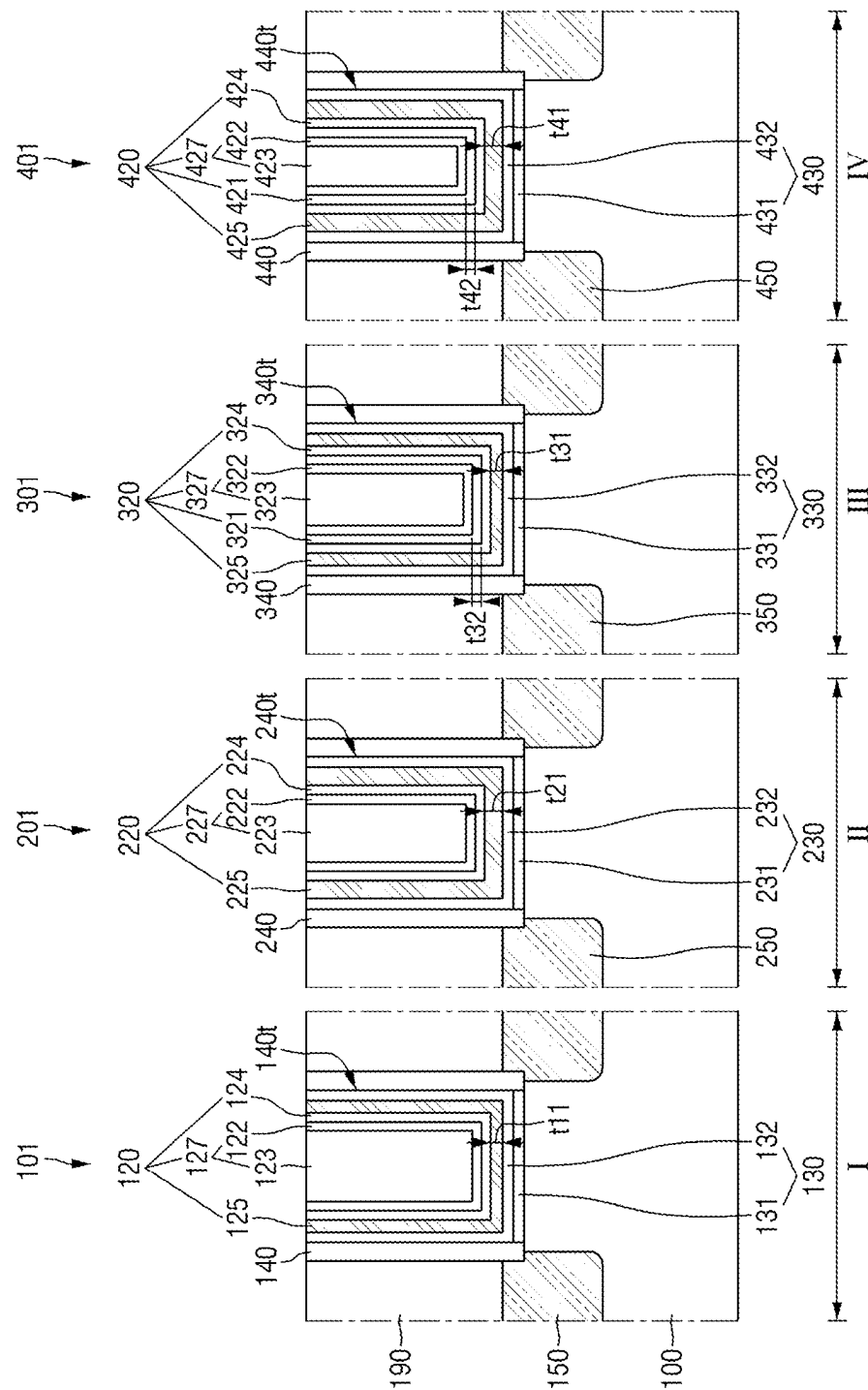


FIG. 5

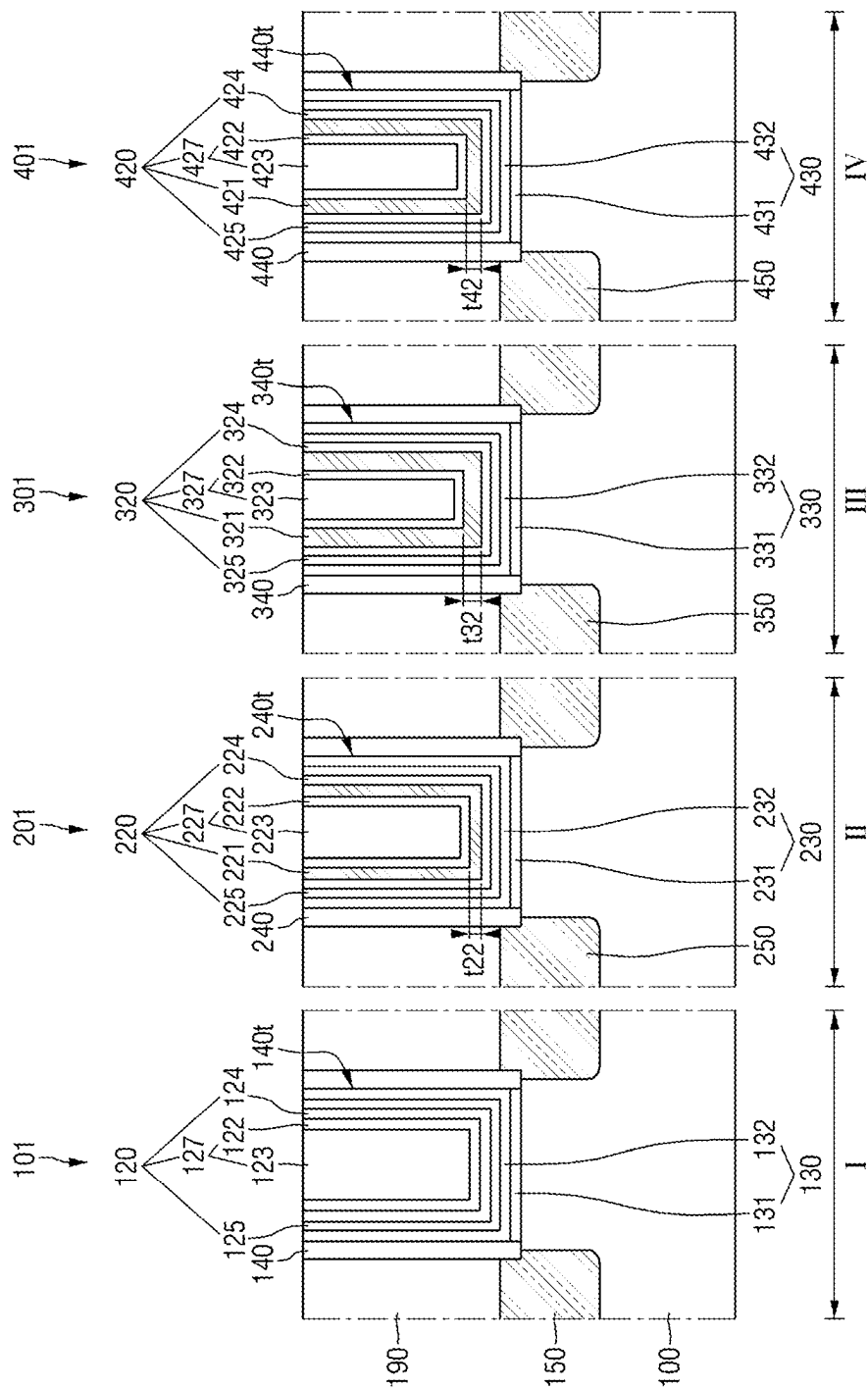


FIG. 6

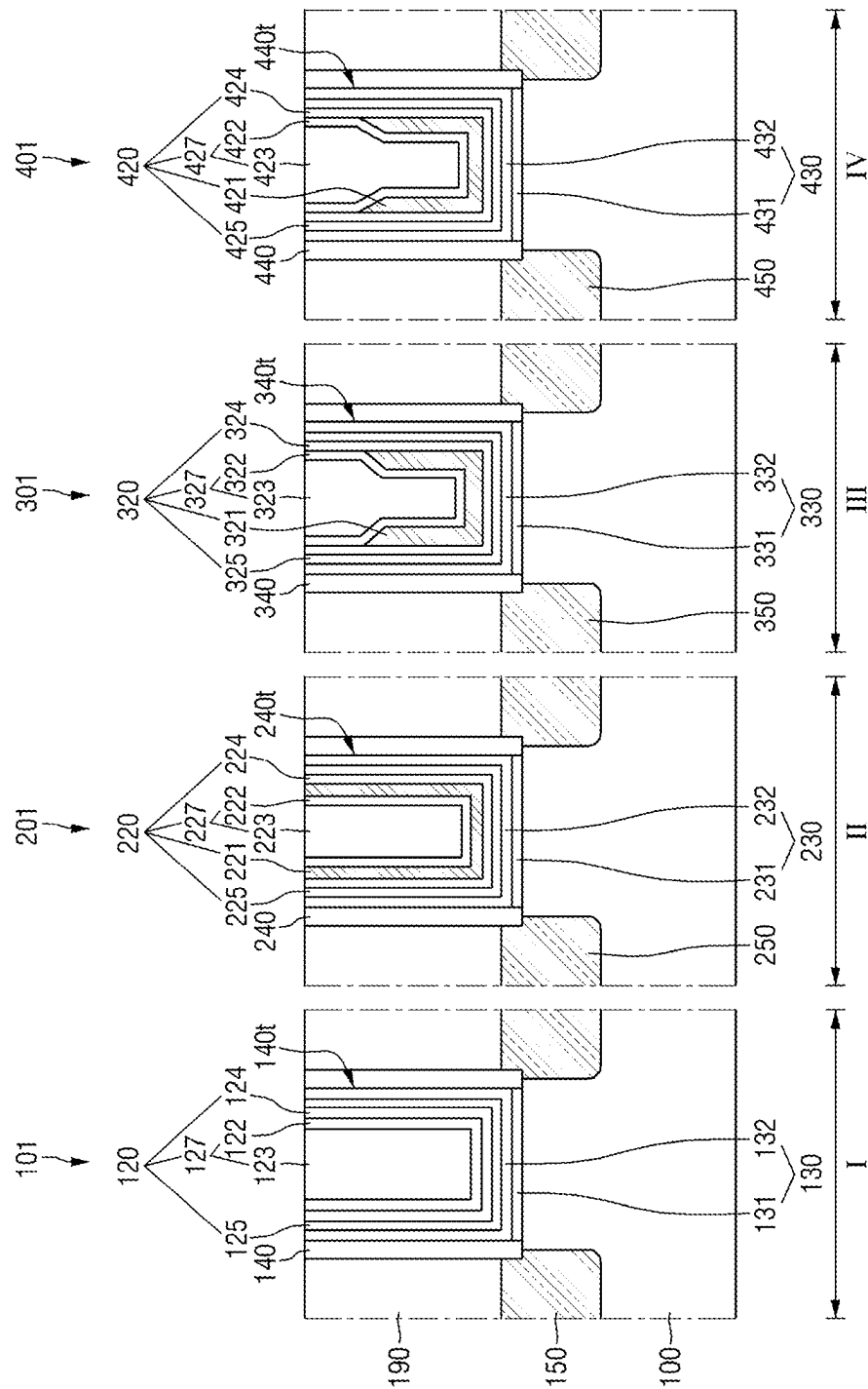


FIG. 7

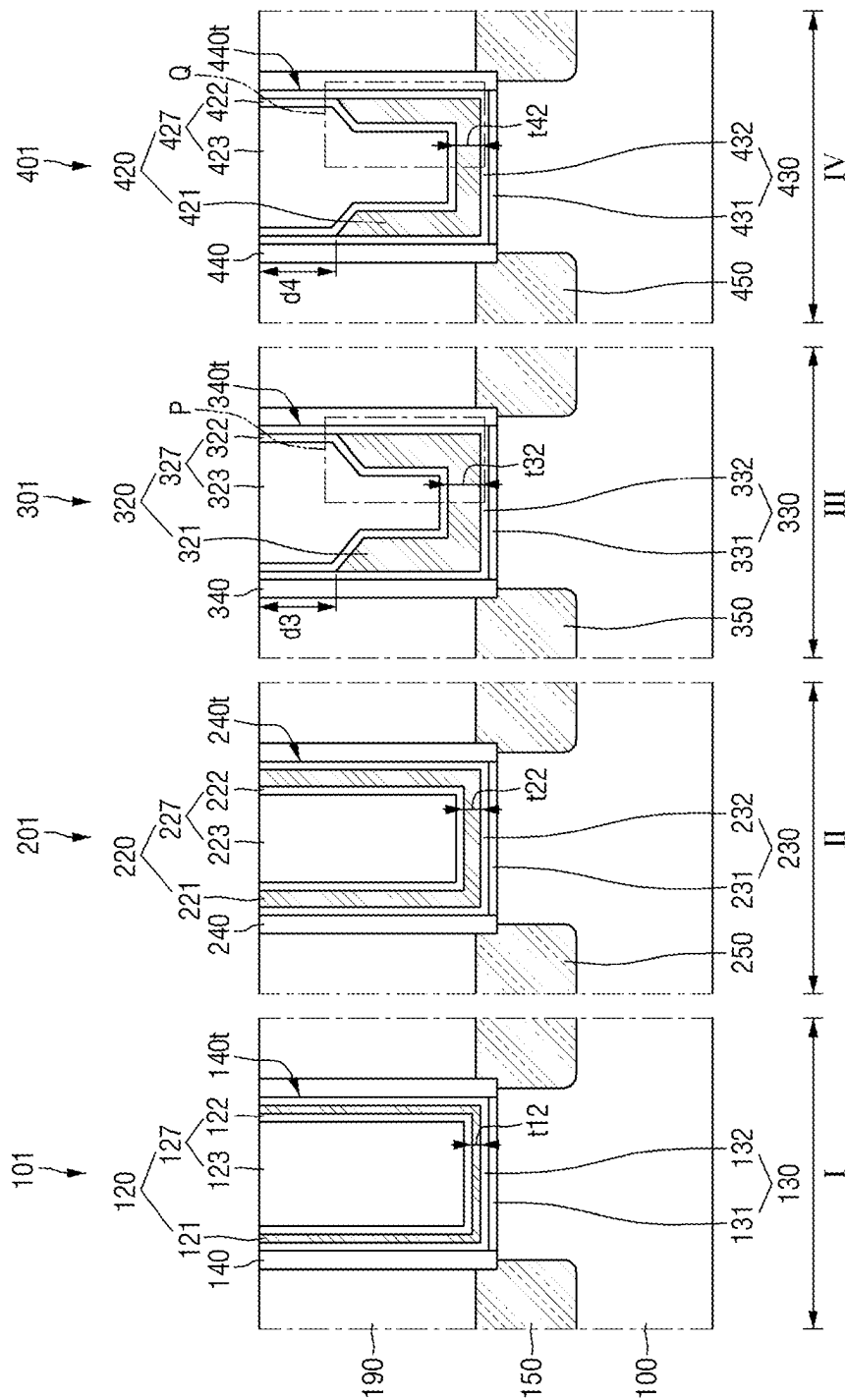


FIG. 8

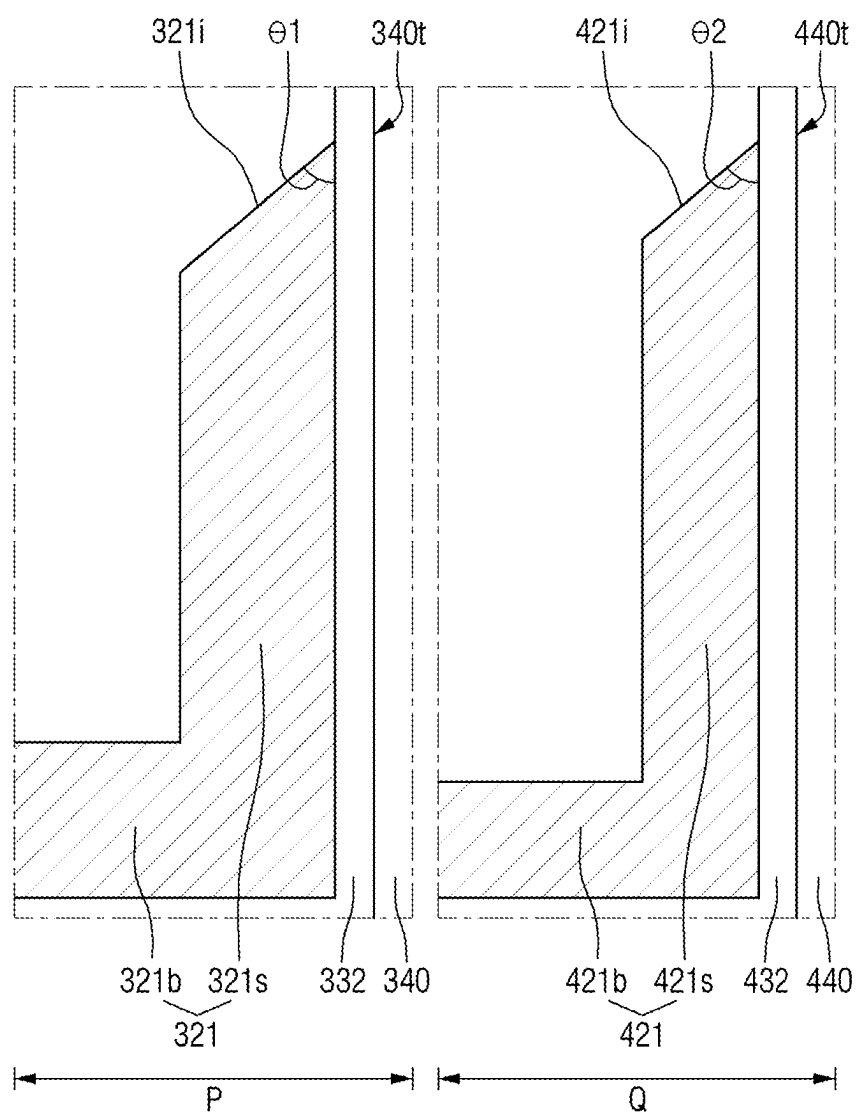


FIG. 9

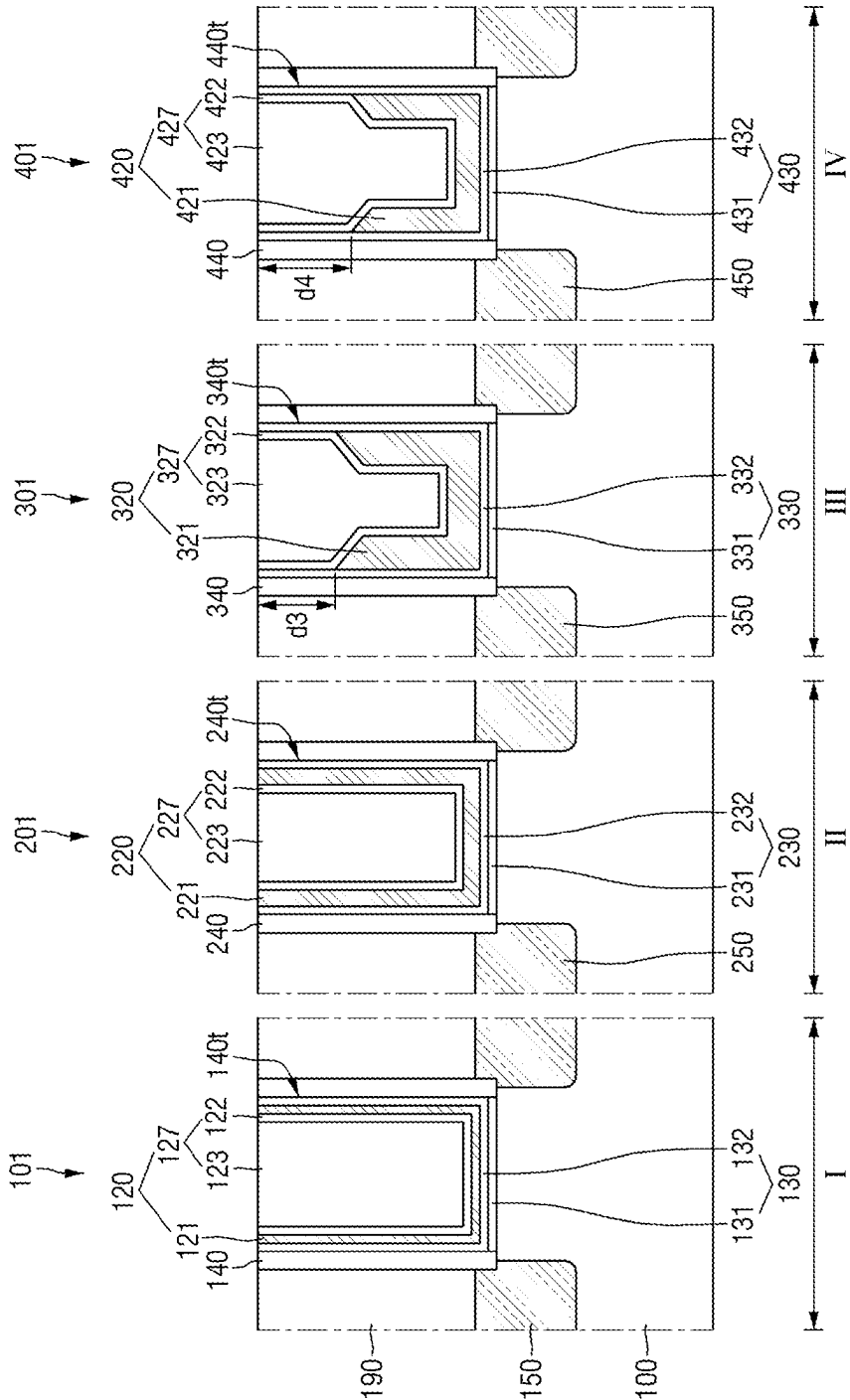


FIG. 10

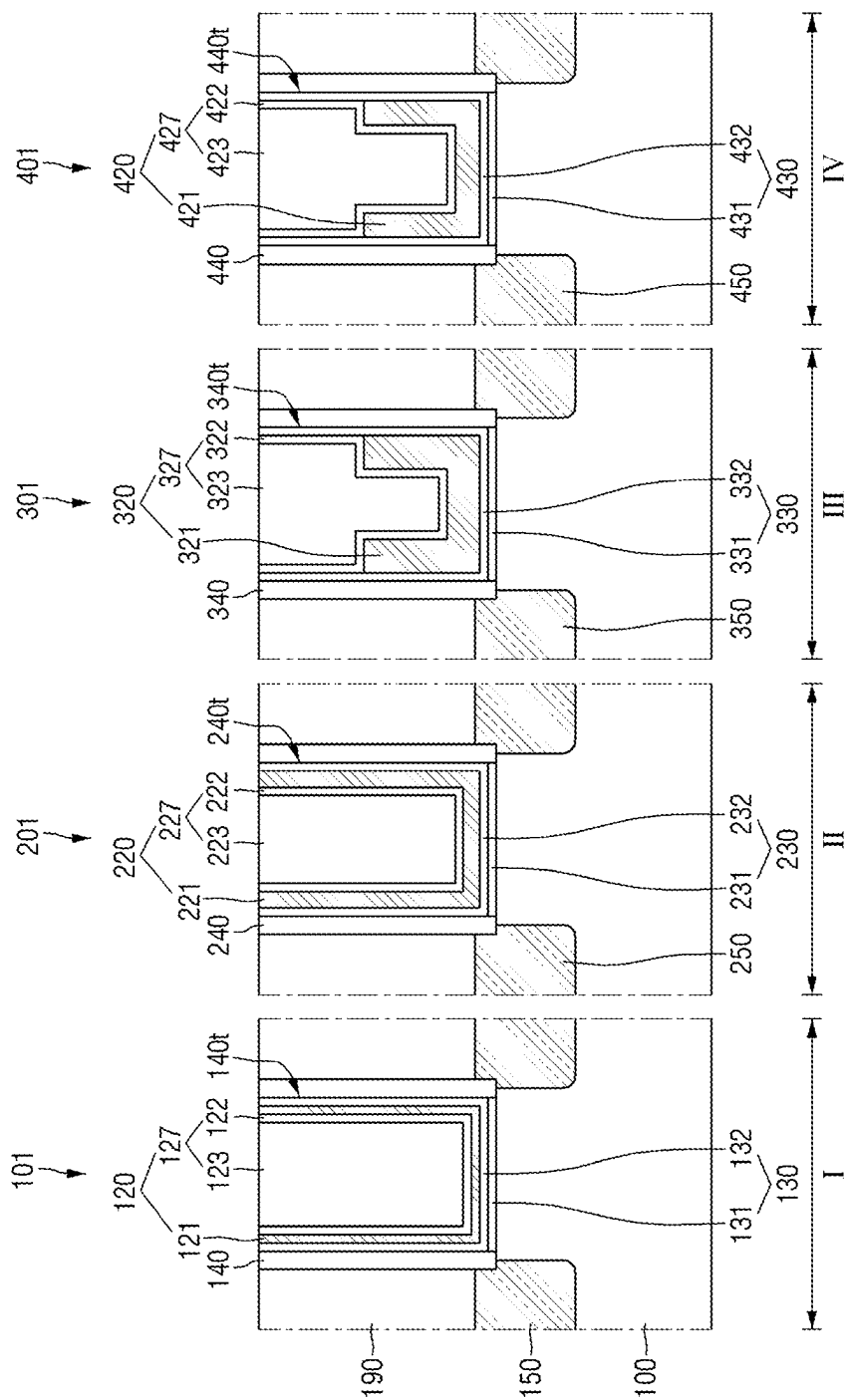


FIG. 11

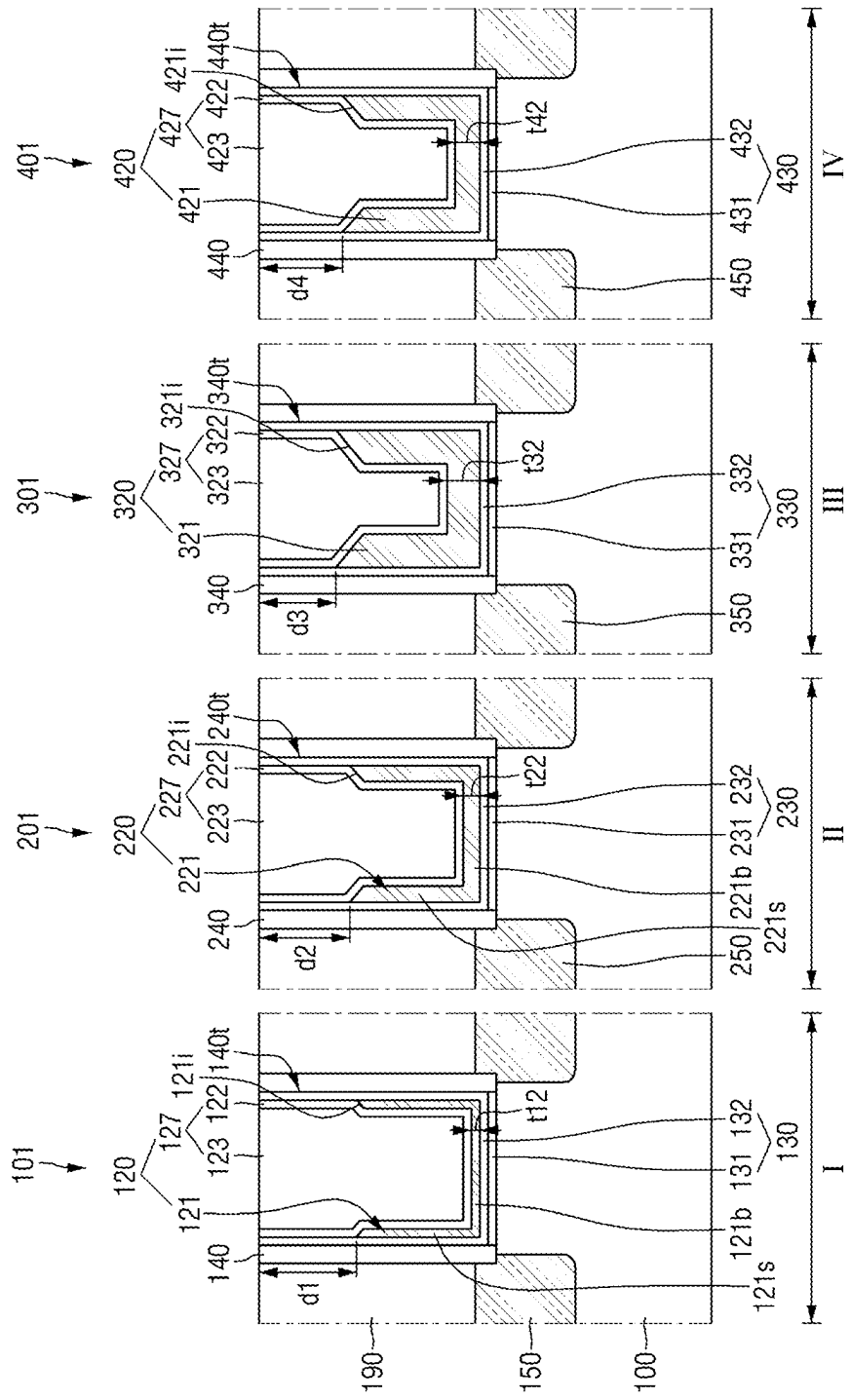


FIG. 12

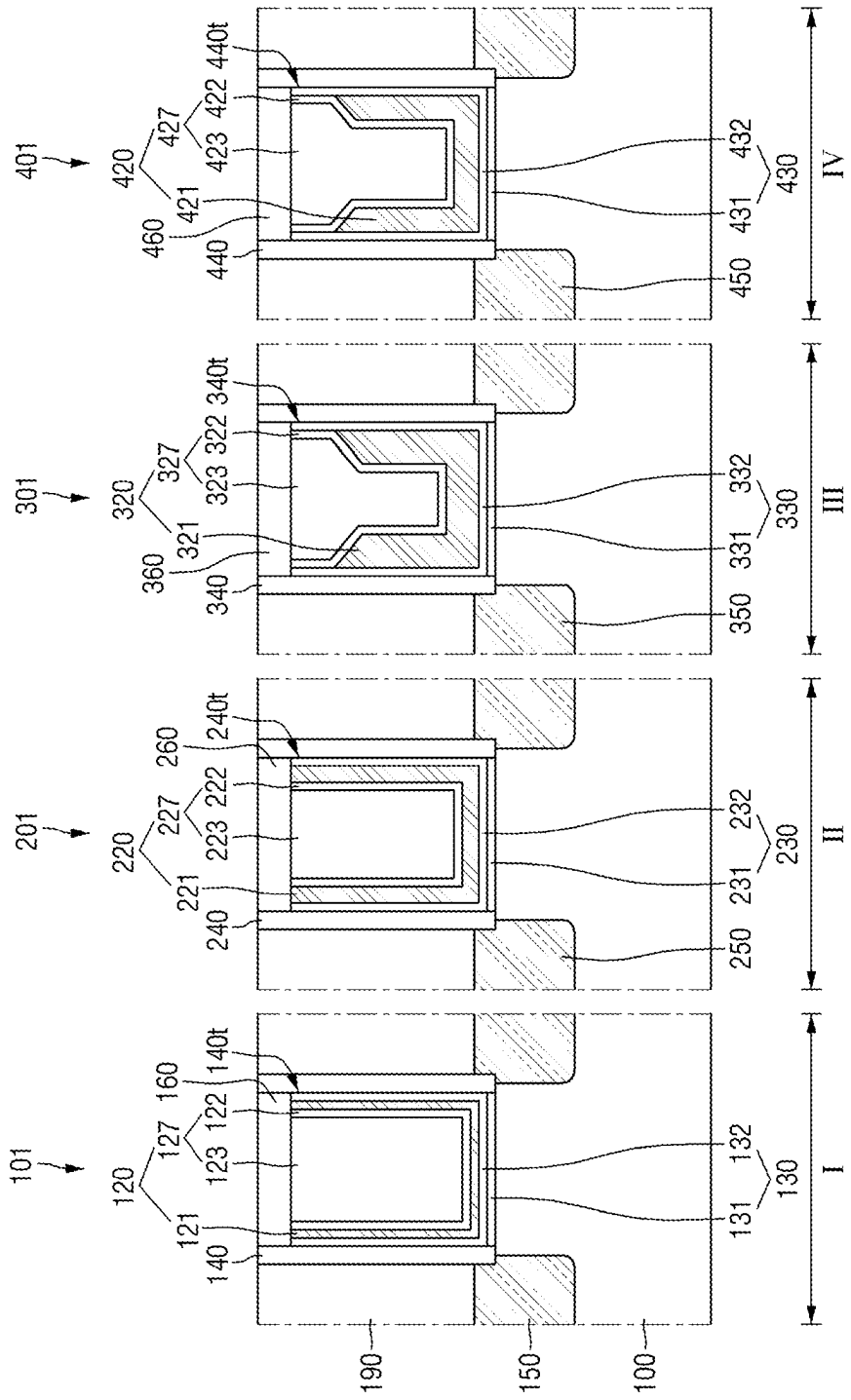


FIG. 13

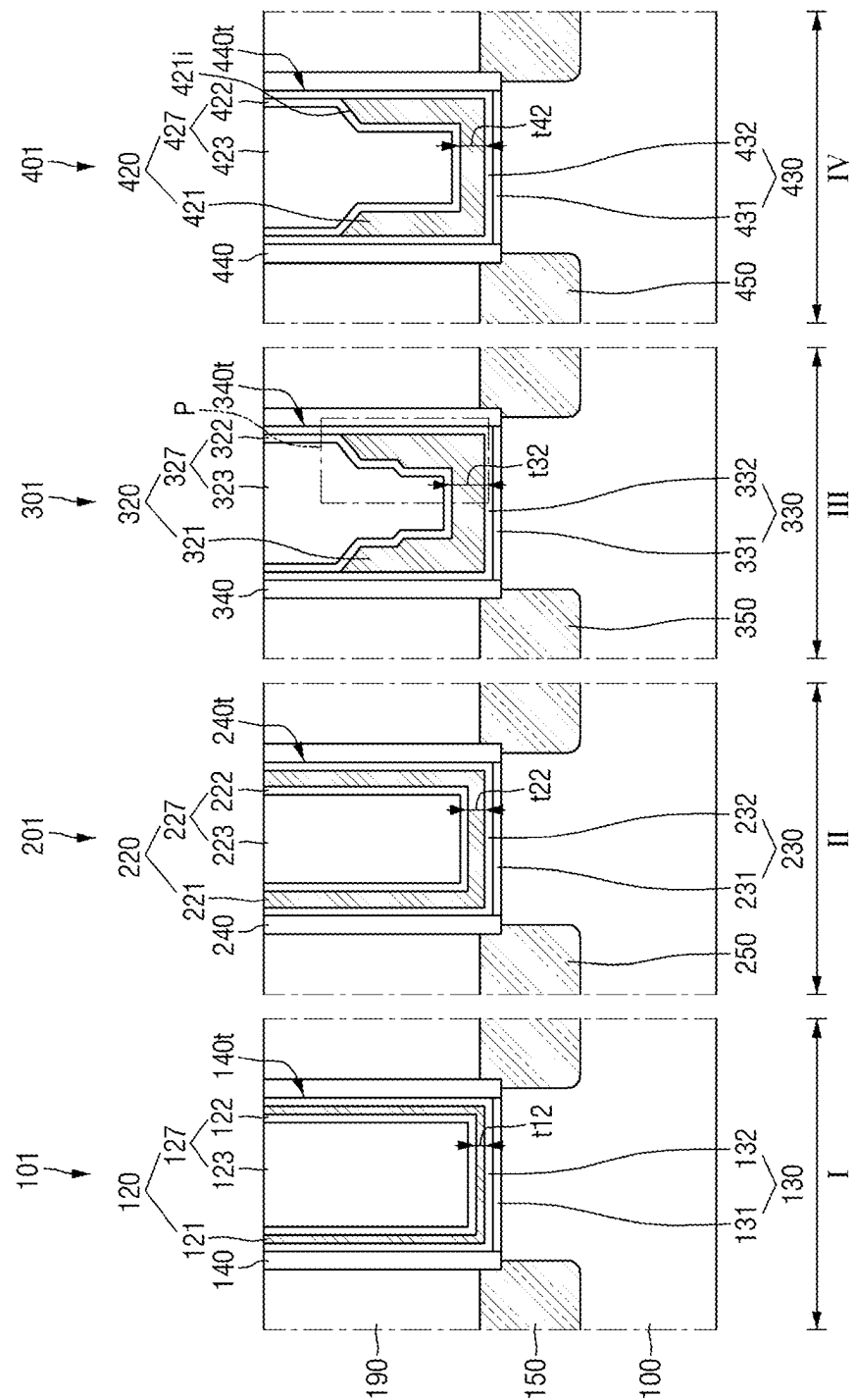


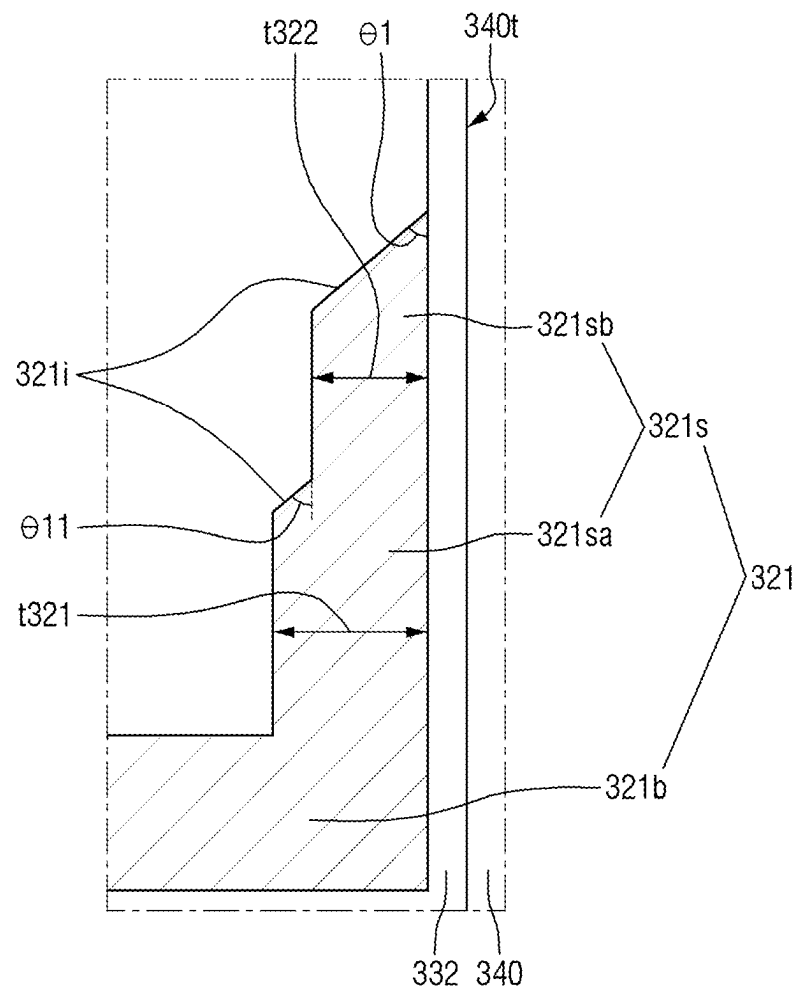
FIG. 14

FIG. 15

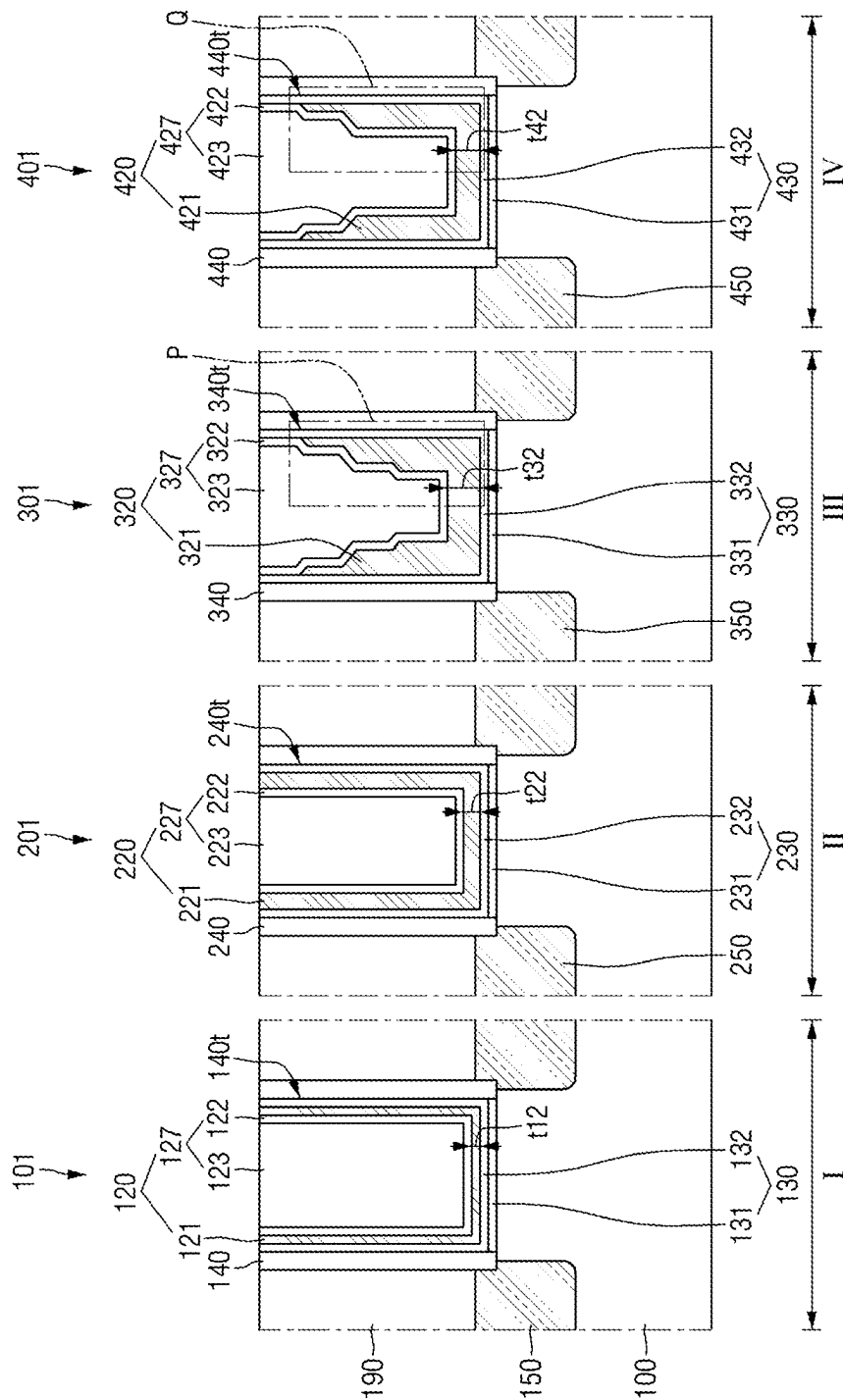


FIG. 16

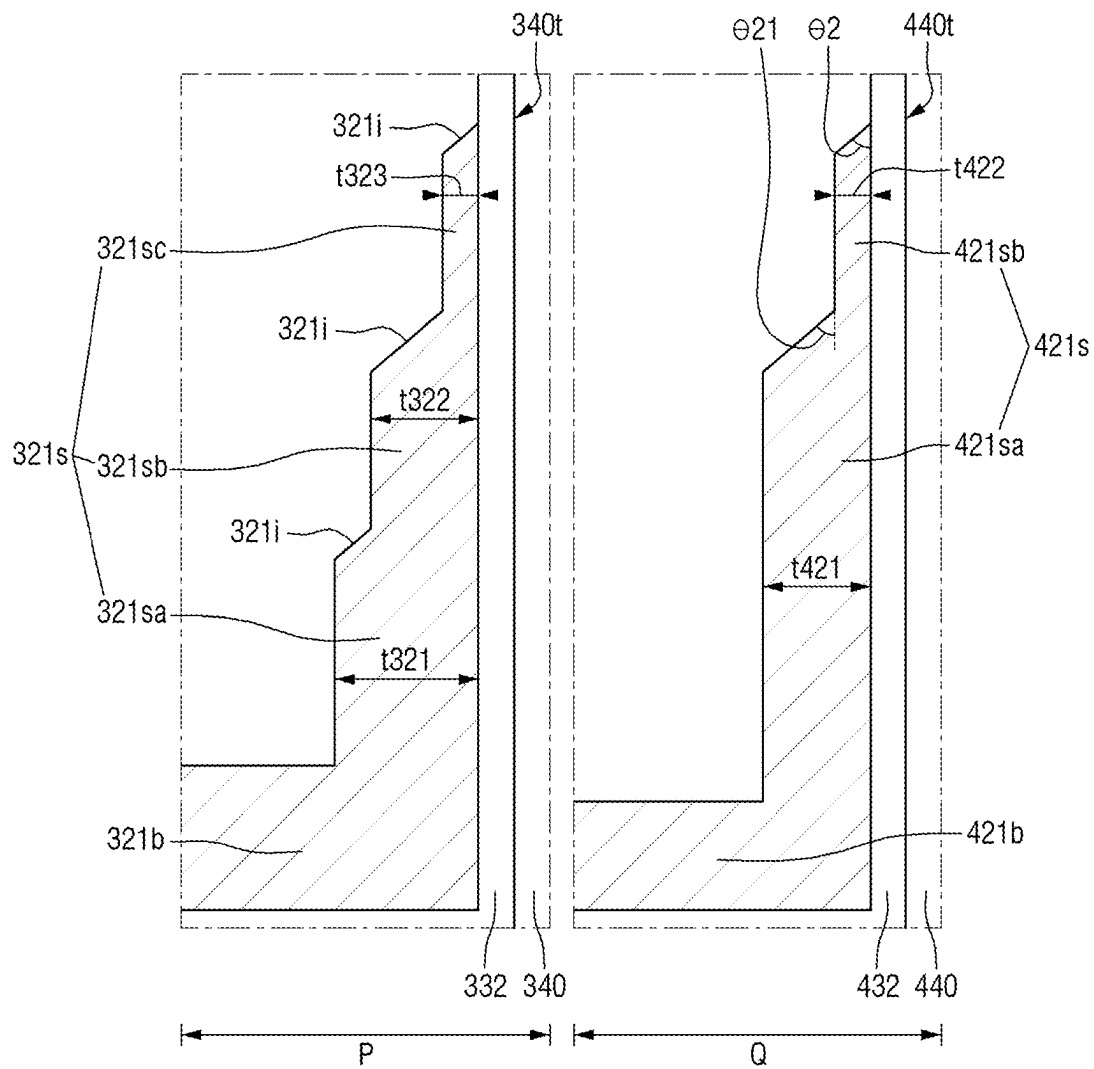


FIG. 17

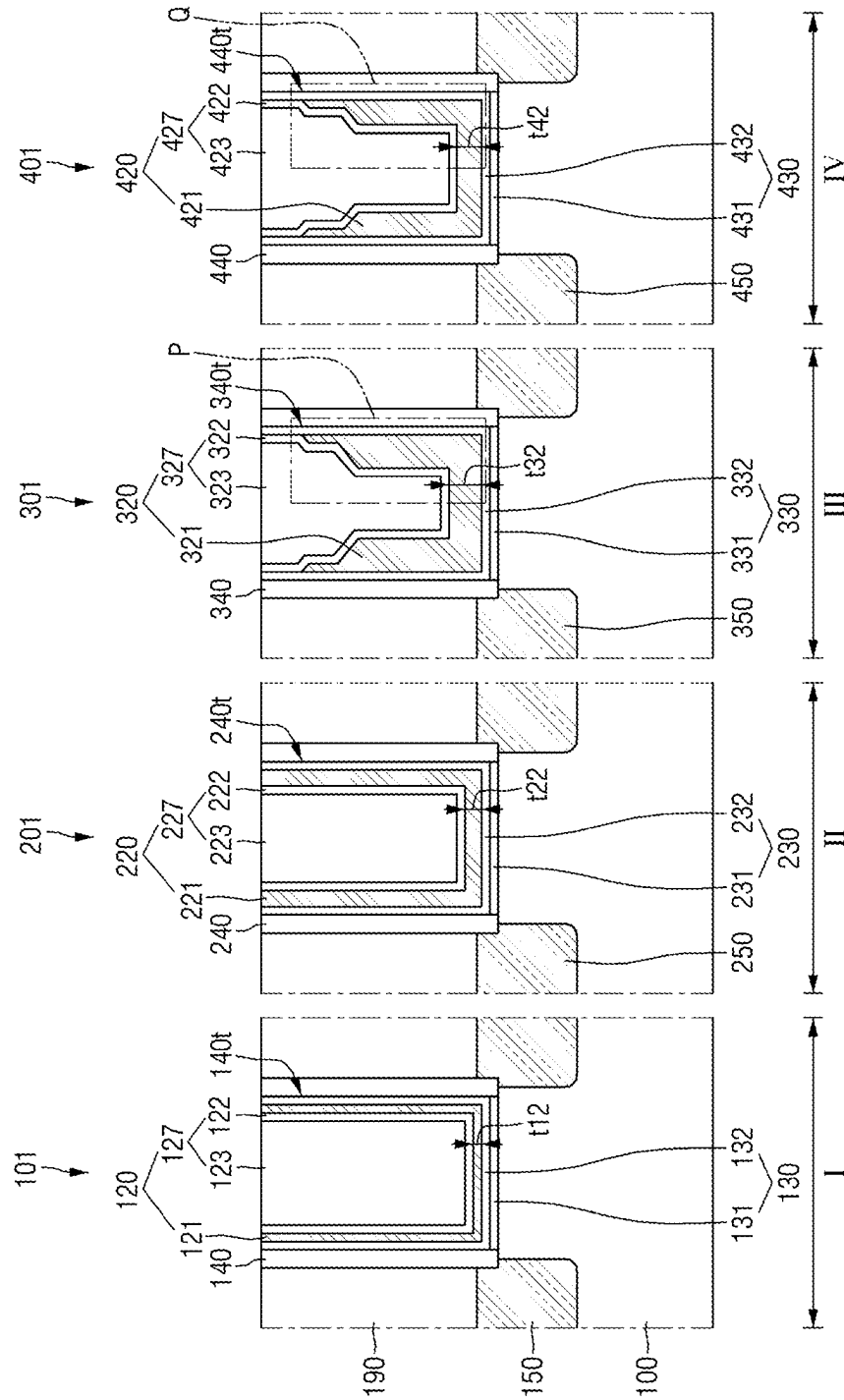


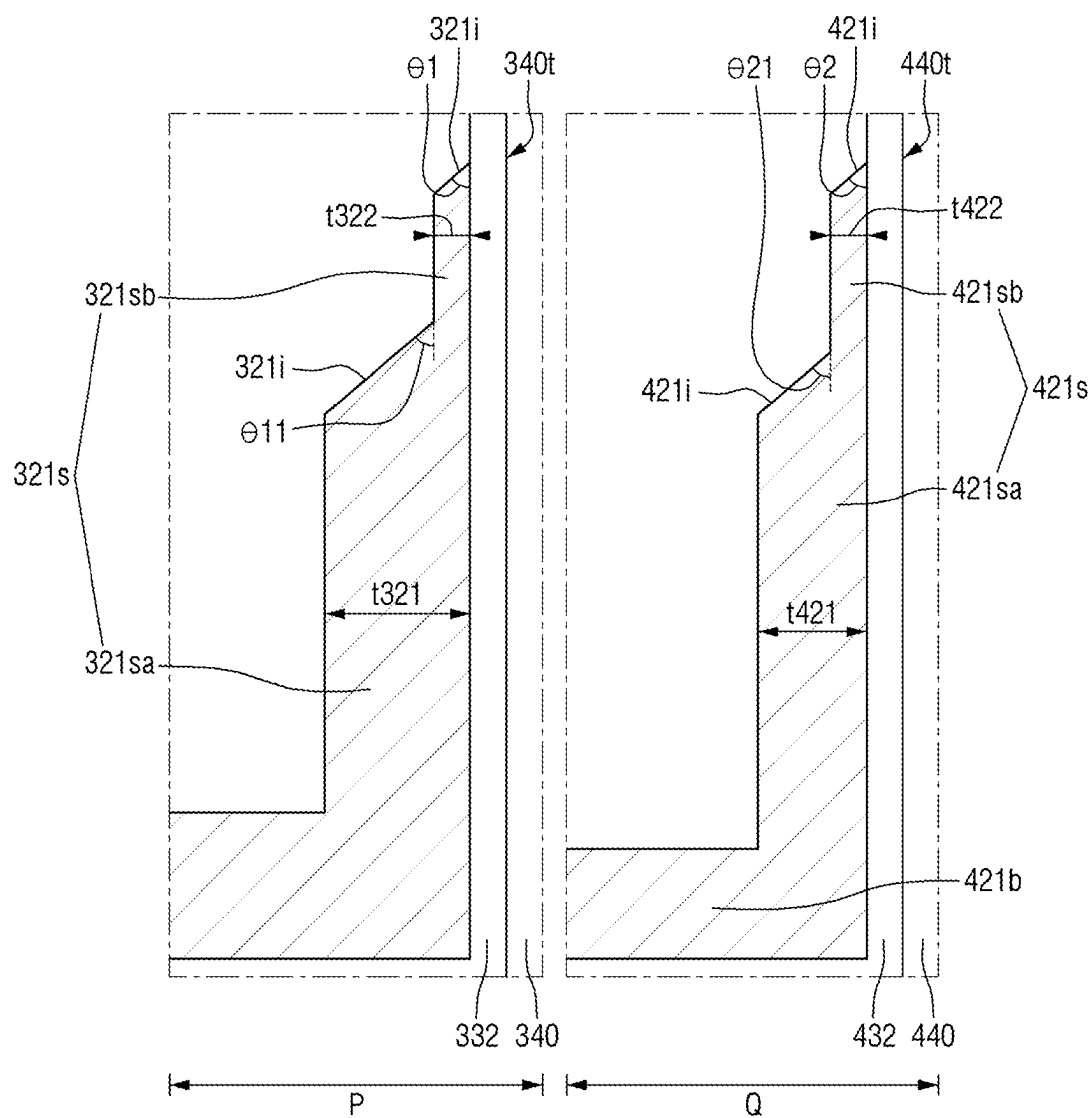
FIG. 18

FIG. 19A

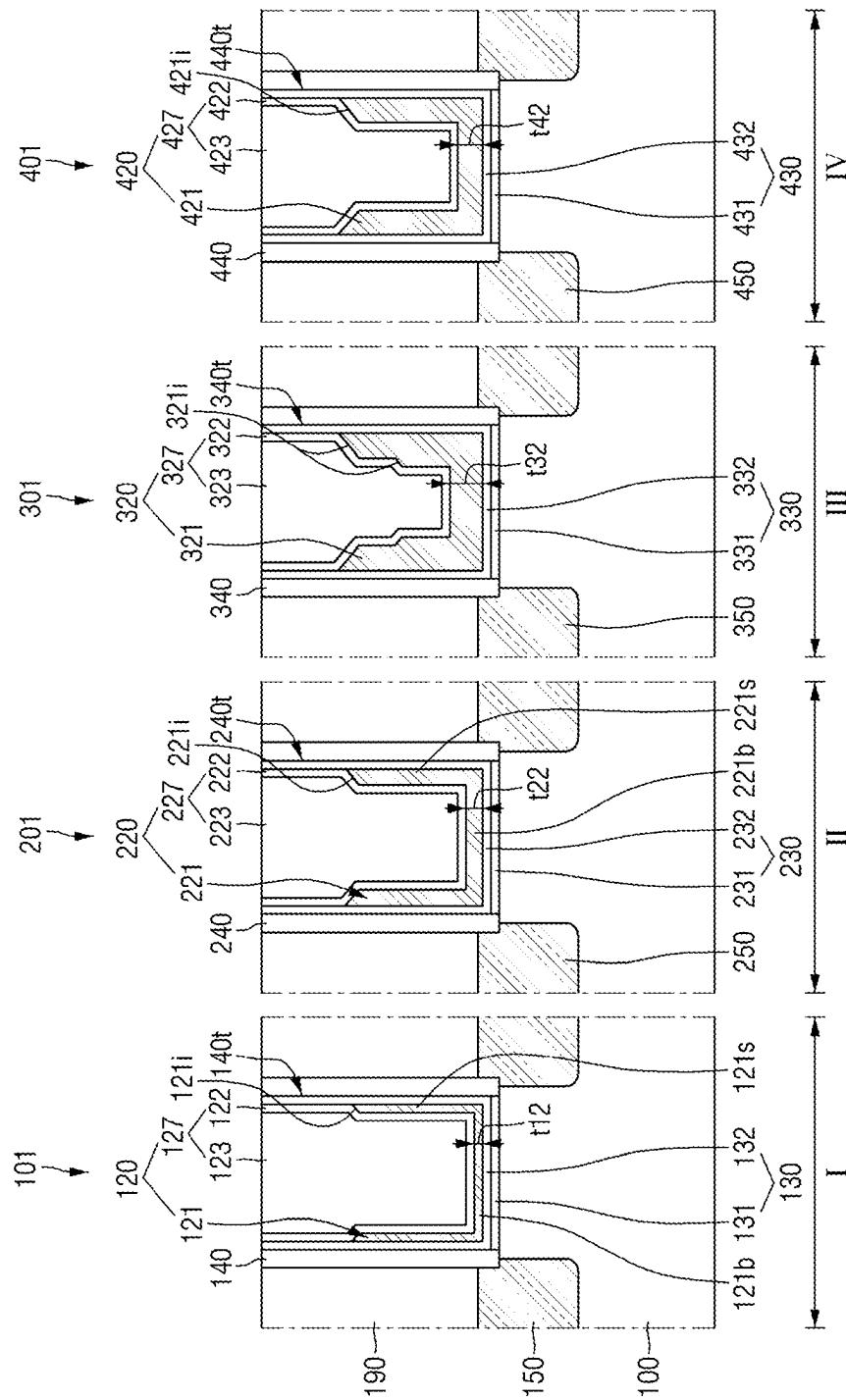


FIG. 19B

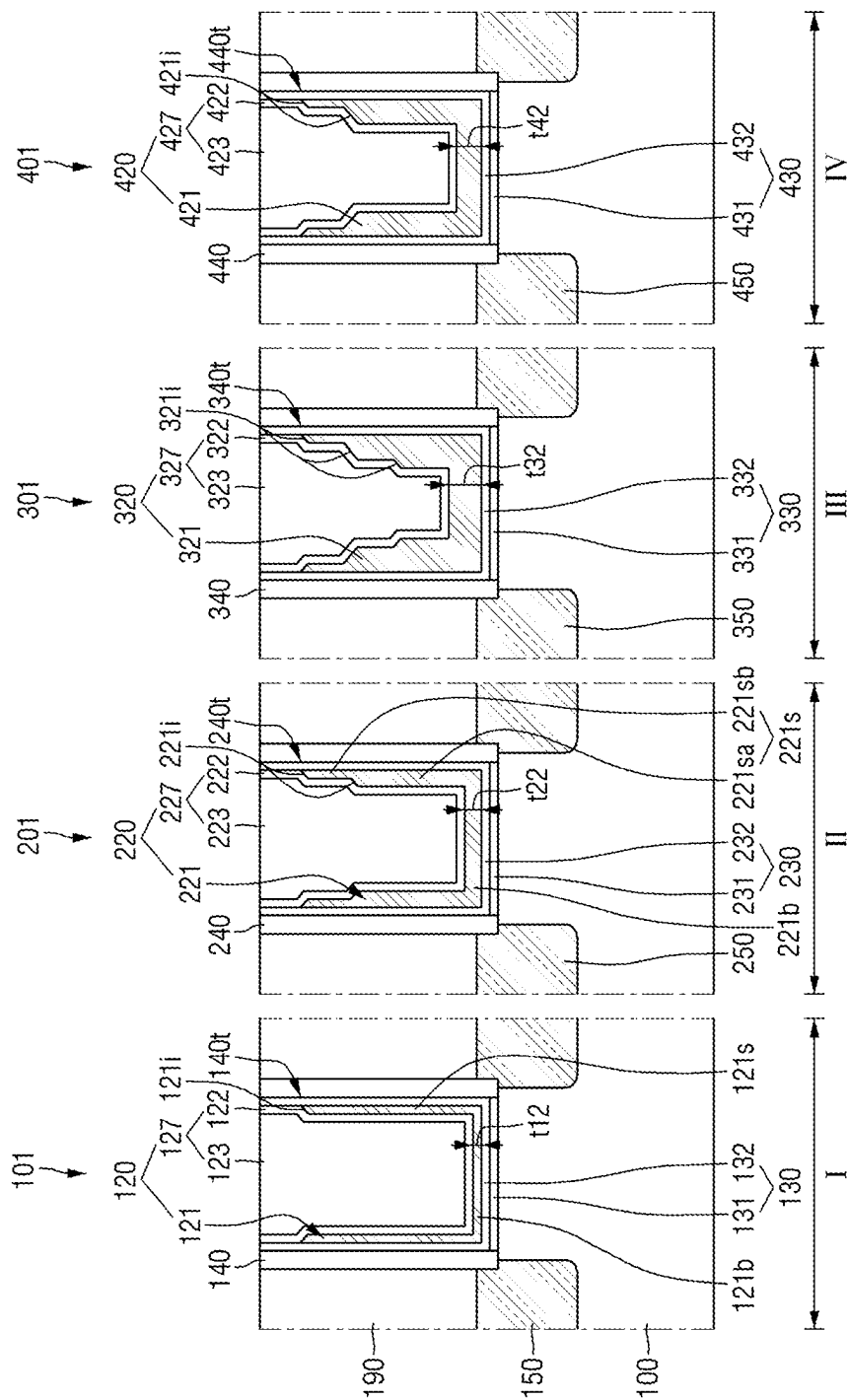


FIG. 20

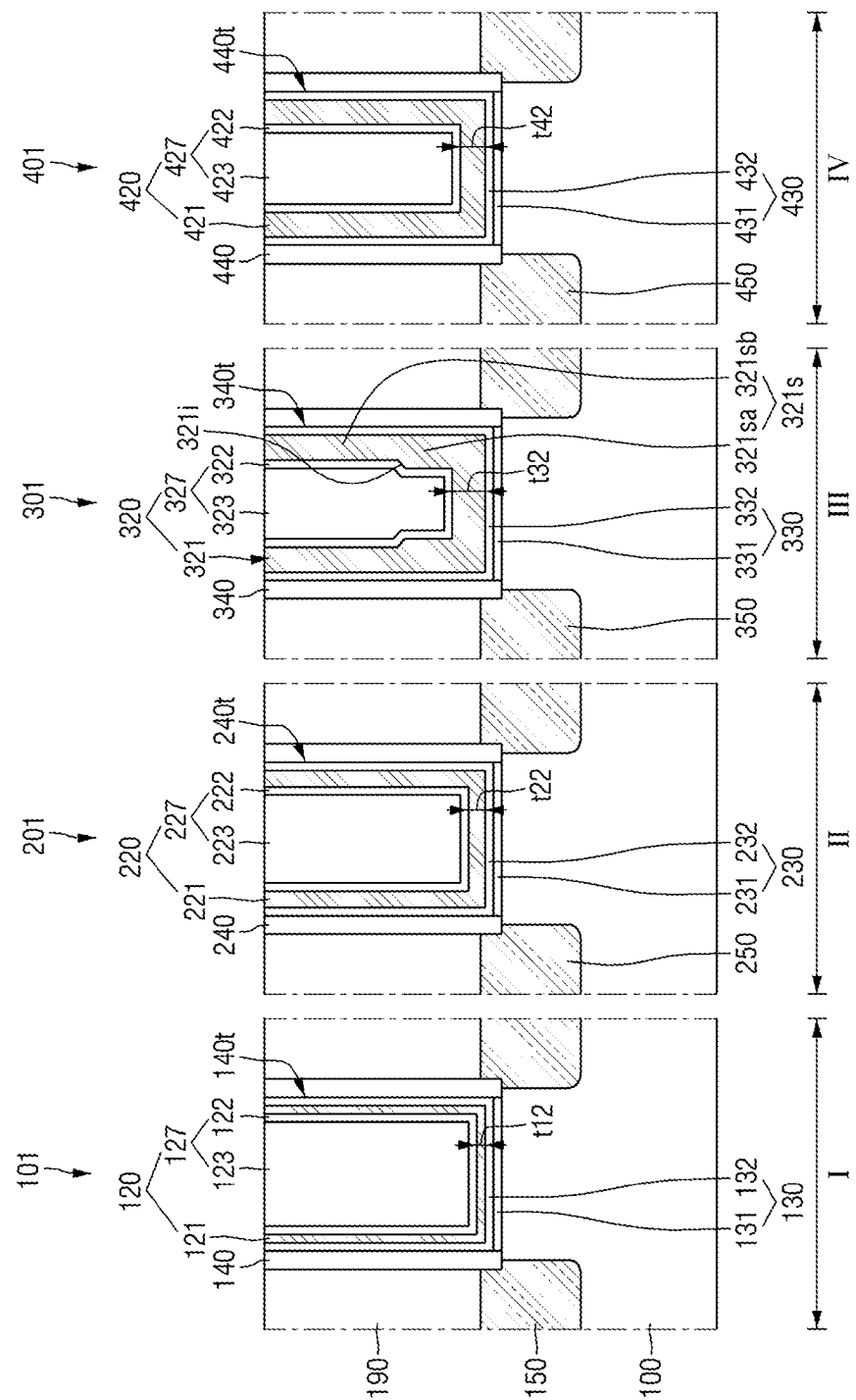
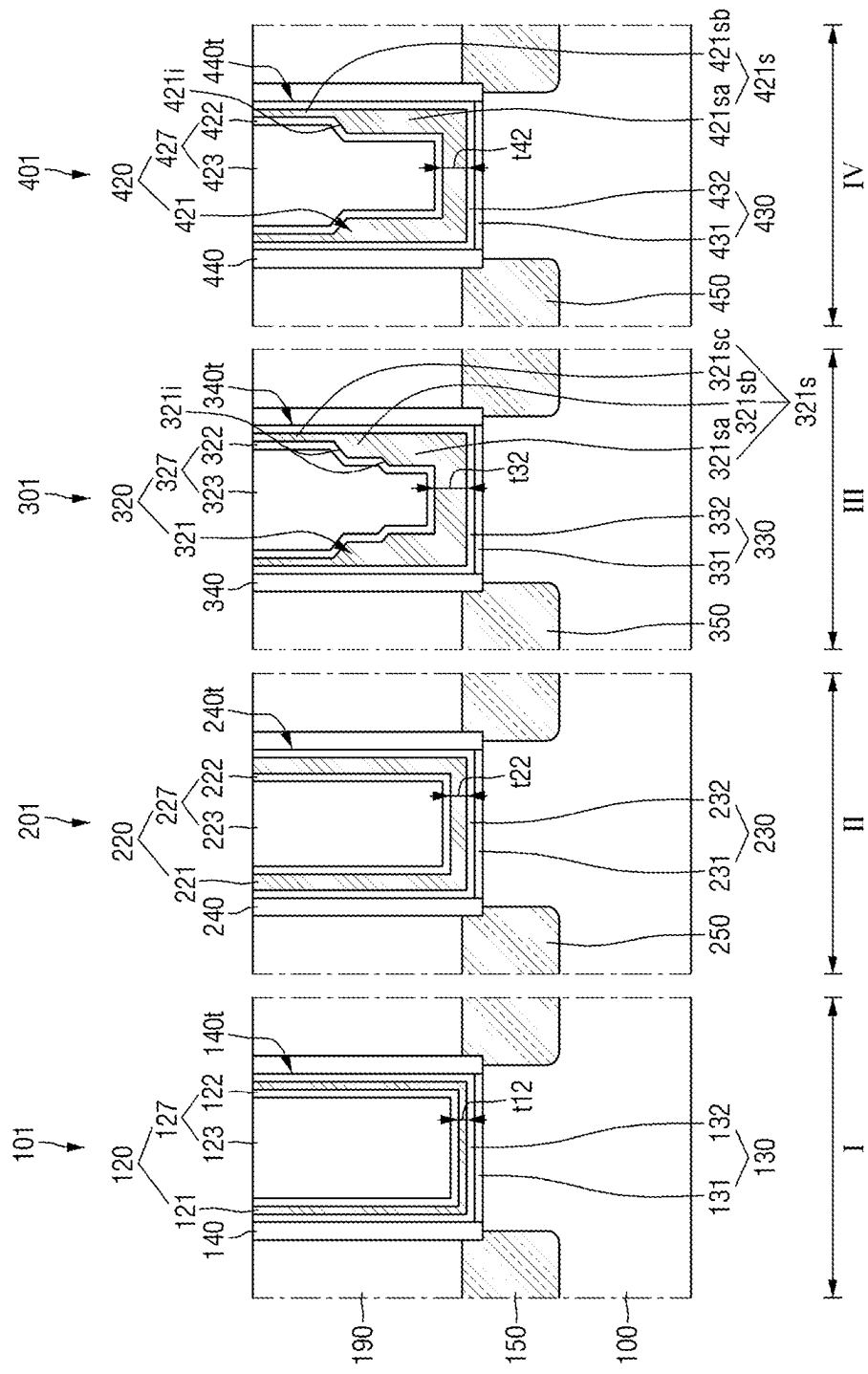


FIG. 21



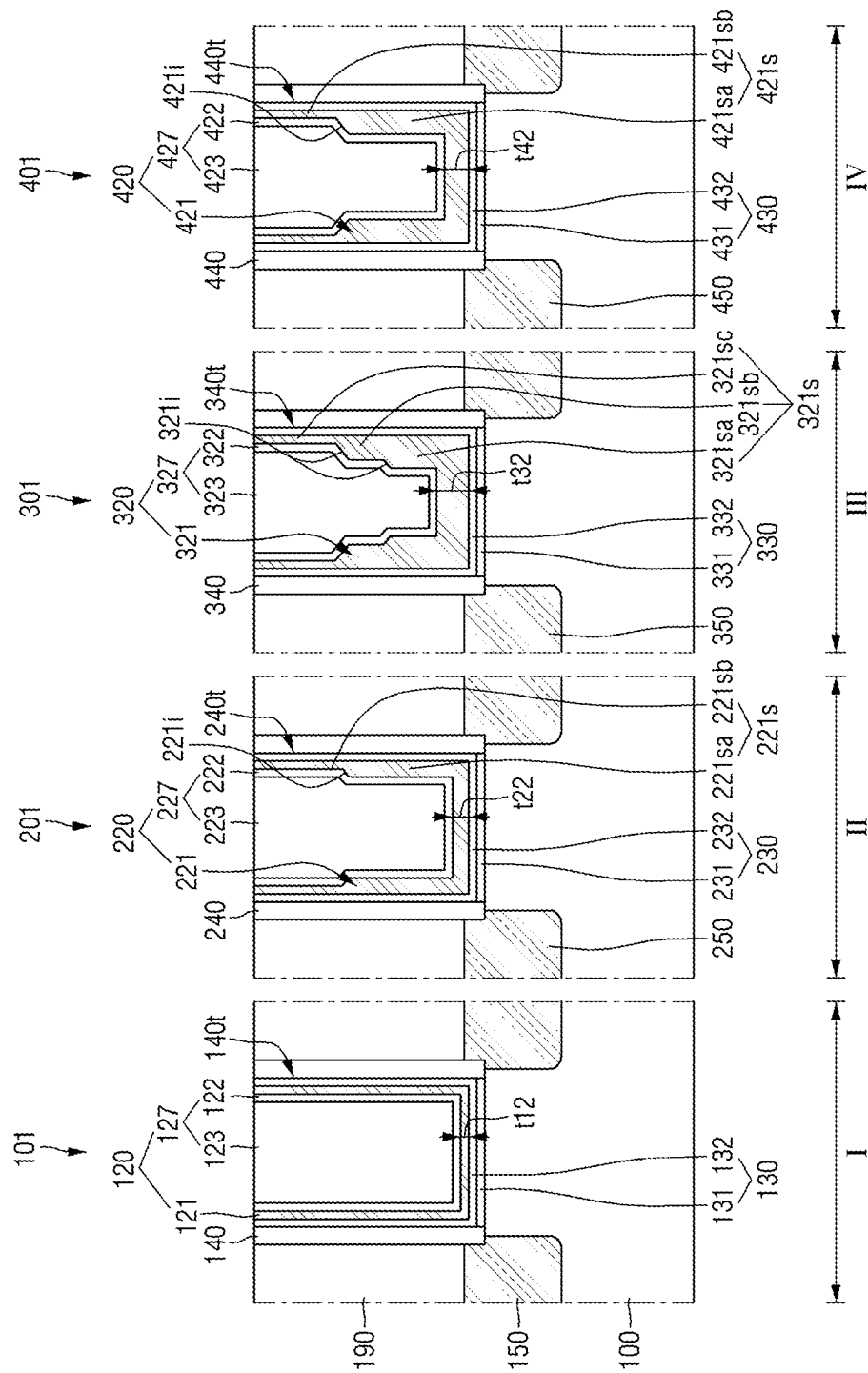


FIG. 23

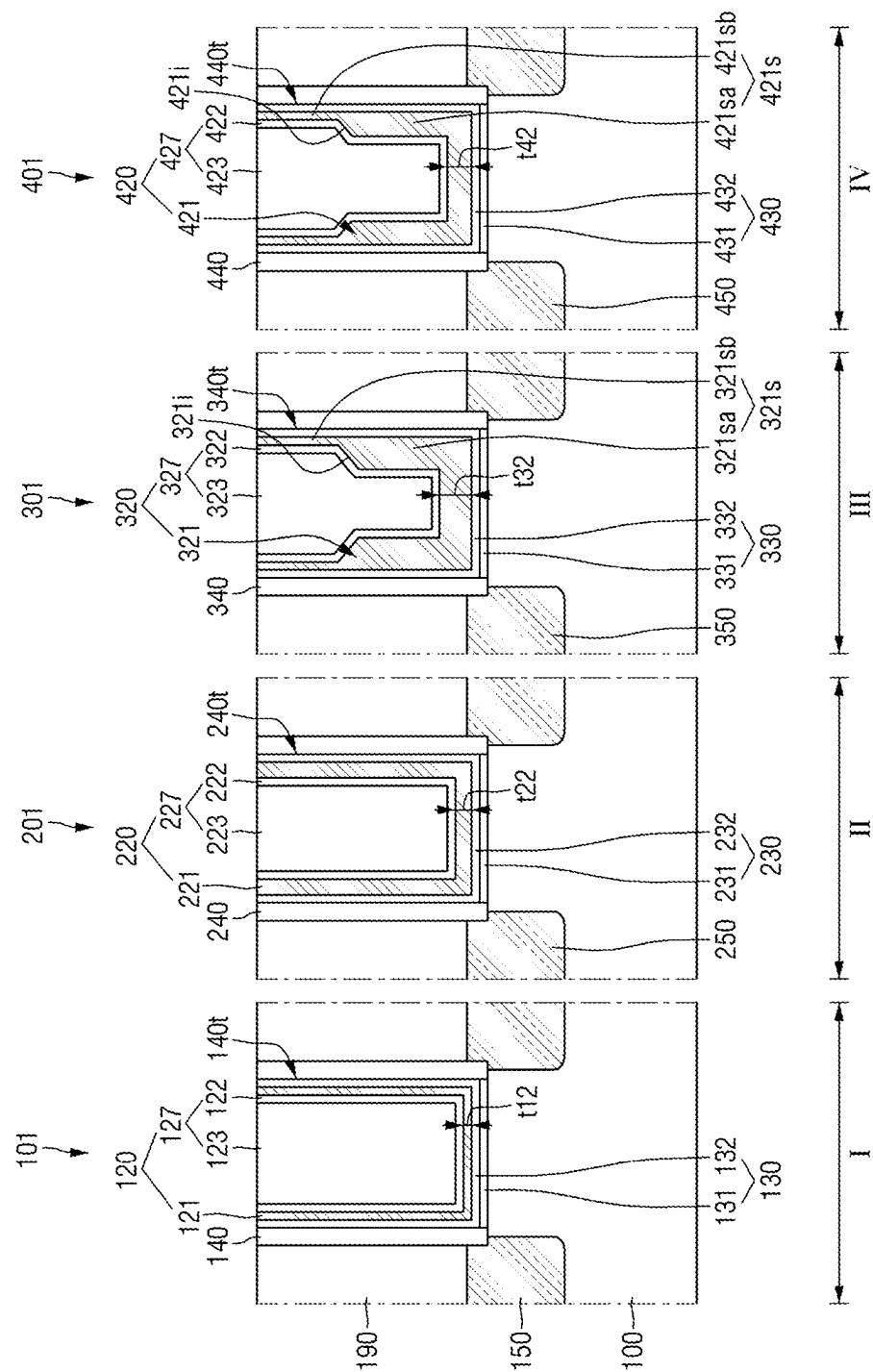


FIG. 24

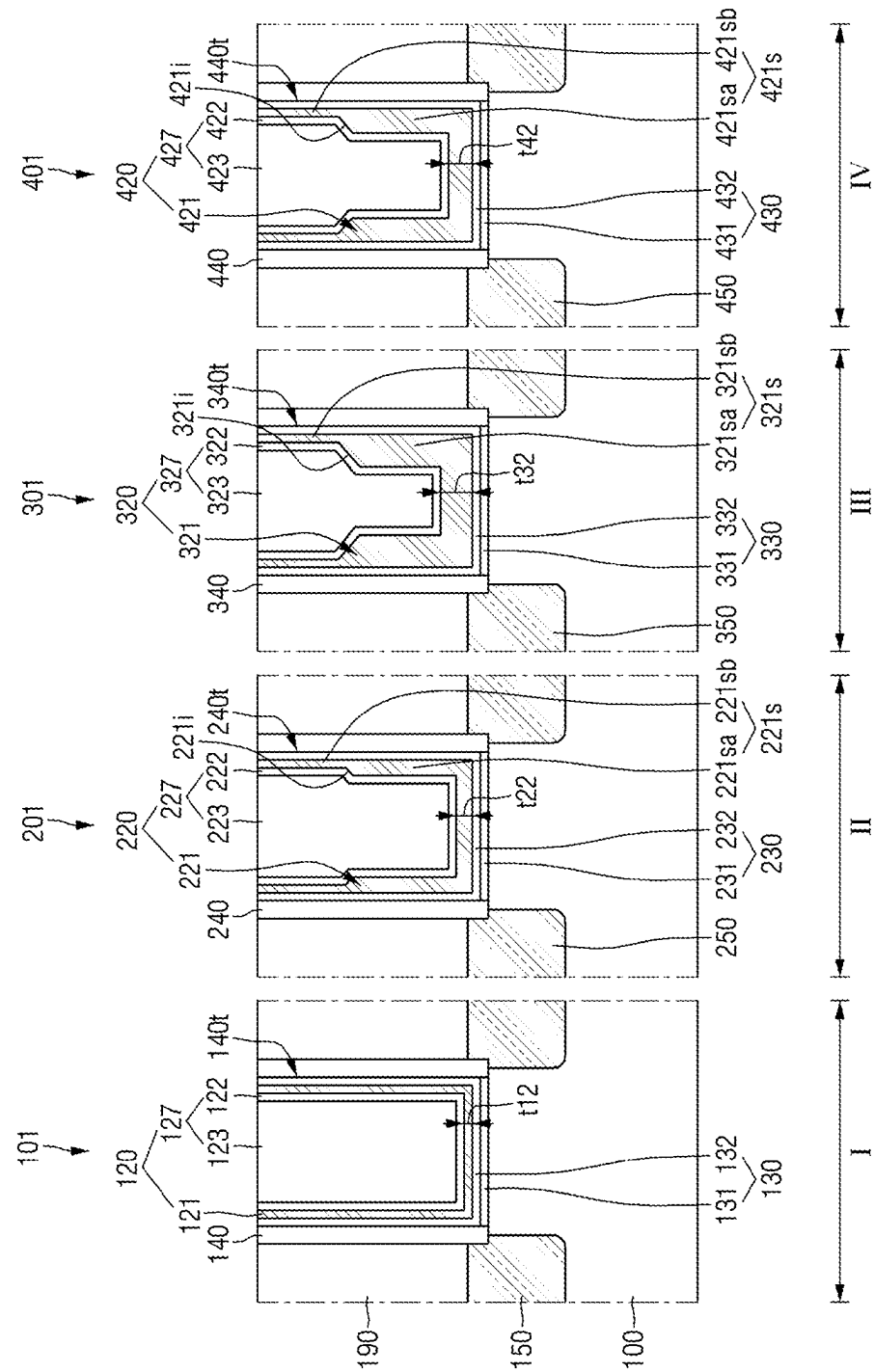


FIG. 25

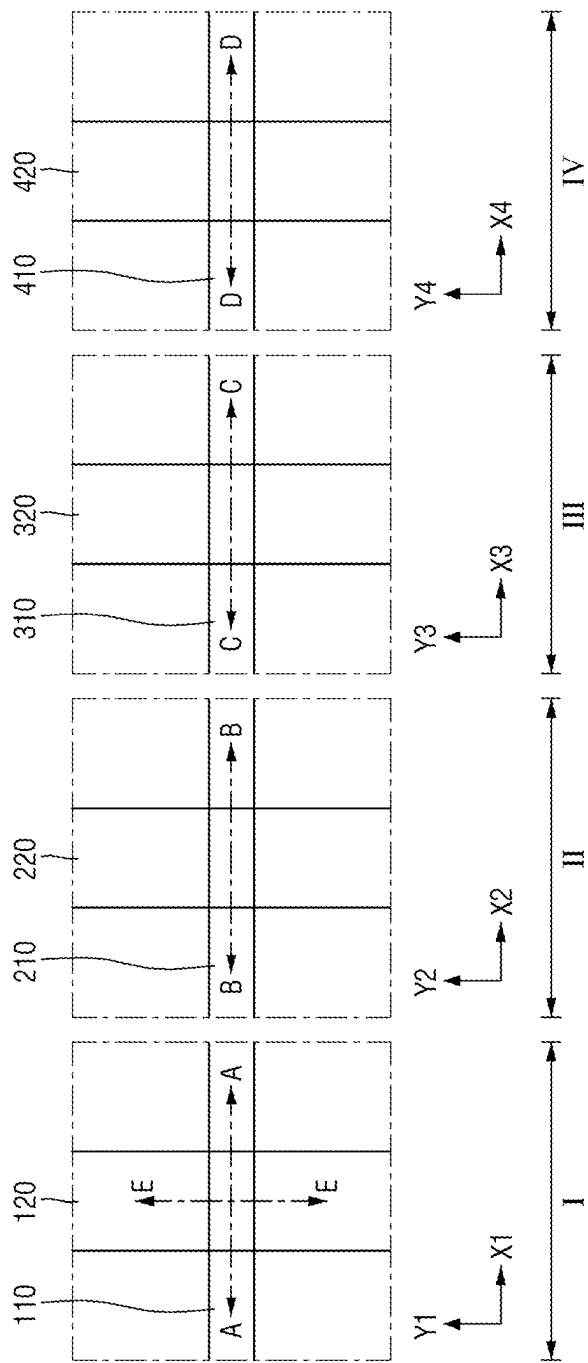


FIG. 26

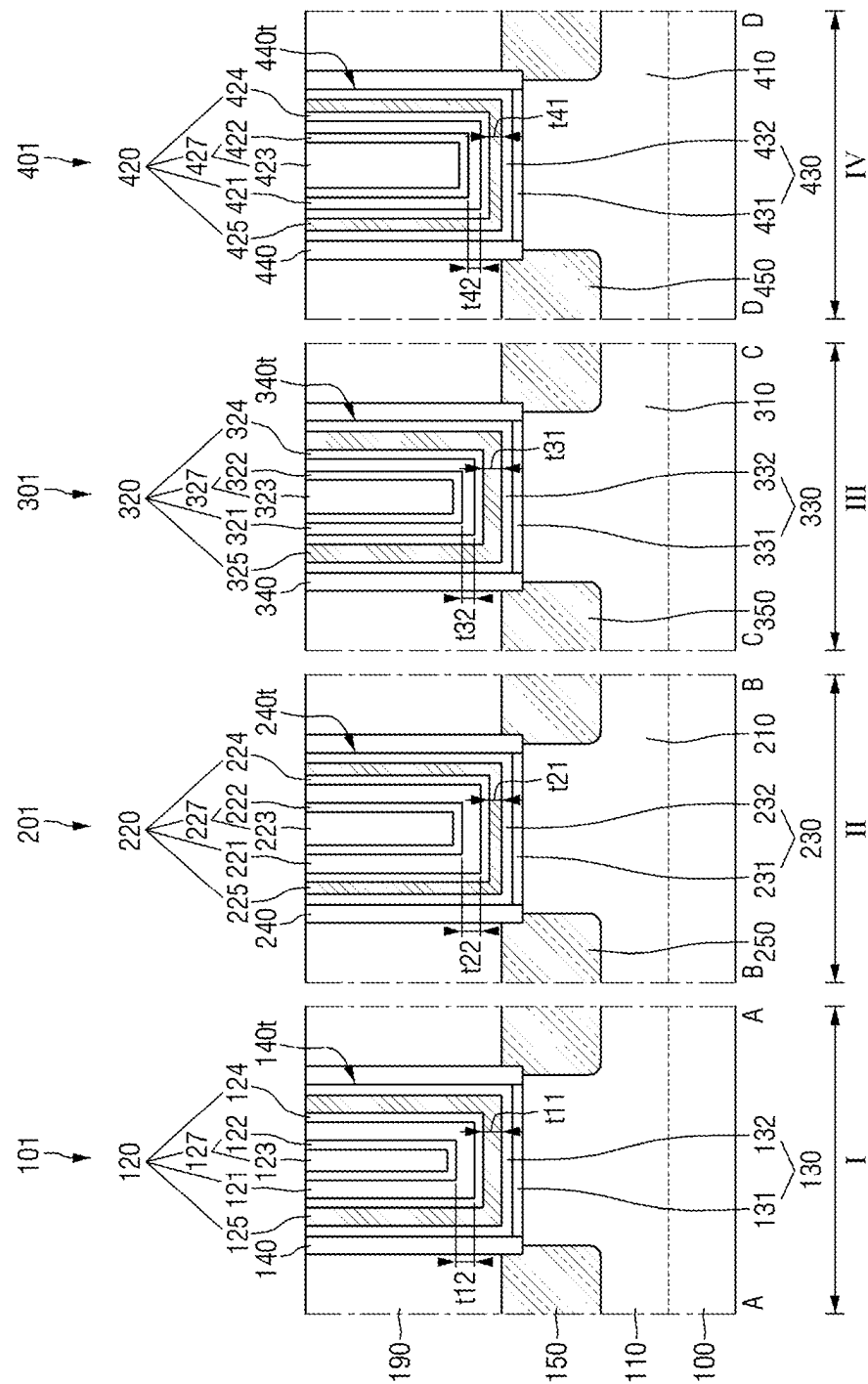


FIG. 27A

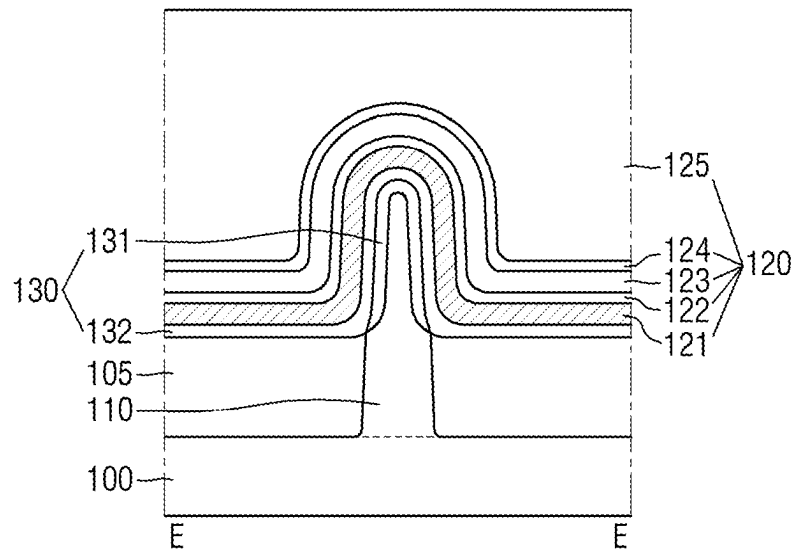


FIG. 27B

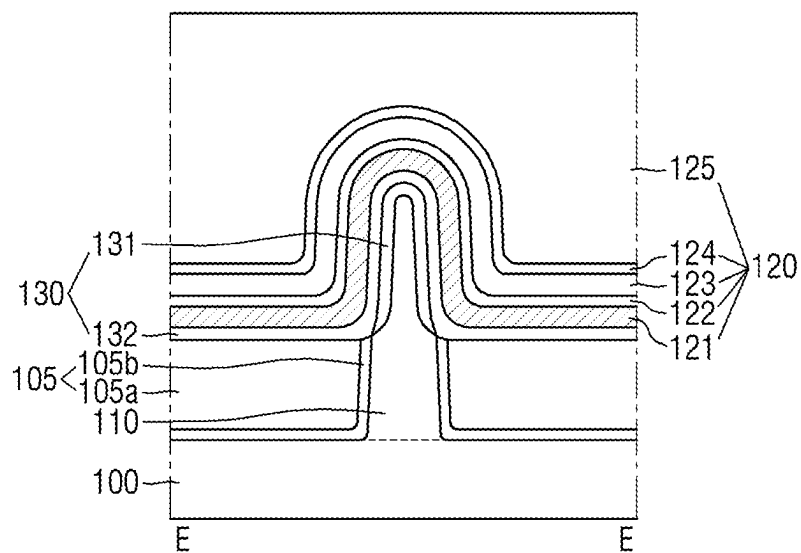


FIG. 27C

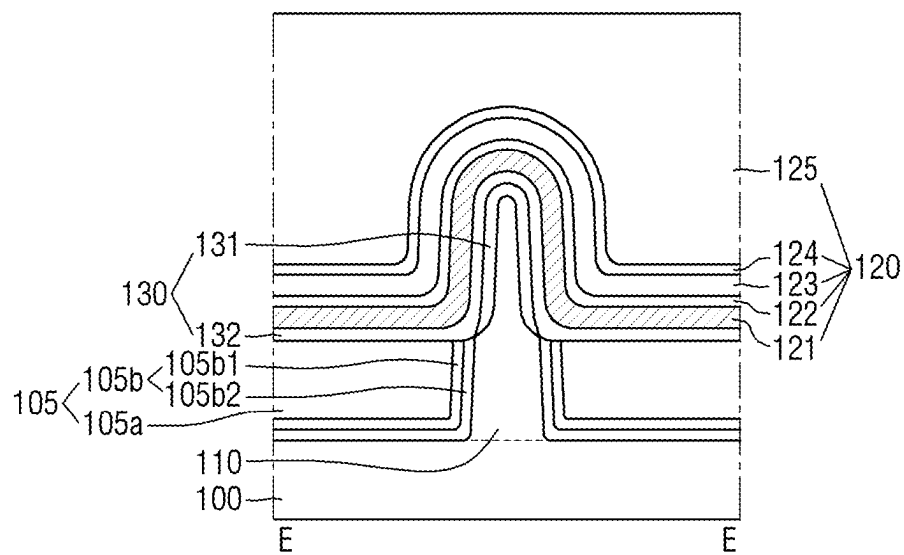


FIG. 28

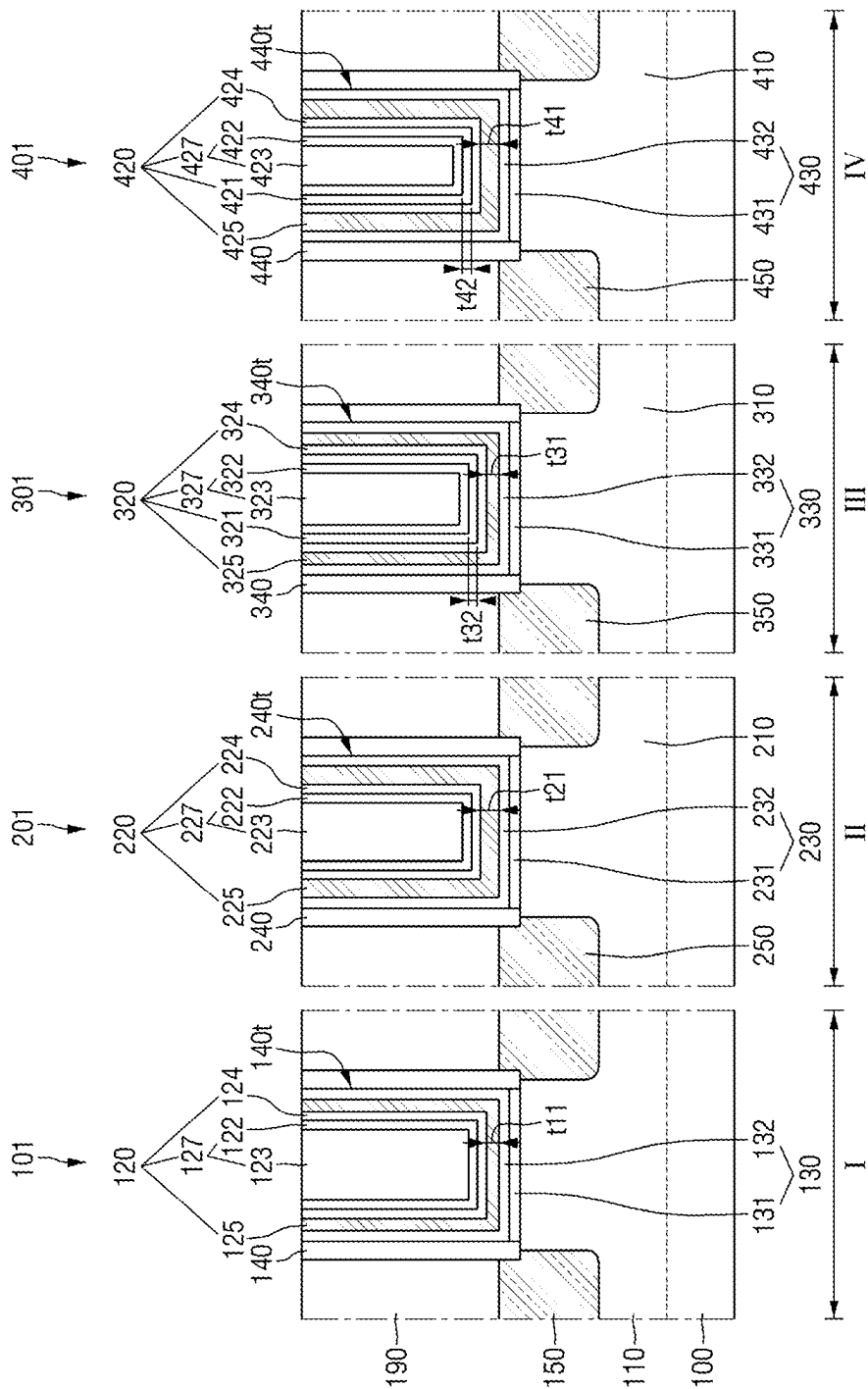


FIG. 29

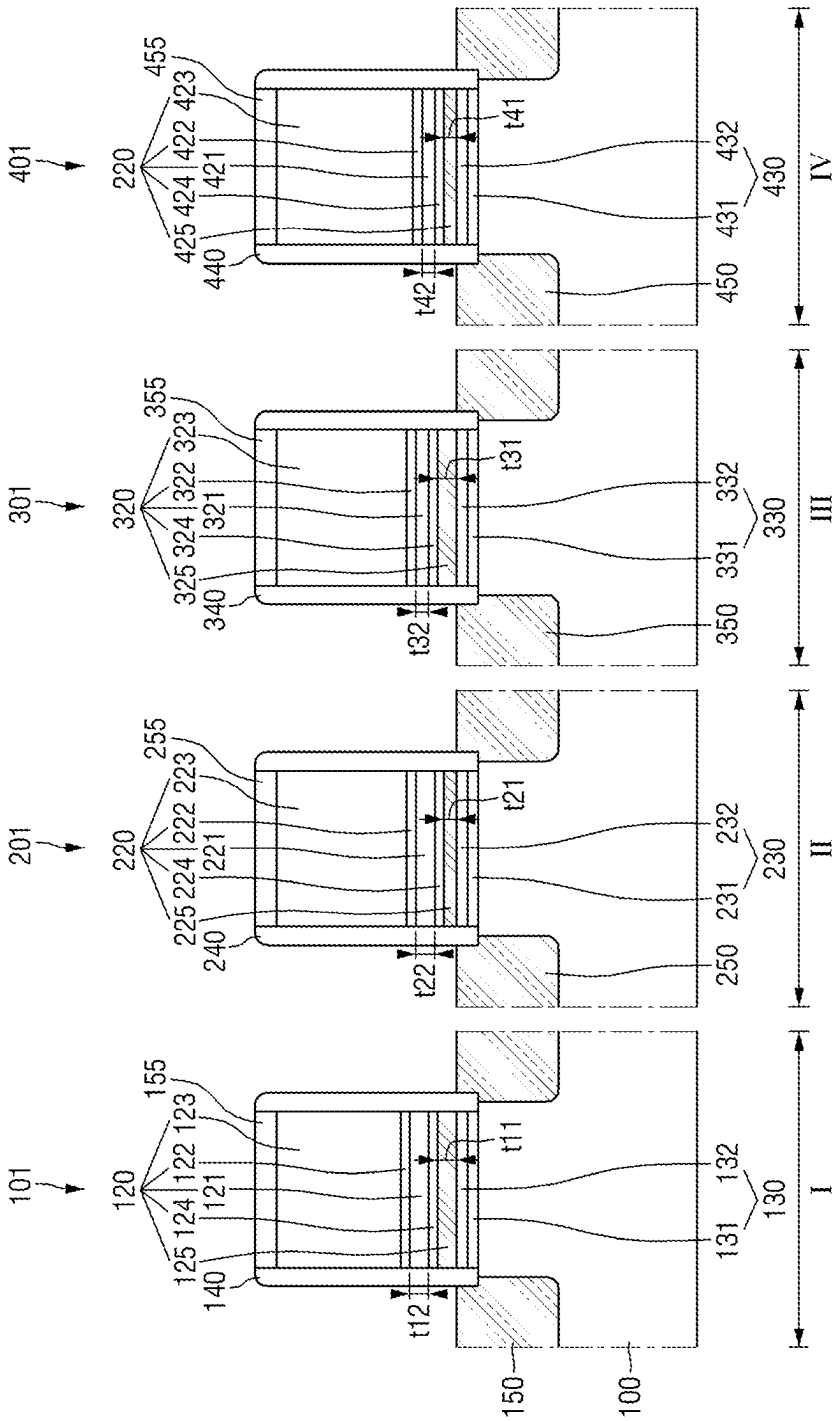


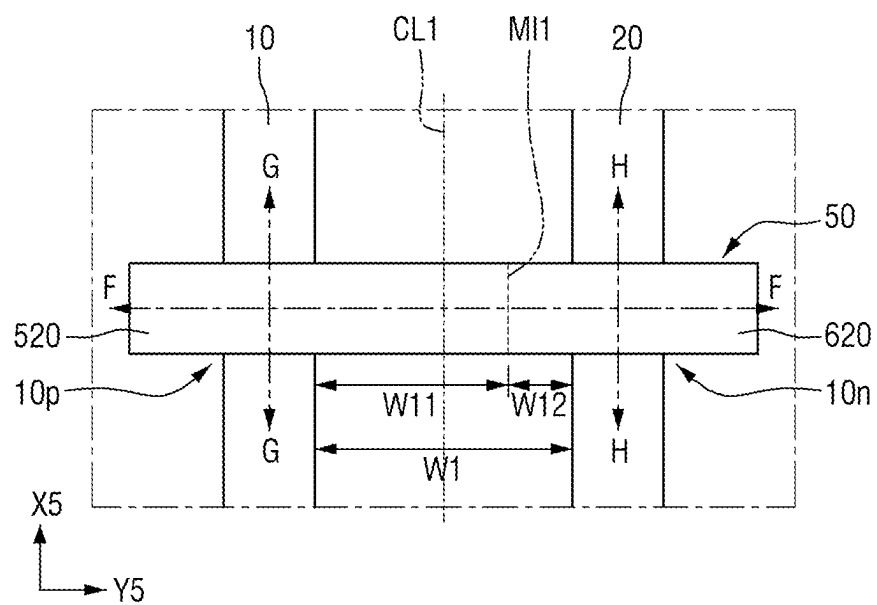
FIG. 30

FIG. 31A

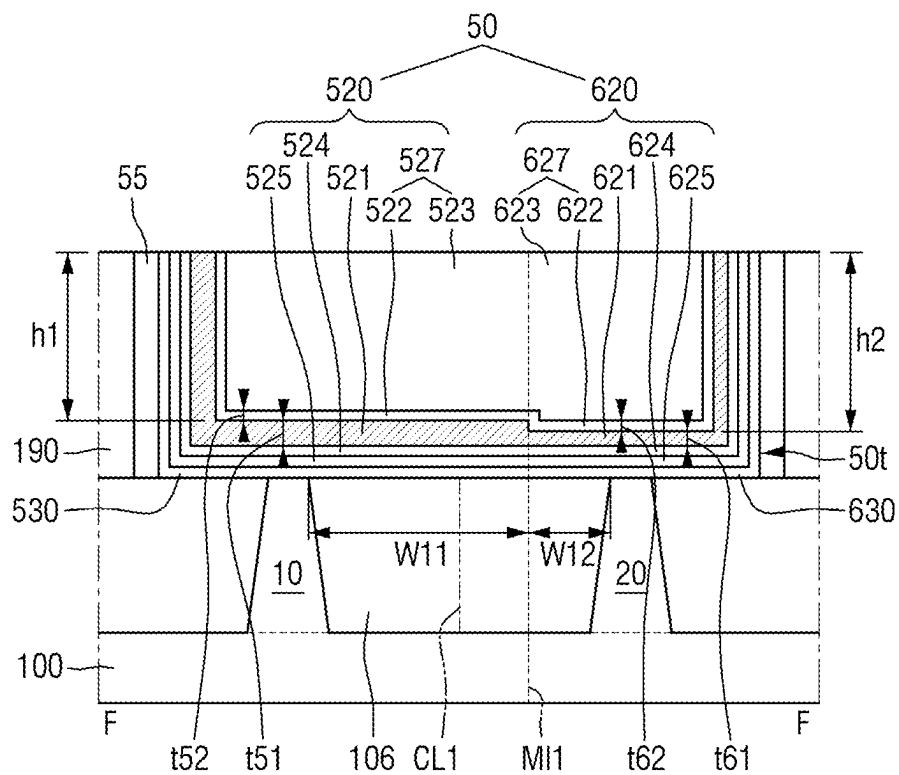


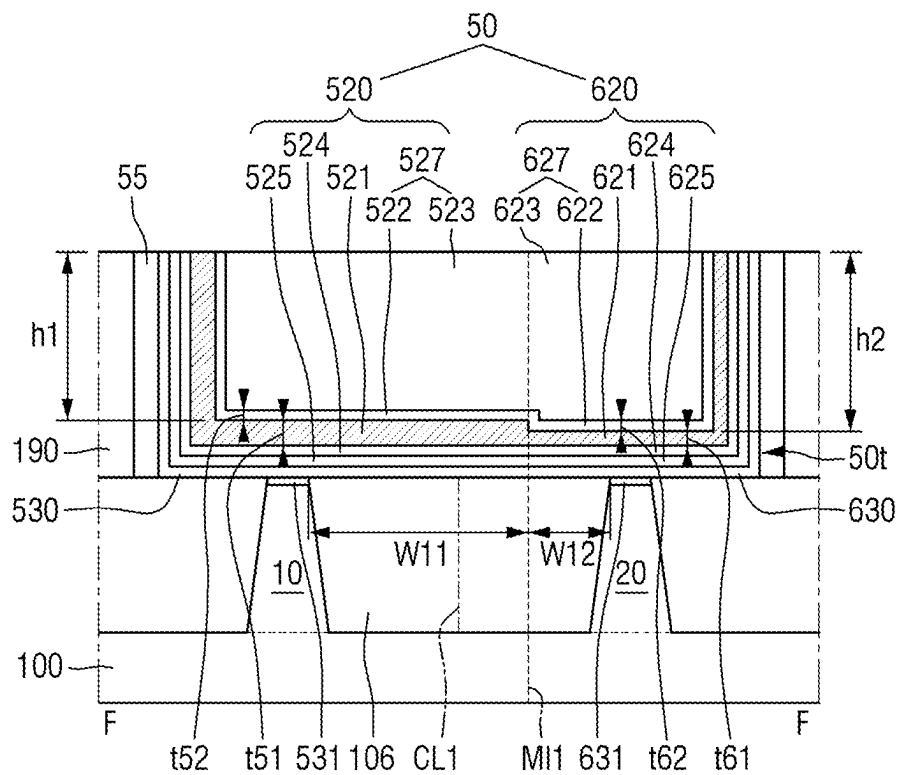
FIG. 31B

FIG. 32A

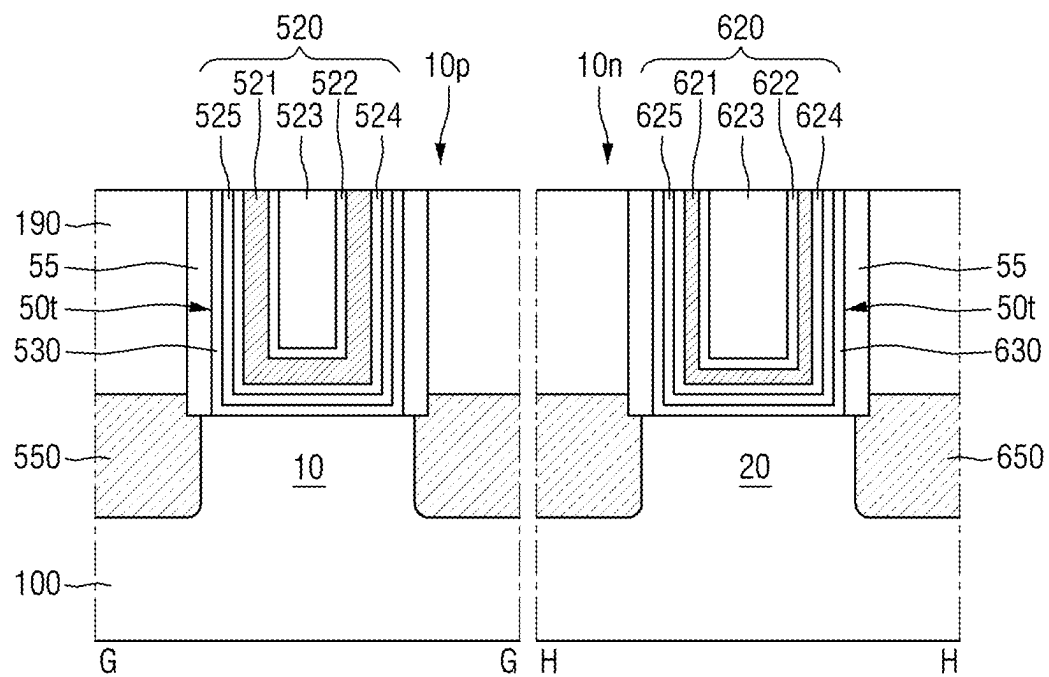


FIG. 32B

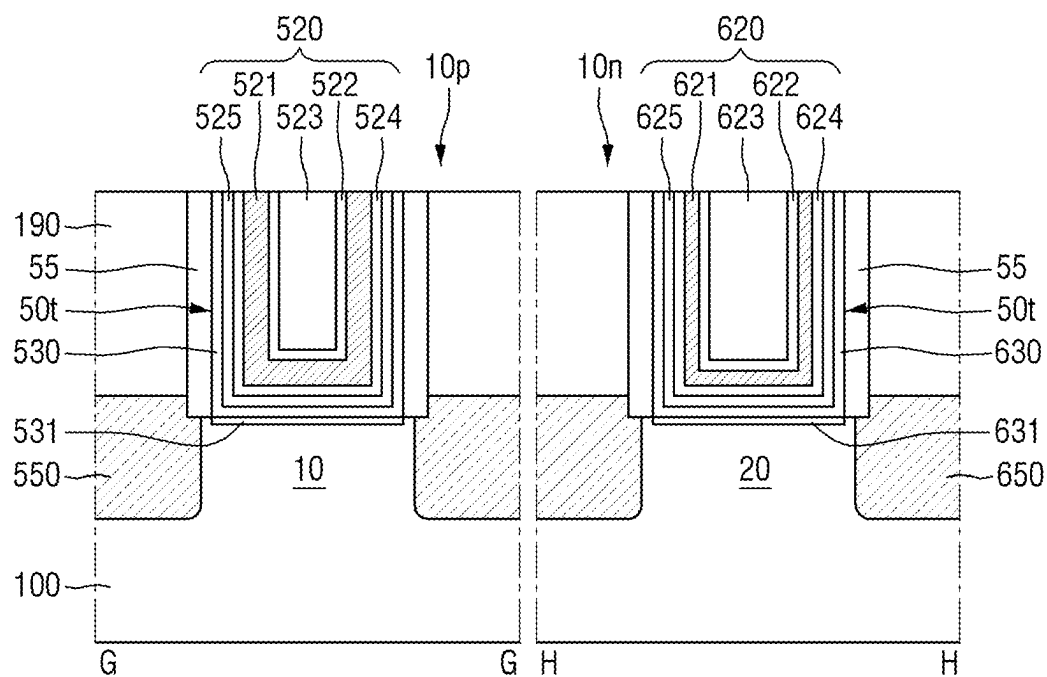


FIG. 33

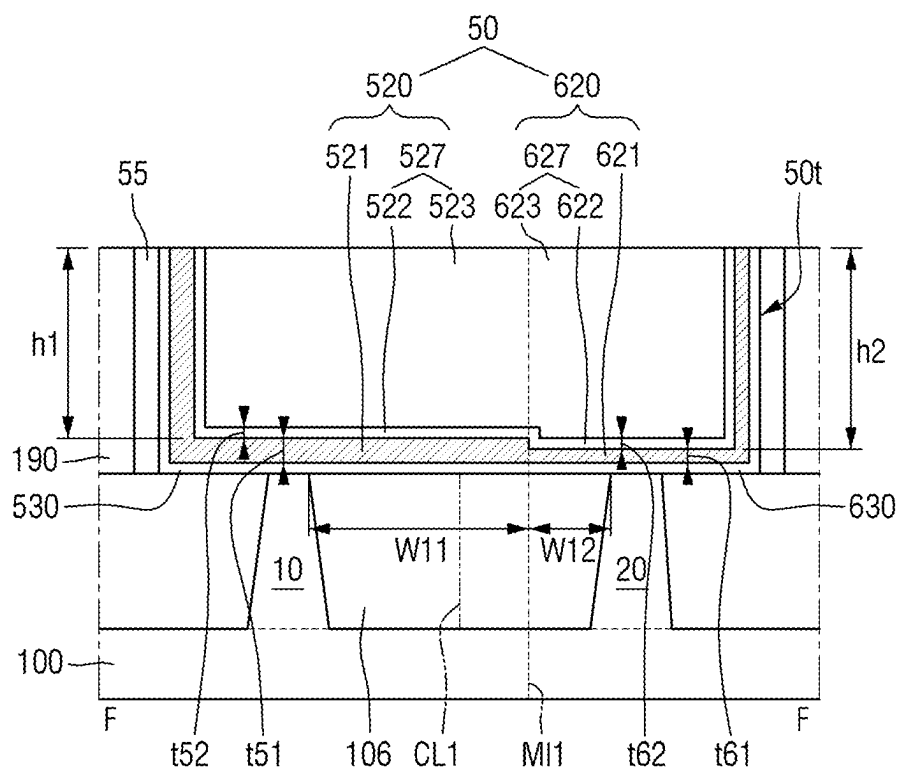


FIG. 34

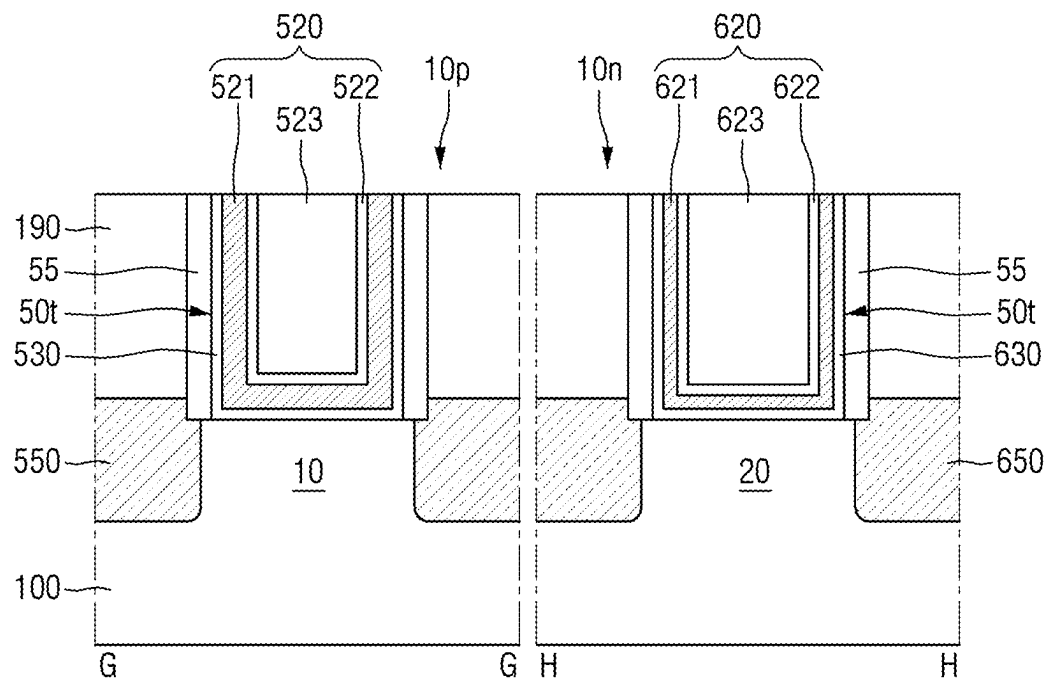


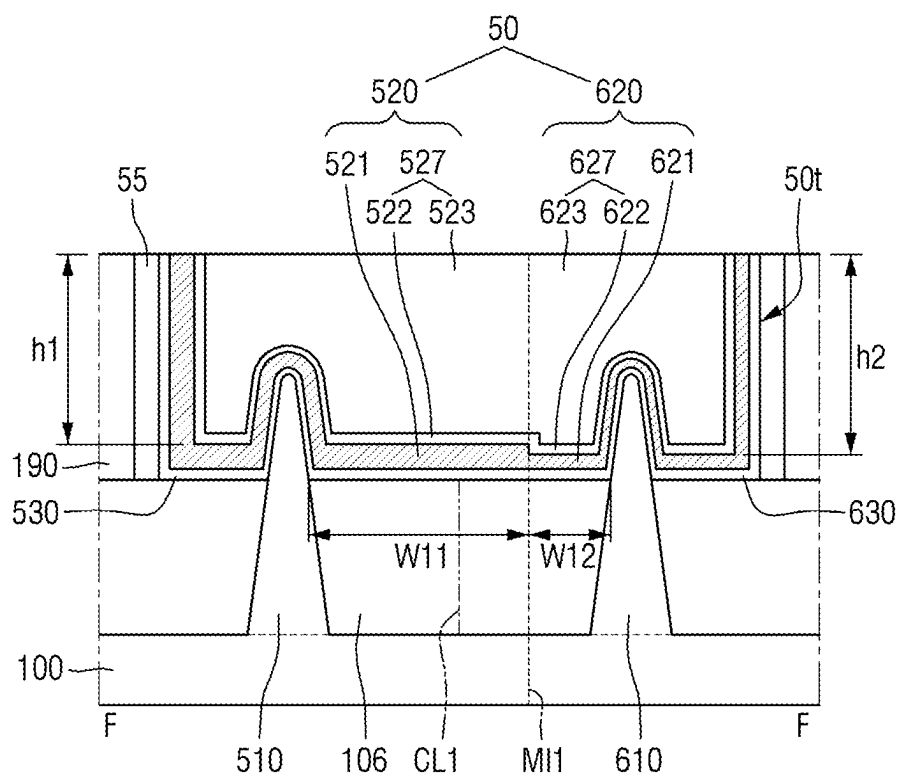
FIG. 39A

FIG. 39B

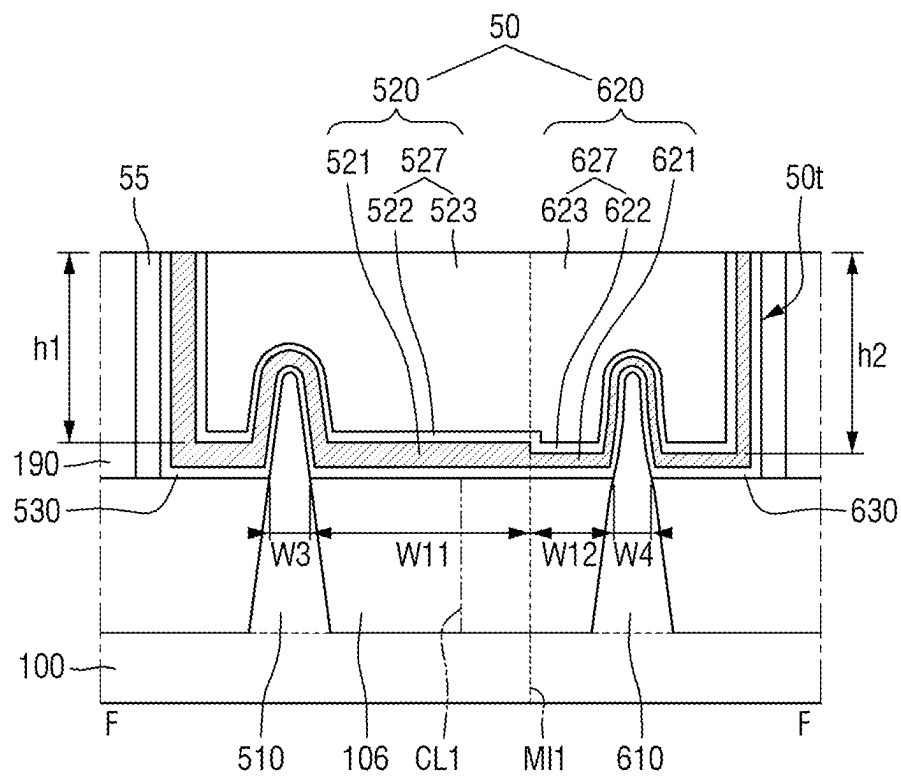


FIG. 40

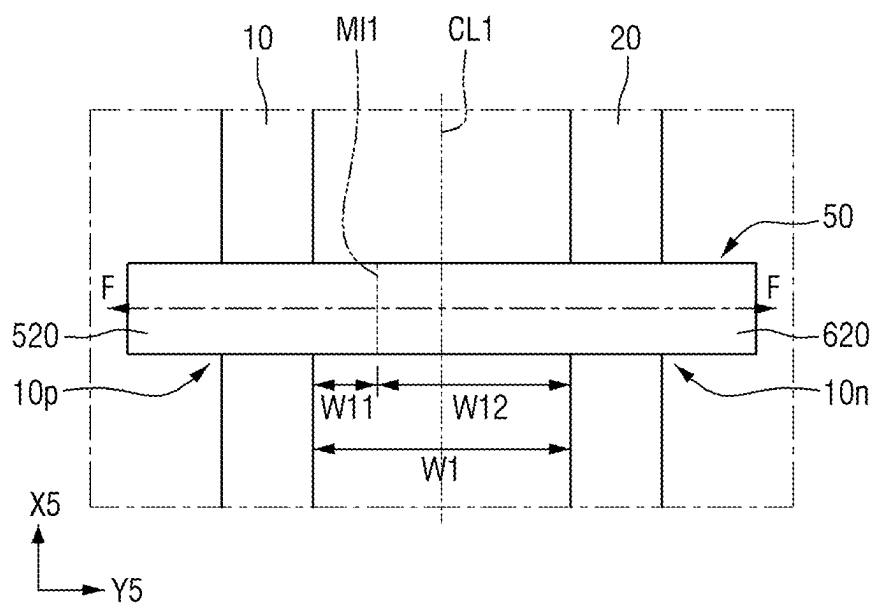


FIG. 41

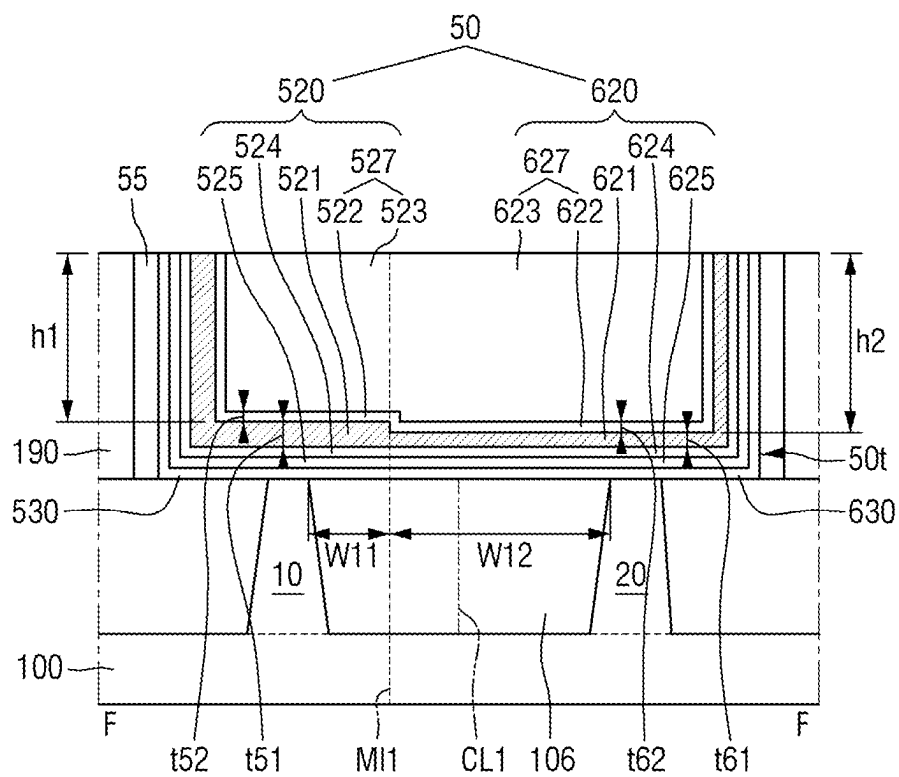


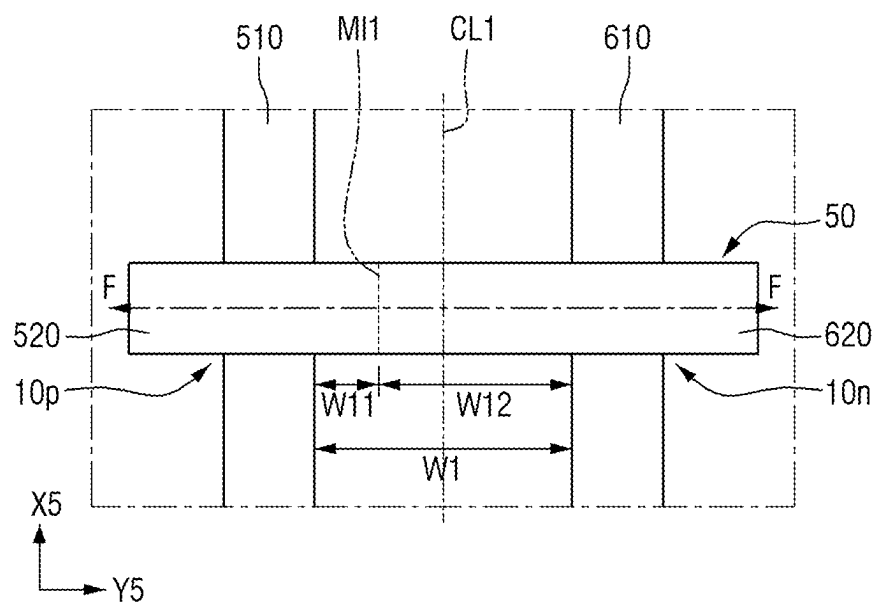
FIG. 43

FIG. 46

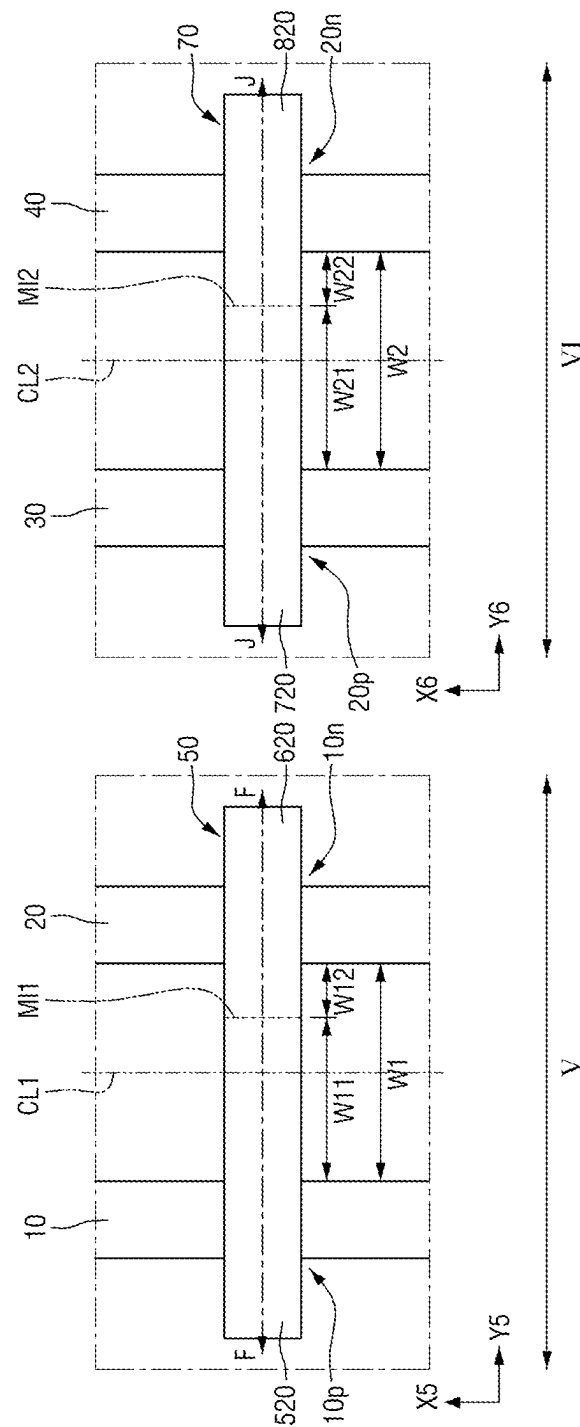


FIG. 50

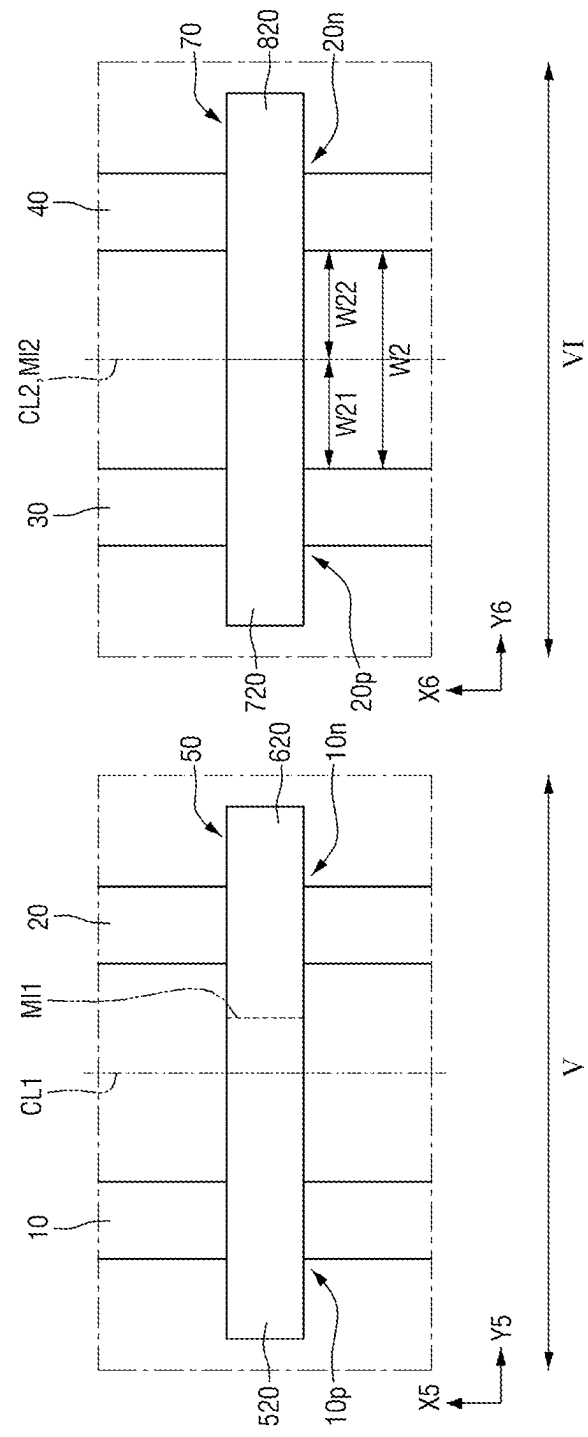


FIG. 51

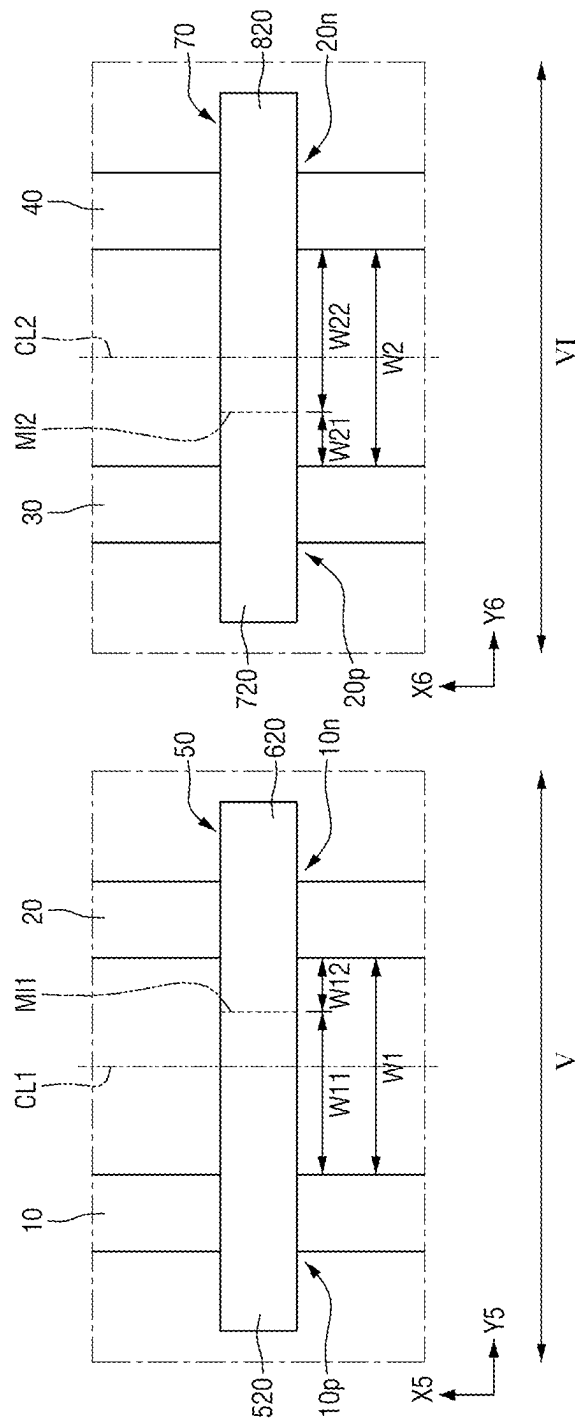


FIG. 52

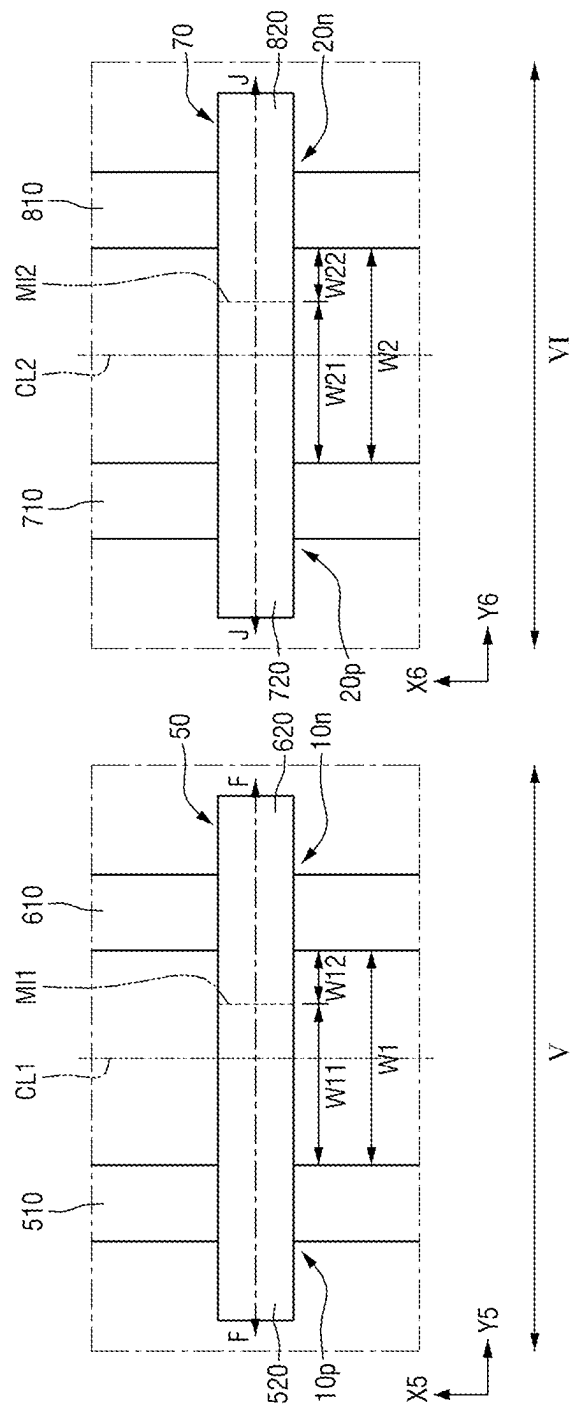


FIG. 53

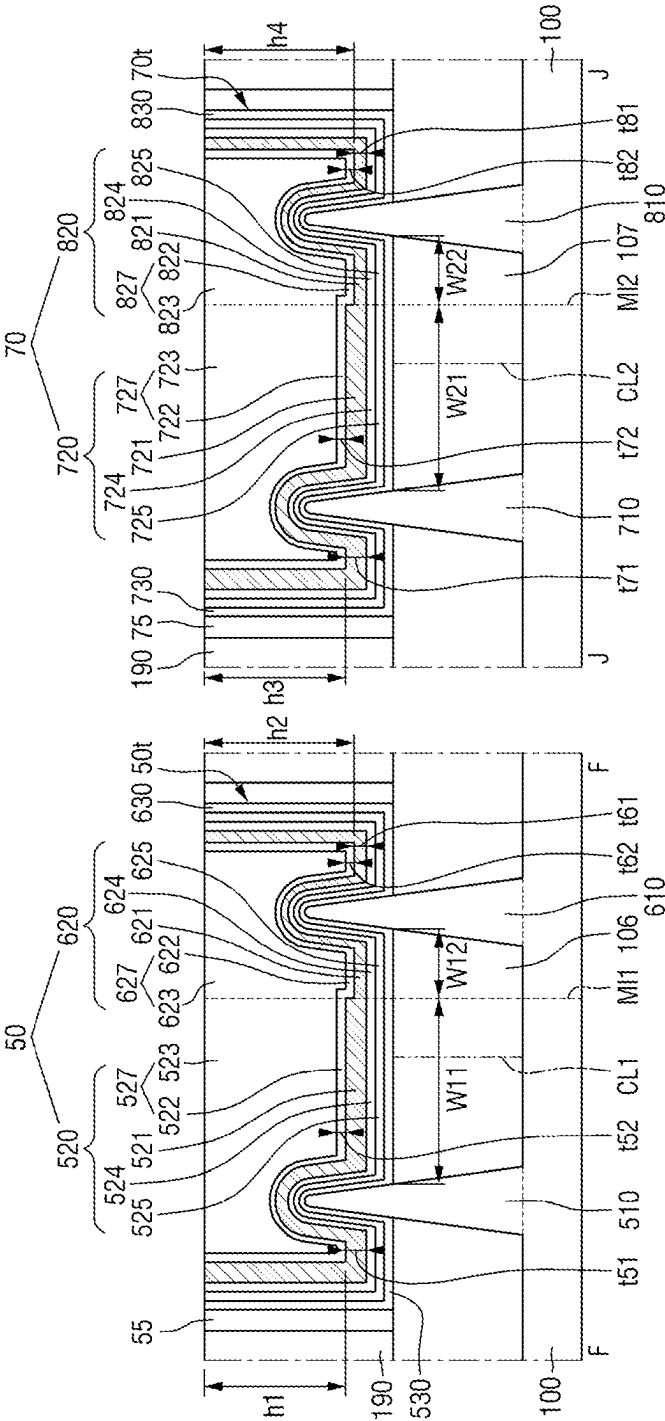


FIG. 54

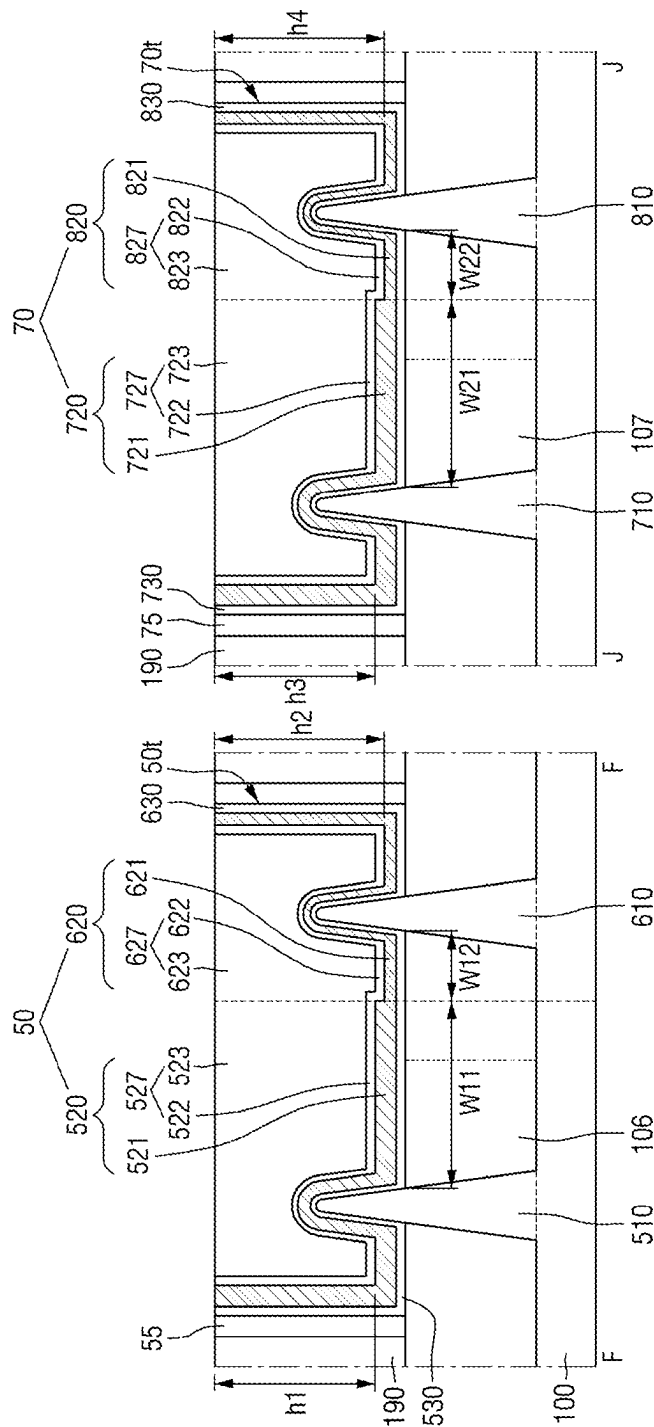


FIG. 55

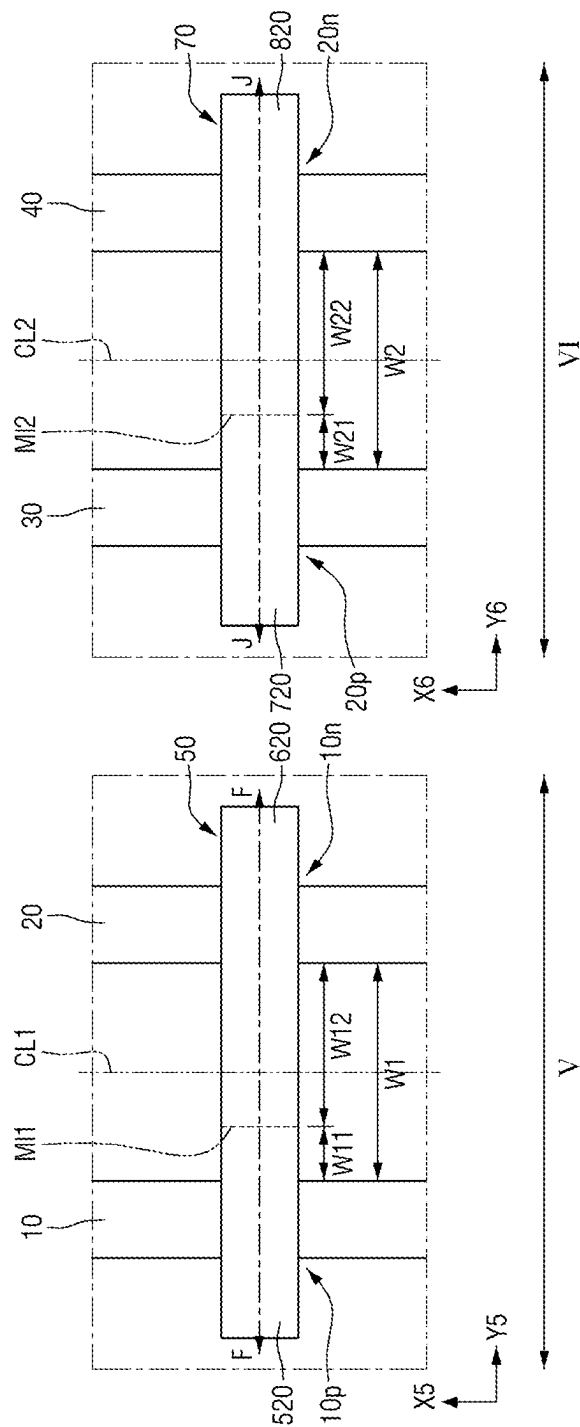


FIG. 56

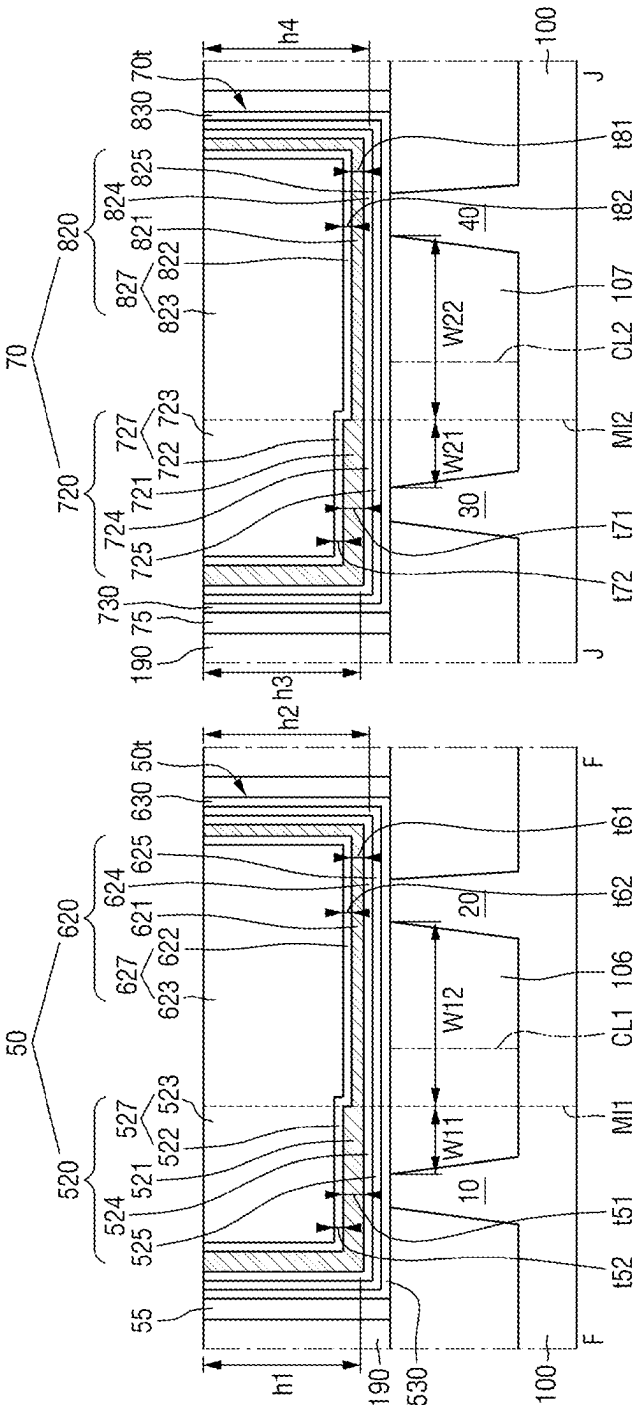


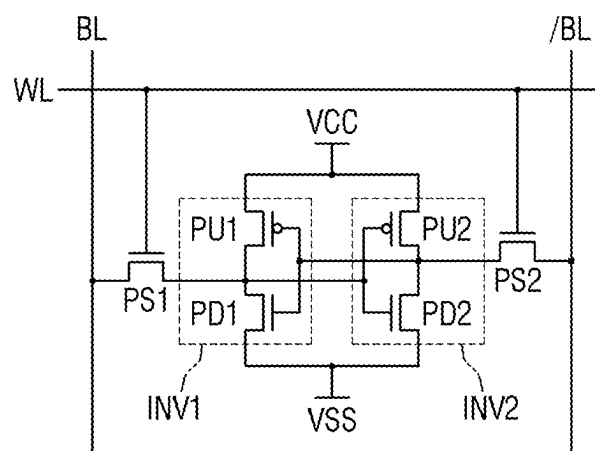
FIG. 57

FIG. 58

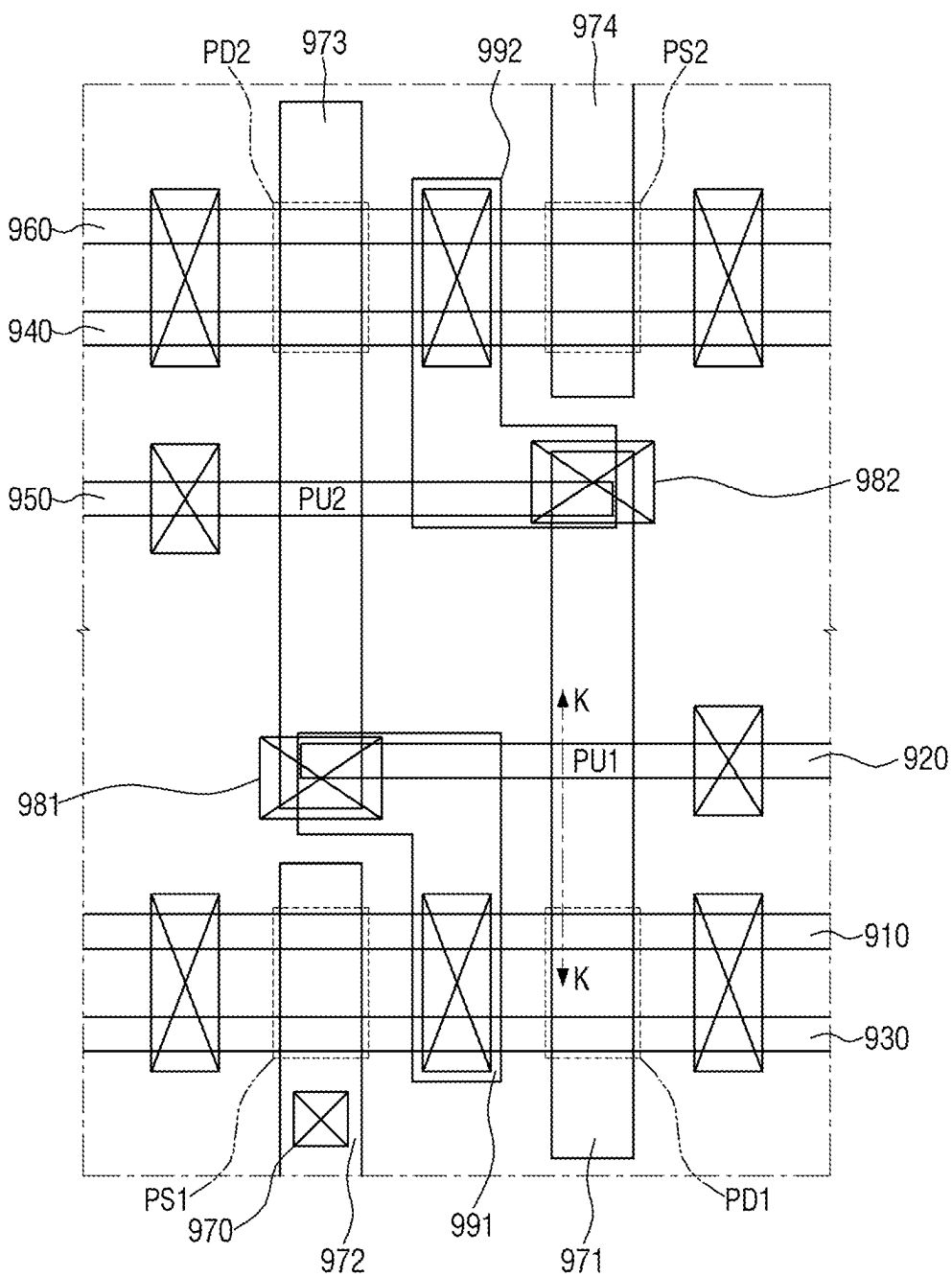


FIG. 60

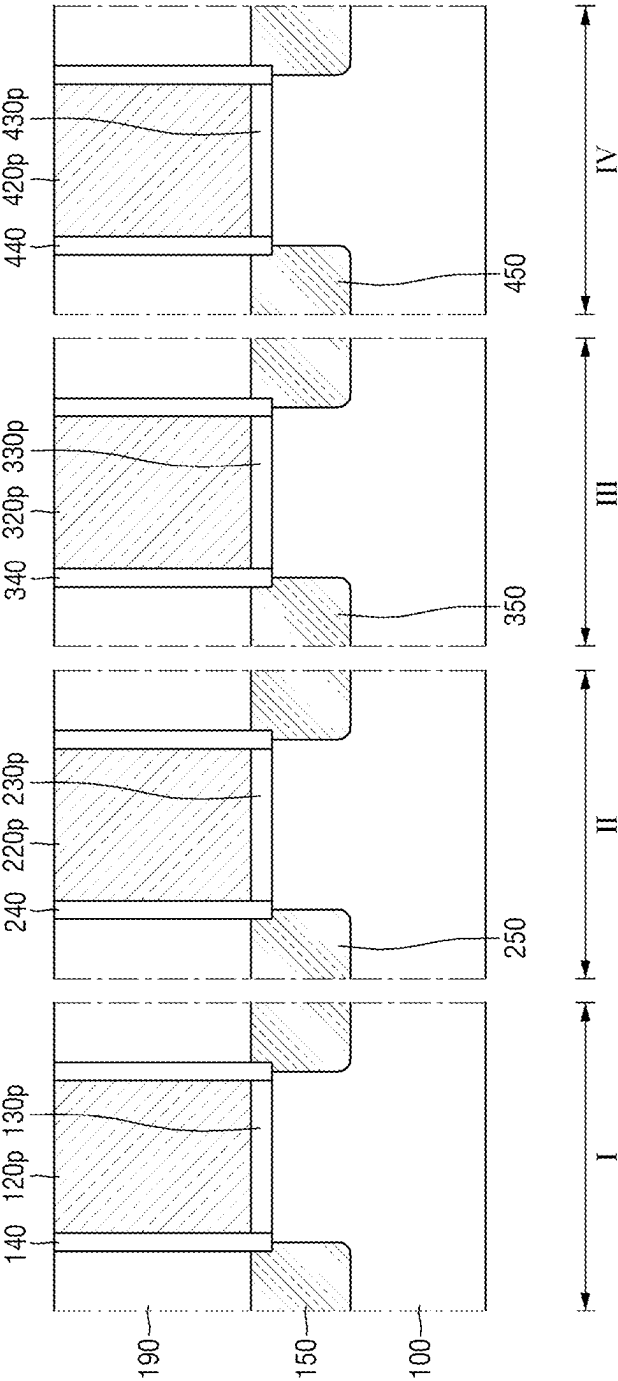


FIG. 61

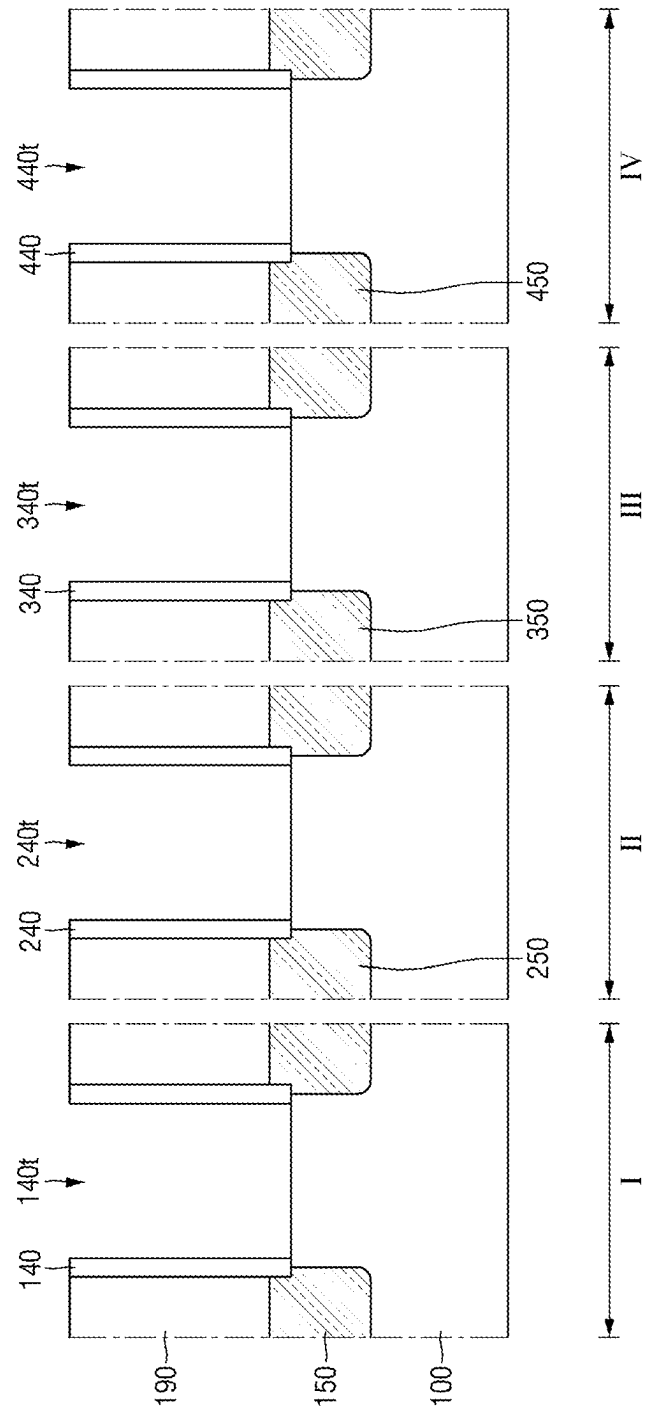


FIG. 62

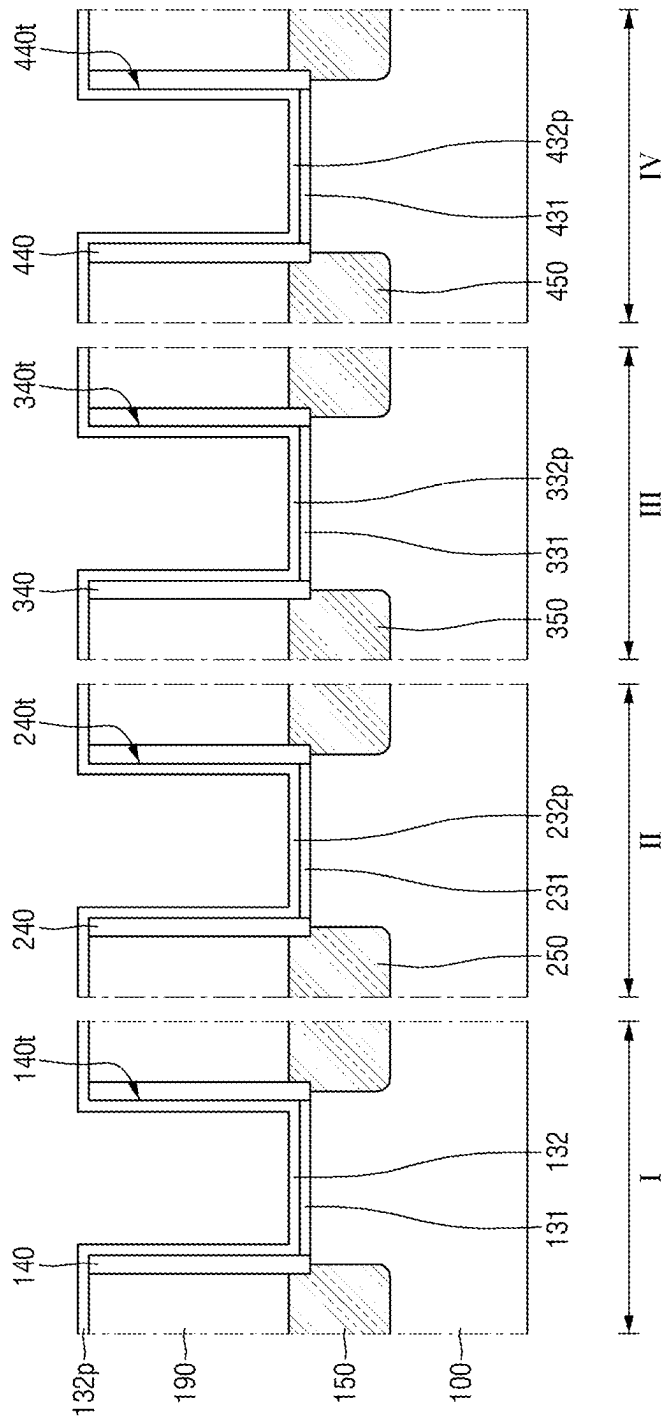


FIG. 63

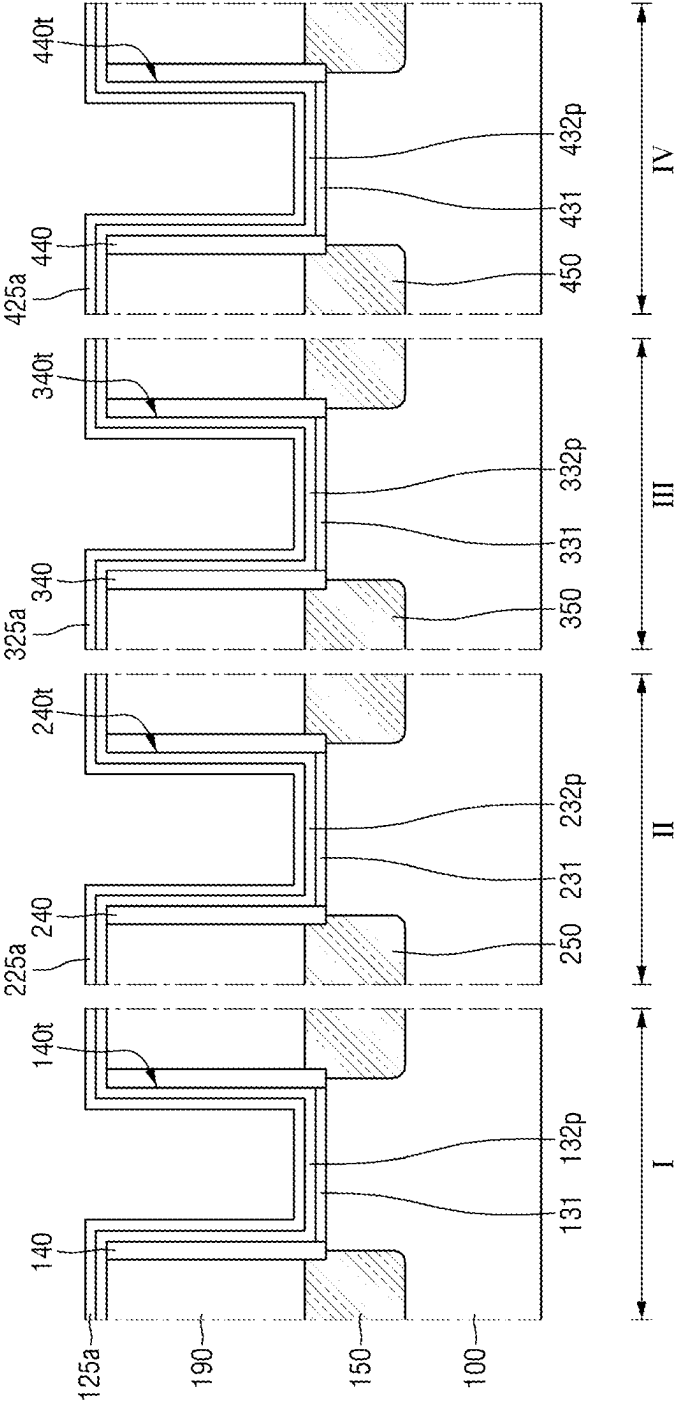


FIG. 64

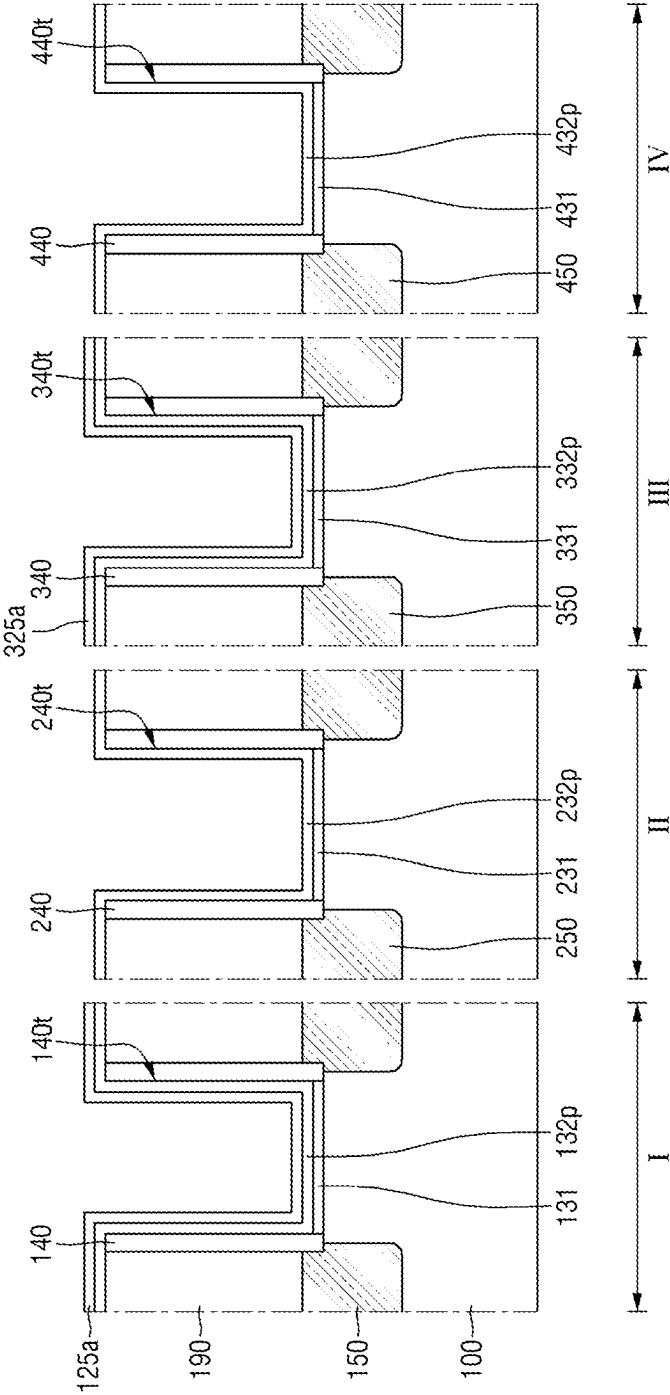


FIG. 65

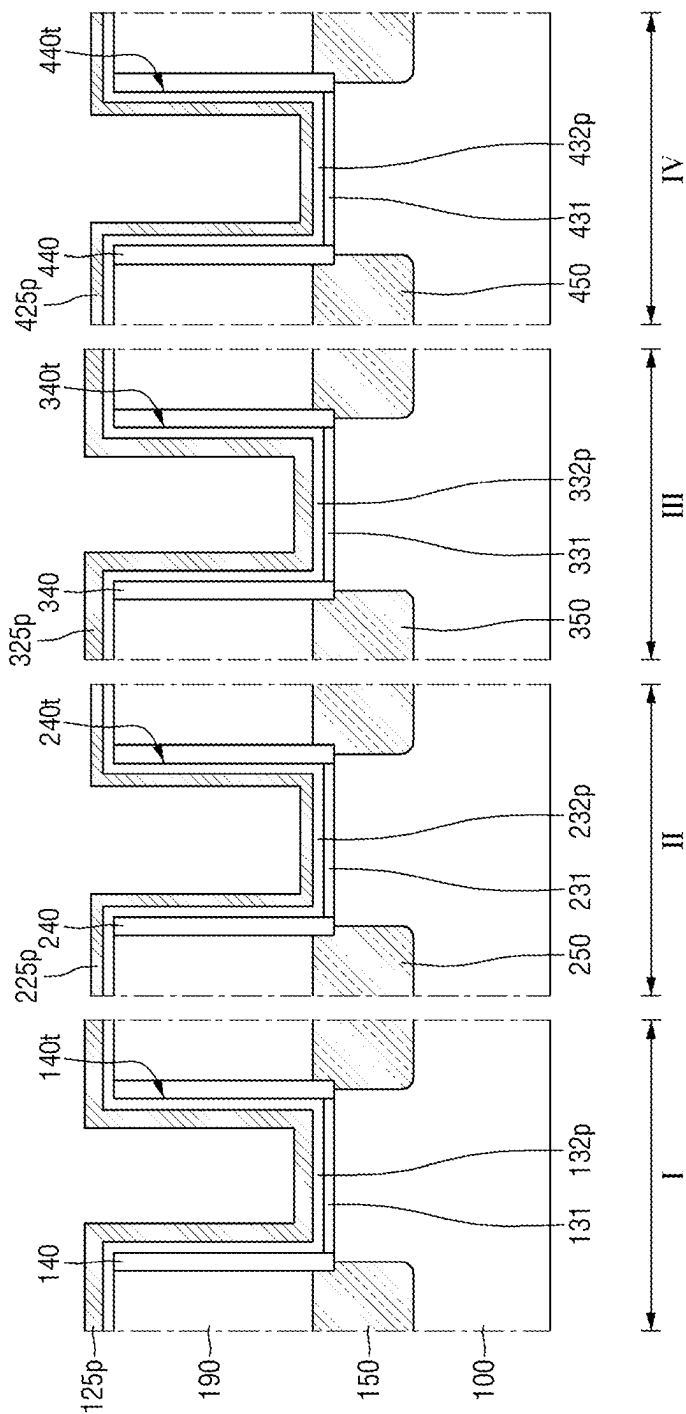


FIG. 66

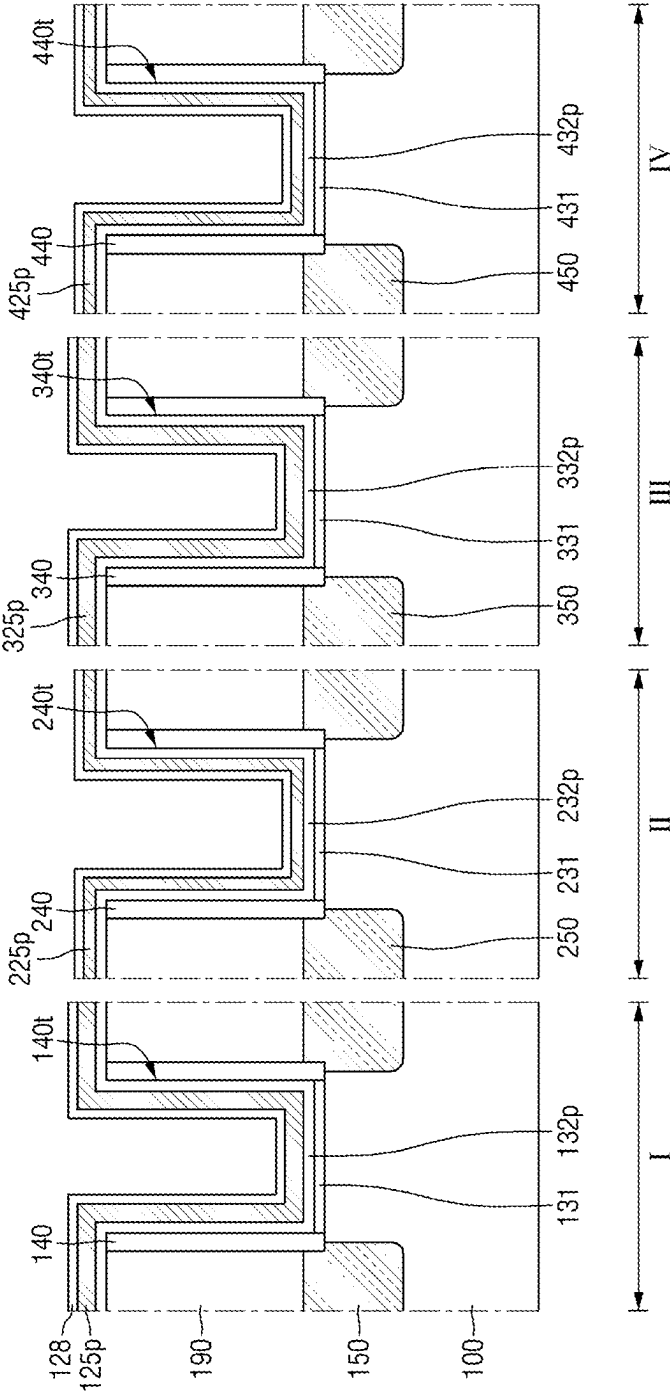


FIG. 67

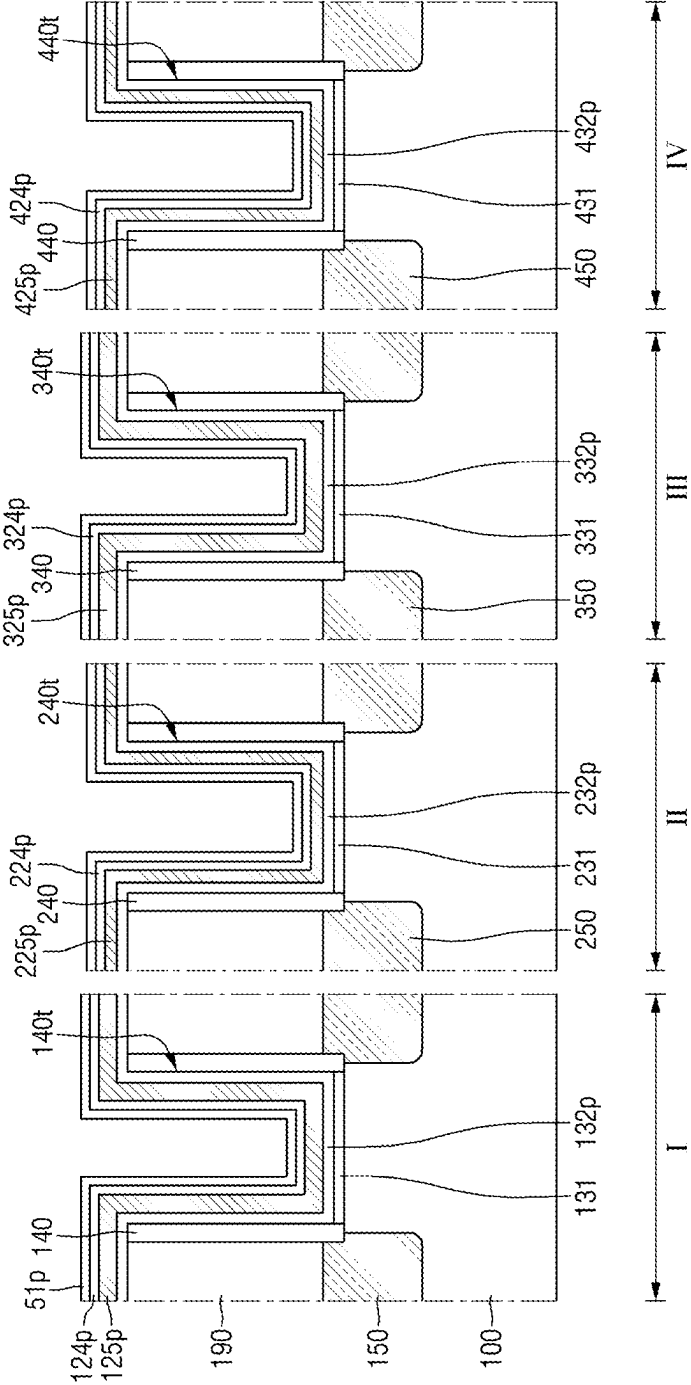


FIG. 68

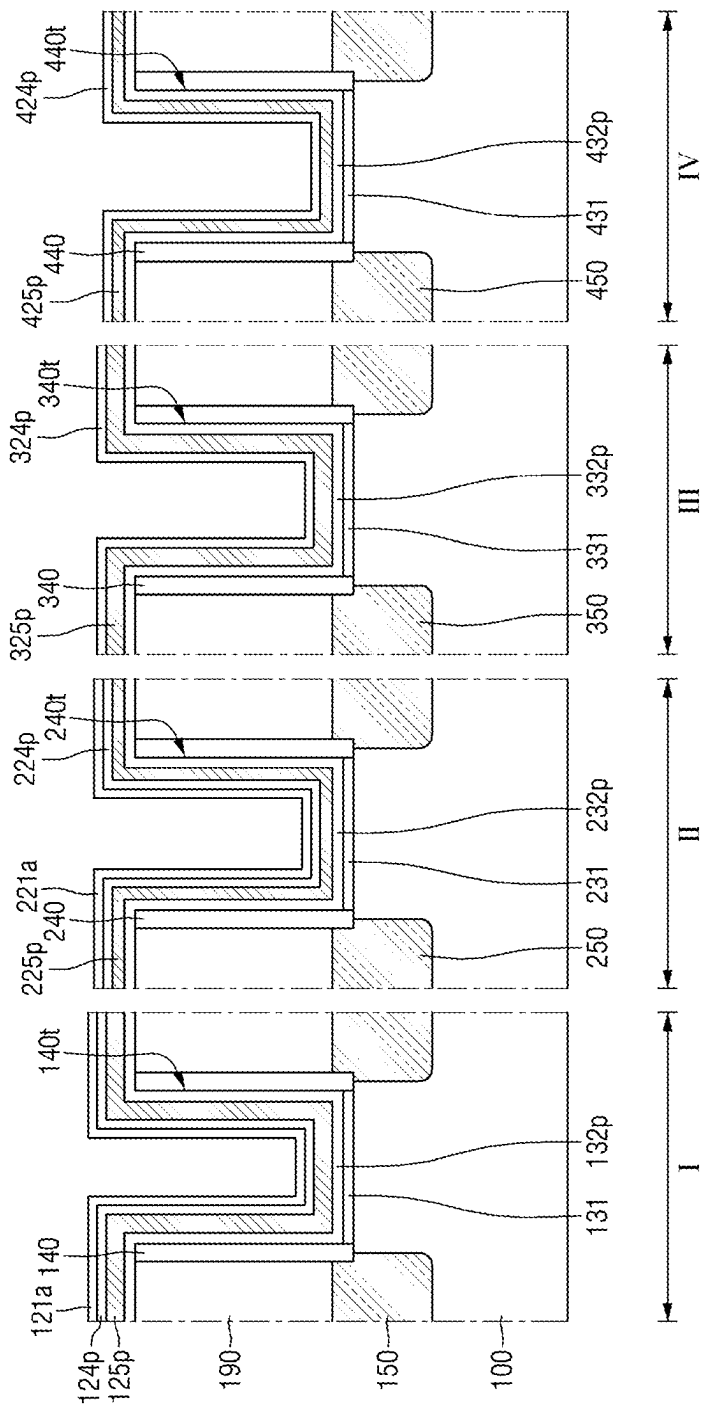


FIG. 69

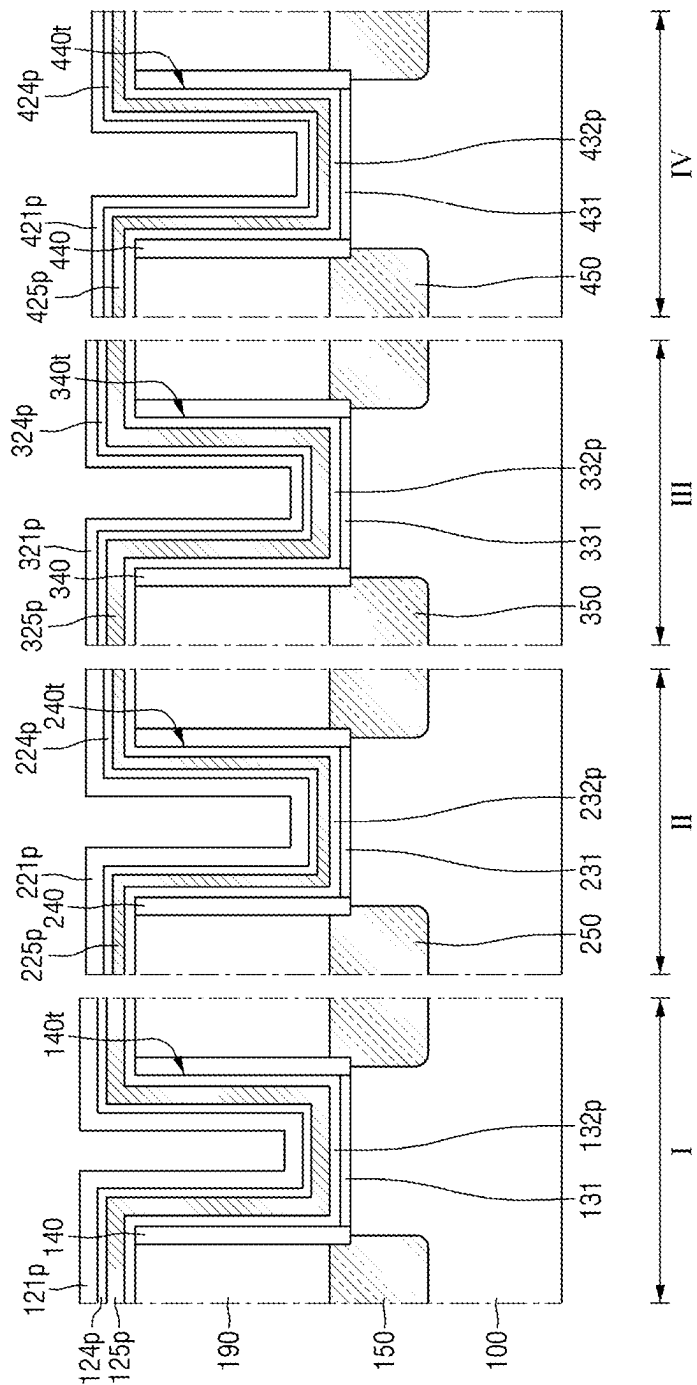


FIG. 70

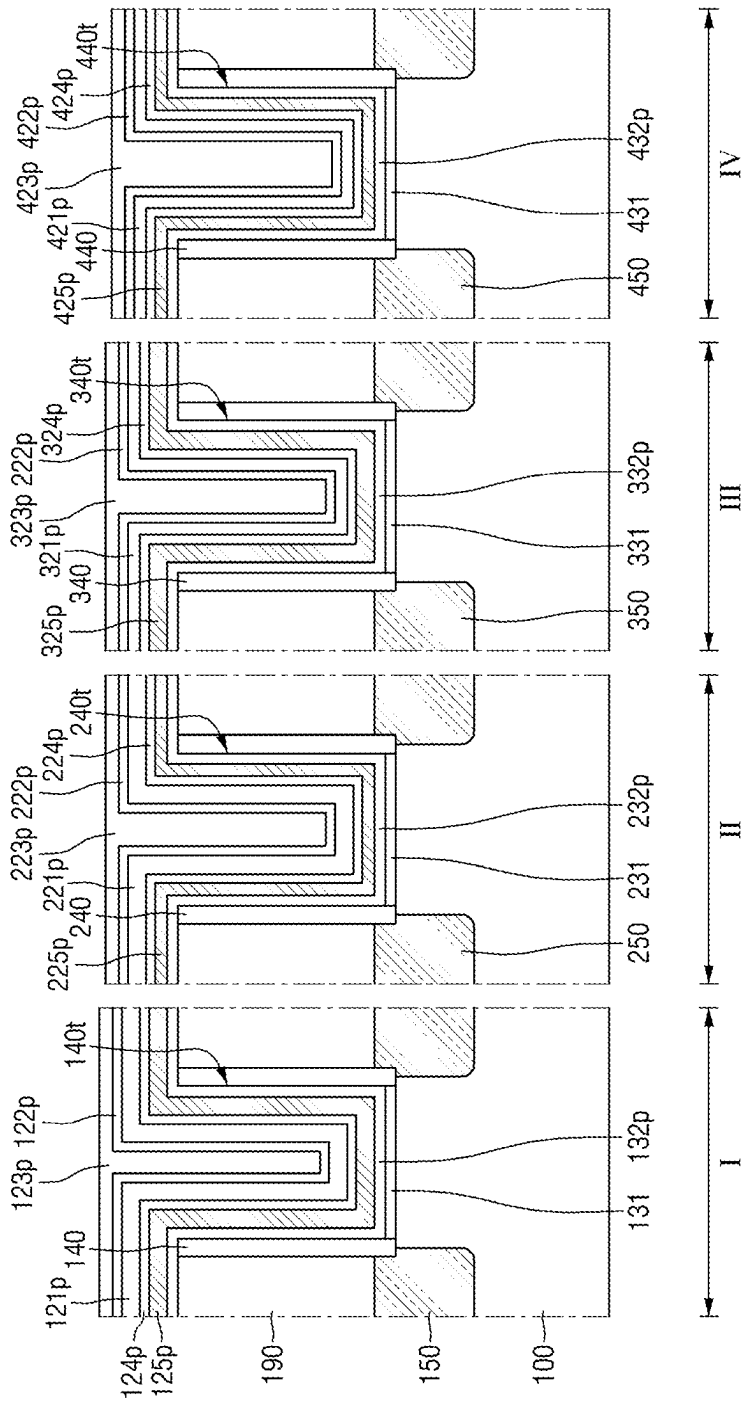


FIG. 71

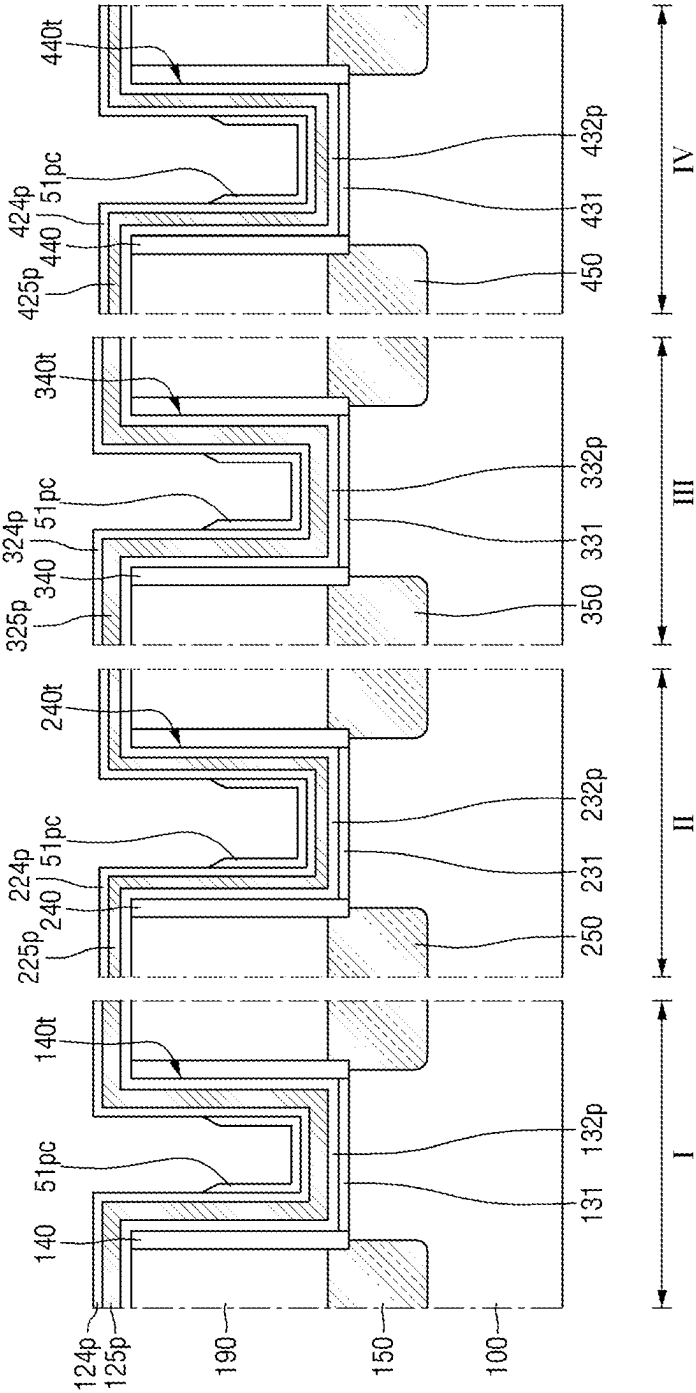


FIG. 72

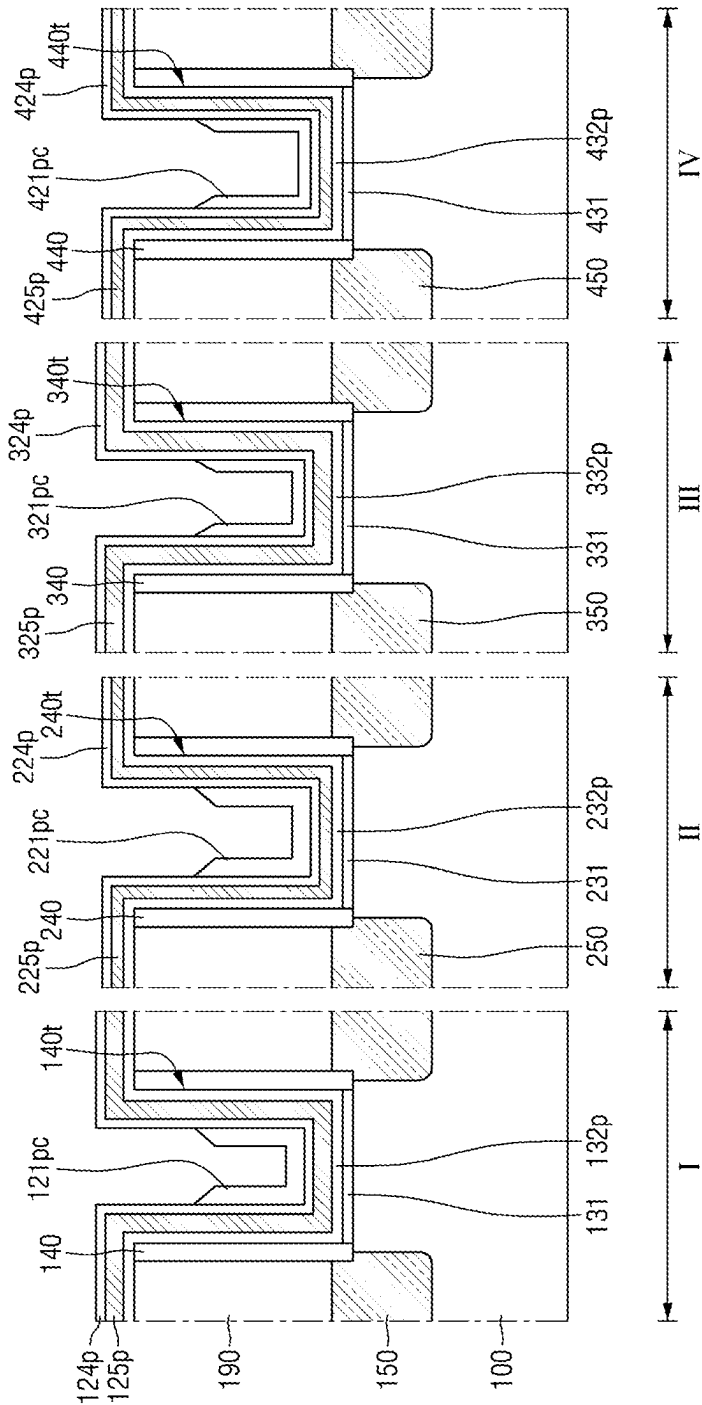


FIG. 73

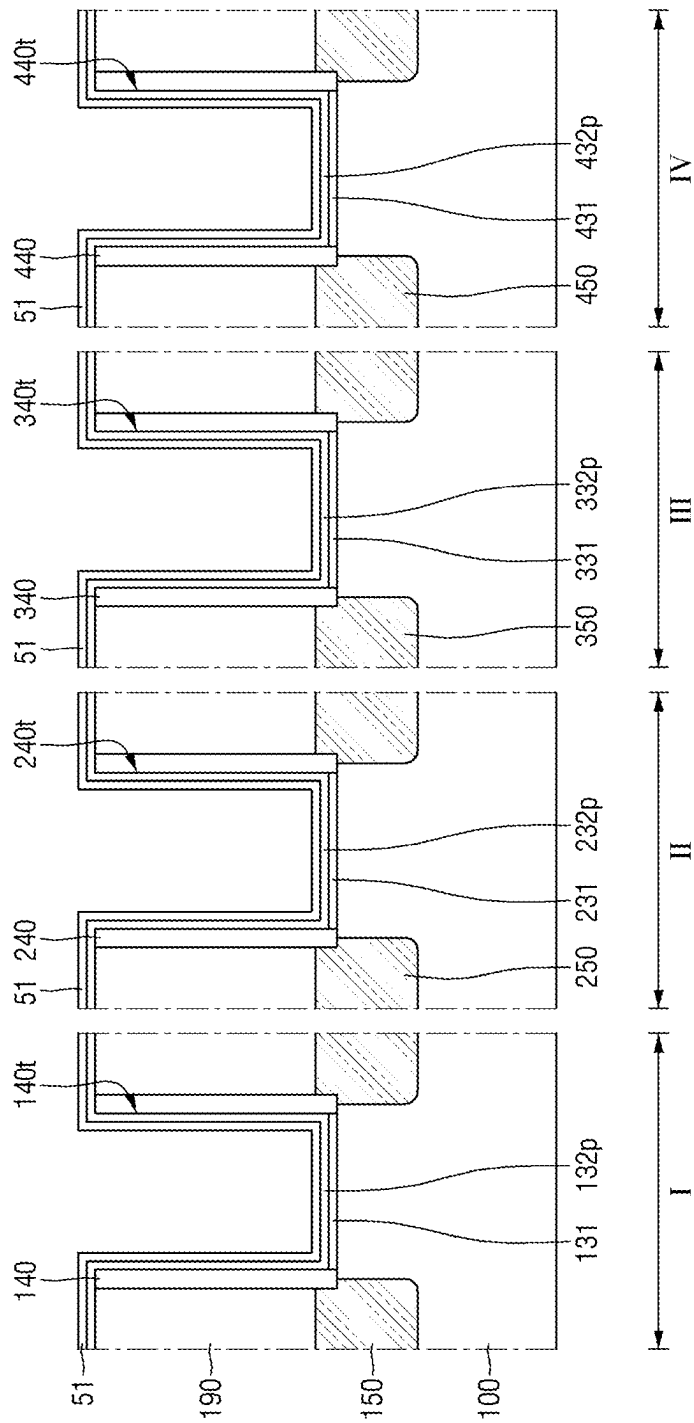


FIG. 74

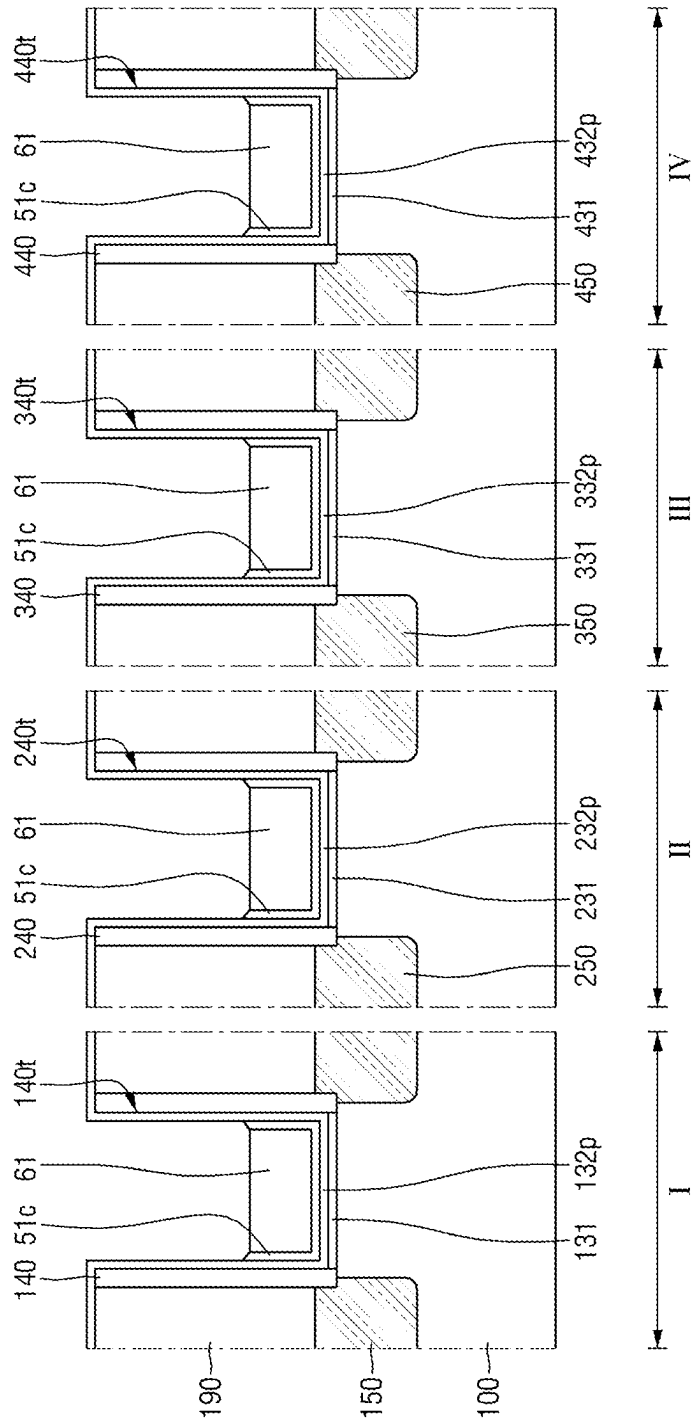


FIG. 75

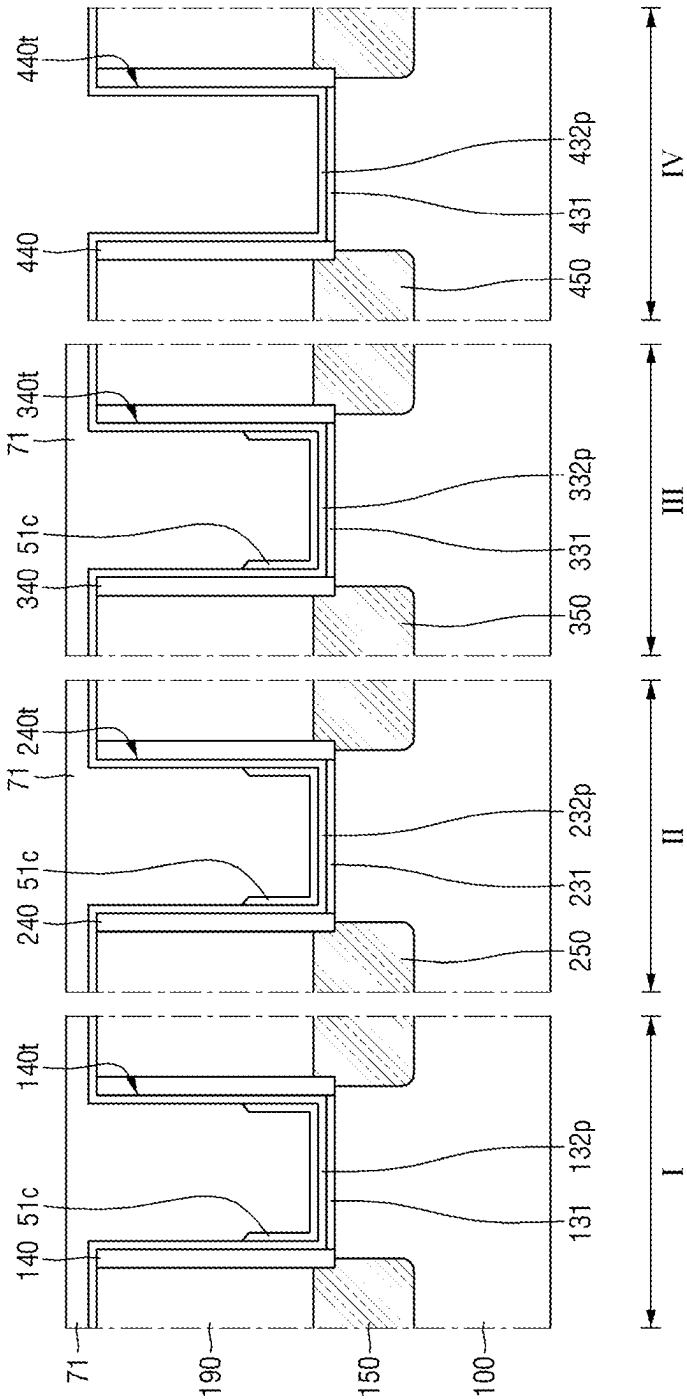


FIG. 76

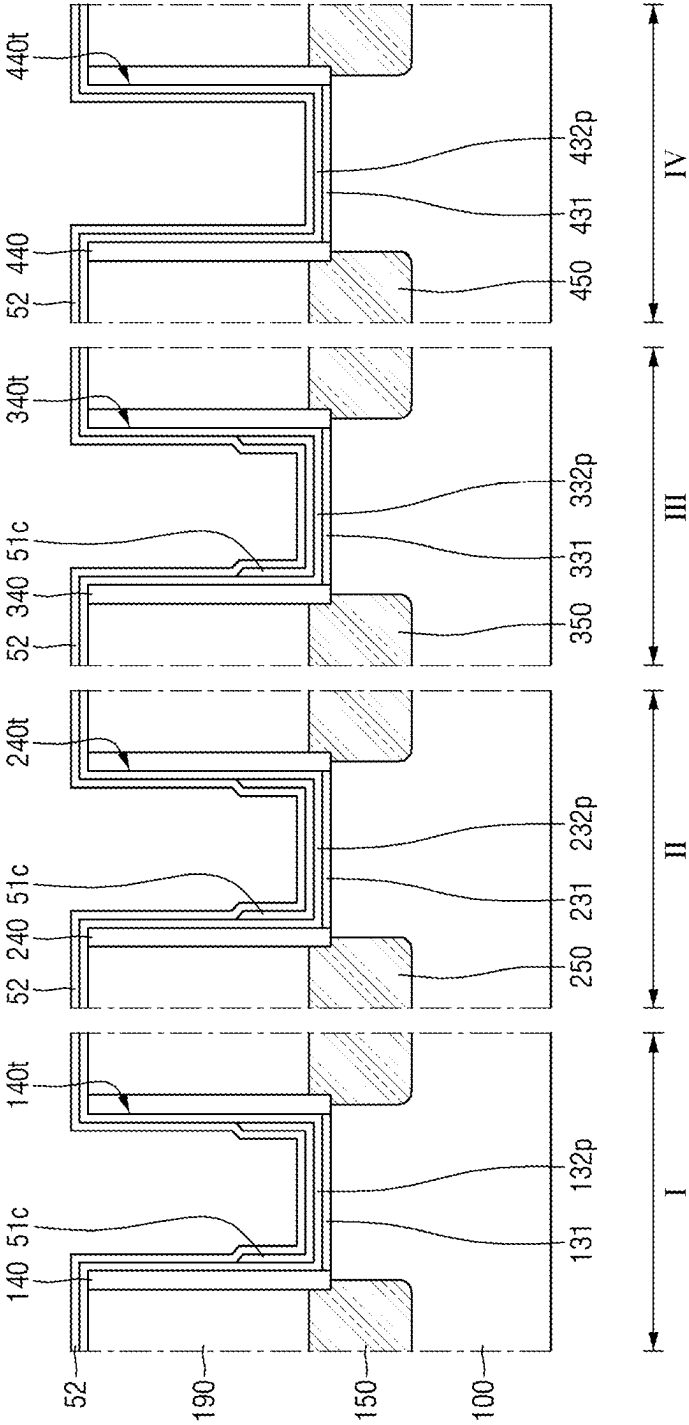


FIG. 77

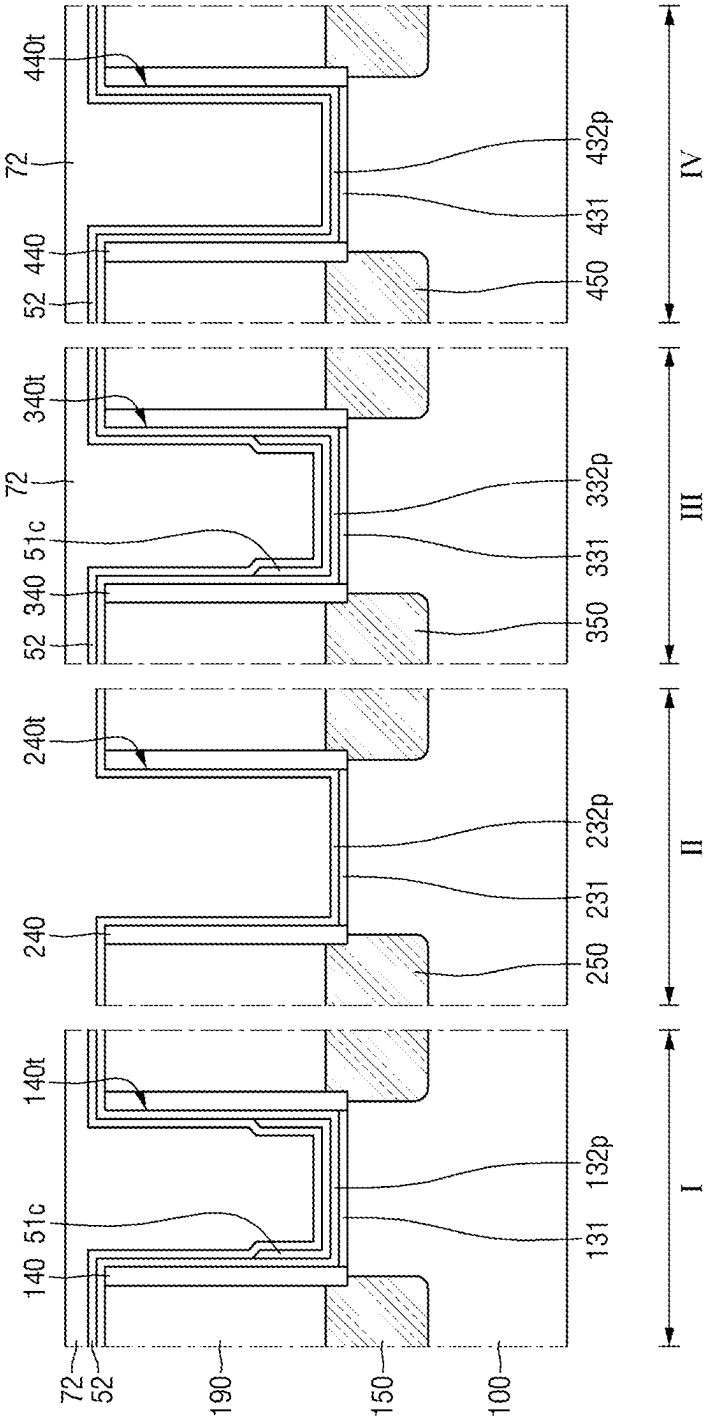


FIG. 78

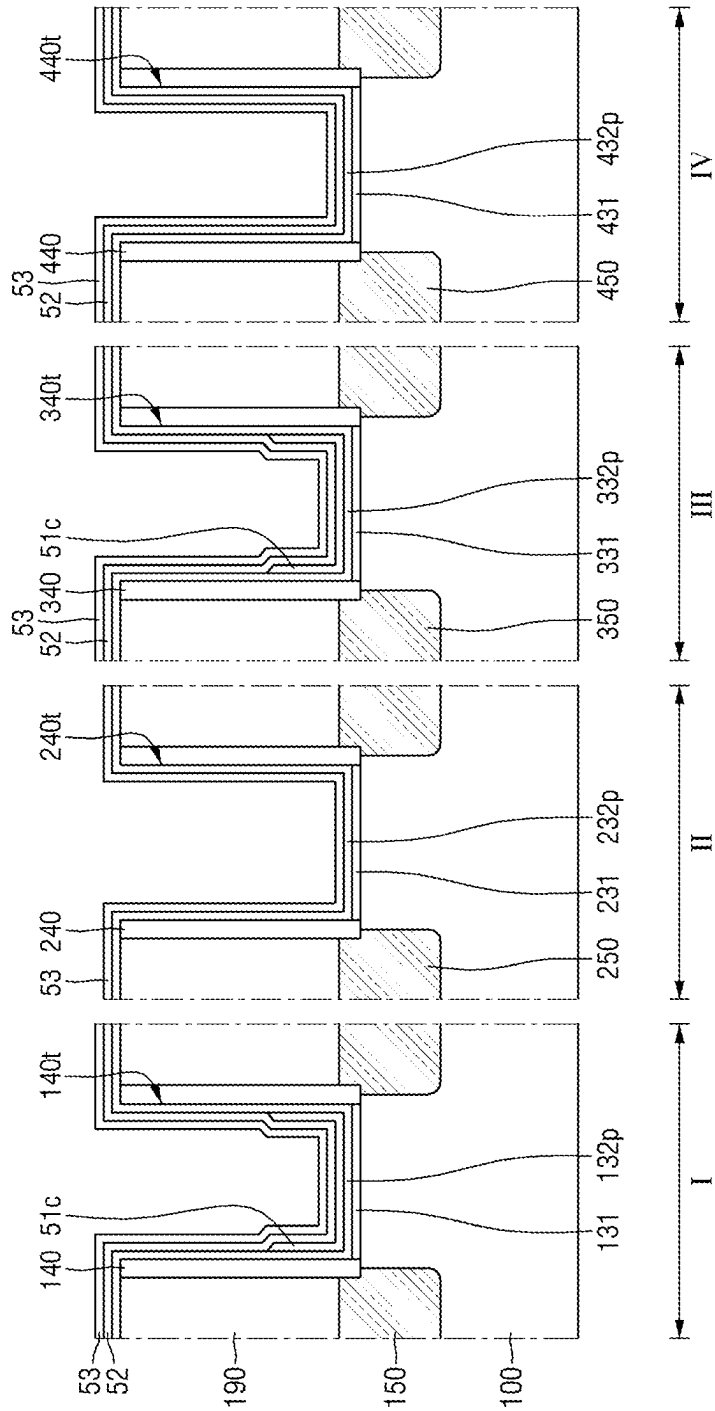


FIG. 79

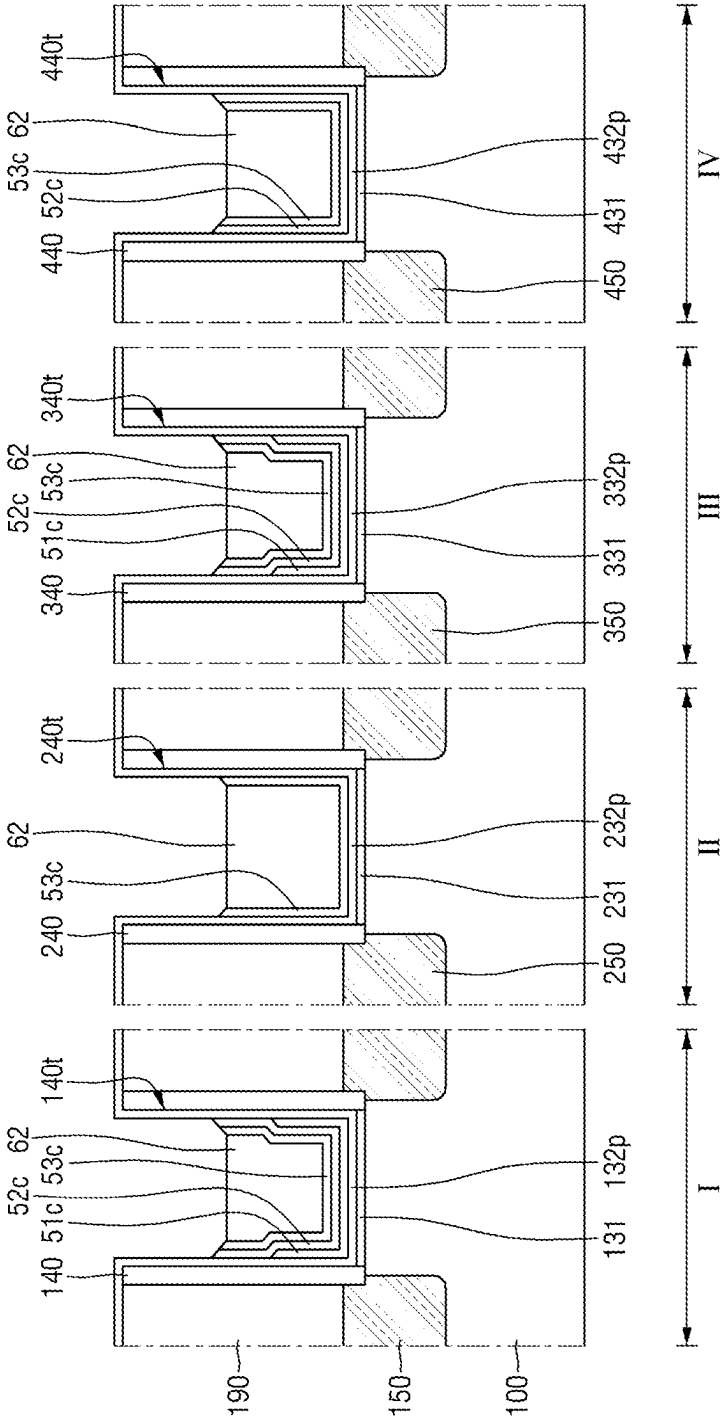


FIG. 80

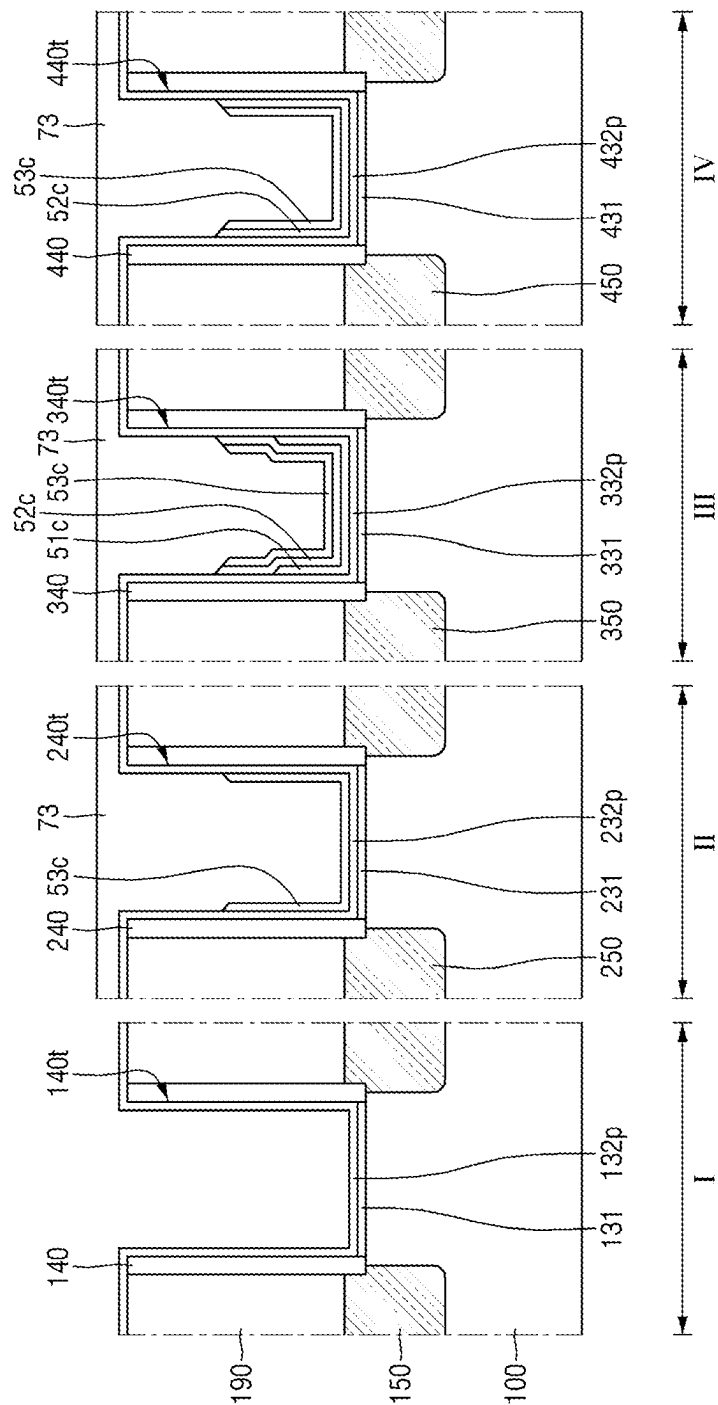


FIG. 81

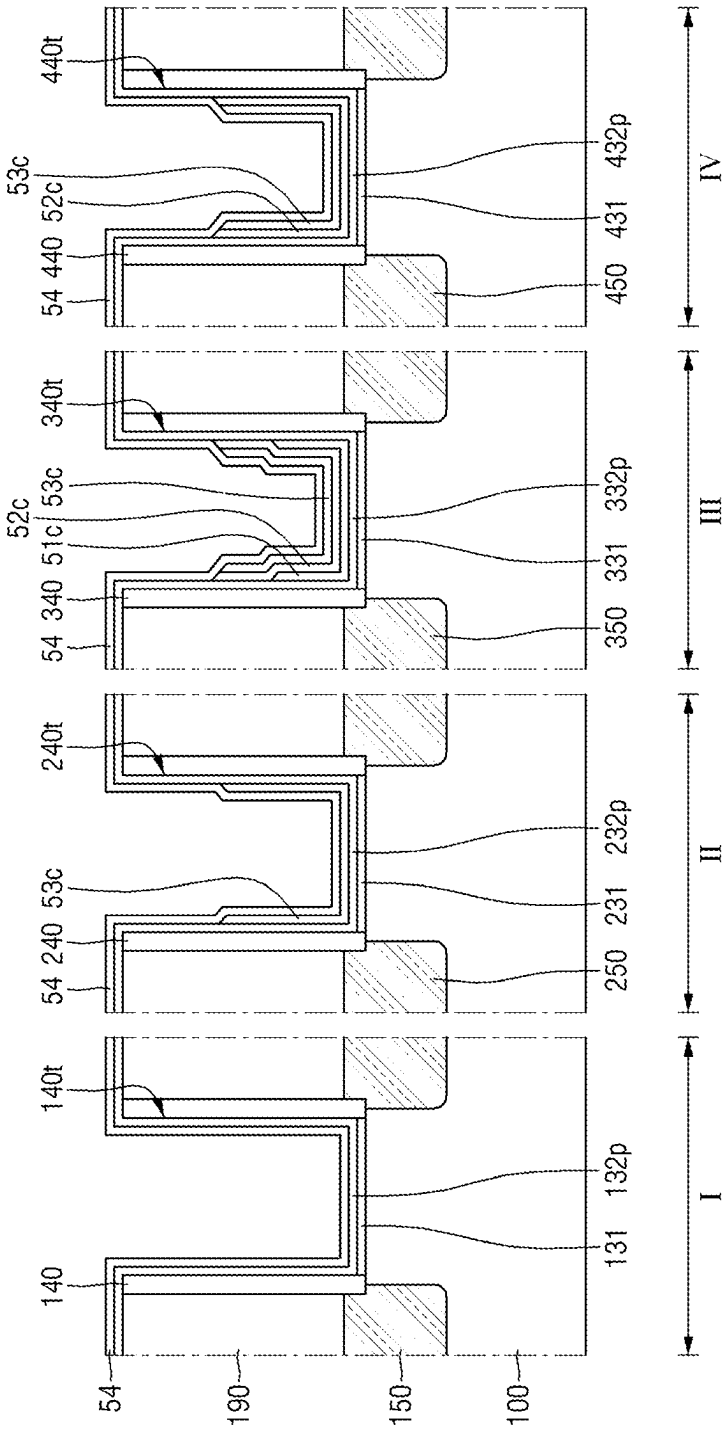


FIG. 82

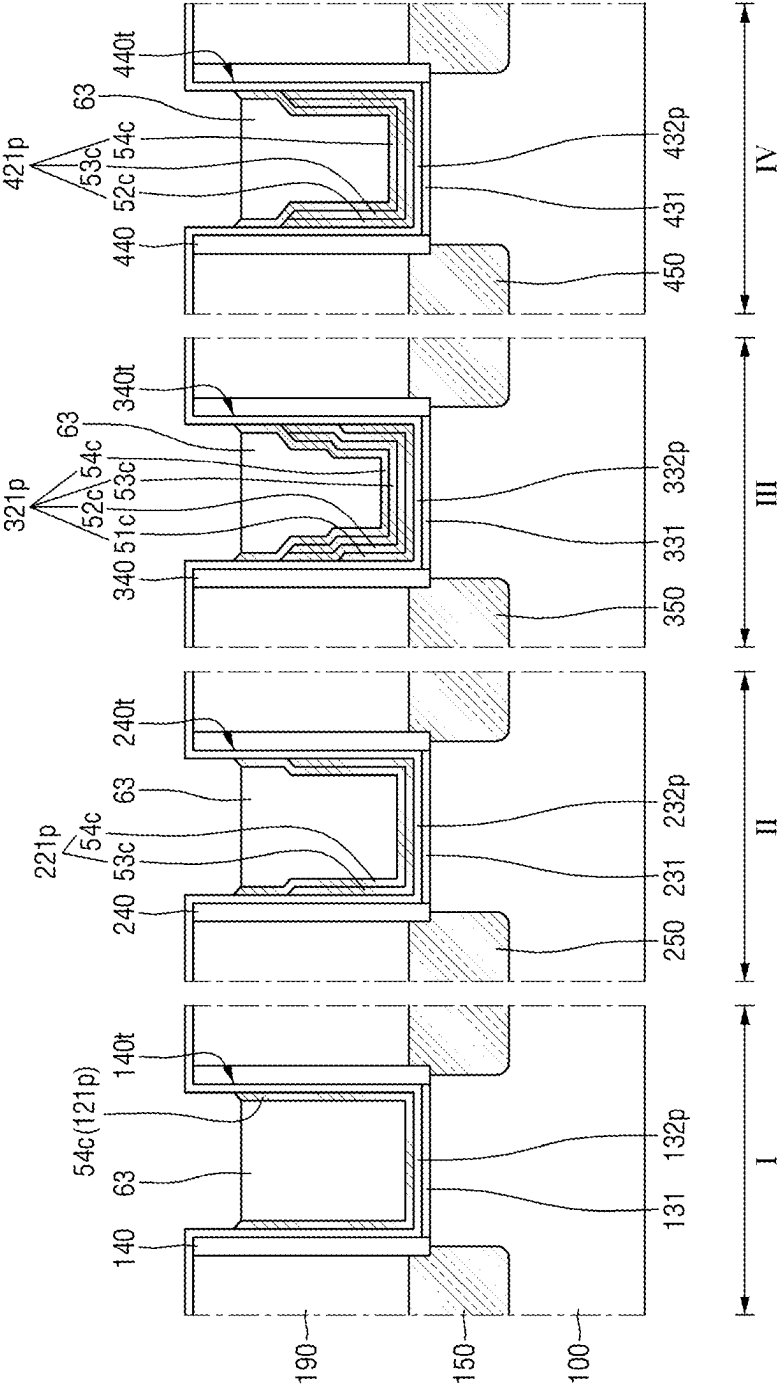
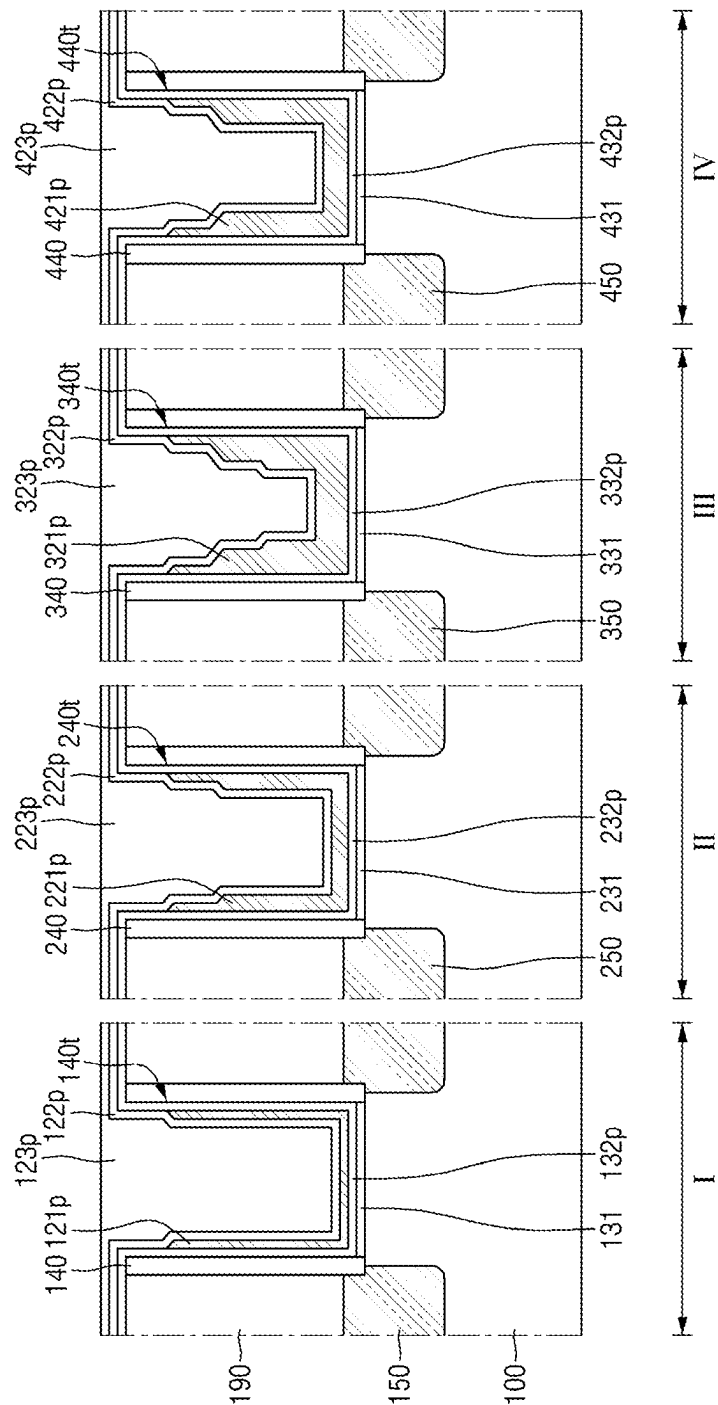


FIG. 83



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SEMICONDUCTOR DEVICE**CROSS-REFERENCE TO RELATED APPLICATIONS**

This is a continuation of Ser. No. 16/117,065 filed Aug. 30, 2018, which is a division of Ser. No. 15/413,680 filed Jan. 24, 2017, now U.S. Pat. No. 10,068,901, which issued Sep. 4, 2018, the entire contents of which are hereby incorporated by reference.

Korean Patent Application No. 10-2016-0008981, filed on Jan. 25, 2016, in the Korean Intellectual Property Office, Korean Patent Application No. 10-2016-0028719, filed on Mar. 10, 2016, in the Korean Intellectual Property Office, Korean Patent Application No. 10-2016-0028822, filed on Mar. 10, 2016, in the Korean Intellectual Property Office, and Korean Patent Application No. 10-2016-0029542, filed on Mar. 11, 2016, in the Korean Intellectual Property Office, are incorporated by reference herein in their entirety.

BACKGROUND**1. Field**

Embodiments relate to a semiconductor device and a method of fabricating the same.

2. Description of the Related Art

A semiconductor device may include transistors having different threshold voltages. Examples of transistors having different threshold voltages include a combination of a logic transistor and a static random access memory (SRAM) transistor or a dynamic random access memory (DRAM) transistor.

Meanwhile, various methods of controlling the threshold voltages of transistors included in a semiconductor device are being studied.

SUMMARY

Embodiments are directed to a semiconductor device including a substrate including a first region and a second region and first and second transistors in the first and second regions, respectively. The first transistor includes a first gate insulating layer on the substrate, a first lower TiN layer on and in contact with the first gate insulating layer, a first etch-stop layer on the first lower TiN layer and a first upper gate electrode on the first etch-stop layer. The second transistor includes a second gate insulating layer on the substrate, a second lower TiN layer on and in contact with the second gate insulating layer, a second etch-stop layer on the second lower TiN layer and a second upper gate electrode on the second etch-stop layer. A thickness of the first lower TiN layer is less than a thickness of the second lower TiN layer.

Embodiments are also directed to a semiconductor device including a substrate including first through fourth regions, and first through fourth transistors of the same conductivity type located in the first through fourth regions, respectively. The first transistor includes a first gate insulating layer on the substrate, a first lower TiN layer on and contacting the first gate insulating layer, a first etch-stop layer on the first lower TiN layer, and a first upper gate electrode including a first work-function control layer on and contacting the first etch-stop layer. The second transistor includes a second gate insulating layer on the substrate, a second lower TiN layer

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on and contacting the second gate insulating layer, a second etch-stop layer on the second lower TiN layer, and a second upper gate electrode including a second work function control layer on and contacting the second etch-stop layer.

The third transistor includes a third gate insulating layer on the substrate, a third lower TiN layer on and contacting the third gate insulating layer, a third etch-stop layer on the third lower TiN layer, and a third upper gate electrode on the third etch-stop layer. The fourth transistor includes a fourth gate insulating layer on the substrate, a fourth lower TiN layer on and contacting the fourth gate insulating layer, a fourth etch-stop layer on the fourth lower TiN layer, and a fourth upper electrode on the fourth etch-stop layer. A thickness of the first work function control layer is substantially equal to a thickness of the second work function control layer. A thickness of the first lower TiN layer is substantially equal to a thickness of the third lower TiN layer, a thickness of the second lower TiN layer is substantially equal to a thickness of the fourth lower TiN layer. The thickness of the first lower TiN layer is greater than a thickness of the second lower TiN layer.

Embodiments are also directed to a semiconductor device including a substrate including a first region and a second region, and first and second transistors formed in the first and second regions, respectively. The first transistor includes a first fin pattern on the substrate, a first gate insulating layer on the first fin pattern, a first lower TiN layer on the first gate insulating layer to intersect the first fin pattern and contact the first gate insulating layer, a first TaN layer on the first lower TiN layer, a first TiAlC layer on the first TaN layer, and a first filling layer on the first TiAlC layer. The second transistor includes a second fin pattern on the substrate, a second gate insulating layer on the second fin pattern, a second lower TiN layer on the second gate insulating layer to intersect the second fin pattern and contact the second gate insulating layer, a second TaN layer on the second lower TiN layer, a second TiAlC layer on the second TaN layer, and a second filling layer on the second TiAlC layer. A thickness of the first lower TiN layer is less than a thickness of the second lower TiN layer.

Embodiments are also directed to a semiconductor device including a substrate including first through fourth regions, an interlayer insulating film on the substrate, the interlayer insulating film including first through fourth trenches in the first through fourth regions, respectively, and first through fourth transistors of the same conductivity type located in the first through fourth trenches, respectively, in the first through fourth regions. Each of the first through fourth transistor includes at least one insulating layer on a bottom and sides of a respective trench of the first through fourth trenches, a lower conductive layer on the insulating layer conforming to the bottom and sides of the trench, an etch-stop layer conforming to a bottom and sides of the lower conductive layer, a work function control layer conforming to a bottom and at least a portion of sides of the etch-stop layer, an insertion layer conforming to a bottom and sides of the work function control layer, and a filling layer filling a remaining space of the trench. In at least one of the first through fourth transistors, the work function control layer is chamfered.

BRIEF DESCRIPTION OF THE DRAWINGS

Features will become apparent to those of ordinary skill in the art by describing in detail exemplary embodiments with reference to the attached drawings in which:

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FIG. 1 illustrates a view of a semiconductor device according to some embodiments;

FIG. 2 illustrates a view of a semiconductor device according to some embodiments;

FIG. 3 illustrates a view of a semiconductor device according to some embodiments;

FIG. 4 illustrates a view of a semiconductor device according to some embodiments;

FIG. 5 illustrates a view of a semiconductor device according to some embodiments;

FIG. 6 illustrates a view of a semiconductor device according to some embodiments;

FIG. 7 illustrates a view of a semiconductor device according to some embodiments;

FIG. 8 illustrates an enlarged view of portions P and Q of FIG. 7;

FIGS. 9 through 12 respectively illustrate views of semiconductor devices according to some embodiments;

FIG. 13 illustrates a view of a semiconductor device according to some embodiments;

FIG. 14 illustrates an enlarged view of a portion P of FIG. 13;

FIG. 15 illustrates a view of a semiconductor device according to some embodiments;

FIG. 16 illustrates an enlarged view of portions P and Q of FIG. 15;

FIG. 17 illustrates a view of a semiconductor device according to some embodiments;

FIG. 18 illustrates an enlarged view of portions P and Q of FIG. 17;

FIGS. 19A and 19B respectively illustrate views of semiconductor devices according to some embodiments;

FIGS. 20 through 24 respectively illustrate views of semiconductor devices according to some embodiments;

FIG. 25 illustrates a layout view of a semiconductor device according to some embodiments;

FIG. 26 illustrates a cross-sectional view taken along the lines A-A, B-B, C-C and D-D of FIG. 25;

FIGS. 27A through 27C illustrate cross-sectional views taken along the line E-E of FIG. 25;

FIG. 28 illustrates a view of a semiconductor device according to some embodiments;

FIG. 29 illustrates a view of a semiconductor device according to some embodiments;

FIG. 30 illustrates a plan view of a semiconductor device according to some embodiments;

FIGS. 31A and 31B illustrate cross-sectional views taken along the line F-F of FIG. 30;

FIGS. 32A and 32B illustrate cross-sectional views taken along the lines G-G and H-H of FIG. 30;

FIGS. 33 and 34 illustrate views of a semiconductor device according to some embodiments;

FIGS. 35 and 36 respectively illustrate views of semiconductor devices according to some embodiments;

FIG. 37 illustrates a plan view of a semiconductor device according to some embodiments;

FIG. 38 illustrates a cross-sectional view taken along the line F-F of FIG. 37;

FIGS. 39A and 39B respectively illustrate views of semiconductor devices according to some embodiments;

FIG. 40 illustrates a plan view of a semiconductor device according to some embodiments;

FIG. 41 illustrates a cross-sectional view taken along the line F-F of FIG. 40;

FIG. 42 illustrates a view of a semiconductor device according to some embodiments;

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FIG. 43 illustrates a plan view of a semiconductor device according to some embodiments;

FIG. 44 illustrates a cross-sectional view taken along the line F-F of FIG. 43;

FIG. 45 illustrates a view of a semiconductor device according to some embodiments;

FIG. 46 illustrates a plan view of a semiconductor device according to some embodiments;

FIG. 47 illustrates a cross-sectional view taken along the lines F-F and J-J of FIG. 46;

FIGS. 48 and 49 respectively illustrate views of semiconductor devices according to some embodiments;

FIGS. 50 and 51 respectively illustrate plan views of semiconductor devices according to some embodiments;

FIG. 52 illustrates a plan view of a semiconductor device according to some embodiments;

FIG. 53 illustrates a cross-sectional view taken along the lines F-F and J-J of FIG. 52;

FIG. 54 illustrates a view of a semiconductor device according to some embodiments;

FIG. 55 illustrates a plan view of a semiconductor device according to some embodiments;

FIG. 56 illustrates a cross-sectional view taken along the lines F-F and J-J of FIG. 55;

FIG. 57 illustrates a circuit diagram of a semiconductor device according to some embodiments;

FIG. 58 illustrates a layout view of the semiconductor device of FIG. 57;

FIG. 59 illustrates a cross-sectional view taken along the line K-K of FIG. 58;

FIGS. 60 through 70 illustrate views illustrating steps of a method of fabricating a semiconductor device according to some embodiments;

FIG. 71 illustrates a view illustrating a step of a method of fabricating a semiconductor device according to some embodiments;

FIG. 72 illustrates a view illustrating a step of a method of fabricating a semiconductor device according to some embodiments; and

FIGS. 73 through 83 illustrate views illustrating steps of a method of fabricating a semiconductor device according to some embodiments.

DETAILED DESCRIPTION

Example embodiments will now be described more fully hereinafter with reference to the accompanying drawings; however, they may be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey exemplary implementations to those skilled in the art.

In the drawing figures, the dimensions of layers and regions may be exaggerated for clarity of illustration. Like reference numerals refer to like elements throughout.

Although the drawings relating to semiconductor devices according to some embodiments show a fin field effect transistor (FinFET) including a channel region of a fin pattern shape, in some implementations, the semiconductor devices according to some embodiments may also include a tunneling FET, a transistor including nanowires, a transistor including a nanosheet, or a three-dimensional (3D) transistor. In addition, the semiconductor devices according to some embodiments may include a bipolar junction transistor, a lateral double diffusion transistor (LDMOS), or the like.

FIG. 1 illustrates a view of a semiconductor device according to embodiments.

Referring to FIG. 1, the semiconductor device may include first through fourth transistors **101** through **401** formed on a substrate **100**.

The substrate **100** may include first through fourth regions I through IV. The first through fourth regions I through IV may be separated from each other or may be connected to each other.

The first through fourth regions I through IV may be included in a portion that performs the same function. For example, in a logic region or an input/output (I/O) region. In some implementations, each of the first through fourth regions I through IV may be included in one of portions that perform different functions from each other, for example, one of a logic region, a static random access memory (SRAM) region and an I/O region.

In the semiconductor device according to some embodiments described with reference to FIG. 1, each of the first through fourth regions I through IV may be a region in which a p-channel metal oxide semiconductor (PMOS) is formed.

The substrate **100** may be a bulk silicon substrate or a silicon-on-insulator (SOI) substrate. In some implementations, the substrate **100** may be a silicon substrate or a substrate made of another material such as silicon germanium, silicon germanium-on-insulator (SGOI), indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide or gallium antimonide.

In the following description, it is assumed, for ease of description, that the substrate **100** is a substrate containing silicon.

The first transistor **101** may be formed in the first region I, the second transistor **201** may be formed in the second region II, the third transistor **301** may be formed in the third region III, and the fourth transistor **401** may be formed in the fourth region IV.

Each of the first through fourth regions I through IV may be a region in which a PMOS is formed, and each of the first through fourth transistors **101** through **401** may be a p-type transistor.

The first transistor **101** may include a first gate insulating layer **130**, a first gate electrode structure **120**, first gate spacers **140**, and first source/drain regions **150**.

The second transistor **201** may include a second gate insulating layer **230**, a second gate electrode structure **220**, second gate spacers **240**, and second source/drain regions **250**.

The third transistor **301** may include a third gate insulating layer **330**, a third gate electrode structure **320**, third gate spacers **340**, and third source/drain regions **350**.

The fourth transistor **401** may include a fourth gate insulating layer **430**, a fourth gate electrode structure **420**, fourth gate spacers **440**, and fourth source/drain regions **450**.

Elements included in each of the first through fourth transistors **101** through **401** will be described in detail below.

An interlayer insulating film **190** may be formed on the substrate **100** of the first through fourth regions I through IV. The interlayer insulating film **190** may include first through fourth trenches **140t** through **440t**.

The first through fourth trenches **140t** through **440t** may correspond to the first through fourth regions I through IV, respectively. For example, the first trench **140t** may be formed on the substrate **100** in the first region I, the second trench **240t** may be formed on the substrate **100** in the second region II, the third trench **340t** may be formed on the

substrate **100** in the third region III, and the fourth trench **440t** may be formed on the substrate **100** in the fourth region IV.

The interlayer insulating film **190** may include at least one of, for example, silicon oxide, silicon nitride, silicon oxynitride, and a low dielectric constant (low-k) material. Examples of the low-k material may include, for example, flowable oxide (FOX), tonen silazene (TOSZ), undoped silicate glass (USG), borosilicate glass (BSG), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), plasma enhanced tetraethylorthosilicate (PETEOS), fluoride silicate glass (FSG), carbon doped silicon oxide (CDO), xerogel, aerogel, amorphous fluorinated carbon, organo silicate glass (OSG), parylene, bis-benzocyclobutene (BCB), SiLK, a polyimide, a porous polymeric material, or combinations of the same.

The first gate spacers **140** may be formed on the substrate **100** of the first region I. The first gate spacers **140** may define the first trench **140t**. The first trench **140t** may have, for example, the first gate spacers **140** as its sidewalls and an upper surface of the substrate **100** as its bottom surface.

The second gate spacers **240** defining the second trench **240t** may be formed on the substrate **100** of the second region II. The third gate spacers **340** defining the third trench **340t** may be formed on the substrate **100** of the third region III. The fourth gate spacers **440** defining the fourth trench **440t** may be formed on the substrate **100** of the fourth region IV.

Each of the first through fourth gate spacers **140** through **440** may include at least one of silicon nitride (SiN), silicon oxynitride (SiON), silicon oxide (SiO₂), silicon oxycarbonitride (SiOCN), and combinations of the same.

Although each of the first through fourth gate spacers **140** through **440** is illustrated as being a single layer, in some implementations, each of the first through fourth gate spacers **140** through **440** may not be a single layer. When each of the first through fourth gate spacers **140** through **440** includes a plurality of layers, at least one of the layers included in each of the first through fourth gate spacers **140** through **440** may include a low-k material such as silicon oxycarbonitride (SiOCN).

When each of the first through fourth gate spacers **140** through **440** includes a plurality of layers, at least one of the layers included in each of the first through fourth gate spacers **140** through **440** may be L-shaped.

In some cases, each of the first through fourth gate spacers **140** through **440** may serve as a guide for forming a self-aligned contact. Accordingly, each of the first through fourth gate spacers **140** through **440** may include a material having an etch selectivity with respect to the interlayer insulating film **190**.

The first gate insulating layer **130** may be formed on the substrate **100** of the first region I. The first gate insulating layer **130** may extend along the sidewalls and bottom surface of the first trench **140t**. The first gate insulating layer **130** may include a first interfacial layer **131** and a first high dielectric constant (high-k) insulating layer **132**.

The first interfacial layer **131** may be formed on the substrate **100**. The first interfacial layer **131** may be formed on the bottom surface of the first trench **140t**.

The first high-k insulating layer **132** may be formed on the first interfacial layer **131**. The first high-k insulating layer **132** may be formed along the bottom and sidewalls of the first trench **140t**.

The second gate insulating layer **230** may be formed on the substrate **100** of the second region II. The second gate insulating layer **230** may extend along sidewalls and a

bottom surface of the second trench **240t**. The second gate insulating layer **230** may include a second interfacial layer **231** and a second high-k insulating layer **232**.

The second interfacial layer **231** may be formed on the substrate **100**. The second interfacial layer **231** may be formed on the bottom surface of the second trench **240t**.

The second high-k insulating layer **232** may be formed on the second interfacial layer **231**. The second high-k insulating layer **232** may be formed along the bottom surface and sidewalls of the second trench **240t**.

The third gate insulating layer **330** may be formed on the substrate **100** of the third region III. The third gate insulating layer **330** may extend along sidewalls and a bottom surface of the third trench **340t**. The third gate insulating layer **330** may include a third interfacial layer **331** and a third high-k insulating layer **332**.

The third interfacial layer **331** may be formed on the substrate **100**. The third interfacial layer **331** may be formed on the bottom surface of the third trench **340t**.

The third high-k insulating layer **332** may be formed on the third interfacial layer **331**. The third high-k insulating layer **332** may be formed along the bottom surface and sidewalls of the third trench **340t**.

The fourth gate insulating layer **430** may be formed on the substrate **100** of the fourth region IV. The fourth gate insulating layer **430** may extend along sidewalls and a bottom surface of the fourth trench **440t**. The fourth gate insulating layer **430** may include a fourth interfacial layer **431** and a fourth high-k insulating layer **432**.

The fourth interfacial layer **431** may be formed on the substrate **100**. The fourth interfacial layer **431** may be formed on the bottom surface of the fourth trench **440t**.

The fourth high-k insulating layer **432** may be formed on the fourth interfacial layer **431**. The fourth high-k insulating layer **432** may be formed along the bottom surface and sidewalls of the fourth trench **440t**.

In some implementations, the first through fourth interfacial layers **131** through **431** may not be formed on the sidewalls of the first through fourth trenches **140t** through **440t**. In some implementations, the first through fourth interfacial layers **131** through **431** may also be formed on the sidewalls of the first through fourth trenches **140t** through **440t**, depending on a method of forming the first through fourth interfacial layers **131** through **431**.

Each of the first through fourth interfacial layers **131** through **431** may include, for example, silicon oxide. In some implementations, each of the first through fourth interfacial layers **131** through **431** may include a different material depending on the type of the substrate **100** or the type of the first, second, third or fourth high-k insulating layer **132**, **232**, **332** or **432**.

The first through fourth high-k insulating layers **132** through **432** may include one or more of hafnium oxide, hafnium silicon oxide, hafnium aluminum oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, and lead zinc niobate.

While the first through fourth high-k insulating layers **132** through **432** including oxides have mainly been described, in some implementations, the first through fourth high-k insulating layers **132** through **432** may also include, for example, one or more of nitrides of the above metallic materials (e.g., hafnium nitride) and oxynitrides (e.g., hafnium oxynitride) of the above metallic materials.

The first gate electrode structure **120** may be formed on the first gate insulating layer **130**. The first gate electrode structure **120** may fill the first trench **140t**.

The first gate electrode structure **120** may include a first lower conductive layer **125**, a first etch-stop layer **124**, a first work function control layer **121**, a first insertion layer **122**, and a first filling layer **123**.

The first lower conductive layer **125** may be formed on the first gate insulating layer **130**. The first lower conductive layer **125** may contact the first gate insulating layer **130**.

The first lower conductive layer **125** may extend along the sidewalls and bottom surface of the first trench **140t**. The first lower conductive layer **125** may be formed along the profile of the first gate insulating layer **130**.

The first etch-stop layer **124** may be formed on the first lower conductive layer **125**.

The first etch-stop layer **124** may extend along the sidewalls and bottom surface of the first trench **140t**. The first etch-stop layer **124** may be formed along the profile of the first lower conductive layer **125**.

The first work function control layer **121** may be formed on the first etch-stop layer **124**. The first work function control layer **121** may contact the first etch-stop layer **124**.

The first work function control layer **121** may extend along the sidewalls and bottom surface of the first trench **140t**. The first work function control layer **121** may be formed along the profile of the first etch-stop layer **124**.

The first insertion layer **122** may be formed on the first work function control layer **121**.

The first insertion layer **122** may contact the first work function control layer **121**.

The first insertion layer **122** may extend along the sidewalls and bottom surface of the first trench **140t**. The first insertion layer **122** may be formed along the profile of the first work function control layer **121**.

The first filling layer **123** may be formed on the first insertion layer **122**. The first filling layer **123** may fill the remaining space of the first trench **140t** in which the first lower conductive layer **125**, the first etch-stop layer **124**, the first work function control layer **121** and the first insertion layer **122** are formed.

The second gate electrode structure **220** may be formed on the second gate insulating layer **230**. The second gate electrode structure **220** may fill the second trench **240t**.

The second gate electrode structure **220** may include a second lower conductive layer **225**, a second etch-stop layer **224**, a second work function control layer **221**, a second insertion layer **222**, and a filling layer **223**.

The second lower conductive layer **225** may be formed on the second gate insulating layer **230**. The second lower conductive layer **225** may contact the second gate insulating layer **230**.

The second lower conductive layer **225** may extend along the sidewalls and bottom surface of the second trench **240t**. The second lower conductive layer **225** may be formed along the profile of the second gate insulating layer **230**.

The second etch-stop layer **224** may be formed on the second lower conductive layer **225**. The second etch-stop layer **224** may extend along the sidewalls and bottom surface of the second trench **240t**. The second etch-stop layer **224** may be formed along the profile of the second lower conductive layer **225**.

The second work function control layer **221** may be formed on the second etch-stop layer **224**. The second work function control layer **221** may contact the second etch-stop layer **224**.

The second work function control layer **221** may extend along the sidewalls and bottom surface of the second trench **240t**. The second work function control layer **221** may be formed along the profile of the second etch-stop layer **224**.

The second insertion layer **222** may be formed on the second work function control layer **221**. The second insertion layer **222** may contact the second work function control layer **221**.

The second insertion layer **222** may extend along the sidewalls and bottom surface of the second trench **240t**. The second insertion layer **222** may be formed along the profile of the second work function control layer **221**.

The second filling layer **223** may be formed on the second insertion layer **222**. The second filling layer **223** may fill the remaining space of the second trench **240t** in which the second lower conductive layer **225**, the second etch-stop layer **224**, the second work function control layer **221** and the second insertion layer **222** are formed.

The third gate electrode structure **320** may be formed on the third gate insulating layer **330**. The third gate electrode structure **320** may fill the third trench **340t**.

The third gate electrode structure **320** may include a third lower conductive layer **325**, a third etch-stop layer **324**, a third work function control layer **321**, a third insertion layer **322**, and a third filling layer **323**.

The third lower conductive layer **325** may be formed on the third gate insulating layer **330**. The third lower conductive layer **325** may contact the third gate insulating layer **330**.

The third lower conductive layer **325** may extend along the sidewalls and bottom surface of the third trench **340t**. The third lower conductive layer **325** may be formed along the profile of the third gate insulating layer **330**.

The third etch-stop layer **324** may be formed on the third lower conductive layer **325**.

The third etch-stop layer **324** may extend along the sidewalls and bottom surface of the third trench **340t**. The third etch-stop layer **324** may be formed along the profile of the third lower conductive layer **325**.

The third work function control layer **321** may be formed on the third etch-stop layer **324**. The third work function control layer **321** may contact the third etch-stop layer **324**.

The third work function control layer **321** may extend along the sidewalls and bottom surface of the third trench **340t**. The third work function control layer **321** may be formed along the profile of the third etch-stop layer **324**.

The third insertion layer **322** may be formed on the third work function control layer **321**. The third insertion layer **322** may contact the third work function control layer **321**.

The third insertion layer **322** may extend along the sidewalls and bottom surface of the third trench **340t**. The third insertion layer **322** may be formed along the profile of the third work function control layer **321**.

The third filling layer **323** may be formed on the third insertion layer **322**. The third filling layer **323** may fill the remaining space of the third trench **340t** in which the third lower conductive layer **325**, the third etch-stop layer **324**, the third work function control layer **321**, and the third insertion layer **322** are formed.

The fourth gate electrode structure **420** may be formed on the fourth gate insulating layer **430**. The fourth gate electrode structure **420** may fill the fourth trench **440t**.

The fourth gate electrode structure **420** may include a fourth lower conductive layer **425**, a fourth etch-stop layer **424**, a fourth work function control layer **421**, a fourth insertion layer **422**, and a fourth filling layer **423**.

The fourth lower conductive layer **425** may be formed on the fourth gate insulating layer **430**. The fourth lower conductive layer **425** may contact the fourth gate insulating layer **430**.

The fourth lower conductive layer **425** may extend along the sidewalls and bottom surface of the fourth trench **440t**. The fourth lower conductive layer **425** may be formed along the profile of the fourth gate insulating layer **430**.

The fourth etch-stop layer **424** may be formed on the fourth lower conductive layer **425**. The fourth etch-stop layer **424** may extend along the sidewalls and bottom surface of the fourth trench **440t**. The fourth etch-stop layer **424** may be formed along the profile of the fourth lower conductive layer **425**.

The fourth work function control layer **421** may be formed on the fourth etch-stop layer **424**. The fourth work function control layer **421** may contact the fourth etch-stop layer **424**.

The fourth work function control layer **421** may extend along the sidewalls and bottom surface of the fourth trench **440t**. The fourth work function control layer **421** may be formed along the profile of the fourth etch-stop layer **424**.

The fourth insertion layer **422** may be formed on the fourth work function control layer **421**. The fourth insertion layer **422** may contact the fourth work function control layer **421**.

The fourth insertion layer **422** may extend along the sidewalls and bottom surface of the fourth trench **440t**. The fourth insertion layer **422** may be formed along the profile of the fourth work function control layer **421**.

The fourth filling layer **423** may be formed on the fourth insertion layer **422**. The fourth filling layer **423** may fill the remaining space of the fourth trench **440t** in which the fourth lower conductive layer **425**, the fourth etch-stop layer **424**, the fourth work function control layer **421**, and the fourth insertion layer **422** are formed.

A work function control layer, an insertion layer, and a filling layer formed on each etch-stop layer may be an upper gate electrode. Of these layers, the insertion layer and the filling layer may be an upper conductive layer. That is, the first through fourth gate electrode structures **120** through **420** respectively include first through fourth upper conductive layers **127** through **427**, each including an insertion layer and a filling layer.

The first through fourth lower conductive layers **125** through **425** may include titanium nitride (TiN). The first through fourth lower conductive layers **125** through **425** may be referred to herein as first through fourth TiN layers.

The first through fourth etch-stop layers **124** through **424** may include the same material. For example, the first through fourth etch-stop layers **124** through **424** may be layers made of the same material.

The first through fourth etch-stop layers **124** through **424** may include, for example, tantalum nitride (Ta₂N₃). The first through fourth etch-stop layers **124** through **424** may be formed at the same level. Here, the term "same level" indicates that the first through fourth etch-stop layers **124** through **424** are formed by the same fabrication process.

For example, the first through fourth etch-stop layers **124** through **424** may have substantially the same thickness.

The first through fourth work function control layers **121** through **421** may include the same material. For example, the first through fourth work function control layers **121** through **421** may be layers made of the same material.

The first through fourth work function control layers **121** through **421** may include, for example, TiN.

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The first through fourth insertion layers **122** through **422** may include the same material. For example, the first through fourth insertion layers **122** through **422** may be layers made of the same material.

Here, the term “layers made of the same material” can be defined as follows. First, if each insertion layer is a single layer, the first through fourth insertion layers **122** through **422** may all be single layers made of the same material.

On the other hand, if each insertion layer includes a plurality of layers, for example, two layers, each of the first through fourth insertion layers **122** through **422** may have a multilayer structure in which a first layer made of material M and a second layer made of material N are sequentially stacked.

The first through fourth insertion layers **122** through **422** may include, for example, one of Ti, TiAl, TiAlN, TiAlC, and TiAlCN. The first through fourth insertion layers **122** through **422** may be formed at the same level.

For example, the first through fourth insertion layers **122** through **422** may have substantially the same thickness.

In the semiconductor device according to some embodiments, the first through fourth insertion layers **122** through **422** are described as layers containing TiAl.

The first through fourth filling layers **123** through **423** may include the same material. The first through fourth filling layers **123** through **423** may include at least one of, for example, W, Al, Co, Cu, Ru, Ni, Pt, Ni—Pt, and TiN.

Each of the first through fourth gate electrode structures **120** through **420** may have an uppermost surface in the same plane as an upper surface of the interlayer insulating film **190**.

The first through fourth source/drain regions **150** through **450** may be formed adjacent to the first through fourth gate electrode structures **120** through **420**.

Although each of the first through fourth source/drain regions **150** through **450** may include, for example, an epitaxial layer formed in the substrate **100**. In some implementations, each of the first through fourth source/drain regions **150** through **450** may be an impurity region formed by implanting impurities into the substrate **100**.

In some implementations, each of the first through fourth source/drain regions **150** through **450** may be an elevated source/drain region having an upper surface protruding above the upper surface of the substrate **100**.

In the embodiment illustrated in FIG. 1, a thickness t_{11} of the first lower conductive layer **125** may be substantially equal to a thickness t_{31} of the third lower conductive layer **325**. A thickness t_{21} of the second lower conductive layer **225** may be substantially equal to a thickness t_{41} of the fourth lower conductive layer **425**.

The thickness t_{11} of the first lower conductive layer **125** may be greater than the thickness t_{21} of the second lower conductive layer **225**.

In some implementations, a thickness t_{12} of the first work function control layer **121** may be substantially equal to a thickness t_{22} of the second work function control layer **221**. A thickness t_{32} of the third work function control layer **321** may be substantially equal to a thickness t_{42} of the fourth work function control layer **421**.

In some implementations, the thickness t_{12} of the first work function control layer **121** may be greater than the thickness t_{32} of the third work function control layer **321**.

A thickness of a lower TiN layer and a thickness of a work function control layer may be different in each of the first through fourth transistors **101** through **401**.

In the following description, unless otherwise specified, the thicknesses of the first through fourth work function

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control layers **121** through **421** and the thicknesses of the first through fourth lower conductive layers **125** through **425** are defined as thicknesses at the bottom surfaces of the first through fourth trenches **140t** through **440t**, respectively.

For example, the thickness t_{12} of the first work function control layer **121** may be the thickness of the first work function control layer **121** formed on the bottom surface of the first trench **140t**, and the thickness t_{22} of the second work function control layer **221** may be the thickness of the second work function control layer **221** formed on the bottom surface of the second trench **240t**. The thickness t_{32} of the third work function control layer **321** may be the thickness of the third work function control layer **321** formed on the bottom surface of the third trench **340t**, and the thickness t_{42} of the fourth work function control layer **421** may be the thickness of the fourth work function control layer **421** formed on the bottom surface of the fourth trench **440t**.

A threshold voltage of each of the first through fourth transistors **101** through **401** may be adjusted using a combination of the thickness of the lower conductive layer and the thickness of the work function control layer.

The respective threshold voltages of the first through fourth transistors **101** through **401** may be different from each other.

The threshold voltage of the second transistor **201** may be greater than the threshold voltage of the first transistor **101** and less than the threshold voltage of the third transistor **301**. The threshold voltage of the third transistor **301** may be less than the threshold voltage of the fourth transistor **401**.

Each of the first through fourth transistors **101** through **401** illustrated in FIG. 1 may be a p-type transistor. Accordingly, the fourth transistor **401** having the highest threshold voltage may be, for example, a p-type high voltage transistor.

The third transistor **301** may be a p-type regular voltage transistor, and the second transistor **201** may be a p-type low voltage transistor. The first transistor **101** having the lowest threshold voltage may be a p-type super low voltage transistor.

For example, in the case of the first transistor **101** and the second transistor **201**, whose respective work function control layers have the same thickness and whose respective lower conductive layers have different thicknesses, the threshold voltage of the second transistor **201** whose lower conductive layer has a lesser thickness may be higher than the threshold voltage of the first transistor **101**.

In the case of the first transistor **101** and the third transistor **301**, whose respective lower conductive layers have the same thickness and whose respective work function control layers have different thicknesses, the threshold voltage of the third transistor **301** whose work function control layer has a lesser thickness may be higher than the threshold voltage of the first transistor **101**.

FIG. 2 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIG. 1 will be mainly described.

Referring to FIG. 2, the semiconductor device may further include first through fourth capping patterns **160** through **460**.

A first gate electrode structure **120** may fill part of a first trench **140t**. An upper surface of a first gate electrode structure **120** may be closer to a substrate **100** than to an upper surface of an interlayer insulating film **190**.

The first capping pattern **160** may be formed on the first gate electrode structure **120**. For example, the first capping

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pattern **160** may be formed on a first upper conductive layer **127**. The first capping pattern **160** may fill the remaining space of the first trench **140t** filled with the first gate electrode structure **120**.

A second gate electrode structure **220** may fill part of a second trench **240t**. An upper surface of the second gate electrode structure **220** may be closer to the substrate **100** than the upper surface of the interlayer insulating film **190**.

The second capping pattern **260** may be formed on the second gate electrode structure **220**. The second capping pattern **260** may be formed on a second upper conductive layer **227**. The second capping pattern **260** may fill the remaining space of the second trench **240t** filled with the second gate electrode structure **220**.

A third gate electrode structure **320** may fill part of a third trench **340t**. An upper surface of the third gate electrode structure **320** may be closer to the substrate **100** than the upper surface of the interlayer insulating film **190**.

The third capping pattern **360** may be formed on the third gate electrode structure **320**. The third capping pattern **360** may be formed on a third upper conductive layer **327**. The third capping pattern **360** may fill the remaining space of the third trench **340t** filled with the third gate electrode structure **320**.

A fourth gate electrode structure **420** may fill part of a fourth trench **440t**. An upper surface of the fourth gate electrode structure **420** may be closer to the substrate **100** than the upper surface of the interlayer insulating film **190**.

The fourth capping pattern **460** may be formed on the fourth gate electrode structure **420**. The fourth capping pattern **460** may be formed on a fourth upper conductive layer **427**. The fourth capping pattern **460** may fill the remaining space of the fourth trench **440t** filled with the fourth gate electrode structure **420**.

When the first through fourth capping patterns **160** through **460** partially fill the first through fourth trenches **140t** through **440t**, respectively, upper surfaces of the first through fourth capping patterns **160** through **460** may lie in the same plane with upper surfaces of first through fourth gate spacers **140** through **440** and the upper surface of the interlayer insulating film **190**.

Each of the first through fourth capping patterns **160** through **460** may serve as a guide for forming a self-aligned contact. Therefore, each of the first through fourth capping patterns **160** through **460** may include a material having an etch selectivity with respect to the interlayer insulating film **190**.

Each of the first through fourth capping patterns **160** through **460** may include at least one of silicon nitride (SiN), silicon oxynitride (SiON), silicon oxide (SiO₂), silicon carbonitride (SiCN), silicon oxycarbonitride (SiOCN), and combinations thereof.

In some implementations, a first gate insulating layer **130** may extend between the first gate spacers **140** and the first capping pattern **160**. For example, a portion of the first gate insulating layer **130** may extend between an inner wall of each first gate spacer **140** and a sidewall of the first capping pattern **160** that faces the inner wall of the first gate spacer **140**.

The degree to which each of second through fourth gate insulating layers **230** through **430** extends may be similar to the degree to which the first gate insulating layer **130** extends.

FIG. 3 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIG. 1 will be mainly described.

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Referring to FIG. 3, in the semiconductor device according to some embodiments, first through fourth work function control layers **121** through **421** may be chamfered.

An uppermost surface of the first work function control layer **121** may be lower than an uppermost surface of a first insertion layer **122** and an uppermost surface of a first filling layer **123** disposed on the first work function control layer **121**. The first work function control layer **121** may not extend up to the top of sidewalls of the first trench **140t**.

Here, the uppermost surface of the first work function control layer **121** may include both ends of the first work function control layer **121** extending along the sidewalls and a bottom surface of the first trench **140t**.

On the portion of the sidewalls of the first trench **140t** to which the first work function control layer **121** does not extend, the first insertion layer **122** and the first etch-stop layer **124** may contact each other.

An uppermost surface of the second work function control layer **221** may be lower than an uppermost surface of a second insertion layer **222** and an uppermost surface of a second filling layer **223** disposed on the second work function control layer **221**. The second work function control layer **221** may not extend up to the top of sidewalls of a second trench **240t**.

On the portion of the sidewalls of the second trench **240t** to which the second work function control layer **221** does not extend, the second insertion layer **222** and the second etch-stop layer **224** may contact each other.

An uppermost surface of the third work function control layer **321** may be lower than an uppermost surface of a third insertion layer **322** and an uppermost surface of a third filling layer **323** on the third work function control layer **321**. The third work function control layer **321** may not extend up to the top of sidewalls of a third trench **340t**.

On the portion of the sidewalls of the third trench **340t** to which the third work function control layer **321** does not extend, the third insertion layer **322** and a third etch-stop layer **324** may contact each other.

An uppermost surface of the fourth work function control layer **421** may be lower than an uppermost surface of a fourth insertion layer **422** and an uppermost surface of a fourth filling layer **423** on the fourth work function control layer **421**. The fourth work function control layer **421** may not extend up to the top of sidewalls of a fourth trench **440t**.

On the portion of the sidewalls of the fourth trench **440t** to which the fourth work function control layer **421** does not extend, the fourth insertion layer **422** and a fourth etch-stop layer **424** may contact each other.

As shown in FIG. 3, the uppermost surfaces of the first through fourth work function control layers **121** through **421** may be inclined surfaces having acute angles to the sidewalls of the first through fourth trenches **140t** through **440t**, respectively.

In some implementations, the uppermost surfaces of the first through fourth work function control layers **121** through **421** may be flat surfaces having right angles to the sidewalls of the first through fourth trenches **140t** through **440t**, respectively.

In addition, as shown in FIG. 3, all of the first through fourth work function control layers **121** through **421** may be chamfered. In some implementations, only some of the first through fourth work function control layers **121** through **421** can be chamfered in consideration of the thickness of the work function control layer and the thickness of the lower conductive layer.

FIG. 4 illustrates a view of a semiconductor device according to embodiments. For ease of description, differ-

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ences from the semiconductor device described above with reference to FIG. 1 will be mainly described.

Referring to FIG. 4, in the semiconductor device according to some embodiments, first through fourth regions I through IV may be regions in which n-channel metal oxide semiconductors (NMOSs) are formed.

For example, each of first through fourth transistors 101 through 401 may be an n-type transistor.

A first gate electrode structure 120 may include a first lower conductive layer 125, a first etch-stop layer 124, a first insertion layer 122, and a first filling layer 123.

The first insertion layer 122 may be formed on the first etch-stop layer 124. The first insertion layer 122 may contact the first etch-stop layer 124. The first insertion layer 122 may be formed along the profile of the first etch-stop layer 124.

A second gate electrode structure 220 may include a second lower conductive layer 225, a second etch-stop layer 224, a second insertion layer 222, and a second filling layer 223.

The second insertion layer 222 may be formed on the second etch-stop layer 224. The second insertion layer 222 may contact the second etch-stop layer 224. The second insertion layer 222 may be formed along the profile of the second etch-stop layer 224.

Unlike a third gate electrode structure 320 and a fourth gate electrode structure 420, the first gate electrode structure 120 and the second gate electrode structure 220 may not include a work function control layer, for example, a TiN layer.

In FIG. 4, a thickness t11 of the first lower conductive layer 125 may be substantially equal to a thickness t31 of a third lower conductive layer 325. In addition, a thickness t21 of the second lower conductive layer 225 may be substantially equal to a thickness t41 of a fourth lower conductive layer 425.

The thickness t11 of the first lower conductive layer 125 may be less than the thickness t21 of the second lower conductive layer 225.

In addition, a thickness t32 of a third work function control layer 321 may be substantially equal to a thickness t42 of a fourth work function control layer 421.

The thickness of the lower conductive layer and the presence or absence of the work function control layer may be different in each of the first through fourth transistors 101 through 401.

A threshold voltage of each of the first through fourth transistors 101 through 401 can be adjusted according to the thickness of the lower TiN layer and the presence or absence of the work function control layer.

The threshold voltages of the first through fourth transistors 101 through 401 may be different from each other.

The threshold voltage of the second transistor 201 may be higher than the threshold voltage of the first transistor 101 and less than the threshold voltage of the third transistor 301. In addition, the threshold voltage of the third transistor 301 may be less than the threshold voltage of the fourth transistor 401.

Each of the first through fourth transistors 101 through 401 illustrated in FIG. 4 may be an n-type transistor. Accordingly, the fourth transistor 401 having the largest threshold voltage may be, for example, an n-type high voltage transistor.

In addition, the third transistor 301 may be an n-type regular voltage transistor, and the second transistor 201 may be an n-type low voltage transistor. Also, the first transistor 101 having the lowest threshold voltage may be an n-type super low voltage transistor.

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In the case of the first transistor 101 and the second transistor 201 which do not have a work function control layer and whose respective lower conductive layers have different thicknesses, the threshold voltage of the second transistor 201 whose lower conductive layer has a greater thickness may be higher than the threshold voltage of the first transistor 101.

Likewise, in the case of the third transistor 301 and the fourth transistor 401 whose respective work function control layers have the same thickness and whose respective lower conductive layers have different thicknesses, the threshold voltage of the fourth transistor 401 whose lower conductive layer has a greater thickness may be higher than the threshold voltage of the third transistor 301.

Furthermore, in the case of the first transistor 101 and the third transistor 301 whose respective lower conductive layers have the same thickness and which are different in terms of the presence or absence of the work function control layer, the threshold voltage of the third transistor 301 having the work function control layer may be higher than the threshold voltage of the first transistor 101.

FIG. 5 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIG. 1 will be mainly described.

Referring to FIG. 5, in the semiconductor device according to some embodiments, a first region I and a second region II may be regions in which NMOSs are formed, and a third region III and a fourth region IV may be regions in which PMOSs are formed.

For example, a first transistor 101 and a second transistor 201 may be n-type transistors, and a third transistor 301 and a fourth transistor 401 may be p-type transistors.

A first gate electrode structure 120 may include a first lower conductive layer 125, a first etch-stop layer 124, a first insertion layer 122, and a first filling layer 123.

The first insertion layer 122 may be formed on the first etch-stop layer 124. The first insertion layer 122 may contact the first etch-stop layer 124. The first insertion layer 122 may be formed along the profile of the first etch-stop layer 124.

For example, unlike second through fourth gate electrode structures 220 through 420, the first gate electrode structure 120 may not include a work function control layer, for example, a TiN layer.

In FIG. 5, the first through fourth lower conductive layers 125 through 425 may have the same or different thicknesses.

A thickness t42 of a fourth work function control layer 421 may be less than a thickness t32 of a third work function control layer 321 and greater than a thickness t22 of a second work function control layer 221.

A threshold voltage of the second transistor 201 including the second work function control layer 221 may be higher than a threshold voltage of the first transistor 101 without a work function control layer.

In the case of the n-type transistors, the threshold voltage of the second transistor 201 including a TiN layer between a TaN layer and a TiAlC layer may be higher than the threshold voltage of the first transistor 101, which does not include a TiN layer between a TaN layer and a TiAlC layer, regardless of the thickness of a lower TiN layer.

On the other hand, a threshold voltage of the third transistor 301 including the third work function control layer 321 may be less than a threshold voltage of the fourth transistor 401 including the fourth work function control layer 421.

In the case of the p-type transistors, the threshold voltage of the third transistor 301 including a thicker TiN layer

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between a TaN layer and a TiAlC layer may be less than the threshold voltage of the fourth transistor **401** including a thinner TiN layer between a TaN layer and a TiAlC layer, regardless of the thickness of a lower TiN layer.

FIG. 6 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIG. 5 will be mainly described.

Referring to FIG. 6, in the semiconductor device according to some embodiments, a third work function control layer **321** and a fourth work function control layer **421** may be chamfered.

For example, the third work function control layer **321** and the fourth work function control layer **421** included in gate electrode structures of p-type transistors may be chamfered.

An uppermost surface of the third work function control layer **321** may be lower than an uppermost surface of a third insertion layer **322** and an uppermost surface of a third filling layer **323** disposed on the third work function control layer **321**. The third work function control layer **321** may not extend up to the top of sidewalls of a third trench **340t**.

An uppermost surface of the fourth work function control layer **421** may be lower than an uppermost surface of a fourth insertion layer **422** and an uppermost surface of a fourth filling layer **423** disposed on the fourth work function control layer **421**. The fourth work function control layer **421** may not extend up to the top of sidewalls of a fourth trench **440t**.

In FIG. 6, the third work function control layer **321** and the fourth work function control layer **421** included in the gate electrode structures of the p-type transistors are chamfered, as examples.

FIG. 7 illustrates a view of a semiconductor device according to embodiments. FIG. 8 illustrates an enlarged view of portions P and Q of FIG. 7. For ease of description, differences from the semiconductor device described above with reference to FIG. 1 will be mainly described.

For reference, only a work function control layer of each gate electrode structure, excluding an insertion layer and a filling layer, is illustrated in FIG. 8.

Referring to FIGS. 7 and 8, in the semiconductor device according to some embodiments, first through fourth gate electrode structures **120** through **420** may not include first through fourth lower conductive layers **125** through **425** and first through fourth etch-stop layers **124** through **424**.

A first region I and a second region II may be regions in which NMOSs are formed, and a third region III and a fourth region IV may be regions in which PMOSs are formed.

The first gate electrode structure **120** may include a first work function control layer **121** and a first upper conductive layer **127**. The first upper conductive layer **127** may include a first insertion layer **122** and a first filling layer **123**.

The first work function control layer **121** may be formed on a first gate insulating layer **130**. The first work function control layer **121** may contact the first gate insulating layer **130**. The first work function control layer **121** may be formed along the profile of the first gate insulating layer **130**.

A height of an uppermost surface of the first work function control layer **121** may be substantially equal to a height of an uppermost surface of the first insertion layer **122** and a height of an uppermost surface of the first filling layer **123** disposed on the first work function control layer **121**. The first work function control layer **121** may extend up to the top of sidewalls of a first trench **140t**. The uppermost surface of the first work function control layer **121** may

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include both ends of the first work function control layer **121** extending along the sidewalls and a bottom surface of the first trench **140t**.

In this embodiment, the first upper conductive layer **127** may not cover the uppermost surface of the first work function control layer **121**.

The second gate electrode structure **220** may include a second work function control layer **221** and a second upper conductive layer **227**. The second upper conductive layer **227** may include a second insertion layer **222** and a second filling layer **223**.

The second work function control layer **221** may be formed on a second gate insulating layer **230**. The second work function control layer **221** may contact the second gate insulating layer **230**. The second work function control layer **221** may be formed along the profile of the second gate insulating layer **230**.

A height of an uppermost surface of the second work function control layer **221** may be substantially equal to a height of an uppermost surface of the second insertion layer **222** and a height of an uppermost surface of the second filling layer **223** disposed on the second work function control layer **221**. The second work function control layer **221** may extend up to the top of sidewalls of a second trench **240t**.

In this embodiment, the second upper conductive layer **227** may not cover the uppermost surface of the second work function control layer **221**.

The third gate electrode structure **320** may include a third work function control layer **321** and a third upper conductive layer **327**. The third upper conductive layer **327** may include a third insertion layer **322** and a third filling layer **323**.

The third work function control layer **321** may be formed on a third gate insulating layer **330**. The third work function control layer **321** may contact the third gate insulating layer **330**.

The third work function control layer **321** may extend along part of sidewalls of a third trench **340t** and a bottom surface of the third trench **340t**. The third work function control layer **321** may not extend up to the top of the sidewalls of the third trench **340t**.

For example, the third work function control layer **321** may not be formed on part of the third gate insulating layer **330** formed on the sidewalls of the third trench **340t**. The third work function control layer **321** may be formed along the profile of the third gate insulating layer **330**.

When the third work function control layer **321** is not formed on part of the sidewalls of the third trench **340t**, the third insertion layer **322** may be formed along the profile of the third work function control layer **321** and the third gate insulating layer **330**.

For example, on portion of the sidewalls of the third trench **340t** to which the third work function control layer **321** does not extend, the third insertion layer **322** and the third gate insulating layer **330** may contact each other.

An uppermost surface of the third work function control layer **321** may be lower than an uppermost surface of the third insertion layer **322** and an uppermost surface of the third filling layer **323**.

The third upper conductive layer **327** may cover the uppermost surface of the third work function control layer **321**. For example, the third insertion layer **322** and the third filling layer **323** may cover the uppermost surface of the third work function control layer **321**.

The fourth gate electrode structure **420** may include a fourth work function control layer **421** and a fourth upper

conductive layer 427. The fourth upper conductive layer 427 may include a fourth insertion layer 422 and a fourth filling layer 423.

The fourth work function control layer 421 may be formed on a fourth gate insulating layer 430. The fourth work function control layer 421 may contact the fourth gate insulating layer 430.

The fourth work function control layer 421 may extend along part of sidewalls of a fourth trench 440t and a bottom surface of the fourth trench 440t. The fourth work function control layer 421 may not extend to the top of the sidewalls of the fourth trench 440t.

For example, the fourth work function control layer 421 may not be formed on part of the fourth gate insulating layer 430 formed on the sidewalls of the fourth trench 440t. The fourth work function control layer 421 may be formed along the profile of the fourth gate insulating layer 430.

When the fourth work function control layer 421 is not formed on part of the sidewalls of the fourth trench 440t, the fourth insertion layer 422 may be formed along the profile of the fourth gate insulating layer 430.

On the portions of the sidewalls of the fourth trench 440t to which the fourth work function control layer 421 does not extend, the fourth insertion layer 422 and the fourth gate insulating layer 430 may contact each other. An uppermost surface of the fourth work function control layer 421 may be lower than an uppermost surface of the fourth insertion layer 422 and an uppermost surface of the fourth filling layer 423.

The fourth upper conductive layer 427 may cover the uppermost surface of the fourth work function control layer 421. For example, the fourth insertion layer 422 and the fourth filling layer 423 may cover the uppermost surface of the fourth work function control layer 421.

The third work function control layer 321 and the fourth work function control layer 421 may be chamfered. The shapes of the third work function control layer 321 and the fourth work function control layer 421 will be described in detail below.

In FIG. 7, a depth d3 from an upper surface of an interlayer insulating film 190 to the third work function control layer 321 may be substantially equal to a depth d4 from the upper surface of the interlayer insulating film 190 to the fourth work function control layer 421.

In FIG. 7, a thickness t12 of the first work function control layer 121, a thickness t22 of the second work function control layer 221, a thickness t32 of the third work function control layer 321, and a thickness t42 of the fourth work function control layer 421 may be different from each other.

For example, the thickness t22 of the second work function control layer 221 may be greater than the thickness t12 of the first work function control layer 121 and less than the thickness t42 of the fourth work function control layer 421. The thickness t32 of the third work function control layer 321 may be greater than the thickness t42 of the fourth work function control layer 421. For example, the third work function control layer 321 may be thickest among the first through fourth work function control layers 121 through 421.

In the semiconductor device according to some embodiments, a threshold voltage of a first transistor 101 may be lower than a threshold voltage of a second transistor 201.

In the case of the n-type transistors, the threshold voltage of the second transistor 201 including the second work function control layer 221, which is thicker than the first work function control layer 121, may be higher than the

threshold voltage of the first transistor 101 including the first work function control layer 121.

In addition, a threshold voltage of a third transistor 301 may be lower than a threshold voltage of a fourth transistor 401.

For example, in the case of the p-type transistors, the threshold voltage of the third transistor 301 including the third work function control layer 321, which is thicker than the fourth work function control layer 421, may be smaller than the threshold voltage of the fourth transistor 401 including the fourth work function control layer 421.

In FIG. 8, the third work function control layer 321 includes a third inclined surface 321i having an acute angle $\theta 1$ with respect to a sidewall of the third trench 340t. For example, the third work function control layer 321 extending along the sidewall of the third trench 340t may have the third inclined surface 321i having the acute angle $\theta 1$ with respect to the sidewall of the third trench 340t.

In addition, the fourth work function control layer 421 may include a fourth inclined surface 421i having an acute angle $\theta 2$ with respect to a sidewall of the fourth trench 440t.

Here, the acute angles $\theta 1$ and $\theta 2$ indicate that angles measured in a clockwise direction with respect to the sidewalls of the third and fourth trenches 340t and 440t do not exceed 90 degrees.

In some implementations, the first work function control layer 121 and the second work function control layer 221 may not include inclined surfaces having acute angles to the sidewalls of the first trench 140t and the second trench 240t.

In FIG. 8, the acute angle $\theta 1$ of the third inclined surface 321i and the acute angle $\theta 2$ of the fourth inclined surface 421i are shown as being substantially equal. In some implementations, the acute angle $\theta 1$ of the third inclined surface 321i and the acute angle $\theta 2$ of the fourth inclined surface 421i may be different from each other. For example, the acute angle $\theta 2$ of the fourth inclined surface 421i may be greater than the acute angle $\theta 1$ of the third inclined surface 321i, or the acute angle $\theta 2$ of the fourth inclined surface 421i may be smaller than the acute angle $\theta 1$ of the third inclined surface 321i.

In FIG. 8, the third inclined surface 321i of the third work function control layer 321 may be the uppermost surface of the third work function control layer 321, and the fourth inclined surface 421i of the fourth work function control layer 421 may be the uppermost surface of the fourth work function control layer 421.

A distance from an upper surface of the substrate 100 to the third inclined surface 321i of the third work function control layer 321 and a distance from the upper surface of the substrate 100 to the fourth inclined surface 421i of the fourth work function control layer 421 may be less than a distance from the upper surface of the substrate 100 to the upper surface of the interlayer insulating film 190.

The above shapes of the third work function control layer 321 and the fourth work function control layer 421 may improve the gap-fill characteristics of the third upper conductive layer 327 and the fourth upper conductive layer 427.

For example, as the size of a semiconductor device becomes ever smaller, various elements (e.g., transistors) included in the semiconductor device also become smaller. Accordingly, the first through fourth trenches 140t through 440t, in each of which a plurality of functional layer patterns for forming a transistor are formed, become narrower.

Unlike the illustration in the drawings, if both ends of the thick third and fourth work function control layers 321 and 421 extend up to the top of the sidewalls of the third and fourth trenches 340t and 440t, the entrance of the third and

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fourth trenches **340t** and **440t**, in which the third and fourth upper conductive layers **327** and **427** are to be formed, may become narrower.

In this case, the metal-fill characteristics of the third and fourth upper conductive layers **327** and **427** may deteriorate.

For this reason, the uppermost surfaces of the third and fourth work function control layers **321** and **421** may not be extended up to the top of the sidewalls of the third and fourth trenches **340t** and **440t** as illustrated in FIG. 7. This may allow the third and fourth upper conductive layers **327** and **427** to have entrance areas sufficient to form the third and fourth upper conductive layers **327** and **427** reliably in a subsequent process.

In FIG. 8, the third work function control layer **321** may include a bottom portion **321b** formed on the bottom surface of the third trench **340t** and a sidewall portion **321s** formed on the sidewalls of the third trench **340t**.

The sidewall portion **321s** of the third work function control layer **321** may protrude from the bottom portion **321b** of the third work function control layer **321**.

Similarly, the fourth work function control layer **421** may include a bottom portion **421b** formed on the bottom surface of the fourth trench **440t** and a sidewall portion **421s** formed on the sidewalls of the fourth trench **440t**.

The sidewall portion **421s** of the fourth work function control layer **421** may protrude from the bottom portion **421b** of the fourth work function control layer **421**.

The sidewall portion **321s** of the third work function control layer **321** may include the uppermost surface of the third work function control layer **321**, and the sidewall portion **421s** of the fourth work function control layer **421** may include the uppermost surface of the fourth work function control layer **421**.

The sidewall portion **321s** of the third work function control layer **321** may include the third inclined surface **321i** having the acute angle $\theta 1$, and the sidewall portion **421s** of the fourth work function control layer **421** may include the fourth inclined surface **421i** having the acute angle $\theta 2$.

The thickness **t32** of the third work function control layer **321** may include the thickness of the bottom portion **321b** of the third work function control layer **321**, and the thickness **t42** of the fourth work function control layer **421** may include the thickness of the bottom portion **421b** of the fourth work function control layer **421**.

In FIG. 7, the third work function control layer **321** and the fourth work function control layer **421** are all chamfered, as an example. In some implementations, one of the third work function control layer **321** and the fourth work function control layer **421** may be chamfered, and the other one may extend up to the top of the sidewalls of a corresponding trench, like the first and second work function control layers **121** and **221**.

For example, the third work function control layer **321** thicker than the fourth work function control layer **421** may be chamfered, and the fourth work function control layer **421** may extend up to the top of the sidewalls of the fourth trench **440t**.

FIG. 9 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. 7 and 8 will be mainly described.

Referring to FIG. 9, in the semiconductor device according to some embodiments, a depth **d3** from an upper surface of an interlayer insulating film **190** to a third work function control layer **321** may be different from a depth **d4** from the upper surface of the interlayer insulating film **190** to a fourth work function control layer **421**.

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For example, the depth **d3** from the upper surface of the interlayer insulating film **190** to the third work function control layer **321** may be less than the depth **d4** from the upper surface of the interlayer insulating film **190** to the fourth work function control layer **421**.

A distance between third gate spacers **340** may be equal to a distance between fourth gate spacers **440**.

When the third work function control layer **321** is thicker than the fourth work function control layer **421**, a distance between portions of the third work function control layer **321** formed on sidewalls of a third trench **340t** may be less than a distance between portions of the fourth work function control layer **421** formed on sidewalls of a fourth trench **440t**.

In the process of chamfering a work function control layer, a distance between portions of the work function control layer formed on sidewalls of a trench may affect the position of an uppermost surface of the chamfered work function control layer.

FIG. 10 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. 7 and 8 will be mainly described.

Referring to FIG. 10, in the semiconductor device according to some embodiments, an uppermost surface of a third work function control layer **321** may not include an inclined surface having an acute angle with respect to sidewalls of a third trench **340t**.

An uppermost surface of a fourth work function control layer **421** may not include an inclined surface having an acute angle with respect to sidewalls of a fourth trench **440t**.

The uppermost surface of the third work function control layer **321** may be a flat surface having a right angle with respect to the sidewalls of the third trench **340t**. Similarly, the uppermost surface of the fourth work function control layer **421** may be a flat surface having a right angle with respect to the sidewalls of the fourth trench **440t**.

FIG. 11 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. 7 and 8 will be mainly described.

Referring to FIG. 11, in the semiconductor device according to some embodiments, a first work function control layer **121** and a second work function control layer **221** may be chamfered.

The first work function control layer **121** may extend along part of sidewalls of a first trench **140t** and a bottom surface of the first trench **140t**. The first work function control layer **121** may not extend up to the top of the sidewalls of the first trench **140t**.

For example, the first work function control layer **121** may not be formed on part of a first gate insulating layer **130** formed on the sidewalls of the first trench **140t**.

When the first work function control layer **121** is not formed on part of the sidewalls of the first trench **140t**, a first insertion layer **122** may be formed along the profile of the first work function control layer **121** and the profile of the first gate insulating layer **130**.

On the portion of the sidewalls of the first trench **140t** to which the first work function control layer **121** does not extend, the first insertion layer **122** and the first gate insulating layer **130** may contact each other.

An uppermost surface of the first work function control layer **121** may be lower than an uppermost surface of the first insertion layer **122** and an uppermost surface of a first filling layer **123**.

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Therefore, a first upper conductive layer **127** may cover the uppermost surface of the first work function control layer **121**. The first insertion layer **122** and the first filling layer **123** may cover the uppermost surface of the first work function control layer **121**.

The second work function control layer **221** may extend along part of sidewalls of a second trench **240t** and a bottom surface of the second trench **240t**. The second work function control layer **221** may not extend up to the top of the sidewalls of the second trench **240t**.

For example, the second work function control layer **221** may not be formed on part of a second gate insulating layer **230** formed on the sidewalls of the second trench **240t**.

Since the second work function control layer **221** is not formed on part of the sidewalls of the second trench **240t**, a second insertion layer **222** may be formed along the profile of the second work function control layer **221** and the profile of the second gate insulating layer **230**.

On the portion of the sidewalls of the second trench **240t** to which the second work function control layer **221** does not extend, the second insertion layer **222** and the second gate insulating layer **230** may contact each other.

An uppermost surface of the second work function control layer **221** may be lower than an uppermost surface of the second insertion layer **222** and an uppermost surface of a second filling layer **223**.

A second upper conductive layer **227** may cover the uppermost surface of the second work function control layer **221**. The second insertion layer **222** and the second filling layer **223** may cover the uppermost surface of the second work function control layer **221**.

For example, the first work function control layer **121** may include a first inclined surface **121i** having an acute angle with respect to the sidewalls of the first trench **140t**. In addition, the second work function control layer **221** may include a second inclined surface **221i** having an acute angle with respect to the sidewalls of the second trench **240t**.

The first work function control layer **121** extending along the sidewalls of the first trench **140t** may include the first inclined surface **121i** having an acute angle with respect to the sidewalls of the first trench **140t**. The second work function control layer **221** extending along the sidewalls of the second trench **240t** may include the second inclined surface **221i** having an acute angle with respect to the sidewalls of the second trench **240t**.

The first inclined surface **121i** of the first work function control layer **121** may be the uppermost surface of the first work function control layer **121**. The second inclined surface **221i** of the second work function control layer **221** may be the uppermost surface of the second work function control layer **221**.

A distance from a substrate **100** to the first inclined surface **121i** of the first work function control layer **121** and a distance from the substrate **100** to the second inclined surface **221i** of the second work function control layer **221** are less than a distance from the substrate **100** to an upper surface of an interlayer insulating film **190**.

In another implementation, the first work function control layer **121** may include a bottom portion **121b** formed on the bottom surface of the first trench **140t** and a sidewall portion **121s** formed on the sidewalls of the first trench **140t**. The second work function control layer **221** may include a bottom portion **221b** formed on the bottom surface of the second trench **240t** and a sidewall portion **221s** formed on the sidewalls of the second trench **240t**.

The sidewall portion **121s** of the first work function control layer **121** may protrude from the bottom portion

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121b of the first work function control layer **121**. The sidewall portion **221s** of the second work function control layer **221** may protrude from the bottom portion **221b** of the second work function control layer **221**.

The sidewall portion **121s** of the first work function control layer **121** may include the uppermost surface of the first work function control layer **121**. The sidewall portion **221s** of the second work function control layer **221** may include the uppermost surface of the second work function control layer **221**.

The sidewall portion **121s** of the first work function control layer **121** may include the first inclined face **121i** having an acute angle. The sidewall portion **221s** of the second work function control layer **221** may include the second inclined surface **221i** having an acute angle.

In FIG. **11**, both the first work function control layer **121** and the second work function control layer **221** are chamfered. In some implementations, one of the first work function control layer **121** and the second work function control layer **221** may be chamfered, and the other one may extend up to the top of the sidewalls of a corresponding trench.

In FIG. **11**, a depth **d2** from the upper surface of the interlayer insulating film **190** to the second work function control layer **221** is smaller than a depth **d1** from the upper surface of the interlayer insulating film **190** to the first work function control layer **121** and is greater than a depth **d4** from the upper surface of the interlayer insulating film **190** to a fourth work function control layer **421**. In some implementations, a depth **d3** from the upper surface of the interlayer insulating film **190** to a third work function control layer **321** may be less than the depth **d4** from the upper surface of the interlayer insulating film **190** to the fourth work function control layer **421**.

In some implementations, the depth **d2** from the upper surface of the interlayer insulating film **190** to the second work function control layer **221** may be substantially equal to the depth **d1** from the upper surface of the interlayer insulating film **190** to the first work function control layer **121**, and the depth **d3** from the upper surface of the interlayer insulating film **190** to the third work function control layer **321** may be substantially equal to the depth **d4** from the upper surface of the interlayer insulating film **190** to the fourth work function control layer **421**. In addition, the depth **d2** from the upper surface of the interlayer insulating film **190** to the second work function control layer **221** may be substantially equal to the depth **d4** from the upper surface of the interlayer insulating film **190** to the fourth work function control layer **421**.

FIG. **12** illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. **7** and **8** will be mainly described.

Referring to FIG. **12**, the semiconductor device according to some embodiments may further include first through fourth capping patterns **160** through **460**.

First through fourth gate electrode structures **120** through **420** may partially fill first through fourth trenches **140t** through **440t**, respectively. Upper surfaces of the first through fourth gate electrode structures **120** through **420** may be closer to a substrate **100** than an upper surface of an interlayer insulating film **190**.

FIG. **13** illustrates a view of a semiconductor device according to embodiments. FIG. **14** is an enlarged view of a portion P of FIG. **13**. For ease of description, differences from the semiconductor device described above with reference to FIGS. **7** and **8** will be mainly described.

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Referring to FIGS. 13 and 14, in the semiconductor device according to some embodiments, a sidewall portion 321s of a third work function control layer 321 may include a first portion 321sa on a bottom portion 321b of the third work function control layer 321 and a second portion 321sb.

The second portion 321sb of the sidewall portion 321s of the third work function control layer 321 may be located farther away from an upper surface of a substrate 100 than the first portion 321sa of the sidewall portion 321s of the third work function control layer 321.

A width t322 of the second portion 321sb of the sidewall portion 321s of the third work function control layer 321 may be different from a width t321 of the first portion 321sa of the sidewall portion 321s of the third work function control layer 321.

For example, the width t321 of the first portion 321sa of the sidewall portion 321s of the third work function control layer 321 may be greater than the width t322 of the second portion 321sb of the sidewall portion 321s of the third work function control layer 321.

In addition, the width t321 of the first portion 321sa of the sidewall portion 321s of the third work function control layer 321 may be substantially equal to a thickness t32 of the third work function control layer 321. For example, the width t322 of the second portion 321sb of the sidewall portion 321s of the third work function control layer 321 may be smaller than the thickness t32 of the third work function control layer 321.

The third work function control layer 321 may include a plurality of third inclined surfaces 321i having acute angles with respect to a sidewall of a third trench 340t. For example, the third work function control layer 321 extending along a sidewall of the third trench 340t may include a plurality of third inclined surfaces 321i having acute angles with respect to the sidewall of the third trench 340t.

Unlike the third work function control layer 321, a fourth work function control layer 421 extending along a sidewall of a fourth trench 440t may include one fourth inclined surface 421i having an acute angle with respect to the sidewall of the fourth trench 440t.

For example, the number of inclined surfaces included in the third work function control layer 321 extending along a sidewall of the third trench 340t may be different from the number of inclined surfaces included in the fourth work function control layer 421 extending along a sidewall of the fourth trench 440t.

For example, when the number of chamfering processes applied to the third work function control layer 321 is different from the number of chamfering processes applied to the fourth work function control layer 421, there may be a difference in shape between the third work function control layer 321 and the fourth work function control layer 421.

In FIG. 14, the first portion 321sa of the sidewall portion 321s of the third work function control layer 321 may be connected to the second portion 321sb of the sidewall portion 321s of the third work function control layer 321 by a third inclined surface 321i having an acute angle θ_{11} with respect to a sidewall of the third trench 340.

More specifically, the first portion 321sa of the sidewall portion 321s of the third work function control layer 321 may include a first sidewall that contacts a third gate insulating layer 330 and a second sidewall that faces a third insertion layer 322. The second portion 321sb of the sidewall portion 321s of the third work function control layer 321 may include a third sidewall that contacts the third gate insulating layer 330 and a fourth sidewall that faces the third insertion layer 322.

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The second sidewall of the first portion 321sa of the sidewall portion 321s of the third work function control layer 321 may be connected to the fourth sidewall of the second portion 321sb of the sidewall portion 321s of the third work function control layer 321 by the third inclined surface 321i having the acute angle θ_{11} with respect to the sidewall of the third trench 340t.

A third inclined surface 321i of the third work function control layer 321 may be an uppermost surface of the third work function control layer 321. The fourth inclined surface 421i of the fourth work function control layer 421 may be an uppermost surface of the fourth work function control layer 421.

In FIG. 14, an acute angle θ_1 of the third inclined surface 321i, which is the uppermost surface of the third work function control layer 321, may be equal to or different from the acute angle θ_{11} of the third inclined surface 321i, which connects the first portion 321sa of the sidewall portion 321s of the third work function control layer 321 and the second portions 321sb of the sidewall portion 321s of the third work function control layer 321.

Further, a connection portion of the second sidewall of the first portion 321sa of the sidewall portion 321s of the third work function control layer 321 with the third inclined surface 321i having the acute angle θ_{11} may be rounded. In addition, a connection portion of the fourth sidewall of the second portion 321sb of the sidewall portion 321s of the third work function control layer 321 with the third inclined surface 321i having the acute angle θ_{11} may be rounded.

FIG. 15 illustrates a view of a semiconductor device according to embodiments. FIG. 16 illustrates an enlarged view of portions P and Q of FIG. 15. For ease of description, differences from the semiconductor device described above with reference to FIGS. 13 and 14 will be mainly described.

Referring to FIGS. 15 and 16, in the semiconductor device according to some embodiments, a sidewall portion 321s of a third work function control layer 321 may include a first portion 321sa on a bottom portion 321b of the third work function control layer 321, a second portion 321sb, and a third portion 321sc.

The second portion 321sb of the sidewall portion 321s of the third work function control layer 321 may be located farther away from an upper surface of a substrate 100 than the first portion 321sa of the sidewall portion 321s of the third work function control layer 321. In addition, the second portion 321sb of the sidewall portion 321s of the third work function control layer 321 may be located closer to the upper surface of the substrate 100 than the third portion 321sc of the sidewall portion 321s of the third work function control layer 321.

A width t321 of the first portion 321sa of the sidewall portion 321s of the third work function control layer 321 may be greater than a width t322 of the second portion 321sb of the sidewall portion 321s of the third work function control layer 321. The width t322 of the second portion 321sb of the sidewall portion 321s of the third work function control layer 321 may be greater than a width t323 of the third portion 321sc of the sidewall portion 321s of the third work function control layer 321.

A sidewall portion 421s of a fourth work function control layer 421 may include a first portion 421sa on a bottom portion 421b of the fourth work function control layer 421 and a second portion 421sb.

The second portion 421sb of the sidewall portion 421s of the fourth work function control layer 421 may be located farther away from the upper surface of the substrate 100 than

the first portion **421sa** of the sidewall portion **421s** of the fourth work function control layer **421**.

A width **t421** of the first portion **421sa** of the sidewall portion **421s** of the fourth work function control layer **421** may be greater than a width **t422** of the second portion **421sb** of the sidewall portion **421s** of the fourth work function control layer **421**.

In addition, the width **t421** of the first portion **421sa** of the sidewall portion **421s** of the fourth work function control layer **421** may be substantially equal to a thickness **t42** of the fourth work function control layer **421**. For example, the width **t422** of the second portion **421sb** of the sidewall portion **421s** of the fourth work function control layer **421** may be smaller than the thickness **t42** of the fourth work function control layer **421**.

The third work function control layer **321** extending along a sidewall of a third trench **340t** may include a plurality of third inclined surfaces **321i** having acute angles with respect to the sidewall of the third trench **340t**. In addition, the fourth work function control layer **421** extending along a sidewall of a fourth trench **440t** may include a plurality of fourth inclined surfaces **421i** having acute angles with respect to the sidewall of the fourth trench **440t**.

In FIG. 16, the number of inclined surfaces included in the third work function control layer **321** extending along a sidewall of the third trench **340t** may be different from the number of inclined surfaces included in the fourth work function control layer **421** extending along a sidewall of the fourth trench **440t**.

A sidewall of the first portion **321sa** of the sidewall portion **321s** of the third work function control layer **321** which faces a third insertion layer **322** may be connected to a sidewall of the second portion **321sb** of the sidewall portion **321s** of the third work function control layer **321** which faces a third insertion layer **322** by a third inclined surface **321i** having an acute angle to a sidewall of the third trench **340t**.

Further, a sidewall of the second portion **321sb** of the sidewall portion **321s** of the third work function control layer **321** which faces the third insertion layer **322** may be connected to a sidewall of the third portion **321sc** of the sidewall portion **321s** of the third work function control layer **321** which faces the third insertion layer **322** by a third inclined surface **321i** having an acute angle to the sidewall of the third trench **340t**.

In FIG. 16, the first portion **421sa** of the sidewall portion **421s** of the fourth work function control layer **421** may be connected to the second portion **421sb** of the sidewall portion **421s** of the fourth work function control layer **421** by a fourth inclined surface **421i** having an acute angle $\theta 21$ with respect to a sidewall of the fourth trench **440t**.

For example, the first portion **421sa** of the sidewall portion **421s** of the fourth work function control layer **421** may include a fifth sidewall that contacts a fourth gate insulating layer **430** and a sixth sidewall that faces a fourth insertion layer **422**. The second portion **421sb** of the sidewall portion **421s** of the fourth work function control layer **421** may include a seventh sidewall that contacts the fourth gate insulating layer **430** and an eighth sidewall that faces the fourth insertion layer **422**.

The sixth sidewall of the first portion **421sa** of the sidewall portion **421s** of the fourth work function control layer **421** may be connected to the eighth sidewall of the second portion **421sb** of the sidewall portion **421s** of the fourth work function control layer **421** by the fourth inclined surface **421i** having the acute angle $\theta 21$ with respect to the sidewall of the fourth trench **440t**.

In FIG. 16, an acute angle $\theta 2$ of a fourth inclined surface **421i**, which is an uppermost surface of the fourth work function control layer **421**, may be equal to or different from the acute angle $\theta 21$ of the fourth inclined surface **421i**, which connects the first portion **421sa** of the sidewall portion **421s** of the fourth work function control layer **421** and the second portion **421sb** of the sidewall portion **421s** of the fourth work function control layer **421**.

Further, a connection portion of the sixth sidewall of the first portion **421sa** of the sidewall portion **421s** of the fourth work function control layer **421** with the fourth inclined surface **421i** having the acute angle $\theta 21$ may be rounded. In addition, a connection portion of the eighth sidewall of the second portion **421sb** of the sidewall portion **421s** of the fourth work function control layer **421** with the fourth inclined surface **421i** having the acute angle $\theta 21$ may be rounded.

FIG. 17 illustrates a view of a semiconductor device according to embodiments. FIG. 18 illustrates an enlarged view of portions P and Q of FIG. 17. For ease of description, differences from the semiconductor device described above with reference to FIGS. 13 and 14 will be mainly described.

Referring to FIGS. 17 and 18, in the semiconductor device according to some embodiments, a sidewall portion **421s** of a fourth work function control layer **421** may include a first portion **421sa** on a bottom portion **421b** of the fourth work function control layer **421** and a second portion **421sb**.

The second portion **421sb** of the sidewall portion **421s** of the fourth work function control layer **421** may be located farther away from an upper surface of a substrate **100** than the first portion **421sa** of the sidewall portion **421s** of the fourth work function control layer **421**.

A width **t421** of the first portion **421sa** of the sidewall portion **421s** of the fourth work function control layer **421** may be greater than a width **t422** of the second portion **421sb** of the sidewall portion **421s** of the fourth work function control layer **421**. The width **t421** of the first portion **421sa** of the sidewall portion **421s** of the fourth work function control layer **421** may be substantially equal to a thickness **t42** of the fourth work function control layer **421**.

The fourth work function control layer **421** may include a plurality of fourth inclined surfaces **421i** having acute angles with respect to a sidewall of a fourth trench **440t**. For example, the fourth work function control layer **421** extending along a sidewall of the fourth trench **440t** may include a plurality of fourth inclined surfaces **421i** having acute angles with respect to the sidewall of the fourth trench **440t**.

The number of inclined surfaces included in a third work function control layer **321** extending along a sidewall of a third trench **340t** may be equal to the number of inclined surfaces included in the fourth work function control layer **421** extending along a sidewall of the fourth trench **440t**.

When the number of chamfering processes applied to the third work function control layer **321** is equal to the number of chamfering processes applied to the fourth work function control layer **421**, the number of the inclined surfaces included in the third work function control layer **321** may be equal to the number of inclined surfaces included in the fourth work function control layer **421**.

FIG. 19A illustrate a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. 13 and 14 will be mainly described.

Referring to FIG. 19A, in the semiconductor device according to some embodiments, a first work function control layer **121** and a second work function control layer **221** may be chamfered.

The first work function control layer **121** may not extend up to the top of sidewalls of a first trench **140t**. The first work function control layer **121** may not be formed on part of a first gate insulating layer **130** formed on the sidewalls of the first trench **140t**.

On portions of the sidewalls of the first trench **140t** to which the first work function control layer **121** does not extend, a first insertion layer **122** and the first gate insulating layer **130** may contact each other. The first work function control layer **121** may include a first inclined surface **121i** having an acute angle with respect to the sidewalls of the first trench **140t**.

The second work function control layer **221** may not extend up to the top of sidewalls of a second trench **240t**. The second work function control layer **221** may not be formed on part of a second gate insulating layer **230** formed on the sidewalls of the second trench **240t**.

On portions of the sidewalls of the second trench **240t** to which the second work function control layer **221** does not extend, a second insertion layer **222** and the second gate insulating layer **230** may contact each other. The second work function control layer **221** may include a second inclined surface **221i** having an acute angle with respect to the sidewalls of the second trench **240t**.

The first inclined surface **121i** of the first work function control layer **121** may be an uppermost surface of the first work function control layer **121**, and the second inclined surface **221i** of the second work function control layer **221** may be an uppermost surface of the second work function control layer **221**.

The first work function control layer **121** may include a bottom portion **121b** formed on a bottom surface of the first trench **140t** and a sidewall portion **121s** formed on the sidewalls of the first trench **140t**. The second work function control layer **221** may include a bottom portion **221b** formed on a bottom surface of the second trench **240t** and a sidewall portion **221s** formed on the sidewalls of the second trench **240t**.

The sidewall portion **121s** of the first work function control layer **121** may include the first inclined face **121i** having an acute angle, and the sidewall portion **221s** of the second work function control layer **221** may include the second inclined surface **221i** having an acute angle.

The first work function control layer **121** extending along a sidewall of the first trench **140t** may include one first inclined surface **121i** having an acute angle with respect to the sidewall of the first trench **140t**. In addition, the second work function control layer **221** extending along a sidewall of the second trench **240t** may include one second inclined surface **221i** having an acute angle with respect to the sidewall of the second trench **240t**.

The number of inclined surfaces included in the third work function control layer **321** extending along a sidewall of a third trench **340t** may be different from the number of inclined surfaces included in the first work function control layer **121** extending along a sidewall of the first trench **140t** and the number of inclined surfaces included in the second work function control layer **221** extending along a sidewall of the second trench **240t**.

FIG. 19B illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. 15 and 16 will be mainly described.

Referring to FIG. 19B, in the semiconductor device according to some embodiments, a first work function control layer **121** may include a first inclined surface **121i**,

and a second work function control layer **221** may include a plurality of second inclined surfaces **221i**.

The first inclined surface **121i** of the first work function control layer **121** may be an uppermost surface of the first work function control layer **121**. One of the second inclined surfaces **221i** of the second work function control layer **221** may be an uppermost surface of the second work function control layer **221**.

A first upper conductive layer **127** may cover the uppermost surface of the first work function control layer **121**, and a second upper conductive layer **227** may cover the uppermost surface of the second work function control layer **221**.

The first work function control layer **121** may include a bottom portion **121b** formed on a bottom surface of a first trench **140t** and a sidewall portion **121s** formed on sidewalls of the first trench **140t**. The second work function control layer **221** may include a bottom portion **221b** formed on a bottom surface of a second trench **240t** and a sidewall portion **221s** formed on sidewalls of the second trench **240t**.

The sidewall portion **221s** of the second work function control layer **221** may include a first portion **221sa** on the bottom portion **221b** of the second work function control layer **221** and a second portion **221sb**.

The second portion **221sb** of the sidewall portion **221s** of the second work function control layer **221** is located farther away from an upper surface of a substrate **100** than the first portion **221sa** of the sidewall portion **221s** of the second work function control layer **221**.

The first portion **221sa** of the sidewall portion **221s** of the second work function control layer **221** may be wider than the second portion **221sb** of the sidewall portion **221s** of the second work function control layer **221**.

The second work function control layer **221** extending along a sidewall of the second trench **240t** may include a plurality of second inclined surfaces **221i** having acute angles with respect to the sidewall of the second trench **240t**.

A sidewall of the first portion **221sa** of the sidewall portion **221s** of the second work function control layer **221** which faces a second insertion layer **222** may be connected to a sidewall of the second portion **221sb** of the sidewall portion **221s** of the second work function control layer **221** which faces a second insertion layer **222** by a second inclined surface **221i** having an acute angle with respect to the sidewall of the second trench **240t**.

In FIG. 19B, the number of inclined surfaces included in a fourth work function control layer **421** extending along a sidewall of a fourth trench **440t** may be equal to the number of inclined surfaces included in the second work function control layer **221** extending along a sidewall of the second trench **240t**.

In addition, the number of inclined surfaces included in the fourth work function control layer **421** extending along a sidewall of the fourth trench **440t** may be less than the number of inclined surfaces included in a third work function control layer **321** extending along a sidewall of a third trench **340t**.

The number of inclined surfaces included in the second work function control layer **221** extending along a sidewall of the second trench **240t** may be greater than the number of inclined surfaces included in the first work function control layer **121** extending along a sidewall of the first trench **140t**.

FIG. 20 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. 7 and 8 will be mainly described.

Referring to FIG. 20, in the semiconductor device according to some embodiments, a third upper conductive layer

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327 may not cover an uppermost surface of a third work function control layer 321. A fourth upper conductive layer 427 may not cover an uppermost surface of a fourth work function control layer 421.

A height of the uppermost surface of the third work function control layer 321 may be substantially equal to a height of an uppermost surface of a third insertion layer 322 and a height of an uppermost surface of a third filling layer 323. The third work function control layer 321 may extend up to the top of sidewalls of a third trench 340t.

The third work function control layer 321 may include a third inclined surface 321i having an acute angle with respect to the sidewalls of the third trench 340t. However, the uppermost surface of the third work function control layer 321 may not be an inclined surface having an acute angle to the sidewalls of the third trench 340t.

A sidewall portion 321s of the third work function control layer 321 may include a first portion 321sa on a bottom portion 321b (see FIG. 8) of the third work function control layer 321 and a second portion 321sb. The second portion 321sb of the sidewall portion 321s of the third work function control layer 321 may be located farther away from an upper surface of a substrate 100 than the first portion 321sa of the sidewall portion 321s of the third work function control layer 321.

The first portion 321sa of the sidewall portion 321s of the third work function control layer 321 may be wider than the second portion 321sb of the sidewall portion 321s of the third work function control layer 321.

A sidewall of the first portion 321sa of the sidewall portion 321s of the third work function control layer 321 which faces the third insertion layer 322 may be connected to a sidewall of the second portion 321sb of the sidewall portion 321s of the third work function control layer 321 which faces the third insertion layer 322 by a third inclined surface 321i having an acute angle with respect to a sidewall of the third trench 340t.

A height of the uppermost surface of the fourth work function control layer 421 may be substantially equal to a height of an uppermost surface of a fourth insertion layer 422 and a height of an uppermost surface of a fourth filling layer 423. The fourth work function control layer 421 may extend up to the top of sidewalls of a fourth trench 440t.

The fourth work function control layer 421 may not include an inclined surface having an acute angle with respect to the sidewalls of the fourth trench 440t.

While the third work function control layer 321 includes the third inclined face 321i having an acute angle with respect to the sidewalls of the third trench 340t, each of a first work function control layer 121, a second work function control layer 221 and the fourth work function control layer 421 may not include an inclined surface having an acute angle with respect to the sidewalls of a corresponding trench.

FIG. 21 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIG. 20 will be mainly described.

Referring to FIG. 21, in the semiconductor device according to some embodiments, a sidewall portion 321s of a third work function control layer 321 may include a first portion 321sa, a second portion 321sb, and a third portion 321sc.

The second portion 321sb of the sidewall portion 321s of the third work function control layer 321 may be located farther away from an upper surface of a substrate 100 than the first portion 321sa of the sidewall portion 321s of the third work function control layer 321. In addition, the

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second portion 321sb of the sidewall portion 321s of the third work function control layer 321 may be located closer to the upper surface of the substrate 100 than the third portion 321sc of the sidewall portion 321s of the third work function control layer 321.

The first portion 321sa of the sidewall portion 321s of the third work function control layer 321 may be wider than the second portion 321sb of the sidewall portion 321s of the third work function control layer 321. The second portion 321sb of the sidewall portion 321s of the third work function control layer 321 may be wider than the third portion 321sc of the sidewall portion 321s of the third work function control layer 321.

A sidewall portion 421s of the fourth work function control layer 421 may include a first portion 421sa on a bottom portion 421b (see FIG. 8) of the fourth work function control layer 421 and a second portion 421sb.

The second portion 421sb of the sidewall portion 421s of the fourth work function control layer 421 may be located farther away from the upper surface of the substrate 100 than the first portion 421sa of the sidewall portion 421s of the fourth work function control layer 421.

A width t421 of the first portion 421sa of the sidewall portion 421s of the fourth work function control layer 421 may be greater than a width t422 of the second portion 421sb of the sidewall portion 421s of the fourth work function control layer 421.

The fourth work function control layer 421 may include a fourth inclined surface 421i having an acute angle with respect to sidewalls of a fourth trench 440t. An uppermost surface of the fourth work function control layer 421 may not be an inclined surface having an acute angle with respect to the sidewalls of the fourth trench 440t.

A sidewall of the first portion 321sa of the sidewall portion 321s of the third work function control layer 321, which faces a third insertion layer 322, may be connected to a sidewall of the second portion 321sb of the sidewall portion 321s of the third work function control layer 321, which faces a third insertion layer 322, by a third inclined surface 321i having an acute angle with respect to a sidewall of a third trench 340t.

In addition, a sidewall of the second portion 321sb of the sidewall portion 321s of the third work function control layer 321, which faces the third insertion layer 322, may be connected to a sidewall of the third portion 321sc of the sidewall portion 321s of the third work function control layer 321, which faces a third insertion layer 322, by a third inclined surface 321i having an acute angle to the sidewall of the third trench 340t.

Further, a sidewall of the first portion 421sa of the sidewall portion 421s of the fourth work function control layer 421, which faces a fourth insertion layer 422, may be connected to a sidewall of the second portion 421sb of the sidewall portion 421s of the fourth work function control layer 421, which faces a fourth insertion layer 422, by the fourth inclined surface 421i having an acute angle with respect to a sidewall of the fourth trench 440t.

The number of inclined surfaces included in the third work function control layer 321 extending along a sidewall of the third trench 340t may be greater than the number of inclined surfaces included in the fourth work function control layer 421 extending along a sidewall of the fourth trench 440t.

FIG. 22 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIG. 21 will be mainly described.

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Referring to FIG. 22, in the semiconductor device according to some embodiments, a sidewall portion 221s of a second work function control layer 221 may include a first portion 221sa on the second work function control layer 221 formed on a bottom surface of a second trench 240t and a second portion 221sb.

The second portion 221sb of the sidewall portion 221s of the second work function control layer 221 may be located farther away from an upper surface of a substrate 100 than the first portion 221sa of the sidewall portion 221s of the second work function control layer 221.

The first portion 221sa of the sidewall portion 221s of the second work function control layer 221 may be wider than the second portion 221sb of the sidewall portion 221s of the second work function control layer 221.

The second work function control layer 221 may include a second inclined surface 221i having an acute angle with respect to the sidewalls of the second trench 240t. However, an uppermost surface of the second work function control layer 221 may not be an inclined surface having an acute angle to the sidewalls of the second trench 240t.

A sidewall of the first portion 221sa of the sidewall portion 221s of the second work function control layer 221, which faces a second insertion layer 222, may be connected to a sidewall of the second portion 221sb of the sidewall portion 221s of the second work function control layer 221, which faces a second insertion layer 222, by the second inclined surface 221i having an acute angle with respect to a sidewall of the second trench 240t.

The number of inclined surfaces included in the second work function control layer 221 extending along a sidewall of the second trench 240t may be equal to the number of inclined surfaces included in a fourth work function control layer 421 extending along a sidewall of a fourth trench 440t.

FIG. 23 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIG. 20 will be mainly described.

Referring to FIG. 23, in the semiconductor device according to some embodiments, a sidewall portion 421s of a fourth work function control layer 421 may include a first portion 421sa on a bottom portion 421b (see FIG. 8) of the fourth work function control layer 421 and a second portion 421sb.

The second portion 421sb of the sidewall portion 421s of the fourth work function control layer 421 may be located farther from an upper surface of a substrate 100 than the first portion 421sa of the sidewall portion 421s of the fourth work function control layer 421.

The first portion 421sa of the sidewall portion 421s of the fourth work function control layer 421 may be wider than the second portion 421sb of the sidewall portion 421s of the fourth work function control layer 421.

The fourth work function control layer 421 extending along a sidewall of a fourth trench 440t may include a fourth inclined surface 421i having an acute angle with respect to the sidewall of the fourth trench 440t. An uppermost surface of the fourth work function control layer 421 may not be an inclined surface having an acute angle with respect to the sidewall of the fourth trench 440t.

The number of inclined surfaces included in a third work function control layer 321 extending along a sidewall of a third trench 340t may be equal to the number of inclined surfaces included in the fourth work function control layer 421 extending along a sidewall of the fourth trench 440t.

FIG. 24 illustrates a view of a semiconductor device according to embodiments. For ease of description, differ-

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ences from the semiconductor device described above with reference to FIG. 23 will be mainly described.

Referring to FIG. 24, in the semiconductor device according to some embodiments, a sidewall portion 221s of a second work function control layer 221 may include a first portion 221sa on the second work function control layer 221 formed on a bottom surface of a second trench 240t and a second portion 221sb.

The second work function control layer 221 may include a second inclined surface 221i having an acute angle with respect to sidewalls of the second trench 240t. An uppermost surface of the second work function control layer 221 may not be an inclined surface having an acute angle to the sidewalls of the second trench 240t.

A sidewall of the first portion 221sa of the sidewall portion 221s of the second work function control layer 221, which faces a second insertion layer 222, may be connected to a sidewall of the second portion 221sb of the sidewall portion 221s of the second work function control layer 221, which faces a second insertion layer 222, by the second inclined surface 221i having an acute angle with respect to a sidewall of the second trench 240t.

The number of inclined surfaces included in the second work function control layer 221 extending along a sidewall of the second trench 240t may be equal to the number of inclined surfaces included in a third work function control layer 321 extending along a sidewall of a third trench 340t and may be equal to the number of inclined surfaces included in a fourth work function control layer 421 extending along a sidewall of a fourth trench 440t.

FIG. 25 illustrates a layout view of a semiconductor device according to embodiments. FIG. 26 illustrates a cross-sectional view taken along the lines A-A, B-B, C-C and D-D of FIG. 25. FIGS. 27A through 27C illustrate cross-sectional views taken along the line E-E of FIG. 25. For ease of description, differences from the semiconductor device described above with reference to FIG. 1 will be mainly described.

For reference, FIG. 26 may be substantially similar to FIG. 1 except for fin patterns. Therefore, a description of elements and features identical to those of FIG. 1 will be given briefly or omitted. However, FIG. 26 can also be substantially similar to FIGS. 2 and 3 except for including fin patterns.

FIGS. 27A through 27C show cross-sectional views of a first region I in a gate direction Y1. However, it is to be understood that cross-sectional views of second through fourth regions II through IV in the gate direction may be similar to FIGS. 27A through 27C.

Referring to FIGS. 25 to 27C, in the semiconductor device according to some embodiments, each of first through fourth transistors 101 through 401 may be a p-type fin transistor.

The first through fourth transistors 101 through 401 may include first through fourth fin patterns 110 through 410, respectively.

The first fin pattern 110 may be formed in the first region I, the second fin pattern 210 may be formed in the second region II, the third fin pattern 310 may be formed in the third region III, and the fourth fin pattern 410 may be formed in the fourth region IV.

Each of the first through fourth fin patterns 110 through 410 may protrude from a substrate 100.

The first fin pattern 110 may extend along a first direction X1. The second fin pattern 210 may extend along a second

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direction X2. The third fin pattern **310** may extend along a third direction X3. The fourth fin pattern **410** may extend along a fourth direction X4.

The first through fourth fin patterns **110** through **410** may be part of the substrate **100** or may include an epitaxial layer grown from the substrate **100**.

Each of the first through fourth fin patterns **110** through **410** may include an elemental semiconductor material such as silicon or germanium. In addition, each of the first through fourth fin patterns **110** through **410** may include a compound semiconductor such as a group IV-IV compound semiconductor or a group III-V compound semiconductor.

For example, the group IV-IV compound semiconductor that forms each of the first through fourth fin patterns **110** through **410** may be, e.g., a binary or ternary compound including two or more of carbon (C), silicon (Si), germanium (Ge) and tin (Sn) or a compound obtained by doping the binary or ternary compound with a group IV element.

The group III-V compound semiconductor that forms each of the first through fourth fin patterns **110** through **410** may be, e.g., a binary, ternary, or quaternary compound composed of at least one of aluminum (Al), gallium (Ga) and indium (In) (i.e., group III elements) bonded with one of phosphorus (P), arsenic (As) and antimony (Sb) (i.e., group V elements).

In the semiconductor device according to some embodiments, the first through fourth fin patterns **110** through **410** are each described as a silicon fin pattern.

For example, when a first field insulating layer **105** partially covers side surfaces of the first fin pattern **110**, the first fin pattern **110** may protrude above the first field insulating layer **105** formed on the substrate **100**.

The first field insulating layer **105** may include, for example, an oxide layer, a nitride layer, an oxynitride layer, or a combination of the same.

In FIG. 27B, unlike in FIG. 27A, the first field insulating layer **105** may include a field liner **105b** and a field filling layer **105a**.

The field liner **105b** may be formed between the field filling layer **105a** and the first fin pattern **110** and between the field filling layer **105a** and the substrate **100**.

The field liner **105b** may include at least one of, for example, polysilicon, amorphous silicon, silicon oxynitride, silicon nitride, and silicon oxide.

In addition, in FIG. 27C, the field liner **105b** may include a first liner layer **105b2** and a second liner layer **105b1**.

The first liner layer **105b2** may be formed along a lower portion of the first fin pattern **110** and an upper surface of the substrate **100**.

The second liner layer **105b1** may be formed on the first liner layer **105b2**. The second liner layer **105b1** may be formed along the first liner layer **105b2**.

The first liner layer **105b2** may include, for example, polysilicon or amorphous silicon.

The second liner layer **105b1** may include, for example, silicon oxide.

First gate spacers **140** may be formed on the first fin pattern **110** protruding above the first field insulating layer **105**. The first gate spacers **140** may extend along a fifth direction Y1 and may intersect the first fin pattern **110**.

A first trench **140t** may be defined by the first gate spacers **140**. Accordingly, the first trench **140t** may extend along the fifth direction Y1.

Similarly, second gate spacers **240** may be formed on the second fin pattern **210** and may extend in a sixth direction Y2. Third gate spacer **340** may be formed on the third fin pattern **310** and may extend in a seventh direction Y3.

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Fourth gate spacers **440** may be formed on the fourth fin pattern **410** and may extend in an eighth direction Y4.

A first gate insulating layer **130** may be formed on the first field insulating layer **105** and the first fin pattern **110**. The first gate insulating layer **130** may be formed along an upper surface of the first field insulating layer **105** and the profile of the first fin pattern **110**.

A first interfacial layer **131** may be formed on the first fin pattern **110**. The first interfacial layer **131** may be formed along the profile of the first fin pattern **110** protruding above the upper surface of the first field insulating layer **105**.

Although the first interfacial layer **131** is shown as being not formed on the upper surface of the first field insulating layer **105**, depending on a method of forming the first interfacial layer **131**, the first interfacial layer **131** may also be formed along the upper surface of the first field insulating layer **105**.

A first high-k insulating layer **132** may be formed on the first interfacial layer **131** and along the profile of the first fin pattern **110** and the upper surface of the first field insulating layer **105**.

A description of second through fourth gate insulating layers **230** through **430** may be substantially the same as that of the first gate insulating layer **130** and thus will not be repeated.

A first gate electrode structure **120** may be formed on the first gate insulating layer **130** and may intersect the first fin pattern **110**. The first gate electrode structure **120** may be formed in the first trench **140t**. Accordingly, the first gate electrode structure may extend in the fifth direction Y1.

A first lower conductive layer **125**, a first etch-stop layer **124**, a first work function control layer **121** and a first insertion layer **122** may be formed along the profile of the first gate insulating layer **130**.

A second gate electrode structure **220** may be formed on the second gate insulating layer **230** and may intersect the second fin pattern **210**. The second gate electrode structure **220** may be formed in a second trench **240t**. Accordingly, the second gate electrode structure **220** may extend in the sixth direction Y2.

A third gate electrode structure **320** may be formed on the third gate insulating layer **330** and may intersect the third fin pattern **310**. The third gate electrode structure **320** may be formed in the third trench **340t**. Accordingly, the third gate electrode structure **320** may extend in the seventh direction Y3.

A fourth gate electrode structure **420** may be formed on the fourth gate insulating layer **430** and may intersect the fourth fin pattern **410**. The fourth gate electrode structure **420** may be formed in a fourth trench **440t**. Accordingly, the fourth gate electrode structure may extend in the eighth direction Y4.

A description of a lower conductive layer, an etch-stop layer, a work function control layer and an insertion layer included in each of the second through fourth gate electrode structures **220** through **420** may be substantially similar to that of the first lower conductive layer **125**, the first etch-stop layer **124**, the first work function control layer **121**, and the first insertion layer **122**.

First source/drain regions **150** may be formed in the first fin pattern **110**, second source/drain regions **250** may be formed in the second fin pattern **210**, third source/drain regions **350** may be formed in the third fin pattern **310**, and fourth source/drain regions **450** may be formed in the fourth fin pattern **410**.

As shown in FIG. 26, a thickness t11 of the first lower conductive layer **125** may be substantially equal to a thick-

ness **t31** of a third lower conductive layer **325**. A thickness **t21** of a second lower conductive layer **225** may be substantially equal to a thickness **t41** of a fourth lower conductive layer **425**.

The thickness **t11** of the first lower conductive layer **125** may be greater than the thickness **t21** of the second lower conductive layer **225**.

Further, a thickness **t12** of the first work function control layer **121** may be substantially equal to a thickness **t22** of a second work function control layer **221**. A thickness **t32** of a third work function control layer **321** may be substantially equal to a thickness **t42** of a fourth work function control layer **421**.

In some implementations, the thickness **t12** of the first work function control layer **121** may be greater than the thickness **t32** of the third work function control layer **321**.

A threshold voltage of the second transistor **201** may be higher than a threshold voltage of the first transistor **101** and lower than a threshold voltage of the third transistor **301**. The threshold voltage of the third transistor **301** may be lower than a threshold voltage of the fourth transistor **401**.

FIG. **28** illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. **25** through **27C** will be mainly described.

For reference, FIG. **28** may be substantially similar to FIG. **4** except for fin patterns. Therefore, a description of elements and features identical to those of FIG. **4** will be given briefly or omitted.

Referring to FIG. **28**, in the semiconductor device according to some embodiments, each of first through fourth transistors **101** through **401** may be an n-type fin transistor.

A first gate electrode structure **120** may include a first lower conductive layer **125**, a first etch-stop layer **124**, a first insertion layer **122**, and a first filling layer **123**.

The first insertion layer **122** may be formed on the first etch-stop layer **124**. The first insertion layer **122** may contact the first etch-stop layer **124**. The first insertion layer **122** may be formed along the profile of the first etch-stop layer **124**.

A second gate electrode structure **220** may include a second lower conductive layer **225**, a second etch-stop layer **224**, a second insertion layer **222**, and a second filling layer **223**.

The second insertion layer **222** may be formed on the second etch-stop layer **224**. The second insertion layer **222** may contact the second etch-stop layer **224**. The second insertion layer **222** may be formed along the profile of the second etch-stop layer **224**.

Unlike a third gate electrode structure **320** and a fourth gate electrode structure **420**, the first gate electrode structure **120** and the second gate electrode structure **220** may not include a work function control layer, for example, a TiN layer.

A thickness **t11** of the first lower conductive layer **125** may be substantially equal to a thickness **t31** of a third lower conductive layer **325**. A thickness **t21** of the second lower conductive layer **225** may be substantially equal to a thickness **t41** of a fourth lower conductive layer **425**.

The thickness **t11** of the first lower conductive layer **125** may be less than the thickness **t21** of the second lower conductive layer **225**.

A thickness **t32** of a third work function control layer **321** may be substantially equal to a thickness **t42** of a fourth work function control layer **421**.

A threshold voltage of the second transistor **201** may be higher than a threshold voltage of the first transistor **101** and lower than a threshold voltage of the third transistor **301**. In

addition, the threshold voltage of the third transistor **301** may be lower than a threshold voltage of the fourth transistor **401**.

Each of the first through fourth transistors **101** through **401** described with reference to FIGS. **5** through **24** may also include a fin pattern.

FIG. **29** illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIG. **1** will be mainly described.

Referring to FIG. **29**, in the semiconductor device according to some embodiments, a first high-k insulating layer **132** may not include a portion extending between a first gate electrode structure **120** and first gate spacers **140**.

In addition, in the first gate electrode structure **120**, a first lower conductive layer **125**, a first etch-stop layer **124**, a first work function control layer **121**, and a first insertion layer **122** may not include portions extending along inner walls of the first gate spacers **140**.

A description of second through fourth high-k insulating layers **232** through **432** may be substantially similar to that of the first high-k insulating layer **132**.

In addition, a description of a lower conductive layer, an etch-stop layer, a work function control layer, and an insertion layer included in each of the second through fourth electrode structures **220** through **420** may be similar to that of the first lower conductive layer **125**, the first etch-stop layer **124**, the first work function control layer **121**, and the first insertion layer **122**.

In FIG. **29**, a thickness **t11** of the first lower conductive layer **125** may be substantially equal to a thickness **t31** of a third lower conductive layer **325**. A thickness **t21** of a second lower conductive layer **225** may be substantially equal to a thickness **t41** of a fourth lower conductive layer **425**.

The thickness **t11** of the first lower conductive layer **125** may be greater than the thickness **t21** of the second lower conductive layer **225**.

Further, a thickness **t12** of the first work function control layer **121** may be substantially equal to a thickness **t22** of a second work function control layer **221**. In addition, a thickness **t32** of a third work function control layer **321** may be substantially equal to a thickness **t42** of a fourth work function control layer **421**.

The thickness **t12** of the first work function control layer **121** may be greater than the thickness **t32** of the third work function control layer **321**.

In FIG. **29**, first through fourth gate hard masks **155**, **255**, **355**, **455** are formed on the first through fourth gate electrode structures **120** through **420**, as an example.

FIG. **30** illustrates a plan view of a semiconductor device according to embodiments.

FIGS. **31A** and **31B** illustrate cross-sectional views taken along the line F-F of FIG. **30**. FIGS. **32A** and **32B** illustrate cross-sectional views taken along the lines G-G and H-H of FIG. **30**.

For reference, FIG. **30** schematically illustrates only a first active region **10**, a second active region **20**, and a fifth gate electrode structure **50** for ease of description.

Referring to FIGS. **30** through **32B**, the semiconductor device according to some embodiments includes a substrate **100** that includes the first active region **10**, the second active region **20** and a second field insulating layer **106** and the fifth gate electrode structure **50** which crosses the first active region **10**, the second active region **20** and the second field insulating layer **106**.

The first active region **10** and the second active region **20** may be defined by the second field insulating layer **106**. The

first active region 10 and the second active region 20 may be spatially separated but adjacent to each other.

Each of the first active region 10 and the second active region 20 may be shaped as a rectangle extending in a ninth direction X5, as an example. The first active region 10 and the second active region 20 may be arranged side by side so as to be adjacent to each other in a long-side direction.

The first active region 10 may be a region in which a PMOS is formed. The second active region 20 may be a region in which an NMOS is formed. For example, the first active region 10 may be a region in which a pull-up transistor of an SRAM is formed, and the second active region 20 may be a region in which a pull-down transistor or a pass transistor of the SRAM is formed.

For example, when a gate voltage is applied by one gate electrode structure, adjacent PMOS and NMOS regions may become the first active region 10 and the second active region 20.

In the semiconductor device according to some embodiments, the first active region 10 and the second active region 20 are described as being formed in an SRAM region.

The second field insulating layer 106 may surround the first active region 10 and the second active region 20. In some implementations, the second field insulating layer 106 may be a portion located between the first active region 10 and the second active region 20.

The second field insulating layer 106 may be disposed between the first active region 10 and the second active region 20 and may be in direct contact with the first active region 10 and the second active region 20.

For example, the second field insulating layer 106 may be in direct contact with the first active region 10 and the second active region 20 in the absence of another active region between the first active region 10 and the second active region 20.

The second field insulating layer 106 may further include at least one field liner layer formed between the first active region 10 and the second field insulating layer 106 and between the second active region 20 and the second field insulating layer 106.

A width of the second field insulating layer 106 located between the first active region 10 and the second active region 20 may be a first width W1. In addition, the second field insulating layer 106 may include a first center line CL1 located at the same distance from the first active region 10 and the second active region 20.

A distance from the first center line CL1 to the first active region 10 may be the same as a distance from the first center line CL1 to the second active region 20 and may be half the width W1 of the second field insulating layer 106. The first center line CL1 of the second field insulating layer 106 may extend parallel to the first active region 10 and the second active region 20.

The fifth gate electrode structure 50 may be formed on the substrate 100. The fifth gate electrode structure 50 may cross the first active region 10, the second active region 20, and the second field insulating layer 106. The fifth gate electrode structure 50 may extend in a tenth direction Y5.

The fifth gate electrode structure 50 may include a first gate electrode 520 and a second gate electrode 620. The first gate electrode 520 and the second gate electrode 620 may be in contact with each other, for example, in direct contact with each other.

The first gate electrode 520 may be a p-type metallic gate electrode and may be formed on the first active region 10 and the second field insulating layer 106. The second gate electrode 620 may be an n-type metallic gate electrode and

may be formed on the second active region 20 and the second field insulating layer 106.

A fifth transistor 10p may be formed in a region in which the first active region 10 and the fifth gate electrode structure 50 intersect each other. A sixth transistor 10n may be formed in a region in which the second active region 20 and the fifth gate electrode structure 50 intersect each other. The fifth transistor 10p may be a p-type transistor, and the sixth transistor 10n may be an n-type transistor.

For example, the fifth transistor 10p and the sixth transistor 10n of different conductivity types may share the fifth gate electrode structure 50.

The first gate electrode 520 may extend onto the second field insulating layer 106. The first gate electrode 520 may overlap not only the first active region 10 but also a portion of the second field insulating layer 106.

The second gate electrode 620 may be in direct contact with the first gate electrode 520. The second gate electrode 620 may overlap not only the second active region 20 but also the other portion of the second field insulating layer 106 that is not overlapped by the first gate electrode 520.

The fifth gate electrode structure 50 may include a first contact surface MI1 at which the first gate electrode 520 and the second gate electrode 620 contact each other. The first contact surface MI1 at which the first gate electrode 520 and the second gate electrode 620 contact each other may be located on the second field insulating layer 106.

The first contact surface MI1 between the first gate electrode 520 and the second gate electrode 620 may not coincide with the first center line CL1 of the second field insulating layer 106. For example, the first contact surface MI1 between the first gate electrode 520 and the second gate electrode 620 may be closer to the second active region 20 than to the first active region 10 or may be closer to the first active region 10 than to the second active region 20.

In the semiconductor device according to some embodiments described with reference to FIG. 30, the first contact surface MI1 between the first gate electrode 520 and the second gate electrode 620 may be located closer to the second active region 20 than to the first active region 10.

In FIG. 30, the first active region 10, the first center line CL1, the first contact surface MI1 and the second active region 20 may be arranged sequentially in this order such that the second gate electrode 620 does not overlap the first center line CL1 of the second field insulating layer 106. The first contact surface MI1 may be located between the second active region 20 and the first center line CL1 of the second field insulating layer 106.

The first active region 10 may include a channel region of a p-type transistor. The second active region 20 may include a channel region of an n-type transistor. The first contact surface MI1 between the first gate electrode 520 and the second gate electrode 620 may be closer to the second active region 20 than to the first active region 10. Accordingly, the first contact surface MI1 may be closer to the channel region of the n-type transistor than to the channel region of the p-type transistor.

A width of a portion of the first gate electrode 520 that extends on the second field insulating layer 106 may be a first overlapping width W11. For example, the width of the first gate electrode 520 from the first contact surface MI1 to a boundary of the first active region 10 may be the first overlapping width W11.

A width of a portion of the second gate electrode 620 that extends on the second field insulating layer 106 may be a second overlapping width W12. The width of the second

gate electrode **620** from the first contact surface **MI1** to a boundary of the second active region **20** may be the second overlapping width **W12**.

The first contact surface **MI1** between the first gate electrode **520** and the second gate electrode **620** may not coincide with the first center line **CL1** of the second field insulating layer **106**. Accordingly, the first overlapping width **W11** may be different from the second overlapping width **W12**.

In the semiconductor devices according to some embodiments described with reference to FIG. **30**, the first overlapping width **W11** may be larger than the second overlapping width **W12** because the first contact surface **MI1** between the first gate electrode **520** and the second gate electrode **620** may be located closer to the second active region **20** than to the first active region **10**.

The width **W11** of the first gate electrode **520** overlapping the second field insulating layer **106** may be greater than the width **W12** of the second gate electrode **620** overlapping the second field insulating layer **106**.

The first gate electrode **520** and the second gate electrode **620** may be in direct contact with each other. Accordingly, the sum of the width **W11** of the first gate electrode **520** overlapping the second field insulating layer **106** and the width **W12** of the second gate electrode **620** overlapping the second field insulating layer **106** may be equal to the width **W1** of the second field insulating layer **106**.

The structures of the first gate electrode **520** and the second gate electrode **620** will be described in detail below.

An interlayer insulating film **190** may be formed on the substrate **100**. The interlayer insulating film **190** may include a fifth trench **50t**.

The fifth trench **50t** may cross the first active region **10**, the second field insulating layer **106**, and the second active region **20**. For example, the fifth trench **50t** may intersect the first active region **10** and the second active region **20**. The fifth trench **50t** may extend in the tenth direction **Y5**.

Fifth gate spacers **55** may be formed on the substrate **100**. The fifth gate spacers **55** may define the fifth trench **50t**. The fifth gate spacers **55** may be formed on sidewalls of the fifth gate electrode structure **50**.

The fifth gate electrode structure **50** may extend in the tenth direction **Y5**. Accordingly, the fifth gate electrode structure **50** may include long sides extending in the tenth direction **Y5** and short sides extending in the ninth direction **X5**.

In FIGS. **31A** through **32B**, the fifth gate spacers **55** may be formed on sidewalls including the long sides of the fifth gate electrode structure **50** and sidewalls including the short sides of the fifth gate electrode structure **50**. In some implementations, the fifth gate spacers **55** may be formed on the sidewalls including the long sides of the fifth gate electrode structure **50** but may not be formed on the sidewalls including the short sides of the fifth gate electrode structure **50**.

In some implementations, thicknesses of the fifth gate spacers **55** formed on the sidewalls including the long sides of the fifth gate electrode structure **50** may be different from thicknesses of the fifth gate spacers **55** formed on the sidewalls including the short sides of the fifth gate electrode structure **50**.

A fifth gate insulating layer **530** and a sixth gate insulating layer **630** may be formed on the substrate **100**. The fifth gate insulating layer **530** may be formed on the first active region **10** and the second field insulating layer **106**. The sixth gate insulating layer **630** may be formed on the second active region **20** and the second field insulating layer **106**.

The fifth gate insulating layer **530** and the sixth gate insulating layer **630** may extend along sidewalls and a bottom surface of the fifth trench **50t**. The fifth and sixth gate insulating layers **530** and **630** extending along the bottom surface of the fifth trench **50t** may traverse the first active region **10**, the second field insulating layer **106**, and the second active region **20**.

The fifth gate insulating layer **530** and the sixth gate insulating layer **630** may be separated by the first contact surface **MI1** of the fifth gate electrode structure **50**. The fifth gate insulating layer **530** and the sixth gate insulating layer **630** may be formed at a same level.

Each of the fifth gate insulating layer **530** and the sixth gate insulating layer **630** may include a high-k insulating layer.

In FIGS. **31B** and **32B**, a fifth interfacial layer **531** and a sixth interfacial layer **631** may be formed between the fifth gate insulating layer **530** and the first active region **10** and between the sixth gate insulating layer **630** and the second active region **20**, respectively.

In FIGS. **31B** and **32B**, upper surfaces of the fifth and sixth interfacial layers **531** and **631** may lie in a same plane as an upper surface of the second field insulating layer **106**, as an example.

The fifth gate electrode structure **50** may be formed on the fifth gate insulating layer **530** and the sixth gate insulating layer **630**. The fifth gate insulating layer **530** and the sixth gate insulating layer **630** may be formed between the fifth gate electrode structure **50** and the substrate **100**. The fifth gate insulating layer **530** and the sixth gate insulating layer **630** may be formed under the fifth gate electrode structure **50**.

The fifth gate electrode structure **50** may fill the fifth trench **50t**. An upper surface of the fifth gate electrode structure **50** may lie in a same plane as upper surfaces of the fifth gate spacers **55** and an upper surface of the interlayer insulating film **190**.

The first gate electrode **520** may include a fifth lower conductive layer **525**, a fifth etch-stop layer **524**, a fifth work function control layer **521**, a fifth insertion layer **522**, and a fifth filling layer **523** formed sequentially on the fifth gate insulating layer **530**.

The second gate electrode **620** may include a sixth lower conductive layer **625**, a sixth etch-stop layer **624**, a sixth work function control layer **621**, a sixth insertion layer **622**, and a sixth filling layer **623** formed sequentially on the sixth gate insulating layer **630**.

The fifth lower conductive layer **525** and the sixth lower conductive layer **625** may be formed on the fifth gate insulating layer **530** and the sixth gate insulating layer **630**. The fifth lower conductive layer **525** may contact the fifth gate insulating layer **530**, and the sixth lower conductive layer **625** may contact the sixth gate insulating layer **630**.

The fifth lower conductive layer **525** may be formed on the first active region **10** and the second field insulating layer **106**. The sixth lower conductive layer **625** may be formed on the second active region **20** and the second field insulating layer **106**.

The fifth lower conductive layer **525** and the sixth lower conductive layer **625** may extend along the sidewalls and bottom surface of the fifth trench **50t**. The fifth lower conductive layer **525** may extend along the profile of the fifth gate insulating layer **530**. The sixth lower conductive layer **625** may extend along the profile of the sixth gate insulating layer **630**.

The fifth lower conductive layer 525 and the sixth lower conductive layer 625 may be separated by the first contact surface M11 of the fifth gate electrode structure 50.

The fifth lower conductive layer 525 and the sixth lower conductive layer 625 may include the same material. The fifth lower conductive layer 525 and the sixth lower conductive layer 625 may include, for example, TiN.

The fifth etch-stop layer 524 and the sixth etch-stop layer 624 may be formed on the fifth lower conductive layer 525 and the sixth lower conductive layer 625. The fifth etch-stop layer 524 may be formed on the first active region 10 and the second field insulating layer 106. The sixth etch-stop layer 624 may be formed on the second active region 20 and the second field insulating layer 106.

The fifth etch-stop layer 524 and the sixth etch-stop layer 624 may extend along the sidewalls and bottom surface of the fifth trench 507. The fifth etch-stop layer 524 may extend along the profile of the fifth lower conductive layer 525, and the sixth etch-stop layer 624 may extend along the profile of the sixth lower conductive layer 625.

The fifth etch-stop layer 524 and the sixth etch-stop layer 624 may be separated by the first contact surface M11 of the fifth gate electrode structure 50. The fifth etch-stop layer 524 and the sixth etch-stop layer 624 may be formed at a same level. A thickness of the fifth etch-stop layer 524 on the first active region 10 may be, for example, substantially equal to a thickness of the sixth etch-stop layer 624 on the second active region 20.

The fifth etch-stop layer 524 and the sixth etch-stop layer 624 may include the same material. The fifth etch-stop layer 524 and the sixth etch-stop layer 624 may include, for example, TaN.

The fifth work function control layer 521 and the sixth work function control layer 621 may be formed on the fifth etch-stop layer 524 and the sixth etch-stop layer 624. The fifth work function control layer 521 may contact the fifth etch-stop layer 524. The sixth work function control layer 621 may contact the sixth etch-stop layer 624.

The fifth work function control layer 521 may be formed on the first active region 10 and the second field insulating layer 106. The sixth work function control layer 621 may be formed on the second active region 20 and the second field insulating layer 106. The fifth work function control layer 521 and the sixth work function control layer 621 may be in direct contact with each other.

The fifth work function control layer 521 and the sixth work function control layer 621 may extend along the sidewalls and bottom surface of the fifth trench 507. The fifth work function control layer 521 may extend along the profile of the fifth gate insulating layer 530 and the fifth etch-stop layer 524, and the sixth work function control layer 621 may extend along the profile of the sixth gate insulating layer 630 and the sixth etch-stop layer 624.

The fifth work function control layer 521 and the sixth work function control layer 621 may include the same material. For example, the fifth work function control layer 521 and the sixth work function control layer 621 may be the same material layer. The fifth work function control layer 521 and the sixth work function control layer 621 may include, for example, TiN.

A thickness t51 of the fifth work function control layer 521 may be different from a thickness t61 of the sixth work function control layer 621. For example, the thickness t51 of the fifth work function control layer 521 may be greater than the thickness t61 of the sixth work function control layer 621.

The thickness t51 of the fifth work function control layer 521 included in a p-type gate electrode may be greater than the thickness t61 of the sixth work function control layer 621 included in an n-type gate electrode. For example, the thickness t51 of the fifth work function control layer 521 may be a thickness on the first active region 10, and the thickness t61 of the sixth work function control layer 621 may be a thickness on the second active region 20.

The first contact surface M11 between the first gate electrode 520 and the second gate electrode 620 may be defined between the fifth and sixth work function control layers 521 and 621 having different thicknesses. For example, if the fifth gate electrode structure 50 were to be cut along a normal to the substrate 100 based on a boundary between the fifth work function control layer 521 and the sixth work function control layer 621 extending on the second field insulating layer 106, the first contact surface M11 between the first gate electrode 520 and the second gate electrode 620 would be formed.

The first contact surface M11 between the first gate electrode 520 and the second gate electrode 620 may be defined as the boundary between the fifth work function control layer 521 and the sixth work function control layer 621. Accordingly, the width W11 by which the fifth work function control layer 521 and the second field insulating layer 106 overlap each other may be different from the width W12 by which the sixth work function control layer 621 and the second field insulating layer 106 overlap each other.

In FIGS. 30 through 31B, the first contact surface M11 between the first gate electrode 520 and the second gate electrode 620 is located closer to the second active region 20 than to the first active region 10. Therefore, the width W11 by which the fifth work function control layer 521 and the second field insulating layer 106 overlap each other may be greater than the width W12 by which the sixth work function control layer 621 and the second field insulating layer 106 overlap each other.

The first active region 10, the first center line CL1, the first contact surface M11 and the second active region 20 may be arranged sequentially in this order. Accordingly, the sixth work function control layer 621 may not overlap the first center line CL1 of the second field insulating layer 106.

The fifth insertion layer 522 and the sixth insertion layer 622 may be formed on the fifth work function control layer 521 and the sixth work function control layer 621. The fifth insertion layer 522 and the sixth insertion layer 622 may be in direct contact with each other.

The fifth insertion layer 522 may be formed on the first active region 10 and the second field insulating layer 106. The sixth insertion layer 622 may be formed on the second active region 20 and the second field insulating layer 106.

The fifth insertion layer 522 and the sixth insertion layer 622 may extend along the sidewalls and bottom surface of the fifth trench 507. The fifth insertion layer 522 and the sixth insertion layer 522 may extend along the profile of the fifth and sixth work function control layers 521 and 621 which are in direct contact with each other.

The fifth insertion layer 522 and the sixth insertion layer 622 may be separated by the first contact surface M11 of the fifth gate electrode structure 50. The fifth insertion layer 522 and the sixth insertion layer 622 may be formed at the same level.

A thickness t52 of the fifth insertion layer 522 may be substantially equal to a thickness t62 of the sixth insertion layer 622. The thickness t52 of the fifth insertion layer 522 may be a thickness on the first active region 10, and the

thickness **t62** of the sixth insertion layer **622** may be a thickness on the second active region **20**.

The fifth insertion layer **522** and the sixth insertion layer **622** may include the same material. The fifth insertion layer **522** and the sixth insertion layer **622** may include, for example, one of Ti, TiAl, TiAlN, TiAlC, and TiAlCN.

In the semiconductor device according to some embodiments, the fifth insertion layer **522** and the sixth insertion layer **622** may be described as layers containing TiAl.

The fifth and sixth filling layers **523** and **623** may be formed on the fifth and sixth insertion layers **522** and **622**. The fifth filling layer **523** and the sixth filling layer **623** may be in direct contact with each other.

The fifth filling layer **523** may be formed on the first active region **10** and the second field insulating layer **106**. The sixth filling layer **623** may be formed on the second active region **20** and the second field insulating layer **106**.

The fifth filling layer **523** and the sixth filling layer **623** may be separated by the first contact surface **M11** of the fifth gate electrode structure **50**. The fifth filling layer **523** and the sixth filling layer **623** may be formed on the same level.

The fifth filling layer **523** and the sixth filling layer **623** may include the same material. The fifth and sixth filling layers **523** and **623** may include at least one of W, Al, Co, Cu, Ru, Ni, Pt, Ni—Pt, and TiN.

The fifth insertion layer **522** and the fifth filling layer **523** disposed on the fifth work function control layer **521** may be a fifth upper conductive layer **527**, and the sixth insertion layer **622** and the sixth filling layer **623** disposed on the sixth work function control layer **621** may be a sixth upper conductive layer **627**.

A thickness **h1** of the fifth upper conductive layer **527** may be a distance from the upper surface of the interlayer insulating film **190** to the fifth work function control layer **521** on the bottom surface of the fifth trench **50t**, and a thickness **h2** of the sixth upper conductive layer **627** may be a distance from the upper surface of the interlayer insulating film **190** to the sixth work function control layer **621** on the bottom surface of the fifth trench **50t**.

The thickness **h1** of the fifth upper conductive layer **527** may be different from the thickness **h2** of the sixth upper conductive layer **627** on the second field insulating layer **106**. For example, the thickness **h1** of the fifth upper conductive layer **527** may be less than the thickness **h2** of the sixth upper conductive layer **627**.

Fifth source/drain regions **550** may respectively be formed on both sides of the first gate electrode **520**, and sixth source/drain regions **650** may respectively be formed on both sides of the second gate electrode **620**.

Although the fifth source/drain regions **550** and the sixth source/drain regions **650** are shown as including an epitaxial layer formed in the substrate **100**, in some embodiments, the fifth source/drain regions **550** and the sixth source/drain regions **650** may be impurity regions formed by implanting impurities into the substrate **100**.

In addition, the fifth source/drain regions **550** and the sixth source/drain regions **650** may be elevated source/drain regions having upper surfaces protruding above an upper surface of the substrate **100**.

FIGS. **33** and **34** illustrate views of a semiconductor device according to embodiments.

For ease of description, differences from the semiconductor device described above with reference to FIGS. **30** through **32B** will be mainly described.

For reference, FIG. **33** illustrates a cross-sectional view taken along the line F-F of FIG. **30**, and FIG. **34** illustrates a cross-sectional view taken along the lines G-G and H-H of FIG. **30**.

Referring to FIGS. **33** and **34**, in the semiconductor device according to some embodiments, a fifth work function control layer **521** may contact a fifth gate insulating layer **530**, and a sixth work function control layer **621** may contact a sixth gate insulating layer **630**.

A first gate electrode **520** may include the fifth work function control layer **521**, a fifth insertion layer **522**, and a fifth filling layer **523** formed sequentially on the fifth gate insulating layer **530**.

A second gate electrode **620** may include the sixth work function control layer **621**, a sixth insertion layer **622** and a sixth filling layer **623** formed sequentially on the sixth gate insulating layer **630**.

A conductive layer may not be interposed between the fifth gate insulating layer **530** and the fifth work function control layer **521**. Similarly, a conductive layer may not be interposed between the sixth gate insulating layer **630** and the sixth work function control layer **621**.

FIG. **35** illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. **30** through **32B** will be mainly described.

Referring to FIG. **35**, the semiconductor device according to some embodiments may further include a fifth capping pattern **60**.

A fifth gate electrode structure **50** may partially fill a fifth trench **50t**. The fifth capping pattern **60** may be formed on the fifth gate electrode structure **50**.

FIG. **36** illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. **30** through **32B** will be mainly described.

Referring to FIG. **36**, a fifth gate insulating layer **530** and a sixth gate insulating layer **630** may include portions extending between a fifth gate electrode structure **50** and fifth gate spacers **55**.

In addition, in a first gate electrode **520**, a fifth lower conductive layer **525**, a fifth etch-stop layer **524**, a fifth work function control layer **521**, and a fifth insertion layer **522** may not include portions extending along an inner wall of a fifth gate spacer **55**.

Similarly, in a second gate electrode **620**, a sixth lower conductive layer **625**, a sixth etch-stop layer **624**, a sixth work function control layer **621**, and a sixth insertion layer **622** may not include portions extending along an inner wall of a fifth gate spacer **55**.

FIG. **37** illustrates a plan view of a semiconductor device according to embodiments. FIG. **38** illustrates a cross-sectional view taken along the line F-F of FIG. **37**. For ease of description, differences from the semiconductor device described above with reference to FIGS. **30** through **32B** will be mainly described.

For reference, FIG. **38** may be substantially the same as FIG. **31** except for including fin patterns. Therefore, a description of elements and features identical to those of FIG. **31** will not be repeated or will be given briefly. For example, a fifth fin pattern **510** may correspond to a first active region **10**, and a sixth fin pattern **610** may correspond to a second active region **20**.

In FIG. **37**, a cross-sectional view taken along the fifth fin pattern **510** and the sixth fin pattern **610** may be substantially the same as in FIG. **32A** except for including fin patterns.

For ease of description, FIG. 37 schematically illustrates only the fifth fin pattern 510, the sixth fin pattern 610, and a fifth gate electrode structure 50.

Referring to FIGS. 37 and 38, the semiconductor device according to some embodiments includes the fifth fin pattern 510, the sixth fin pattern 610 disposed adjacent to the fifth fin pattern 510, a second field insulating layer 106 located between the fifth fin pattern 510 and the sixth fin pattern 610, and the fifth gate electrode structure 50 crossing the fifth fin pattern 510, the second field insulating layer 106 and the sixth fin pattern 610.

The fifth fin pattern 510 and the sixth fin pattern 610 may protrude from a substrate 100. The fifth fin pattern 510 and the sixth fin pattern 610 may extend in a ninth direction X5.

The fifth fin pattern 510 may be a region in which a PMOS is formed. The sixth fin type pattern 610 may be a region in which an NMOS is formed. For example, the fifth fin pattern 510 and the sixth fin pattern 610 may be formed in an SRAM region.

The fifth fin pattern 510 and the sixth fin pattern 610 may be part of the substrate 100 or may include an epitaxial layer grown from the substrate 100.

Each of the fifth fin pattern 510 and sixth fin pattern 610 may include an elemental semiconductor material such as silicon or germanium. In addition, each of the fifth fin pattern 510 and the sixth fin pattern 610 may include a compound semiconductor such as a group IV-IV compound semiconductor or a group III-V compound semiconductor.

The second field insulating layer 106 may partially cover sidewalls of the fifth fin pattern 510 and sidewalls of the sixth fin pattern 610. Accordingly, the fifth fin pattern 510 and the sixth fin pattern 610 may protrude above an upper surface of the second field insulating layer 106 formed on the substrate 100.

The fifth fin pattern 510 and the sixth fin pattern 610 may be defined by the second field insulating layer 106. The fifth fin pattern 510 and the sixth fin pattern 610 may be spatially separated but adjacent to each other.

The second field insulating layer 106 may be disposed between the fifth and the sixth fin patterns 510 and 610 and may be in direct contact with the fifth and the sixth fin patterns 510 and 610.

The second field insulating layer 106 may be in direct contact with the fifth and the sixth fin patterns 510 and 610 in the absence of a fin pattern that protrudes above the upper surface of the second field insulating layer 106 between the fifth and sixth fin patterns 510 and 610.

In some implementations, the second field insulating layer 106 may further include at least one field liner layer between the fifth fin pattern 510 and the second field insulating layer 106 and between the sixth fin pattern 610 and the second field insulating layer 106.

A distance from a first center line CL1 to the fifth fin pattern 510 may be equal to a distance from the first center line CL1 to the sixth fin pattern 610.

The fifth gate electrode structure 50 may cross the fifth fin pattern 510, the sixth fin pattern 610, and the second field insulating layer 106. The fifth gate electrode structure 50 may extend in a tenth direction Y5.

A first gate electrode 520 may be formed on the fifth fin pattern 510 and the second field insulating layer 106. A second gate electrode 620 may be formed on the sixth fin pattern 610 and the second field insulating layer 106.

A fifth transistor 10p formed in a region where the fifth fin pattern 510 and the fifth gate electrode structure 50 intersect each other may be a p-type fin transistor. A sixth transistor

10n formed in a region where the sixth fin pattern 610 and the fifth gate electrode structure 50 intersect each other may be an n-type fin transistor.

A first contact surface MI1 between the first gate electrode 520 and the second gate electrode 620 may be closer to the sixth fin pattern 610 than to the fifth fin pattern 510 or may be closer to the fifth fin pattern 510 than to the sixth fin pattern 610.

In the semiconductor device according to the embodiment described with reference to FIG. 37, the first contact surface MI1 between the first gate electrode 520 and the second gate electrode 620 may be located closer to the sixth fin pattern 610 than to the fifth fin pattern 510.

Accordingly, a width W11 of the first gate electrode 520 overlapping the second field insulating layer 106 may be greater than a width W12 of the second gate electrode 620 overlapping the second field insulating layer 106.

A fifth trench 50t defined by fifth gate spacers 55 may traverse the fifth fin pattern 510, the second field insulating layer 106, and the sixth fin pattern 610. For example, the fifth trench 50t may intersect the fifth fin pattern 510 and the sixth fin pattern 610.

A fifth gate insulating layer 530 may be formed on the second field insulating layer 106 and the fifth fin pattern 510. The fifth gate insulating layer 530 may be formed along the upper surface of the second field insulating layer 106 and the profile of the fifth fin pattern 510.

A sixth gate insulating layer 630 may be formed on the second field insulating layer 106 and the sixth fin pattern 610. The sixth gate insulating layer 630 may be formed along the upper surface of the second field insulating layer 106 and the profile of the sixth fin pattern 610.

The fifth and sixth gate insulating layers 530 and 630 extending along a bottom surface of the fifth trench 50t may be formed along the profile of the fifth fin pattern 510, the upper surface of the second field insulating layer 106, and the profile of the sixth fin pattern 610.

The fifth gate electrode structure 50 may be formed on the fifth and sixth gate insulating layers 530 and 630.

The first gate electrode 520 may be formed on the fifth gate insulating layer 530 and may intersect the fifth fin pattern 510. The second gate electrode 620 may be formed on the sixth gate insulating layer 630 and may intersect the sixth fin pattern 610.

A fifth lower conductive layer 525, a fifth etch-stop layer 524, a fifth work function control layer 521, and a fifth insertion layer 522 may be formed along the profile of the fifth gate insulating layer 530.

For example, the fifth lower conductive layer 525, the fifth etch-stop layer 524, the fifth work function control layer 521, and the fifth insertion layer 522 may extend along the profile of the fifth fin pattern 510 and the upper surface of the second field insulating layer 106.

A sixth lower conductive layer 625, a sixth etch-stop layer 624, a sixth work function control layer 621, and a sixth insertion layer 622 may be formed along the profile of the sixth gate insulating layer 630.

The sixth lower conductive layer 625, the sixth etch-stop layer 624, the sixth work function control layer 621, and the sixth insertion layer 622 may extend along the profile of the sixth fin pattern 610 and the upper surface of the second field insulating layer 106.

The fifth and sixth work function control layers 521 and 621 extending along the bottom surface of the fifth trench 50t may extend continuously along the profile of the fifth fin pattern 510, the upper surface of the second field insulating layer 106 and the profile of the sixth fin pattern 610.

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In FIG. 38, a thickness t51 of the fifth work function control layer 521 may be greater than a thickness t61 of the sixth work function control layer 621. A thickness t52 of the fifth insertion layer 522 may be substantially equal to a thickness t62 of the sixth insertion layer 622.

A thickness h1 of a fifth upper conductive layer 527 from an upper surface of an interlayer insulating film 190 to the fifth work function control layer 521 may be less than a thickness h2 of a sixth upper conductive layer 627 from the upper surface of the interlayer insulating film 190 to the sixth work function control layer 621.

FIG. 39A illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. 37 and 38 will mainly be described.

Referring to FIG. 39A, in the semiconductor device according to some embodiments, a first gate electrode 520 may include a fifth work function control layer 521, a fifth insertion layer 522, and a fifth filling layer 523 formed sequentially on a fifth gate insulating layer 530.

A second gate electrode 620 may include a sixth work function control layer 621, a sixth insertion layer 622, and a sixth filling layer 623 formed sequentially on a sixth gate insulating layer 630.

The fifth work function control layer 521 may contact the fifth gate insulating layer 530. The sixth work function control layer 621 may contact the sixth gate insulating layer 630.

FIG. 39B illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIG. 39A will be mainly described.

Referring to FIG. 39B, in the semiconductor device according to some embodiments, a width W3 of a fifth fin pattern 510 may be different from a width W4 of a sixth fin pattern 610.

For example, the width W3 of the fifth fin pattern 510 may be greater than the width W4 of the sixth fin pattern 610.

The width of a fin pattern may denote the width of the fin pattern at a portion where the fin pattern meets an upper surface of a second field insulating layer 106. For example, if the number of processes for adjusting the shape of the fifth fin pattern 510 is different from the number of processes for adjusting the shape of the sixth fin pattern 610, the width of the fifth fin pattern 510 may be different from that of the sixth fin pattern 610.

In some implementations, the width W3 of the fifth fin pattern 510 may be smaller than the width W4 of the sixth fin pattern 610.

FIG. 40 illustrates a plan view of a semiconductor device according to embodiments. FIG. 41 illustrates a cross-sectional view taken along the line F-F of FIG. 40. For ease of description, differences from the semiconductor device described above with reference to FIGS. 30 through 32B will be mainly described.

In addition, in FIG. 40, a cross-sectional view taken along a first active region 10 and a second active region 20 may be substantially the same as that of FIG. 32A.

Referring to FIGS. 40 and 41, in the semiconductor device according to some embodiments, a first contact surface MI1 between a first gate electrode 520 and a second gate electrode 620 may be located closer to the first active region 10 than to the second active region 20.

The first active region 10, the first contact surface MI1, a first center line CL1, and the second active region 20 may be arranged sequentially in this order. Accordingly, the first gate electrode 520 may not overlap the first center line CL1

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of a second field insulating layer 106. For example, the first contact surface MI1 may be located between the first active region 10 and the first center line CL1 of the second field insulating layer 106.

The first active region 10 may include a channel region of a p-type transistor, and the second active region 20 includes a channel region of an n-type transistor. The first contact surface MI1 between the first gate electrode 520 and the second gate electrode 620 may be located closer to the first active region 10 than to the second active region 20. Accordingly, the first contact surface MI1 may be closer to the channel region of the p-type transistor than to the channel region of the n-type transistor.

The first contact surface MI1 between the first gate electrode 520 and the second gate electrode 620 may be located closer to the first active region 10 than to the second active region 20. Accordingly, a width W11 of the first gate electrode 520 overlapping the second field insulating layer 106 may be smaller than a width W12 of the second gate electrode 620 overlapping the second field insulating layer 106.

The first gate electrode 520 may include a fifth lower conductive layer 525, a fifth etch-stop layer 524, a fifth work function control layer 521, a fifth insertion layer 522, and a fifth filling layer 523 formed sequentially on a fifth gate insulating layer 530.

The second gate electrode 620 may include a sixth lower conductive layer 625, a sixth etch-stop layer 624, a sixth work function control layer 621, a sixth insertion layer 622, and a sixth filling layer 623 formed sequentially on a sixth gate insulating layer 630.

The first contact surface MI1 between the first gate electrode 520 and the second gate electrode 620 may be located closer to the first active region 10 than to the second active region 20. Accordingly, the width W11 by which the fifth work function control layer 521 and the second field insulating layer 106 overlap each other may be smaller than the width W12 by which the sixth work function control layer 621 and the second field insulating layer 106 overlap each other.

In addition, the first active region 10, the first contact surface MI1, the first center line CL1, and the second active region 20 may be arranged sequentially in this order. Accordingly the fifth work function control layer 521 may not overlap the first center line CL1 of the second field insulating layer 106.

FIG. 42 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. 40 and 41 will be mainly described.

Referring to FIG. 42, in the semiconductor device according to some embodiments, a fifth work function control layer 521 may contact a fifth gate insulating layer 530, and a sixth work function control layer 621 may contact a sixth gate insulating layer 630.

FIG. 43 illustrates a plan view of a semiconductor device according to embodiments.

FIG. 44 illustrates a cross-sectional view taken along the line F-F of FIG. 43.

For ease of description, differences from the semiconductor device described above with reference to FIGS. 40 and 41 will be mainly described.

Referring to FIGS. 43 and 44, the semiconductor device according to some embodiments may include a fifth fin pattern 510 and a sixth fin pattern 610 adjacent to the fifth fin pattern 510.

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A first contact surface M11 of a fifth gate electrode structure 50, which intersects the fifth and sixth fin patterns 510 and 610, may be closer to the fifth fin pattern 510 than to the sixth fin pattern 610.

A width W11 by which a first gate electrode 520 formed on the fifth fin pattern 510 overlaps a second field insulating layer 106 may be smaller than a width W12 by which a second gate electrode 620 formed on the sixth fin pattern 610 overlaps the second field insulating layer 106.

The first contact surface M11 of the fifth gate electrode structure 50 may be defined at a boundary between the fifth work function control layer 521 and the sixth work function control layer 621. The width W11 by which the fifth work function control layer 521 formed along the profile of the fifth fin pattern 510 overlaps the second field insulating layer 106 may be smaller than the width W12 by which the sixth work function control layer 621 formed along the profile of the sixth fin pattern 610 overlaps the second field insulating layer 106.

FIG. 45 illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. 43 and 44 will be mainly described.

Referring to FIG. 45, in the semiconductor device according to some embodiments, a first gate electrode 520 may include a fifth work function control layer 521 formed on a fifth gate insulating layer 530 to be in contact with the fifth gate insulating layer 530, a fifth insertion layer 522 formed on the fifth work function control layer 521, and a fifth filling layer 523 formed on the fifth insertion layer 522.

A second gate electrode 620 may include a sixth work function control layer 621 formed on a sixth gate insulating layer 630 to be in contact with the sixth gate insulating layer 630, a sixth insertion layer 622 formed on the sixth work function control layer 621, and a sixth filling layer 623 formed on the sixth insertion layer 622.

FIG. 46 illustrates a plan view of a semiconductor device according to embodiments. FIG. 47 illustrates a cross-sectional view taken along the lines F-F and J-J of FIG. 46.

A first active region 10, a second active region 20, and a fifth gate electrode structure 50 shown in a fifth region V of FIGS. 46 and 47 may be substantially the same as those described above with reference to FIGS. 30 through 32B. Therefore, FIGS. 46 and 47 will be described, focusing on elements of a sixth region VI.

Referring to FIGS. 46 and 47, the semiconductor device according to some embodiments may include a substrate 100 including the fifth region V and the sixth region VI, the fifth gate electrode structure 50 formed in the fifth region V, and a sixth gate electrode structure 70 formed in the sixth region VI.

The substrate 100 may include the first active region 10, the second active region 20, a third active region 30, a fourth active region 40, a second field insulating layer 106, and a third field insulating layer 107.

The fifth region V and the sixth region VI may be regions in which elements having different functions are formed. For example, the fifth region V may be an SRAM region, and the sixth region VI may be a logic region or an I/O region.

The substrate 100 of the fifth region V may include the first active region 10, the second active region 20 and the second field insulating layer 106.

The substrate 100 of the sixth region VI may include the third active region 30, the fourth active region 40 and the third field insulating layer 107.

The third active region 30 and the fourth active region 40 may be defined by the third field insulating layer 107. The

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third active region 30 and the fourth active region 40 may be spatially separated but adjacent to each other.

Each of the third active region 30 and the fourth active region 40 may be shaped as a rectangle extending in an eleventh direction X6, as examples. The third active region 30 and the fourth active region 40 may be arranged side by side so as to be adjacent to each other in a long-side direction.

The third active region 30 may be a region in which a PMOS is formed, and the fourth active region 40 may be a region in which an NMOS is formed.

The third field insulating layer 107 may surround the third active region 30 and the fourth active region 40. In some implementations, the third field insulating layer 107 may be a portion located between the third active region 30 and the fourth active region 40.

The third field insulating layer 107 may be disposed between the third active region 30 and the fourth active region 40 and may be in direct contact with the third active region 30 and the fourth active region 40.

For example, the third field insulating layer 107 may directly contact the third active region 30 and the fourth active region 40 due to an absence of another active region between the third field insulating layer 107 and the third active region 30 and between the third field insulating layer 107 and the fourth active region 40.

A width of the third field insulating layer 107 located between the third active region 30 and the fourth active region 40 may be a second width W2. The third field insulating layer 107 may include a second center line CL2 located at a same distance from the third active region 30 and the fourth active region 40.

For example, a distance from the second center line CL2 to the third active region 30 may be equal to a distance from the second center line CL2 to the fourth active region 40 and may be half of the second width W2. The second center line CL2 of the third field insulating layer 107 may extend parallel to the third active region 30 and the fourth active region 40.

The fifth gate electrode structure 50 may be formed on the substrate 100 of the fifth region V.

The sixth gate electrode structure 70 may be formed on the substrate 100 of the sixth region VI. The sixth gate electrode structure 70 may cross the third active region 30, the fourth active region 40, and the third field insulating layer 107. The sixth gate electrode structure 70 may extend in a twelfth direction Y6.

The sixth gate electrode structure 70 may include a third gate electrode 720 and a fourth gate electrode 820. The third gate electrode 720 and the fourth gate electrode 820 may be in direct contact with each other.

The third gate electrode 720 may be a p-type metallic gate electrode and may be formed on the third active region 30 and the third field insulating layer 107. The fourth gate electrode 820 may be an n-type metal gate electrode and may be formed on the fourth active region 40 and the third field insulating layer 107.

A seventh transistor 20p may be formed in a region in which the third active region 30 and the sixth gate electrode structure 70 intersect each other. An eighth transistor 20n may be formed in a region in which the fourth active region 40 and the sixth gate electrode structure 70 intersect each other. The seventh transistor 20p may be a p-type transistor, and the eighth transistor 20n may be an n-type transistor.

The third gate electrode 720 may extend onto the third field insulating layer 107. The third gate electrode 720 may

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overlap not only the third active region 30 but also a portion of the third field insulating layer 107.

The fourth gate electrode 820 may be in direct contact with the third gate electrode 720. The fourth gate electrode 820 may overlap not only the fourth active region 40 but also the other portion of the third field insulating layer 107 that is not overlapped by the third gate electrode 720.

The sixth gate electrode structure 70 may include a second contact surface MI2 at which the third gate electrode 720 and the fourth gate electrode 820 contact each other. The second contact surface MI2 at which the third gate electrode 720 and the fourth gate electrode 820 contact each other may be located on the third field insulating layer 107.

In FIG. 46, a first contact surface MI1 between a first gate electrode 520 and a second gate electrode 620 may not coincide with a first center line CL1 of the second field insulating layer 106. In addition, the second contact surface MI2 between the third gate electrode 720 and the fourth gate electrode 820 may not coincide with the second center line CL2 of the third field insulating layer 107.

The first contact surface MI1 between the first gate electrode 520 and the second gate electrode 620 may be located closer to the second active region 20 than to the first active region 10. The second contact surface MI2 between the third gate electrode 720 and the fourth gate electrode 820 may be located closer to the fourth active region 40 than to the third active region 30.

The second gate electrode 620 may not overlap the first center line CL1 of the second field insulating layer 106. The fourth gate electrode 820 may not overlap the second center line CL2 of the third field insulating layer 107.

The third active region 30 may include a channel region of a p-type transistor. The fourth active region 40 may include a channel region of an n-type transistor. The second contact surface MI2 between the third gate electrode 720 and the fourth gate electrode 820 may be located closer to the fourth active region 40 than to the third active region 30. Accordingly, the second contact surface MI2 may be closer to the channel region of the n-type transistor than to the channel region of the p-type transistor.

A width of a portion of the third gate electrode 720 that extends on the third field insulating layer 107 may be a third overlapping width W21. The width of the third gate electrode 720 from the second contact surface MI2 to a boundary of the third active region 30 may be the third overlapping width W21.

A width of a portion of the fourth gate electrode 820 that extends on the third field insulating layer 107 may be a fourth overlapping width W22. The width of the fourth gate electrode 820 from the second contact surface MI2 to a boundary of the fourth active region 40 may be the fourth overlapping width W42.

In FIG. 46, the first contact surface MI1 between the first gate electrode 520 and the second gate electrode 620 may be located closer to the second active region 20 than to the first active region 10. Accordingly, the first overlapping width W11 may be greater than second overlapping width W12. The second contact surface MI2 between the third gate electrode 720 and the fourth gate electrode 820 may be located closer to the fourth active region 40 than to the third active region 30. Accordingly, the third overlapping width W21 may be greater than the fourth overlapping width W22.

The width W11 of the first gate electrode 520 overlapping the second field insulating layer 106 may be greater than the width W12 of the second gate electrode 620 overlapping the second field insulating layer 106. The width W21 of the third gate electrode 720 overlapping the third field insulating

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layer 107 may be greater than the width W22 of the fourth gate electrode 820 overlapping the third field insulating layer 107.

The structures of the third gate electrode 720 and the fourth gate electrode 820 will be described in detail below.

An interlayer insulating film 190 may include a fifth trench 50t formed in the fifth region V and a sixth trench 70t included in the sixth region VI.

The sixth trench 70t may cross the third active region 30, the third field insulating layer 107, and the fourth active region 40. For example, the sixth trench 70t may intersect the third active region 30 and the fourth active region 40. The sixth trench 70t may extend in the twelfth direction Y6.

Fifth gate spacers 55 formed in the fifth region V may define the fifth trench 50t. Sixth gate spacers 75 formed in the sixth region VI may define the sixth trench 70t. The sixth gate spacers 75 may be formed on the substrate 100. The sixth gate spacers 75 may be formed on sidewalls of the sixth gate electrode structure 70.

The sixth gate electrode structure 70 may extend in the twelfth direction Y6. Accordingly, the sixth gate electrode structure 70 may include long sides extending in the twelfth direction Y6 and short sides extending in the eleventh direction X6.

The sixth gate spacers 75 are shown as being formed on sidewalls including the long sides of the sixth gate electrode structure 70 and sidewalls including the short sides of the sixth gate electrode structure 70, as examples.

A description of the sixth gate spacers 75 may be substantially the same as the description of the fifth gate spacers 55 and thus will not be repeated.

A seventh gate insulating layer 730 and an eighth gate insulating layer 830 may be formed on the substrate 100. The seventh gate insulating layer 730 may be formed on the third active region 30 and the third field insulating layer 107. The eighth gate insulating layer 830 may be formed on the fourth active region 40 and the third field insulating layer 107.

The seventh gate insulating layer 730 and the eighth gate insulating layer 830 may extend along sidewalls and a bottom surface of the sixth trench 70t. The seventh and eighth gate insulating layers 730 and 830 extending along the bottom surface of the sixth trench 70t may traverse the third active region 30, the third field insulating layer 107 and the fourth active region 40.

The seventh gate insulating layer 730 and the eighth gate insulating layer 830 may be separated by the second contact surface MI2 of the sixth gate electrode structure 70. The fifth through eighth gate insulating layers 530 through 830 may be formed at a same level.

Each of the seventh gate insulating layer 730 and the eighth gate insulating layer 830 may include a high-k insulating layer.

The sixth gate electrode structure 70 may be formed on the seventh gate insulating layer 730 and the eighth gate insulating layer 830. The seventh gate insulating layer 730 and the eighth gate insulating layer 830 may be formed between the sixth gate electrode structure 70 and the substrate 100. The seventh gate insulating layer 730 and the eighth gate insulating layer 830 may be formed under the sixth gate electrode structure 70.

The sixth gate electrode structure 70 may fill the sixth trench 70t. An upper surface of the sixth gate electrode structure 70 may lie in the same plane with upper surfaces of the sixth gate spacers 75 and an upper surface of the interlayer insulating film 190.

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The third gate electrode 720 may include a seventh lower conductive layer 725, a seventh etch-stop layer 724, a seventh work function control layer 721, a seventh insertion layer 722, and a seventh filling layer 723 formed sequentially on the seventh gate insulating layer 730.

The fourth gate electrode 820 may include an eighth lower conductive layer 825, an eighth etch-stop layer 824, an eighth work function control layer 821, an eighth insertion layer 822, and an eighth filling layer 823 formed sequentially on the eighth gate insulating layer 830.

The seventh lower conductive layer 725 and the eighth lower conductive layer 825 may be formed on the seventh and eighth gate insulating layers 730 and 830. The seventh lower conductive layer 725 may contact the seventh gate insulating layer 730, and the eighth lower conductive layer 825 may contact the eighth gate insulating layer 830.

The seventh lower conductive layer 725 may be formed on the third active region 30 and the third field insulating layer 107. The eighth lower conductive layer 825 may be formed on the fourth active region 40 and the third field insulating layer 107.

The seventh lower conductive layer 725 and the eighth lower conductive layer 825 may extend along the sidewalls and bottom surface of the sixth trench 70t. The seventh lower conductive layer 725 may extend along the profile of the seventh gate insulating layer 730, and the eighth lower conductive layer 825 may extend along the profile of the eighth gate insulating layer 830.

The seventh lower conductive layer 725 and the eighth lower conductive layer 825 may be separated by the second contact surface MI1 of the sixth gate electrode structure 70.

The fifth through eighth lower conductive layers 525 through 825 may include the same material.

The seventh etch-stop layer 724 and the eighth etch-stop layer 824 may be formed on the seventh and eighth lower conductive layers 725 and 825. The seventh etch-stop layer 724 may be formed on the third active region 30 and the third field insulating layer 107. The eighth etch-stop layer 824 may be formed on the fourth active region 40 and the third field insulating layer 107.

The seventh etch-stop layer 724 and the eighth etch-stop layer 824 may extend along the sidewalls and bottom surface of the sixth trench 70t. The seventh etch-stop layer 724 may extend along the profile of the seventh lower conductive layer 725, and the eighth etch-stop layer 824 may extend along the profile of the eighth lower conductive layer 825.

The seventh etch-stop layer 724 and the eighth etch-stop layer 824 may be separated by the second contact surface MI2 of the sixth gate electrode structure 70. The fifth through eighth etch-stop layers 524 through 824 may be formed at the same level. The fifth through eighth etch-stop layers 524 through 824 may have, for example, substantially the same thickness.

The fifth through eighth etch-stop layers 524 through 824 may include the same material.

The seventh work function control layer 721 and the eighth work function control layer 821 may be formed on the seventh and eighth etch-stop layers 724 and 824. The seventh work function control layer 721 may contact the seventh etch-stop layer 724, and the eighth work function control layer 821 may contact the eighth etch-stop layer 824.

The seventh work function control layer 721 may be formed on the third active region 30 and the third field insulating layer 107. The eighth work function control layer 821 may be formed on the fourth active region 40 and the third field insulating layer 107. The seventh work function

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control layer 721 and the eighth work function control layer 821 may be in direct contact with each other.

The seventh work function control layer 721 and the eighth work function control layer 821 may extend along the sidewalls and bottom surface of the sixth trench 70t. The seventh work function control layer 721 may extend along the profile of the seventh gate insulating layer 730 and the seventh etch-stop layer 724, and the eighth work function control layer 821 may extend along the profile of the eighth gate insulating layer 830 and the eighth etch-stop layer 824.

The fifth through eighth work function control layers 521 through 821 may include the same material. For example, the fifth through eighth work function control layers 521 through 821 may be the same material layer.

A thickness t71 of the seventh work function control layer 721 may be different from a thickness t81 of the eighth work function control layer 821. For example, the thickness t71 of the seventh work function control layer 721 may be greater than the thickness t81 of the eighth work function control layer 821.

The thickness t71 of the seventh work function control layer 721 included in a p-type gate electrode may be greater than the thickness t81 of the eighth work function control layer 821 included in an n-type gate electrode. For example, the thickness t71 of the seventh work function control layer 721 may be a thickness on the third active region 30, and the thickness t81 of the eighth work function control layer 821 may be a thickness on the fourth active region 40.

In the semiconductor device according to some embodiments, a difference between the thickness t71 of the seventh work function control layer 721 and the thickness t81 of the eighth work function control layer 821 may be greater than or equal to a difference between a thickness t51 of the fifth work function control layer 521 and a thickness t61 of the sixth work function control layer 621.

The second contact surface MI2 between the third gate electrode 720 and the fourth gate electrode 820 may be defined between the seventh and eighth work function control layers 721 and 821 having different thicknesses. For example, if the sixth gate electrode structure 70 were to be cut along a normal to the substrate 100 based on a boundary between the seventh work function control layer 721 and the eighth work function control layer 821 extending on the third field insulating layer 107, the second contact surface MI2 between the third gate electrode 720 and the fourth gate electrode 820 would be formed.

In FIG. 47, the first contact surface MI1 between the first gate electrode 520 and the second gate electrode 620 may be located closer to the second active region 20 than to the first active region 10. Accordingly a width W11 by which the fifth work function control layer 521 and the second field insulating layer 106 overlap each other may be greater than a width W12 by which the sixth work function control layer 621 and the second field insulating layer 106 overlap each other.

In addition, the second contact surface MI2 between the third gate electrode 720 and the fourth gate electrode 820 may be located closer to the fourth active region 40 than to the third active region 30. Accordingly, the width W21 by which the seventh work function control layer 721 and the third field insulating layer 107 overlap each other may be greater than the width W22 by which the eighth work function control layer 821 and the third field insulating layer 107 overlap each other.

Further, the first active region 10, the first center line CL1, the first contact surface MI1 and the second active region 20 may be arranged sequentially in this order. Accordingly the

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sixth work function control layer **621** may not overlap the first center line **CL1** of the second field insulating layer **106**. In addition, the third active region **30**, the second center line **CL2**, the second contact surface **MI2** and the fourth active region **40** may be arranged sequentially in this order. Accordingly, the eighth work function control layer **821** may not overlap the second center line **CL2** of the third field insulating layer **107**.

The seventh insertion layer **722** and the eighth insertion layer **822** may be formed on the seventh and eighth work function control layers **721** and **821**. The seventh insertion layer **722** and the eighth insertion layer **822** may be in direct contact with each other.

The seventh insertion layer **722** may be formed on the third active region **30** and the third field insulating layer **107**. The eighth insertion layer **822** may be formed on the fourth active region **40** and the third field insulating layer **107**.

The seventh insertion layer **722** and the eighth insertion layer **822** may extend along the sidewalls and bottom surface of the sixth trench **70t**. The seventh insertion layer **722** and the eighth insertion layer **822** may extend along the profile of the seventh and eighth work function control layers **721** and **821** which are in direct contact with each other.

The seventh insertion layer **722** and the eighth insertion layer **822** may be separated by the second contact surface **MI2** of the sixth gate electrode structure **70**. The fifth through eighth insertion layers **522** through **822** may be formed at the same level.

A thickness **t52** of the fifth insertion layer **522** may be substantially equal to a thickness **t62** of the sixth insertion layer **622**. A thickness **t72** of the seventh insertion layer **722** may be substantially equal to a thickness **t82** of the eighth insertion layer **822**. The thickness **t52** of the fifth insertion layer **522** may be substantially equal to the thickness **t72** of the seventh insertion layer **722**.

The fifth through eighth insertion layers **522** through **822** may include the same material.

In the semiconductor device according to some embodiments, the fifth through eighth insertion layers **522** through **822** may be described as layers containing **TiAl**.

The seventh and eighth filling layers **723** and **823** may be formed on the seventh and eighth insertion layers **722** and **822**. The seventh filling layer **723** and the eighth filling layer **823** may be in direct contact with each other.

The seventh filling layer **723** may be formed on the third active region **30** and the third field insulating layer **107**. The eighth filling layer **823** may be formed on the fourth active region **40** and the third field insulating layer **107**.

The seventh filling layer **723** and the eighth filling layer **823** may be separated by the second contact surface **MI2** of the sixth gate electrode structure **70**. The fifth through eighth filling layers **523** through **823** may be formed at the same level.

The fifth through eighth filling layers **523** through **823** may include the same material.

The seventh insertion layer **722** and the seventh filling layer **723** on the seventh work function control layer **721** may be a seventh upper conductive layer **727**. The eighth insertion layer **822** and the eighth filling layer **823** on the eighth work function control layer **821** may be an eighth upper conductive layer **827**.

A thickness **h3** of the seventh upper conductive layer **727** may be a distance from the upper surface of the interlayer insulating film **190** to the seventh work function control layer **721** on the bottom surface of the sixth trench **70t**. A thickness **h4** of the eighth upper conductive layer **827** may

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be a distance from the upper surface of the interlayer insulating film **190** to the eighth work function control layer **821** on the bottom surface of the sixth trench **70t**.

The thickness **h3** of the seventh upper conductive layer **727** may be different from the thickness **h4** of the eighth upper conductive layer **827** on the third field insulating layer **107**. For example, the thickness **h3** of the seventh upper conductive layer **727** may be less than the thickness **h4** of the eighth upper conductive layer **827**.

FIG. **48** illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. **46** and **47** will be mainly described.

Referring to FIG. **48**, in the semiconductor device according to some embodiments, a fourth gate electrode **820** may include an eighth lower conductive layer **825**, an eighth etch-stop layer **824**, an eighth insertion layer **822**, and an eighth filling layer **823** formed sequentially on an eighth gate insulating layer **830**.

The eighth etch-stop layer **824** may contact the eighth insertion layer **822** disposed on the eighth etch-stop layer **824**.

In addition, in a third gate electrode **720**, a seventh work function control layer **721** may contact a seventh insertion layer **722** formed on the seventh work function control layer **721**.

The fourth gate electrode **820** may not include a **TiN**-containing work function control layer between the eighth etch-stop layer **824** and the eighth insertion layer **822**. Accordingly, a second contact surface **MI2** between the third gate electrode **720** and the fourth gate electrode **820** may be located at an end of the seventh work function control layer **721** extending onto a third field insulating layer **107**.

FIG. **49** illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. **46** and **47** will be mainly described.

Referring to FIG. **49**, in the semiconductor device according to some embodiments, a fifth work function control layer **521** may contact a fifth gate insulating layer **530**, and a sixth work function control layer **621** may contact a sixth gate insulating layer **630**.

In addition, a seventh work function control layer **721** may contact a seventh gate insulating layer **730**, and an eighth work function control layer **821** may contact an eighth gate insulating layer **830**.

FIG. **50** illustrates a plan view of a semiconductor device according to embodiments.

FIG. **51** illustrates a plan view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. **46** and **47** will be mainly described.

Referring to FIG. **50**, in the semiconductor device according to some embodiments, a width **W21** by which a third gate electrode **720** and a third field insulating layer **107** overlap each other may be substantially equal to a width **W22** by which a fourth gate electrode **820** and the third field insulating layer **107** overlap each other.

For example, a second contact surface **MI2** between the third gate electrode **720** and the fourth gate electrode **820** may coincide with a second center line **CL2** of the third field insulating layer **107**.

The second contact surface **MI2** between the third gate electrode **720** and the fourth gate electrode **820** may be separated from a third active region **30** and a fourth active region **40** by the same distance.

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Since the second contact surface MI2 between the third gate electrode 720 and the fourth gate electrode 820 is located at the same distance from the third active region 30 and the fourth active region 40, the second contact surface MI2 may be separated from a channel region of a p-type transistor and a channel region of an n-type transistor by the same distance.

Referring to FIG. 51, a width W21 by which a third gate electrode 720 and a third field insulating layer 107 overlap each other may be less than a width W22 by which a fourth gate electrode 820 and the third field insulating layer 107 overlap each other.

A second contact surface MI2 between the third gate electrode 720 and the fourth gate electrode 820 may be located closer to a third active region 30 than to a fourth active region 40.

When the second contact surface MI2 between the third gate electrode 720 and the fourth gate electrode 820 is located closer to the third active region 30 than to the fourth active region 40, the second contact surface MI2 may be closer to a channel region of a p-type transistor than to a channel region of an n-type transistor.

For example, a first contact surface MI1 between a first gate electrode 520 and a second gate electrode 620 in a fifth region V may be closer to a channel region of an n-type transistor than to a channel region of a p-type transistor. The second contact surface MI2 between the third gate electrode 720 and the fourth gate electrode 820 in a sixth region VI may be closer to the channel region of the p-type transistor than to the channel region of the n-type transistor.

FIG. 52 illustrates a plan view of a semiconductor device according to embodiments.

FIG. 53 illustrates a cross-sectional view taken along the lines F-F and J-J of FIG. 52.

A fifth fin pattern 510, a sixth fin pattern 610 and a fifth gate electrode structure 50 shown in a fifth region V of FIGS. 52 and 53 may be substantially the same as those described above with reference to FIGS. 30 through 32B, 37 and 38. Therefore, FIGS. 52 and 53 will be described, focusing on elements of a sixth region VI.

A sixth gate electrode structure 70 shown in the sixth region VI may be substantially the same as that described above with reference to FIGS. 46 and 47. Therefore, a redundant description of the sixth gate electrode structure 70 will not be repeated or will be given briefly.

Referring to FIGS. 52 and 53, the semiconductor device according to some embodiments may include the fifth fin pattern 510 and the sixth fin pattern 610, which are formed in the fifth region V and adjacent to each other, and a seventh fin pattern 710 and an eighth fin pattern 810, which are formed in the sixth region VI and adjacent to each other.

For example, the fifth fin pattern 510 and the sixth fin pattern 610 may be formed in an SRAM region, and the seventh fin pattern 710 and the eighth fin pattern 810 may be formed in a logic region or an I/O region.

The seventh fin pattern 710 and the eighth fin pattern 810 may protrude from a substrate 100. The seventh fin pattern 710 and the eighth fin pattern 810 may extend in an eleventh direction X6.

The seventh fin pattern 710 may be used as a channel region of a PMOS, and the eighth fin pattern 810 may be used as a channel region of an NMOS.

The seventh fin pattern 710 and the eighth fin pattern 810 may be part of the substrate 100 or may include an epitaxial layer grown from the substrate 100.

Each of the seventh and eighth fin patterns 710 and 810 may include an elemental semiconductor material such as

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silicon or germanium. In addition, each of the seventh fin pattern 710 and the eighth fin pattern 810 may include a compound semiconductor such as a group IV-IV compound semiconductor or a group compound semiconductor.

A third field insulating layer 107 may partially cover sidewalls of the seventh fin pattern 710 and sidewalls of the eighth fin pattern 810. Accordingly, the seventh fin pattern 710 and the eighth fin pattern 810 may protrude above an upper surface of the third field insulating layer 107 formed on the substrate 100.

A fin pattern protruding above the upper surface of the third field insulating layer 107 may not be formed between the seventh and eighth fin patterns 710 and 810.

The sixth gate electrode structure 70 may cross the seventh fin pattern 710, the eighth fin pattern 810, and the third field insulating layer 107. The sixth gate electrode structure 70 may extend in a twelfth direction Y6.

A second contact surface MI2 of the sixth gate electrode structure 70, which intersects the seventh and eighth fin patterns 710 and 810, may be closer to the eighth fin pattern 810 than to the seventh fin pattern 710.

A width W21 by which a third gate electrode 720 formed on the seventh fin pattern 710 overlaps the third field insulating layer 107 may be greater than a width W22 by which a fourth gate electrode 820 formed on the eighth fin pattern 810 overlaps the third field insulating layer 107.

A seventh gate insulating layer 730 may be formed on the third field insulating layer 107 and the seventh fin pattern 710. The seventh gate insulating layer 730 may be formed along the upper surface of the third field insulating layer 107 and the profile of the seventh fin pattern 710.

An eighth gate insulating layer 830 may be formed on the third field insulating layer 107 and the eighth fin pattern 810. The eighth gate insulating layer 830 may be formed along the upper surface of the third field insulating layer 107 and the profile of the eighth fin pattern 810.

The seventh and eighth gate insulating layers 730 and 830 extending along a bottom surface of a sixth trench 70t may be formed along the profile of the seventh fin pattern 710, the upper surface of the third field insulating layer 107, and the profile of the eighth fin pattern 810.

The sixth gate electrode structure 70 may be formed on the seventh and eighth gate insulating layers 730 and 830.

The third gate electrode 720 may be formed on the seventh gate insulating layer 730 and may intersect the seventh fin pattern 710. The fourth gate electrode 820 may be formed on the eighth gate insulating layer 830 and may intersect the eighth fin pattern 810.

A seventh lower conductive layer 725, a seventh etch-stop layer 724, a seventh work function control layer 721 and a seventh insertion layer 722 may be formed along the profile of the seventh gate insulating layer 730.

For example, the seventh lower conductive layer 725, the seventh etch-stop layer 724, the seventh work function control layer 721 and the seventh insertion layer 722 may extend along the profile of the seventh fin pattern 710 and the upper surface of the third field insulating layer 107.

An eighth lower conductive layer 825, an eighth etch-stop layer 824, an eighth work function control layer 821 and an eighth insertion layer 822 may be formed along the profile of the eighth gate insulating layer 830.

The eighth lower conductive layer 825, the eighth etch-stop layer 824, the eighth work function control layer 821 and the eighth insertion layer 822 may extend along the profile of the eighth fin pattern 810 and the upper surface of the third field insulating layer 107.

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The seventh and eighth work function control layers **721** and **821** extending along the bottom surface of the sixth trench **70t** may extend continuously along the profile of the seventh fin pattern **710**, the upper surface of the third field insulating layer **107**, and the profile of the eighth fin pattern **810**.

In FIG. **53**, a thickness **t71** of the seventh work function control layer **721** is shown as being greater than a thickness **t81** of the eighth work function control layer **821**. In some implementations, a thickness **t72** of the seventh insertion layer **722** may be substantially equal to a thickness **t82** of the eighth insertion layer **822**.

The second contact surface **M12** of the sixth gate electrode structure **70** is defined at a boundary between the seventh and eighth work function control layers **721** and **821**. The width **W21** by which the seventh work function control layer **721** formed along the profile of the seventh fin pattern **710** overlaps the third field insulating layer **107** may be greater than the width **W22** by which the eighth work function control layer **821** formed along the profile of the eighth fin pattern **810** overlaps the third field insulating layer **107**.

A thickness **h3** of a seventh upper conductive layer **727** from an upper surface of an interlayer insulating film **190** to the seventh work function control layer **721** may be less than a thickness **h4** of an eighth upper conductive layer **827** from the upper surface of the interlayer insulating film **190** to the eighth work function control layer **821**.

FIG. **54** illustrates a view of a semiconductor device according to embodiments. For ease of description, differences from the semiconductor device described above with reference to FIGS. **52** and **53** will be mainly described.

Referring to FIG. **54**, a third gate electrode **720** may include a seventh work function control layer **721** formed on a seventh gate insulating layer **730** to be in contact with the seventh gate insulating layer **730**, a seventh insertion layer **722** formed on the seventh work function control layer **721**, and a seventh filling layer **723** formed on the seventh insertion layer **722**.

A fourth gate electrode **820** may include an eighth work function control layer **821** formed on an eighth gate insulating layer **830** to be in contact with the eighth gate insulating layer **830**, an eighth insertion layer **822** formed on the eighth work function control layer **821**, and an eighth filling layer **823** formed on the eighth insertion layer **822**.

FIG. **55** illustrates a plan view of a semiconductor device according to embodiments.

FIG. **56** illustrates a cross-sectional view taken along the lines F-F and J-J of FIG. **55**. For ease of description, differences from the semiconductor device described above with reference to FIGS. **40** and **41** will be mainly described.

A first active region **10**, a second active region **20** and a fifth gate electrode structure **50** shown in a fifth region V of FIGS. **55** and **56** may be substantially the same as those described above with reference to FIGS. **40** and **41**. Therefore, FIGS. **55** and **56** will be described, focusing on elements of a sixth region VI.

Referring to FIGS. **55** and **56**, the semiconductor device according to some embodiments may include a substrate **100** including the fifth region V and the sixth region VI, the fifth gate electrode structure **50** formed in the fifth region V, and a sixth gate electrode structure **70** formed in the sixth region VI.

The substrate **100** may include the first active region **10**, the second active region **20**, a third active region **30**, a fourth active region **40**, a second field insulating layer **106**, and a third field insulating layer **107**.

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The fifth region V and the sixth region VI may be regions in which elements having different functions are formed. For example, the fifth region V may be an SRAM region, and the sixth region VI may be a logic region or an I/O region.

The substrate **100** of the fifth region V may include the first active region **10**, the second active region **20** and the second field insulating layer **106**.

The substrate **100** of the sixth region VI may include the third active region **30**, the fourth active region **40** and the third field insulating layer **107**.

The third active region **30** may be a region in which a PMOS is formed, and the fourth active region **40** may be a region in which an NMOS is formed.

The third field insulating layer **107** may be disposed between the third active region **30** and the fourth active region **40** and may be in direct contact with the third active region **30** and the fourth active region **40**. The third field insulating layer **107** may include a second center line **CL2** located at the same distance from the third active region **30** and the fourth active region **40**.

The sixth gate electrode structure **70** may be formed on the substrate **100** of the sixth region VI. The sixth gate electrode structure **70** may cross the third active region **30**, the fourth active region **40**, and the third field insulating layer **107**. The sixth gate electrode structure **70** may extend in a twelfth direction **Y6**.

A second contact surface **M12** between a third gate electrode **720** and a fourth gate electrode **820** may be located closer to the third active region **30** than to the fourth active region **40**.

The third active region **30**, the second contact surface **M12**, the second center line **CL2** and the fourth active region **40** may be arranged sequentially in this order. Accordingly, the third gate electrode **720** may not overlap the second center line **CL2** of the third field insulating layer **107**.

The second contact surface **M12** between the third gate electrode **720** and the fourth gate electrode **820** may be located closer to the third active region **30** than to the fourth active region **40**. Accordingly, a width **W21** of the third gate electrode **720** overlapping the third field insulating layer **107** may be smaller than a width **W22** of the fourth gate electrode **820** overlapping the third field insulating layer **107**.

The third gate electrode **720** may include a seventh lower conductive layer **725**, a seventh etch-stop layer **724**, a seventh work function control layer **721**, a seventh insertion layer **722**, and a seventh filling layer **723** formed sequentially on a seventh gate insulating layer **730**.

The fourth gate electrode **820** may include an eighth lower conductive layer **825**, an eighth etch-stop layer **824**, an eighth work function control layer **821**, an eighth insertion layer **822**, and an eighth filling layer **823** formed sequentially on an eighth gate insulating layer **830**.

A description of the third gate electrode **720** and the fourth gate electrode **820** may be substantially the same as that of the first gate electrode **520** and the second gate electrode **620** and thus, a description thereof will not be repeated.

The first through fourth active regions **10** through **40** may be multi-channel active patterns such as fin patterns.

In some implementations, the second contact surface **M12** between the third gate electrode **720** and the fourth gate electrode **820** may be located closer to the fourth active region **40** than to the third active region **30**. In some implementations, the second contact surface **M12** between the third gate electrode **720** and the fourth gate electrode **820** may be defined at a position separated from the third active region **30** and the fourth active region **40** by the same distance.

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In FIGS. 30 through 56, fifth and sixth work function control layers 521 and 621 having different thicknesses and the seventh and eighth work function control layers 721 and 821 having different thicknesses may be formed by patterning a TiN layer at least once.

The fifth region V including the fifth work function control layer 521 and the sixth work function control layer 621 and the sixth region V including the seventh work function control layer 721 and the eighth work function control layer 821 may be regions having different functions.

For example, the fifth region V may be an SRAM region, and the sixth region VI may be a logic region.

In this case, according to a process of forming a transistor included in the fifth region V, when a boundary between an n-type gate electrode and a p-type gate electrode is close to a channel region of an n-type transistor, threshold voltages of the n-type transistor and a p-type transistor, which share different gate electrode structures, can be improved.

In some implementations, when the boundary between the n-type gate electrode and the p-type gate electrode is close to a channel region of the p-type transistor, the threshold voltages of the n-type transistor and the p-type transistor, which share different gate electrode structures, may be improved.

According to a process of forming a transistor included in the sixth region VI in which a logic device is formed, when the boundary between the n-type gate electrode and the p-type gate electrode is close to the channel region of the n-type transistor, the threshold voltage of the n-type transistor and the p-type transistor, which share different gate electrode structures, may be improved.

In some implementations, when the boundary between the n-type gate electrode and the p-type gate electrode is close to the channel region of the p-type transistor, the threshold voltages of the n-type transistor and the p-type transistor, which share different gate electrode structures, may be improved.

In some implementations, when the boundary between the n-type gate electrode and the p-type gate electrode is half-way between the channel region of the p-type transistor and the channel region of the n-type transistor, the threshold voltages of the n-type transistor and the p-type transistor, which share different gate electrode structures, may be improved.

The boundary between the n-type gate electrode and the p-type gate electrode in regions having different functions may vary according to materials contained in the n-type gate electrode and the p-type gate electrode.

In some implementations, as the distance between the channel region of the p-type transistor and the channel region of the n-type transistor increases or decreases, the boundary between the n-type gate electrode and the p-type gate electrode in the regions having different functions may vary.

In addition, the boundary between the n-type gate electrode and the p-type gate electrode in the regions having different functions may vary according to a method of manufacturing the n-type gate electrode and the p-type gate electrode.

FIG. 57 illustrates a circuit diagram of a semiconductor device according to embodiments. FIG. 58 illustrates a layout view of the semiconductor device of FIG. 57. FIG. 59 illustrates a cross-sectional view taken along the line K-K of FIG. 58.

Referring to FIG. 57, the semiconductor device may include a pair of first and second inverters INV1 and INV2 connected in parallel between a power supply node Vcc and

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a ground node Vss and first and second pass transistors PS1 and PS2 respectively connected to output nodes of the first and second inverters INV1 and INV2. The first and second pass transistors PS1 and PS2 may be connected to a bit line BL and a complementary bit line/BL, respectively. Gates of the first and second pass transistors PS1 and PS2 may be connected to a word line WL.

The first inverter INV1 may include a first pull-up transistor PU1 and a first pull-down transistor PD1 connected in series. The second inverter INV2 may include a second pull-up transistor PU2 and a second pull-down transistor PD2 connected in series. The first and second pull-up transistors PU1 and PU2 may be p-channel field effect transistors (PFETs), and the first and second pull-down transistors PD1 and PD2 may be n-channel field effect transistors (NFETs).

In addition, an input node of the first inverter INV1 may be connected to the output node of the second inverter INV2, and an input node of the second inverter INV2 may be connected to the output node of the first inverter INV1 such that the first and second inverters INV1 and INV2 form a single latch circuit.

Referring to FIGS. 57 and 58, a fifth active region 910, a sixth active region 920, a seventh active region 930, an eighth active region 940, a ninth active region 950, and a tenth active region 960 may extend in a direction (e.g., a horizontal direction in FIG. 58) and may be spaced apart from each other. The sixth active region 920 and the ninth active region 950 may be shorter than the fifth active region 910, the seventh active region 930, the eighth active region 940, and the tenth active region 960.

In addition, a first gate line 971, a second gate line 972, a third gate line 973 and a fourth gate line 974 may extend in the other direction (e.g., a vertical direction in FIG. 58) to intersect the fifth through tenth active regions 910 through 960.

For example, the first gate line 971 may completely intersect the fifth through seventh active regions 910 through 930 and may partially overlap an end of the ninth active region 950. The third gate line 973 may completely intersect the eighth through tenth active regions 940 through 960 and may partially overlap an end of the sixth active region 920. The second gate line 972 may intersect the fifth active region 910 and the seventh active region 930. The fourth gate line 974 may intersect the eighth active region 940 and the tenth active region 960.

As illustrated in the drawings, the first pull-up transistor PU1 may be defined around a region where the first gate line 971 intersects the sixth active region 920. The first pull-down transistor PD1 may be defined around a region where the first gate line 971 intersects the fifth active region 910 and the seventh active region 930. The first pass transistor PS1 may be defined around a region where the second gate line 972 intersects the fifth active region 910 and the seventh active region 930.

The second pull-up transistor PU2 may be defined around a region where the third gate line 973 intersects the ninth active region 950. The second pull-down transistor PD2 may be defined around a region where the third gate line 973 intersects the eighth active region 940 and the tenth active region 960. The second pass transistor PS2 may be defined around a region where the fourth gate line 974 intersects the eighth active region 940 and the tenth active region 960.

Source/drain regions may be formed on both sides of each of the intersections between the first through fourth gate lines 971 through 974 and the fifth through tenth active

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regions 910 through 960, respectively. A plurality of contacts 970 may also be formed.

In addition, a first shared contact 981 may simultaneously connect the sixth active region 920, the third gate line 973, and a wiring layer 991. A second shared contact 982 may simultaneously connect the ninth active region 950, the first gate line 971, and a wiring layer 992.

In FIG. 58, the pull-down transistors PD1 and PD2 and the pass transistors PS1 and PS2, which may be n-type transistors, are shown as being defined around a plurality of active regions, as examples.

In FIG. 58, one of the fifth active region 910 and the seventh active region 930 may be omitted, and one of the eighth active region 940 and the tenth active region 960 may be omitted.

In FIGS. 58 and 59, a substrate 100 may include the fifth active region 910, the sixth active region 920, and a first field insulating layer 105 disposed between the fifth active region 910 and the sixth active region 920.

Seventh gate spacers 1640 may be formed on the substrate 100 and may define a seventh trench 1640t. The seventh trench 1640t may cross the fifth active region 910, the first field insulating layer 105, and the sixth active region 920.

The first gate line 971 may be formed on the substrate 100. The first gate line 971 may be formed in the seventh trench 1640t. Accordingly, the first gate line 971 may cross the fifth active region 910, the first field insulating layer 105, and the sixth active region 920.

The first gate line 971 may include a fifth gate electrode 1620 and a sixth gate electrode 1670 that are in direct contact with each other. The fifth gate electrode 1620 and the sixth gate electrode 1670 may meet each other on the first field insulating layer 105.

The first pull-down transistor PD1 may include a ninth gate insulating layer 1630 and the fifth gate electrode 1620. The first pull-up transistor PU1 may include a tenth gate insulating layer 1680 and the sixth gate electrode 1670.

The ninth gate insulating layer 1630 and the tenth gate insulating layer 1680 may be formed along sidewalls and a bottom surface of the seventh trench 1640t. The ninth gate insulating layer 1630 may extend along an upper surface of the fifth active region 910 and an upper surface of the first field insulating layer 105. The tenth gate insulating layer 1680 may extend along an upper surface of the sixth active region 920 and the upper surface of the first field insulating layer 105.

The ninth gate insulating layer 1630 may include a seventh interfacial layer 1631 and a fifth high-k insulating layer 1632. The tenth gate insulating layer 1680 may include an eighth interfacial layer 1681 and a sixth high-k insulating layer 1682.

The seventh interfacial layer 1631 and the eighth interfacial layer 1681 are shown as being formed only in the fifth active region 910 and the sixth active region 920 as examples. The fifth gate electrode 1620 may be formed on the ninth gate insulating layer 1630. The fifth gate electrode 1620 may include a ninth lower conductive layer 1625, a ninth etch-stop layer 1624, a ninth work function control layer 1621, a ninth insertion layer 1622, and a ninth filling layer 1623.

The ninth lower conductive layer 1625 may be formed on the ninth gate insulating layer 1630. The ninth lower conductive layer 1625 may contact the ninth gate insulating layer 1630. The ninth lower conductive layer 1625 may be formed along the profile of the ninth gate insulating layer 1630.

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The ninth etch-stop layer 1624 may be formed on the ninth lower conductive layer 1625. The ninth etch-stop layer 1624 may be formed along the profile of the ninth lower conductive layer 1625.

The ninth work function control layer 1621 may be formed on the ninth etch-stop layer 1624. The ninth work function control layer 1621 may contact the ninth etch-stop layer 1624. The ninth work function control layer 1621 may be formed along the profile of the ninth etch-stop layer 1624.

The ninth insertion layer 1622 may be formed on the ninth work function control layer 1621. The ninth insertion layer 1622 may contact the ninth function control layer 1621. The ninth insertion layer 1622 may be formed along the profile of the ninth work function control layer 1621.

The ninth filling layer 1623 may be formed on the ninth insertion layer 1622.

The sixth gate electrode 1670 may be formed on the tenth gate insulating layer 1680. The sixth gate electrode 1670 may include a tenth lower conductive layer 1675, a tenth etch-stop layer 1674, a tenth work function control layer 1671, a tenth insertion layer 1672, and a tenth filling layer 1673.

The tenth lower conductive layer 1675 may be formed on the tenth gate insulating layer 1680. The tenth lower conductive layer 1675 may contact the tenth gate insulating layer 1680. The tenth lower conductive layer 1675 may be formed along the profile of the tenth gate insulating layer 1680.

The tenth etch-stop layer 1674 may be formed on the tenth lower conductive layer 1675. The tenth etch-stop layer 1674 may be formed along the profile of the tenth lower conductive layer 1675.

The tenth work function control layer 1671 may be formed on the tenth etch-stop layer 1674. The tenth work function control layer 1671 may contact the tenth etch-stop layer 1674. The tenth work function control layer 1671 may be formed along the profile of the tenth etch-stop layer 1674.

The tenth insertion layer 1672 may be formed on the tenth work function control layer 1671. The tenth insertion layer 1672 may contact the tenth work function control layer 1671. The tenth insertion layer 1672 may be formed along the profile of the tenth work function control layer 1671.

The tenth filling layer 1673 may be formed on the tenth insertion layer 1672.

A work function control layer, an insertion layer, and a filling layer formed on each etch-stop layer may be an upper gate electrode.

The ninth lower conductive layer 1625 and the tenth lower conductive layer 1675 may include, for example, TiN.

The ninth etch-stop layer 1624 and the tenth etch-stop layer 1674 may include the same material. The ninth etch-stop layer 1624 and the tenth etch-stop layer 1674 may include, for example, TaN.

The ninth etch-stop layer 1624 and the tenth etch-stop layer 1674 may be in direct contact with each other on the first field insulating layer 105.

The ninth work function control layer 1621 and the tenth work function control layer 1671 may include the same material. The ninth work function control layer 1621 and the tenth work function control layer 1671 may include, for example, TiN.

The ninth work function control layer 1621 and the tenth work function control layer 1671 may be in direct contact with each other on the first field insulating layer 105.

The ninth insertion layer 1622 and the tenth insertion layer 1672 may include the same material. The ninth inser-

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tion layer **1622** and the tenth insertion layer **1672** may include, for example, one of Ti, TiAl, TiAlN, TiAlC, and TiAlCN.

The ninth insertion layer **1622** and the tenth insertion layer **1672** may be in direct contact with each other on the first field insulating layer **105**.

The ninth filling layer **1623** and the tenth filling layer **1673** may include the same material. The ninth filling layer **1623** and the tenth filling layer **1673** may include at least one of W, Al, Co, Cu, Ru, Ni, Pt, Ni—Pt, and TiN.

The ninth filling layer **1623** and the tenth filling layer **1673** may be in direct contact with each other on the first field insulating layer **105**.

In FIG. **59**, a thickness **t91** of the ninth lower conductive layer **1625** may be less than a thickness **t93** of the tenth lower conductive layer **1675**. In addition, a thickness **t92** of the ninth work function control layer **1621** may be less than a thickness **t94** of the tenth work function control layer **1671**.

For example, the tenth lower conductive layer **1675** and the tenth work function control layer **1671** included in a PMOS may be thicker than the ninth lower conductive layer **1625** and the ninth work function control layer **1621** included in an NMOS.

In FIG. **59**, the fifth active region **910** and the sixth active region **920** are illustrated as substrate **100** having a flat upper surface. In some implementations, the fifth active region **910** and the sixth active region **920** may be also be multi-channel active patterns (such as fin patterns) including a plurality of channel regions.

A method of fabricating a semiconductor device according to embodiments will be described with reference to FIGS. **1** and **60** through **70**.

FIGS. **60** through **70** illustrate views depicting stages of a method of fabricating a semiconductor device according to embodiments.

Referring to FIG. **60**, a first dummy gate insulating layer **130p** and a first dummy gate electrode **120p** may be sequentially stacked on a substrate **100** of a first region I. A second dummy gate insulating layer **230p** and a second dummy gate electrode **220p** may be sequentially stacked on the substrate **100** of a second region II.

In addition, a third dummy gate insulating layer **330p** and a third dummy gate electrode **320p** may be sequentially stacked on the substrate **100** of a third region III. A fourth dummy gate insulating layer **430p** and a fourth dummy gate electrode **420p** may be sequentially stacked on the substrate **100** of a fourth region IV.

The first through fourth dummy gate insulating layers **130p** through **430p** may include silicon oxide, silicon oxynitride, or a combination of the same. Each of the first through fourth dummy gate electrodes **120p** through **420p** may be silicon. For example, each may include one of polycrystalline silicon (poly Si), amorphous silicon (a-Si), and a combination of the same. The first through fourth dummy gate electrodes **120p** through **420p** may not be doped with impurities or may be doped with impurities.

First through fourth gate spacers **140** through **440** may be formed on sidewalls of the first through fourth dummy gate electrodes **120p** through **420p**.

After the formation of the first through fourth gate spacers **140** through **440**, first through fourth source/drain regions **150** through **450** may be formed adjacent to the first through fourth dummy gate electrodes **120p** through **420p**, respectively.

An interlayer insulating film **190** may be formed on the substrate **100** to cover the first through fourth dummy gate electrodes **120p** through **420p**.

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The interlayer insulating film **190** may be planarized to expose upper surfaces of the first through fourth dummy gate electrodes **120p** through **420p**.

Referring to FIG. **61**, the first through fourth dummy gate electrodes **120p** through **420p** may be removed.

After the removal of the first through fourth dummy gate electrodes **120p** through **420p**, the first through fourth dummy gate insulating layers **130p** through **430p** may be removed. As a result, first through fourth trenches **140t** through **440t** may be formed.

The first through fourth dummy gate electrodes **120p** through **420p** may be removed using a wet-etching process or a dry-etching process. Taking wet etching as an example, the first through fourth dummy gate electrodes **120p** through **420p** may be substantially removed through exposure to an aqueous solution containing a hydroxide source at a sufficient temperature for a sufficient time. The hydroxide source may include, for example, ammonium hydroxide or tetra alkyl ammonium hydroxide, for example, tetra methyl ammonium hydroxide (TMAH).

The first through fourth dummy gate insulating layers **130p** through **430p** may be removed by wet etching, dry etching, or a combination of the same. An etchant or an etching gas may be varied according to the material of the first through fourth dummy gate insulating layers **130p** through **430p**.

Referring to FIG. **62**, first through fourth interfacial layers **131** through **431** may be formed on the substrate **100**.

The first through fourth interfacial layers **131** through **431** may be formed on bottom surfaces of the first through fourth trenches **140t** through **440t**.

First through fourth pre-high-k insulating layers **132p**, **232p**, **332p**, **432p** may be formed on the first through fourth interfacial layers **131** through **431**.

For example, the first pre-high-k insulating layer **132p** may extend along sidewalls and the bottom surface of the first trench **140t** and an upper surface of the interlayer insulating film **190**.

Referring to FIG. **63**, first through fourth pre-TiN layers **125a**, **225a**, **325a**, **425a** may be formed on the first through fourth pre-high-k insulating layers **132p** through **432p**.

For example, the first pre-TiN layer **125a** may extend along the sidewalls and bottom surface of the first trench **140t** and the upper surface of the interlayer insulating film **190**. The first pre-TiN layer **125a** may be formed along the profile of the first pre-high-k insulating layer **132p**.

Referring to FIG. **64**, the second pre-TiN layer **225a** of the second region II may be removed to expose the second pre-high-k insulating layer **232p**.

The fourth pre-TiN layer **425a** of the fourth region IV may be removed to expose the fourth pre-high-k insulating layer **432p**.

Referring to FIG. **65**, an additional TiN layer may be formed along the sidewalls and bottom surfaces of the first through fourth trenches **140t** through **440t**.

As a result, a first pre-lower conductive layer **125p** may be formed in the first region I along the sidewalls and bottom surface of the first trench **140t** and the upper surface of the interlayer insulating film **190**.

Second through fourth pre-lower conductive layers **225p** through **425p** may also be formed in the second through fourth region II through IV.

The first pre-lower conductive layer **125p** may be made up of the first pre-TiN layer **125a** and the additional TiN layer formed on the first pre-TiN layer **125a**. Accordingly, the first pre-lower conductive layer **125p** may be thicker than the second pre-lower conductive layer **225p**.

Similarly, the third pre-lower conductive layer **325p** may be thicker than the fourth pre-lower conductive layer **425p**.

Referring to FIG. 66, a capping layer **128** may be formed on each of the first through fourth pre-lower conductive layers **125p** through **425p**. After the formation of the capping layer **128**, a heat treatment may be performed.

The capping layer **128** may include, for example, amorphous silicon, polysilicon, or a combination of the same. During the heat treatment, the capping layer **128** may prevent thicknesses of the first through fourth interfacial layers **131** through **431** from increasing.

After the heat treatment, the capping layer **128** may be removed.

Referring to FIG. 67, first through fourth pre-etch-stop layers **124p**, **224p**, **324p**, **424p** may be formed on the first through fourth pre-lower conductive layers **125p** through **425p**.

For example, the first pre-etch-stop layer **124p** may extend along the sidewalls and bottom surface of the first trench **140t** and the upper surface of the interlayer insulating film **190**. The first pre-etch-stop layer **124p** may be formed along the profile of the first pre-lower conductive layer **125p**.

Next, a pre-conductive layer **51p** may be formed on the first through fourth pre-etch-stop layers **124p** through **424p**.

The pre-conductive layer **51p** may be formed along the profile of each of the first through fourth pre-etch-stop layers **124p** through **424p**.

The pre-conductive layer **51p** may include, for example, a TiN layer.

Referring to FIG. 68, the pre-conductive layer **51p** on the third and fourth pre-etch-stop layers **324p** and **424p** may be removed to expose the third and fourth pre-etch-stop layers **324p** and **424p**.

The removal of the pre-conductive layer **51p** disposed on the third and fourth pre-etch-stop layers **324p** and **424p** may result in the formation of a first pre-upper TiN layer **121a** on the first pre-etch-stop layer **124p** and the formation of a second pre-upper TiN layer **221a** on the second pre-etch-stop layer **224p**.

Referring to FIG. 69, an additional TiN layer may be formed along the sidewalls and bottom surfaces of the first through fourth trenches **140t** through **440t**.

As a result, a first pre-work function control layer **121p** may be formed along the sidewalls and bottom surface of the first trench **140t** and the upper surface of the interlayer insulating film **190** in the first region I.

Second through fourth pre-work function control layers **221p** through **421p** may also be formed in the second through fourth regions II through IV.

The first pre-work function control layer **121p** may be made of the first pre-upper TiN layer **121a** and the additional TiN layer formed on the first pre-upper TiN layer **121a**. Accordingly, the first pre-work function control layer **121p** may be thicker than the third pre-work function control layer **321p** and the fourth pre-work function control layer **421p**.

Similarly, the second pre-work function control layer **221p** may be thicker than the third pre-work function control layer **321p** and the fourth pre-work function control layer **421p**.

Referring to FIG. 70, first through fourth pre-insertion layers **122p**, **222p**, **322p**, **422p** may be formed on the first through fourth pre-work function control layers **121p** through **421p**.

For example, the first pre-insertion layer **122p** may extend along the sidewalls and bottom surface of the first trench **140t** and the upper surface of the interlayer insulating film

190. The first pre-insertion layer **122p** may be formed along the profile of the first pre-work function control layer **121p**.

First through fourth pre-filling layers **123p**, **223p**, **323p**, **423p** may be formed on the first through fourth pre-insertion layers **122p** through **422p** to fill the first through fourth trenches **140t** through **440t**.

Referring to FIG. 1, the first through fourth pre-filling layers **123p** through **423p**, the first through fourth pre-insertion layers **122p** through **422p**, the first through fourth pre-work function control layers **121p** through **421p**, the first through fourth pre-etch-stop layers **124p** through **424p**, and the first through fourth pre-lower conductive layers **125p** through **425p** formed on the upper surface of the interlayer insulating film **190** may be removed to form first through fourth gate electrode structures **120** through **420** and first through fourth gate insulating layers **130** through **430**.

FIG. 71 illustrates a view depicting a stage of a method of fabricating a semiconductor device according to embodiments.

For reference, FIG. 71 may be a process performed after the process illustrated in FIG. 67.

Referring to FIG. 71, a pre-conductive layer **51p** formed on sidewalls of first through fourth trenches **140t** through **440t** may be partially removed to form a chamfered pre-conductive layer **51pc** on each of first through fourth pre-etch-stop layers **124p** through **424p**.

In some implementations, while the chamfered pre-conductive layer **51pc** is being formed, the pre-conductive layer **51p** formed on an upper surface of an interlayer insulating film **190** may also be removed.

FIG. 72 illustrates a view depicting a stage of a method of fabricating a semiconductor device according to embodiments.

For reference, FIG. 72 may be a process performed after the process illustrated in FIG. 69.

Referring to FIG. 72, first through fourth pre-work function control layers **121p** through **421p** formed on sidewalls of first through fourth trenches **140t** through **440t** may be partially removed.

As a result, chamfered first through fourth pre-work function control layers **121pc**, **221pc**, **321pc**, **421pc** may be formed on first through fourth pre-etch-stop layers **124p** through **424p**, respectively.

In some implementations, while the chamfered first through fourth pre-work function control layers **121pc** through **421pc** are being formed, the first through fourth pre-work function control layers **121p** through **421p** formed on an upper surface of an interlayer insulating film **190** may be removed.

FIGS. 73 through 83 are views illustrating stages of a method of fabricating a semiconductor device according to embodiments.

For reference, FIG. 73 may be a process performed after the process illustrated in FIG. 62.

Referring to FIG. 73, a first conductive layer **51** may be formed on first through fourth pre-high-k insulating layers **132p** through **432p**.

For example, the first conductive layer **51** may extend along sidewalls and bottom surfaces of first through fourth trenches **140t** through **440t** and an upper surface of an interlayer insulating film **190**.

The first conductive layer **51** may contact the first through fourth pre-high-k insulating layers **132p** through **432p**. The first conductive layer **51** may include, for example, TiN.

Referring to FIG. 74, a first sacrificial pattern **61** may be formed on the first conductive layer **51** to partially fill each of the first through fourth trenches **140t** through **440t**.

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For example, a first sacrificial layer may be formed on the first conductive layer **51** to fill the first through fourth trenches **140t** through **440t**. The first sacrificial layer may also be formed on the upper surface of the interlayer insulating film **190**. The first sacrificial layer formed on the upper surface of the interlayer insulating film **190** and part of the first sacrificial layer filling the first through fourth trenches **140t** through **440t** may be removed to form the first sacrificial pattern **61**.

The first conductive layer **51** formed on the sidewalls of the first through fourth trenches **140t** through **440t** may be partially exposed by the first sacrificial pattern **61**.

Next, the first conductive layer **51** formed on the sidewalls of the first through fourth trenches **140t** through **440t** may be partially removed using the first sacrificial pattern **61** as a mask.

As a result, a chamfered first conductive layer **51c** may be formed in each of the first through fourth trenches **140t** through **440t**.

Then, the first sacrificial pattern **61** formed in the first through fourth trenches **140t** through **440t** may be removed.

Referring to FIG. **75**, a first mask pattern **71** may be formed on the chamfered first conductive layer **51c**.

The first mask pattern **71** may be formed on a substrate **100** of first through third region I through III and may not be formed on the substrate **100** of a fourth region IV.

The first mask pattern **71** may cover the first through third pre-high-k insulating layers **132p** through **332p** and the chamfered first conductive layer **51c** formed in the first through third trenches **140t** through **340t**. The fourth pre-high-k insulating layer **432p** and the chamfered first conductive layer **51c** formed in the fourth trench **440t** may be exposed by the first mask pattern **71**.

The chamfered first conductive layer **51c** formed in the fourth trench **440t** may be removed using the first mask pattern **71** as a mask.

Then, the first mask pattern **71** may be removed.

In some implementations, the first conductive layer **51** formed in the fourth region IV may be removed using the first mask pattern **71** as a mask without performing a chamfering process on the first conductive layer **51**.

Referring to FIG. **76**, a second conductive layer **52** may be formed on the first through fourth pre-high-k insulating layers **132p** through **432p** and the chamfered first conductive layer **51c**.

For example, the second conductive layer **52** may extend along the sidewalls and bottom surfaces of the first through fourth trenches **140t** through **440t** and the upper surface of the interlayer insulating film **190**.

The second conductive layer **52** may contact the first through fourth pre-high-k insulating layers **132p** through **432p** and the chamfered first conductive layer **51c**. The second conductive layer **52** may include, for example, TiN.

Referring to FIG. **77**, a second mask pattern **72** may be formed on the second conductive layer **52**.

The second mask pattern **72** may be formed on the substrate **100** of the first region I, the third region III and the fourth region IV and may not be formed on the substrate **100** of the second region II.

The second mask pattern **72** may cover the second conductive layer **52** formed in the first region I, the third region III and the fourth region IV. The second conductive layer **52** formed in the second region II may be exposed by the second mask pattern **72**.

Next, the chamfered first conductive layer **51c** formed in the second trench **240t** and the second conductive layer **52**

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extending along the sidewalls and bottom surface of the second trench **240t** may be removed using the second mask pattern **72** as a mask.

Then, the second mask pattern **72** may be removed.

Referring to FIG. **78**, a third conductive layer **53** may be formed to extend along the profile of the second conductive layer **52** formed in the first region I, the third region III and the fourth region IV and along the sidewalls and bottom surface of the second trench **240t**.

The third conductive layer **53** may contact the second conductive layer **52** formed in the first region I, the third region III and the fourth region IV and contact the second pre-high-k insulating layer **232p**. The third conductive layer **53** may include, for example, TiN.

Referring to FIG. **79**, a second sacrificial pattern **62** may be formed on the third conductive layer **53** to partially fill the first through fourth trenches **140t** through **440t**.

For example, a second sacrificial layer may be formed on the third conductive layer **53** to fill the first through fourth trenches **140t** through **440t**. The second sacrificial layer may also be formed on the upper surface of the interlayer insulating film **190**. The second sacrificial layer formed on the upper surface of the interlayer insulating film **190** and part of the second sacrificial layer filling the first through fourth trenches **140t** through **440t** may be removed to form the second sacrificial pattern **62**.

The third conductive layer **53** formed on the sidewalls of the first through fourth trenches **140t** through **440t** may be partially exposed by the second sacrificial pattern **62**.

As shown in FIG. **79**, in some implementations, an upper surface of the second sacrificial pattern **62** formed in the first region I and the third region III may be higher than an uppermost portion of the chamfered first conductive layer **51c** formed in the first trench **140t** and the third trench **340t**.

Next, the second conductive layer **52** and the third conductive layer **53** formed on the sidewalls of the first through fourth trenches **140t** through **440t** may be partially removed using the second sacrificial pattern **62** as a mask.

As a result, a chamfered second conductive layer **52c** and a chamfered third conductive layer **53c** may be formed in the first trench **140t**, the third trench **340t** and the fourth trench **440t**. In addition, the chamfered third conductive layer **53c** may be formed in the second trench **240t**.

The chamfered first through third conductive layers **51c** through **53c** sequentially formed on the substrate **100** may be disposed in the first trench **140t** and the third trench **340t**. The chamfered second and third conductive layers **52c** and **53c** sequentially formed on the substrate **100** may be disposed in the fourth trench **440t**.

Next, the second sacrificial pattern **62** formed in the first through fourth trenches **140t** through **440t** may be removed.

Referring to FIG. **80**, a third mask pattern **73** may be formed on the chamfered third conductive layer **53c**.

The third mask pattern **73** may be formed on the substrate **100** of the second through fourth regions II through IV and may not be formed on the substrate **100** of the first region I.

The third mask pattern **73** may cover the second through fourth pre-high-k insulating layers **232p** through **432p** and the chamfered third conductive layer **53c** formed in the second through fourth trenches **240t** through **440t**. The first pre-high-k insulating layer **132** and the chamfered first through third conductive layers **51c** through **53c** formed in the first trench **140t** may be exposed by the third mask pattern **73**.

Next, the chamfered first through third conductive layers **51c** through **53c** formed in the first trench **140t** may be removed using the third mask pattern **73** as a mask.

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Then, the third mask pattern 73 may be removed.

In some implementations, the chamfered first conductive layer 51c and the second and third conductive layers 52 and 53 formed in the first region may be removed using the third mask pattern 73 as a mask without performing a chamfering process on the second conductive layer 52 and the third conductive layer 53.

Referring to FIG. 81, a fourth conductive layer 54 may be formed on the first through fourth pre-high-k insulating layers 132p through 432p and the chamfered first through third conductive layers 51c through 53c.

For example, the fourth conductive layer 54 may extend along on the exposed sidewalls of the first through fourth trenches 140t through 440t, the profile of the chamfered third conductive layer 53c, and the upper surface of the interlayer insulating film 190.

The fourth conductive layer 54 may contact the first through fourth pre-high-k insulating layers 132p through 432p and the chamfered second and third conductive layers 52c and 53c. The fourth conductive layer 54 may include, for example, TiN.

Referring to FIG. 82, a third sacrificial pattern 63 may be formed on the fourth conductive layer 54 to partially fill the first through fourth trenches 140t through 440t.

For example, a third sacrificial layer may be formed on the fourth conductive layer 54 to fill the first through fourth trenches 140t through 440t. The third sacrificial layer may also be formed on the upper surface of the interlayer insulating film 190. The third sacrificial layer formed on the upper surface of the interlayer insulating film 190 and part of the third sacrificial layer filling the first through fourth trenches 140t through 440t may be removed to form the third sacrificial pattern 63.

The fourth conductive layer 54 formed on the sidewalls of the first through fourth trenches 140t through 440t may be partially exposed by the third sacrificial pattern 63.

As illustrated in FIG. 82, in some implementations, an upper surface of the third sacrificial pattern 63 formed in the second through fourth regions II through IV may be higher than an uppermost portion of the chamfered second conductive layer 52c and/or an uppermost portion of the chamfered third conductive layer 53c formed in the second through fourth trenches 240t through 440t.

Next, the fourth conductive layer 54 formed on the sidewalls of the first through fourth trenches 140t through 440t may be partially removed using the third sacrificial pattern 63 as a mask.

Accordingly, a chamfered fourth conductive layer 54c may be formed in the first through fourth trenches 140t through 440t.

Consequently, a first pre-work function control layer 121p including the chamfered fourth conductive layer 54c may be formed in the first trench 140t. A second pre-work function control layer 221p including the chamfered third and fourth conductive layers 53c and 54c sequentially formed on the substrate 100 may be formed in the second trench 240t. A third pre-work function control layer 321p including the chamfered first through fourth conductive layers 51c through 54c sequentially formed on the substrate 100 may be formed in the third trench 340t. A fourth pre-work function control layer 421p including the chamfered second through fourth conductive layers 52c through 54c sequentially formed on the substrate 100 may be formed in the fourth trench 440t.

Referring to FIG. 83, first through fourth pre-insertion layers 122p through 422p may be formed on the first through fourth pre-work function control layers 121p through 421p.

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For example, the first pre-insertion layer 122p may extend along the sidewalls and bottom surface of the first trench 140t and the upper surface of the interlayer insulating film 190. The first pre-insertion layer 122p may be formed along the profile of the first pre-work function control layer 121p. The second through fourth pre-insertion layers 222p through 422p may be formed similarly to the first pre-insertion layer 122p.

Next, first through fourth pre-filling layers 123p through 423p may be formed on the first through fourth pre-insertion layers 122p through 422p to fill the first through fourth trenches 140t through 440t.

In some implementations, the first through fourth pre-insertion layers 122p through 422p and the first through fourth pre-filling layers 123p through 423p may be formed without performing a chamfering process on the fourth conductive layer 54.

Referring to FIG. 19B, the first through fourth pre-filling layers 123p through 423p, the first through fourth pre-insertion layers 122p through 422p and the first through fourth pre-high-k insulating layers 132p through 432p formed on the upper surface of the interlayer insulating film 190 may be removed to form first through fourth gate electrodes 120 through 420 and first through fourth gate insulating layers 130 through 430.

In the method of fabricating a semiconductor device according to some embodiments, at least one chamfering process may be performed during the formation of the first through fourth pre-work function control layers 121p through 421p.

By way of summation and review, Aspects provide a semiconductor device including a plurality of transistors having different threshold voltages.

Aspects also provide a semiconductor device including a plurality of transistors having different threshold voltages and capable of improving gap-fill characteristics of a metallic gate electrode.

Aspects also provide a semiconductor device capable of improving threshold voltages of transistors.

Aspects also provide a method of fabricating a semiconductor device which can vary threshold voltages of a plurality of transistors.

Aspects also provide a method of fabricating a semiconductor device which can vary threshold voltages of a plurality of transistors while improving gap-fill characteristics of a metallic gate electrode.

Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope thereof the present invention as set forth in the following claims.

What is claimed is:

1. A semiconductor device, comprising:

a substrate including a first active region, a second active region, and a first field insulating layer between the first active region and the second active regions, the first field insulating layer directly contacting the first active region and the second active regions; and

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a first gate structure on the substrate, the first gate structure crossing the first active region, the second active region, and the first field insulating layer, wherein the first gate structure includes a first p-type gate electrode and a first n-type gate electrode directly contacting each other, wherein the first p-type gate electrode includes a first work function layer on the first active region and the first field insulating layer, and a first upper conductive layer on the first work function layer, wherein the first n-type gate electrode includes a second work function layer on the second active region and the first field insulating layer, and a second upper conductive layer on the second work function layer, wherein the first work function layer and the second work function layer are in direct contact with each other and are formed of the same material layer, wherein a thickness of the first work function layer is different from a thickness of the second work function layer, and wherein an overlapping width of the first p-type gate electrode and the first field insulating layer is different from an overlapping width of the first n-type gate electrode and the first field insulating layer.

2. The semiconductor device as claimed in claim 1, wherein the thickness of the first work function layer is greater than the thickness of the second work function layer.

3. The semiconductor device as claimed in claim 1, wherein the overlapping width of the first p-type gate electrode and the first field insulating layer is greater than the overlapping width of the first n-type gate electrode and the first field insulating layer.

4. The semiconductor device as claimed in claim 1, wherein the overlapping width of the first p-type gate electrode and the first field insulating layer is smaller than the overlapping width of the first n-type gate electrode and the first field insulating layer.

5. The semiconductor device as claimed in claim 1, wherein each of the first active region and the second active region is included in a static random-access memory (SRAM) region.

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6. The semiconductor device as claimed in claim 1, wherein:

the substrate includes a static random-access memory (SRAM) region and a logic region,

the first active region and the second active region are in the SRAM region,

a portion of the substrate including the logic region has a third active region, a fourth active region, and a second field insulating layer between and directly contacting the third and fourth active regions,

the semiconductor device further includes a second gate structure on the substrate, the second gate structure crossing the third active region, the fourth active region, and the second field insulating layer,

the first gate structure includes a second p-type gate electrode and a second n-type gate electrode directly contacting each other,

the second p-type gate electrode is on the third active region and the second field insulating layer, and

the second n-type gate electrode is on the fourth active region and the second field insulating layer.

7. The semiconductor device as claimed in claim 6, wherein a contact surface of the second p-type gate electrode and the second n-type gate electrode is closer to the fourth active region than to the third active region.

8. The semiconductor device as claimed in claim 6, wherein an overlapping width of the second p-type gate electrode and the second field insulating layer is substantially the same as an overlapping width of the second n-type gate electrode and the second field insulating layer.

9. The semiconductor device as claimed in claim 6, wherein a contact surface of the second p-type gate electrode and the second n-type gate electrode is closer to the third active region than to the fourth active region.

10. The semiconductor device as claimed in claim 1, wherein each of the first work function layer and the second work function layer is a TiN layer.

11. The semiconductor device as claimed in claim 1, wherein each of the first active region and the second active region includes a corresponding fin type pattern.

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